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ABCPMX SOFTWARE V1.10 COMPOSITE BEAMS DESIGN

TECHNICAL REPORT

« Etablissement certifié qualité ISO 9001, le CTICM assure un suivi de
chaque étude dans le plus strict respect de ses procédures qualité »

Date :	19/05/2025	Auteur :	POM	Nombre total de pages : (y compris celle-ci)	170
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	Project: ABCPMX-II Software - DAL-22-003		
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Révision	Date	Auteur	Objet
A	06/02/2025	POM	First release for V1.00
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C			
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	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	1	Rev.: B

CONTENT

1	General	8
1.1	Object.....	8
1.2	Scope of application.....	8
1.3	Assumptions.....	8
1.4	Axes reference.....	8
1.5	Terms.....	9
2	References	9
2.1	First generation of Eurocodes.....	9
2.2	Drafts for the second generation of Eurocodes.....	10
2.3	Other references.....	10
3	Use of ABC-PMX.....	11
3.1	General description	11
3.1.1	Main window	11
3.1.2	Menus.....	11
3.1.3	Project toolbar	12
3.1.4	Beam toolbar.....	12
3.1.5	Selection of a beam.....	13
3.1.6	View of the selected beam and graphical toolbar	13
3.2	Configuration of the software	14
3.2.1	Toolbar and menu.....	14
3.2.2	General parameters.....	14
3.2.3	Directories	14
3.2.4	Units.....	14
3.2.5	Databases	15
3.2.6	Preliminary design report.....	15
3.3	Calculation options	15
3.3.1	Toolbar and menu.....	15
3.3.2	Default national factors.....	16
3.3.3	Scope of application	17
3.3.4	Calculation options.....	17
3.3.5	Connectors.....	19
3.3.6	Slab and slim-floors.....	19
3.3.7	Fire design.....	20
4	Definition of a beam.....	21
4.1	Type of the beam	21
4.1.1	Types of cross-sections.....	21
4.1.2	Definition of the type in ABC-PMX.....	24
4.2	Identification of the beam	25
4.3	Geometrical description of the beam.....	26
4.4	Concrete slab above steel profiles.....	28
4.4.1	General.....	28

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	2	Rev.:	B

4.4.2	Definition of the slab	28
4.4.3	Configurations of composite slabs at beam support	30
4.4.4	Concrete	31
4.4.5	Definition of the longitudinal reinforcement in the slab	31
4.4.6	Requirements for the concrete cover	32
4.4.7	Definition of the transverse reinforcement	33
4.5	Profiled steel sheeting.....	33
4.5.1	General.....	33
4.5.2	Geometry	34
4.5.3	Other parameters.....	35
4.6	Steel cross-sections	35
4.6.1	Definition of the steel profile	35
4.6.2	General description of a steel cross-section.....	35
4.6.3	Hot rolled profile.....	36
4.6.4	Bi-symmetrical welded section	37
4.6.5	Mono-symmetrical welded section	37
4.6.6	Steel	38
4.7	Connection for usual composite beams	39
4.7.1	General.....	39
4.7.2	Welded studs	39
4.7.3	Connection arrangement.....	40
4.7.4	Detailing of connectors	40
4.8	Compatibility of the connectors and the profiled steel sheeting	41
4.8.1	First generation of Eurocodes.....	41
4.8.2	Second generation of Eurocodes	41
4.9	Partial concrete encasement.....	42
4.9.1	General.....	42
4.9.2	Parameters	42
4.10	Lateral restraint of steel beams.....	44
4.10.1	General.....	44
4.10.2	Point lateral restraints	44
4.11	Lateral restraints of composite beams during the construction phase.....	44
4.11.1	General.....	44
4.11.2	Parameters of the restraint by the steel profiled sheeting	45
4.12	Propping	48
4.12.1	General.....	48
4.12.2	Effect of the propping on the beam design.....	48
4.12.3	Definition of the point propping.....	49
4.13	Load cases.....	50
4.13.1	General.....	50
4.13.2	Type of loading	50
4.13.3	Definition of the loadings.....	51
4.14	Nationally Determined Parameters	51

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	3	Rev.: B

4.14.1	General.....	51
4.14.2	Partial factors.....	52
4.14.3	Combination factors.....	52
4.15	Combination of loads.....	53
4.15.1	General.....	53
4.15.2	Ultimate Limit States (ULS) – Persistent or transient situations	53
4.15.3	Serviceability Limit States (SLS)	54
4.15.4	Ultimate limit state – Fire situation	54
4.16	Vibratory comfort assessment.....	55
4.16.1	General.....	55
4.16.2	Parameters	55
4.17	Calculation options.....	56
4.17.1	General.....	56
4.17.2	Options	57
4.18	Fire design options	58
4.18.1	General.....	58
4.18.2	Beam parameters	58
4.18.3	Calculation options.....	60
4.18.4	Fire design parameters	60
5	Material properties.....	61
5.1	Structural steel.....	61
5.1.1	Structural steel for hot rolled profiles	61
5.1.2	Structural steel for welded sections	61
5.1.3	Structural steel for plates	61
5.1.4	Other structural steel properties	62
5.2	Concrete.....	62
5.3	Steel for reinforcements.....	63
6	Basic calculations.....	64
6.1	Effective width of the concrete flange	64
6.1.1	General.....	64
6.1.2	Effective width of the concrete flange in the main span	64
6.1.3	Effective width of the concrete flange in the cantilever spans.....	65
6.1.4	Simplified models for the effective width.....	65
6.2	Active part of the concrete slab for composite sections.....	67
6.2.1	General.....	67
6.2.2	Usual sections	67
6.3	Modular ratio.....	67
6.3.1	Eurocode generation 1	67
6.3.2	Notional size of a member	69
6.4	Classification of cross-sections.....	69
6.4.1	General.....	69
6.4.2	Usual steel sections.....	69
6.4.3	Usual steel sections with partial concrete encasement	71
6.4.4	Usual composite sections.....	71

		Project: ABCPMX-II Software - DAL-22-003					
		Product reference: DRV/PMX/MT/001					
	Date:	19/05/2025	Authors:	POM	Page:	4	Rev.: B

6.4.5	Usual composite sections with partial concrete encasement	72
6.5	Section bending properties	72
6.5.1	General.....	72
6.5.2	Steel sections.....	72
6.5.3	Simplified methods for symmetrical steel sections.....	74
6.5.4	Usual steel sections with partial concrete encasement	74
6.5.5	Usual composite sections.....	75
6.5.6	Usual composite sections with partial concrete encasement	75
6.5.7	Reduction of the plastic resistance for composite section with deep plastic neutral axis	75
6.6	Other properties of sections.....	76
6.6.1	Shear area of usual steel and composite sections.....	76
6.6.2	Torsion and warping of steel profiles	76
6.7	Shear resistance of welded studs	77
6.7.1	General.....	77
6.7.2	First generation of Eurocodes.....	77
6.8	Calculation of elastic stresses – Usual Sections	79
6.8.1	Principles.....	79
6.8.2	Usual steel sections.....	80
6.8.3	Usual steel sections with partial concrete encasement	81
6.8.4	Usual composite sections.....	82
6.8.5	Usual composite section with partial concrete encasement.....	84
6.8.6	Equivalent stresses in the steel profile.....	85
6.8.7	Effect of concrete cracking.....	85
7	Global analysis.....	86
7.1	General.....	86
7.2	Design sections.....	86
7.3	Critical sections.....	86
7.4	Cracked and uncracked segments of the slab	86
7.5	Cracked parts of the encasement concrete	87
7.6	Load cases.....	87
7.7	Specific load cases.....	87
7.7.1	Permanent load case G1	87
7.7.2	Self-weight	88
7.7.3	Shrinkage load case	89
7.7.4	Live load arrangements	89
7.8	Calculation of internal forces, moments and deflections.....	90
8	Verifications of beams at ultimate limit states	90
8.1	General.....	90
8.2	Verification of the resistance to bending moment	90
8.2.1	Usual steel sections.....	90
8.2.2	Usual steel sections with partial concrete encasement	91
8.2.3	Usual composite sections.....	91
8.2.4	Usual composite section with partial concrete encasement.....	92
8.3	Verification of the resistance to shear force	93
8.3.1	General criterion.....	93

		Project: ABCPMX-II Software - DAL-22-003					
		Product reference: DRV/PMX/MT/001					
	Date:	19/05/2025	Authors:	POM	Page:	5	Rev.: B

8.3.2	Usual cross-sections without partial concrete encasement.....	93
8.3.3	Usual cross-sections with partial concrete encasement.....	94
8.4	Verification of the resistance to MV interaction – First generation of EN	94
8.4.1	General.....	94
8.4.2	General criterion for usual sections with compact webs.....	94
8.4.3	General criterion for usual section with slender webs.....	95
8.4.4	Elastic design.....	95
8.5	Resistance to lateral torsional buckling.....	95
8.5.1	General.....	95
8.5.2	Beams with usual steel sections.....	95
8.5.3	Beams with usual steel sections and partial concrete encasement.....	97
8.5.4	Beams with usual composite sections – composite stage – First generation of EN	97
8.5.5	Beams with usual composite sections – construction stage	99
8.5.6	Beams with usual composite sections and partial concrete encasement.....	99
8.5.7	Verification of the profiled steel sheeting used as lateral restraint in the construction phase	99
8.6	Reinforcement of usual composite cross-section under negative bending moment.....	103
8.7	Design of welds for welded sections	104
9	Design of the connection between the slab and the steel profile	105
9.1	Introduction.....	105
9.2	Plastic design of the connection	105
9.2.1	General.....	105
9.2.2	Degree of shear connection.....	105
9.2.3	Limitation of partial shear connection.....	106
9.3	Elastic design of the connection	107
9.4	Automatic design of the connection.....	108
9.5	Design of transverse reinforcement for usual composite sections	109
9.5.1	General and assumptions.....	109
9.5.2	Minimum transverse reinforcement.....	109
9.5.3	Configurations of slabs and potential areas of shear failure.....	109
9.5.4	Longitudinal shear force.....	110
9.5.5	Verification of the concrete crushing	112
9.5.6	Design of transverse reinforcement.....	112
9.5.7	Design of transverse reinforcement with continuous profiled steel sheetings	113
10	Fire design	114
10.1	General.....	114
10.2	Principles for beams without partial concrete encasement.....	114
10.3	Principles for beams with partial concrete encasement.....	115
10.4	Section properties - Classification	116
10.5	Heating of the section	116
10.5.1	Usual unprotected steel sections	116
10.5.2	Usual protected steel sections	118
10.5.3	Usual unprotected composite sections.....	118
10.5.4	Slab – Simplified method – First generation of EN	119
10.5.5	Slab – Simplified method – Second generation of EN.....	120
10.5.6	Slab – Numerical method.....	121

		Project: ABCPMX-II Software - DAL-22-003					
		Product reference: DRV/PMX/MT/001					
	Date:	19/05/2025	Authors:	POM	Page:	6	Rev.: B

10.5.7	Usual protected composite sections.....	121
10.6	Section factors.....	122
10.6.1	Usual unprotected steel or composite sections.....	122
10.6.2	Usual protected steel or composite sections.....	123
10.7	Reduction factors for mechanical properties.....	123
10.7.1	Construction steel.....	123
10.7.2	Concrete.....	125
10.7.3	Reinforcing steel.....	126
10.8	Fire resistance criteria.....	126
10.8.1	Principles.....	126
10.8.2	Resistance to bending moment.....	127
10.8.3	Resistance to shear force.....	127
10.8.4	Resistance to MV interaction.....	127
10.8.5	Resistance to shear buckling.....	127
10.8.6	Resistance to lateral torsional buckling.....	128
10.9	Verifications in fire situation.....	129
10.9.1	Usual steel sections.....	129
10.9.2	Usual composite sections.....	129
10.9.3	Profiles with partial concrete encasement.....	130
10.10	Connection.....	130
10.10.1	Shear resistance of connectors.....	130
10.10.2	Design of the connection.....	131
11	Verifications at serviceability limit states.....	132
11.1	Deflections.....	132
11.2	Eigen frequencies.....	133
11.3	Comfort to vibrations.....	133
11.3.1	General.....	133
11.3.2	Dynamic properties of the floor.....	133
11.3.3	Determination of the OS-RMS ₉₀ class.....	134
11.3.4	Vibration performance assessment.....	134
11.4	Cracking of concrete for beams with usual composite sections.....	134
11.4.1	General.....	134
11.4.2	Minimum reinforcement for simply supported beams.....	135
11.4.3	Minimum reinforcement for crack width limitation without direct loading.....	135
11.4.4	Control of cracking due to direct loading.....	136
11.4.5	Minimum reinforcement for crack width limitation in partial concrete encasement.....	136
Annex A	Specific configuration of composite slabs with Cofraplus 220.....	137
Annex B	Procedure for the calculation of section bending properties.....	138
B.1	General.....	138
B.2	Section properties.....	138
	Elastic properties.....	138
	Plastic properties.....	139
Annex C	Procedure for the Finite Element calculations.....	140
C.1	Principles.....	140

		Project: ABCPMX-II Software - DAL-22-003					
		Product reference: DRV/PMX/MT/001					
	Date:	19/05/2025	Authors:	POM	Page:	7	Rev.: B

C.2	Numerical modelling.....	140
Annex D	Critical moments for LT Buckling	142
D.1	Scope of application.....	142
D.2	Critical moments.....	142
Annex E	Critical moment for LT Buckling taking account of the steel deck	143
E.1	Scope of application.....	143
E.2	Calculation of the critical moment.....	143
E.3	Shear rigidity of the steel profiled sheeting.....	143
	General	143
	Flexibilities due to the sheet deformation	143
	Flexibilities due to the fasteners deformation	144
	Slip of fasteners	145
E.4	Stiffness of the torsional restraint.....	145
Annex F	Bending moment resistance in fire situation of partially encased profiles.....	152
F.1	Scope of application.....	152
F.2	Bending resistance under positive moment.....	152
F.3	Longitudinal shear in the connection under positive moment.....	158
F.4	Bending resistance under negative moment.....	159
F.5	Longitudinal shear in the connection under negative moment	161
F.6	Resistance to shear force.....	161
Annex G	Numerical procedure for the assessment of the temperature in a solid slab in fire situation.....	162
G.1	Scope of application.....	162
G.2	Assumptions.....	162
G.3	Properties of concrete	163
G.4	Temperature in the slab under the effect of an ISO fire.....	164
Annex H	Temperature in a steel profile with perpendicular composite slab, when the voids are unfilled.....	166
H.1	Scope of application.....	166
H.2	Temperature of gages in the voids.....	166
H.3	Temperature of the upper flange	167
H.4	Temperature of the lower flange	167
H.5	Temperature of the web	168

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	8	Rev.: B

1 GENERAL

1.1 Object

- (1) **ABC-PMX** software aims at the calculation of composite beams according to EN 1994-1-1. Its development is co-funded by CTICM (**PMX** software) and ArcelorMittal (**ABC+** software).

This document provides the technical description of the software. It describes:

- the parameters,
- the assumptions and methods applied for the section and global analyses,
- the calculations of loads and combinations of loads,
- the design checks applied for the verification of the beams at ultimate limit state (final stage and construction stage),
- the design checks at serviceability limit states.

It also gives a description of the main features of the User Interface of the software.

1.2 Scope of application

- (1) The software deals with composite beams obtained by the connection of the slab to the upper flange of a steel profile (standard beams) or by a steel profile embedded in the concrete slab (slim floors). It also covers the case of partially encased steel profiles.
- (2) It applies for buildings only, according to EN 1994-1-1 (first and second generations). Applications for bridges are not covered.



Except when otherwise stated in this document, it is assumed that the same rules apply for both generations of Eurocodes. Version V1.00 of the software is available only for the first generation of Eurocodes.

- (3) **ABC-PMX** does not cover the possibility to deal with specific National Annexes. The User has the possibility to set the parameters of the software to reproduce the choices of any National Annex.
- (4) Single span beams are considered. It is also possible to add cantilever spans at each support, except for slim floor beams. Continuous beams with multiple spans are not covered in this version of the software.
- (5) The software covers primary as well as secondary beams.
- (6) Large web openings are not covered in the software. Slab openings are possible.
- (7) Class 4 cross-sections are not covered.



Slim floors beams, fire design and second generation of Eurocodes are not included in version V1.10 of ABC-PMX. They will be released in later versions.

1.3 Assumptions

- (1) All beams have a uniform section along the span length. The same cross-section is considered in the cantilever spans, if any.
- (2) In presence of cantilever spans, the joints which provide the continuity at supports are assumed to be rigid and resistant enough. The software will not provide any calculations for these joints.
- (3) Torsion effects are not considered, even for asymmetrical configurations of slim floors. In the latter case, specific devices are assumed to be present to prevent torsion in the beam.

1.4 Axes reference

- (1) For steel sections below the slab, with or without connection, the location of centroids is measured from the interface between concrete and steel, i.e. from the top fibre of the upper flange. Values are negative below the interface and positive above (**Figure 1**).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	9	Rev.: B

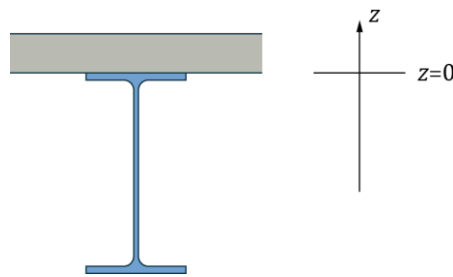


Figure 1: Axis z for usual sections



The term “usual sections” is used here for all the cross-sections defined in § 4.1.1 (2) and (3).

- (2) For steel and composite slim floor sections, the location of centroids is measured from the bottom of the slab. Values are negative below the interface and positive above (**Figure 2**).

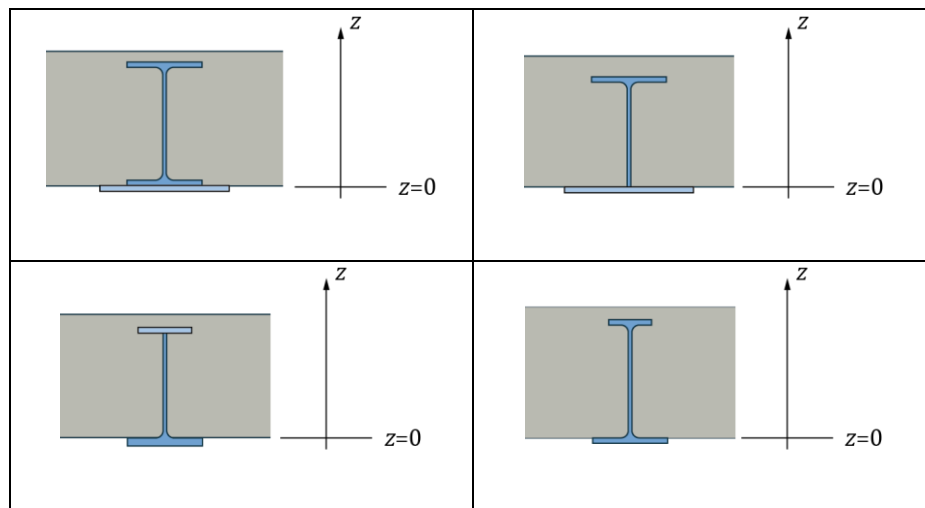


Figure 2: Axis z for slim floor sections

1.5 Terms

- (1) The term “composite slabs” is used for a concrete slab where a steel profiled sheeting is used initially as permanent formwork and subsequently combines structurally with the hardened concrete to act as a tensile reinforcement in the finished floor.

The software does not cover the design of the slab, so that it can be also used for slabs with steel profiled sheeting which acts only as lost formwork.

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	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	10	Rev.: B

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	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	11	Rev.: B

3 USE OF ABC-PMX

3.1 General description

3.1.1 Main window

- (1) The main window provides the following elements (**Figure 3**):
- ① a menu toolbar (§ 3.1.2);
 - ② a toolbar for the management of the project (§ 3.1.3);
 - ③ a toolbar for the definition of the selected beam (§ 3.1.4 and § 4);
 - ④ an area to select the beam of the project;
 - ⑤ a graphical area where the selected beam is displayed;
 - ⑥ a toolbar for the options of the graphical area (§ 3.1.6);
 - ⑦ a toolbar for the setting of the software (§ 3.2 and § 3.3).

ABC-PMX deals with a project which can consists of at least one or more beams. Each beam can be defined independently.

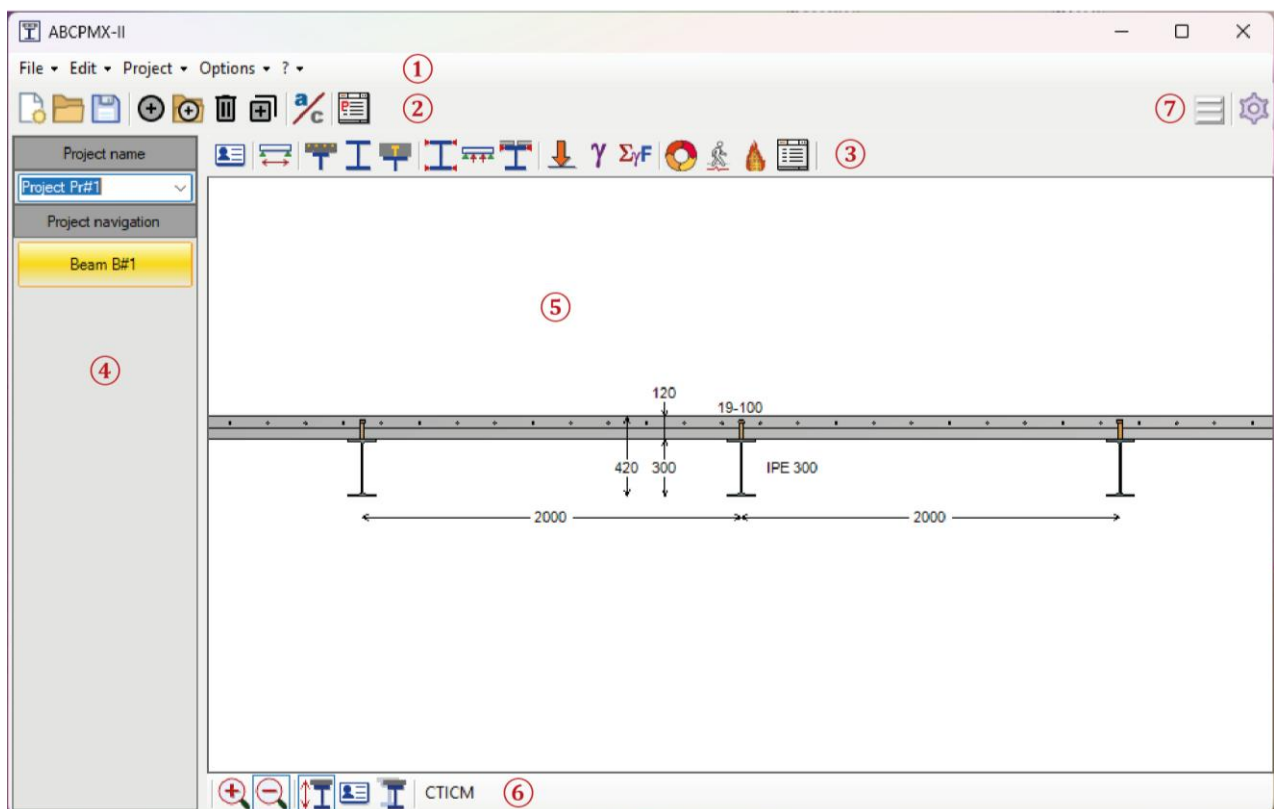


Figure 3: General layout of the main window

3.1.2 Menus

- (1) The following menus and sub-menus are available:

Menu File:

- New: creation of a new project;
- Open: opening a project file;
- Save: saving the project (with all its beams) in the current file;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	12	Rev.: B

- Save as: saving the project in another file;
- a direct access to the ten last opened beams.

Menu Edit:

- Studs: edition of the database for welded studs (it is possible to add new studs);
- Steel sheeting: edition of the database of steel profiled sheetings (it is possible to add new sheets);
- Profiles: edition of the hot rolled profile database; for each profile, the dimensions and the section properties are provided.

Menu Project:

- Add beam: add a new beam in the project;
- Delete beam: delete the selected beam from the project;
- Duplicate beam: add a new beam in the project which is a copy of the selected beam;
- Project report: edit the design report summarising the results of the project beams.

Menu Options:

- Software configuration: setting of the software parameters;
- Calculation options: setting of the calculation parameters.

Menu “?”:

- Contact support: possibility to send an email for help or support about **ABC-PMX**;
- About: information about the software;
- Technical Report: edition of the present file;
- Validation Report: edition of the validation report [24].

3.1.3 Project toolbar

- (1) The project toolbar gives the features to manage a project.

Following buttons are available in the toolbar:

-  open a project file
-  save the current project
-  add a new beam in the project
-  import beams in the project from the file of another project
-  delete the selected beam
-  duplicate the selected beam
-  open the modular ratio tool
-  edit the project design report

3.1.4 Beam toolbar

- (1) The beam toolbar allows the user to open one of the different modules for the definition and the treatment of the selected beam. The logical order to go through these modules is from left to right. However, the piece of software allows a return to a previous module. In this case, the user should be aware that a modification of some data in a module might have some consequences on data defined in a module located more on the right in the tool bar.
- (2) Following buttons are available in the toolbar (according to the type of the selected beam):

-  beam references
-  definition of the beam geometry
-  definition of the slab

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	13	Rev.: B

	definition of the steel profile
	definition of the connection (for composite beams only)
	definition of the lateral restraints (for the construction phase and the non-composite beams)
	definition of the propping conditions (for composite beams only)
	definition of the lateral restraint by the profiled steel sheeting in the construction phase of composite beams
	definition of the loads
	definition of the combination and partial factors
	definition of the combinations of loads
	definition of the calculation parameters
	parameters for the vibratory assessment
	parameters for the fire analysis
	edition of the design report

3.1.5 Selection of a beam

- (1) Several beams may be defined in a project. The selected beam is chosen by clicking on the associated button in the selection area ④ (**Figure 3**).

3.1.6 View of the selected beam and graphical toolbar

- (1) A set of tools is proposed for the drawing of the selected beam in the graphical zone. The available tools are the following ones:

	increase of the zooming
	decrease of the zooming
	display or hide the dimensions
	display or hide the beam references
	copy the drawing

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	14	Rev.: B

3.2 Configuration of the software

3.2.1 Toolbar and menu

- (1) The window for the configuration of the software can be opened either by the menu “options/software options” or directly by clicking on the associated button in the setting toolbar (⑦ in **Figure 3**).

3.2.2 General parameters

- (1) Language settings:
- language of the software interface;
 - default language of the preliminary design report (two different languages may be chosen for the interface and for the report).
- (2) Default identification:
The User may define the default names of the firm and of the User, that will be used for each beam. These identification parameters may be modified individually for each beam.
- (3) Internet Update:
ACB-PMX is provided with a feature which automatically checks if more recent versions of the software or of the profiles and steel databases are available. This feature requires an internet access. It may be deactivated by the User.

3.2.3 Directories

- (1) Directories:
The software uses an installation directory which may be chosen by the User at installation only. It also needs a configuration directory which is automatically defined.
A working directory may be defined by the User which will be used by default to save and to open files. To change the default working directory, the User should click on the associated textbox.
- (2) Initial working directory:
When the function “open” and “save” are used, the software may either:
- always use the default working directory;
 - or use the directory that has been selected at the last “open” or “save” action.

3.2.4 Units

- (1) The User may define the units to be used for:
- dimensions;
 - lengths;
 - bending modulus;
 - second moment of areas;
 - forces;
 - moments;
 - stresses.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	15	Rev.: B

3.2.5 Databases

- (1) **ABC-PMX** uses the following databases:

Databases:

- a base for welded studs;
- a base for steel profiled sheetings;
- a base for hot-rolled sections;
- a base for prefabricated floors;
- a base for steel properties.

None of these databases may be modified by the User. For welded studs and steel profiled sheeting, it is nevertheless possible to add custom items in the databases.

Databases may be edited by clicking on their names.

The version of the databases for steel profiles and for steel properties is displayed.

Databases from internet updates:

When available, it is possible to directly update the databases for steel profiles and for the steel properties from internet. The number of updates is displayed.



This function is not implemented in this version.

3.2.6 Preliminary design report

- (1) It is possible to define the content of the preliminary design report as follows:

- (2) Analysis of the beam, forces and moments:

The User may define for which situation the internal forces are to be displayed, in the following list:

- for each load case,
- for each combination of loads at ultimate limit states,
- for each combination of loads at serviceability limit states,
- for each combination of loads at fire limit states.

The internal forces and moments may be displayed in the calculation report as follows:

- by tables with the values at each node of the model,
- by diagrams showing the distribution of internal forces and moments along the beam,
- by tables giving the envelope values.

- (3) Elastic moments:

The User may include the values of the elastic moments of composite beams in the calculation report. Elastic moments are calculated according to EN 1994-1-1: 2004, formula (6.4).

- (4) Serviceability limit states, vibratory assessment:

When a vibratory assessment of the beam is performed (see § 11.3), the latter is done according to the Hivoss method. It is possible to include in the calculation report the diagram used to assess the category of comfort.

3.3 Calculation options

3.3.1 Toolbar and menu

- (1) The window for the definition of the default calculation options of **ABC-PMX** can be opened either by the menu "options/calculation options" or directly by clicking on the associated button in the setting toolbar (⑦ in **Figure 3**).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	16	Rev.: B

3.3.2 Default national factors

- (1) The default values of the partial factors may be defined by the User. These values will be used as default values for the definition of each new beam. It is nevertheless possible to define specific values of the partial factors for each beam (see § 4.14).
- (2) The default setting values of the software are as given in **Table 1** and in **Table 2**. for the partial factors.

Partial factor	Recommended values	French NAs	Partial factor	Recommended values	French NAs
$\gamma_{G,sup}$	1,35	1,35	γ_{M2}	1,25	1,25
$\gamma_{G,inf}$	1,0	1,0	γ_c	1,5	1,5
γ_Q	1,5	1,5	γ_s	1,15	1,15
γ_{M0}	1,0	1,0	γ_v	1,25	1,25
γ_{M1}	1,0	1,0	γ_p	1,0	1,0

Table 1: Default setting values of the partial factors

Partial factor	Recommended values	French NAs
$\gamma_{M,fi}$	1,0	1,0
$\gamma_{M,fi,a}$	1,0	1,0
$\gamma_{M,fi,c}$	1,0	1,0
$\gamma_{M,fi,v}$	1,0	1,5
$\gamma_{M,fi,s}$	1,0	1,0

Table 2: Default setting values of the partial factors for fire design

- (3) The default setting values of the software are as given in **Table 3** for the combination factors.

Comb. factor	Q_1		Q_2	
	Recommended values	French NAs	Recommended values	French NAs
ψ_0	$\psi_{0,1} = 0,7$	$\psi_{0,1} = 0,7$	$\psi_{0,2} = 0,7$	$\psi_{0,2} = 0,7$
ψ_1	$\psi_{1,1} = 0,5$	$\psi_{1,1} = 0,5$	$\psi_{1,2} = 0,5$	$\psi_{1,2} = 0,5$
ψ_2	$\psi_{2,1} = 0,3$	$\psi_{2,1} = 0,3$	$\psi_{2,1} = 0,3$	$\psi_{2,1} = 0,3$

Table 3: Default setting values of the combination factors

3.3.3 Scope of application

- (1) This window gives the following pieces of information (**Figure 4**):
- the minimum and maximum values of the span length of a beam,
 - the minimum and maximum values of the length of cantilever spans,
 - the minimum dimensions of welded sections,
 - the minimum total thickness of a slab,
 - the maximum thickness of a haunch,
 - the minimum thickness of a slab above the ribs of a steel profiled sheeting ($t_{c,min}$),
 - the minimum and maximum values of the density of light weight concrete.
- (2) The User may adjust the value of the angle of inclination θ_h of the sides of concrete haunches. This inclination is defined about the vertical axis (cf. § 4.4.2 and **Figure 16**). The value should be between 0 and 45°. The default value is 30°.

The inclination of haunches sides is used for the drawing and to assess the volume and mass of concrete. Haunches are ignored in the calculation of resistances.

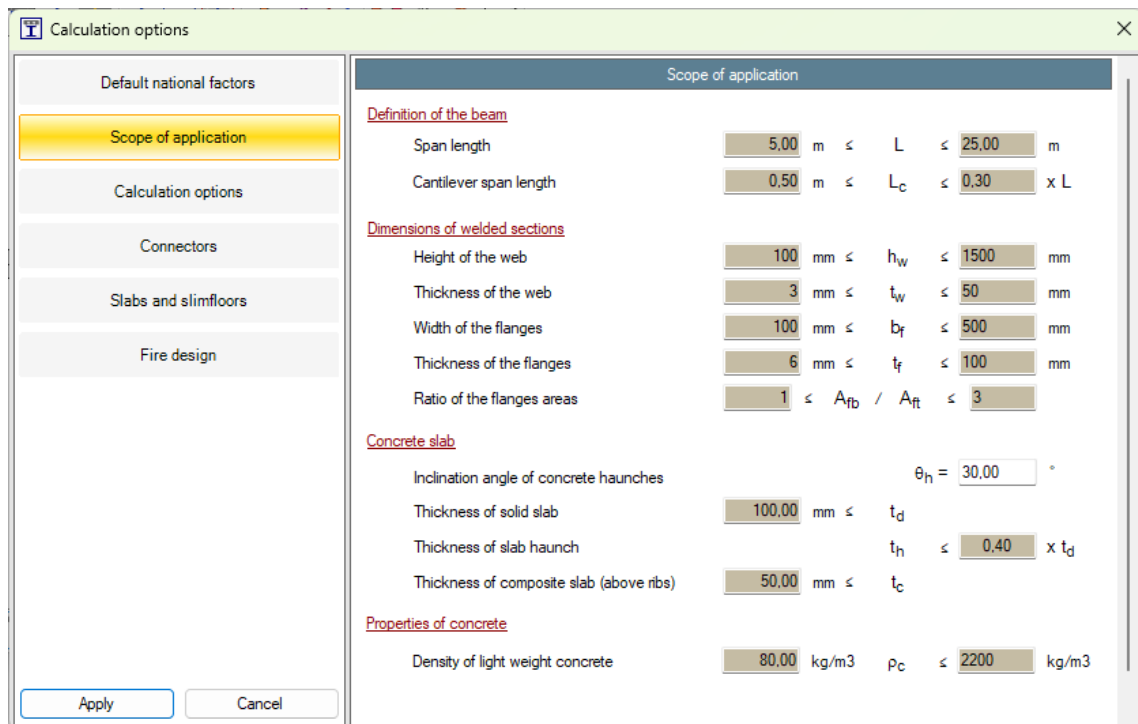


Figure 4: Definition of the scope of application

3.3.4 Calculation options

- (1) The User may define or read the value of the following parameters (**Figure 5**):
- (2) Properties of cross-sections:
- For the calculation of the section properties and the check of cross-sections, the User may choose to consider the simplified distribution of the effective width of the concrete flange along the beam, according to clause 6.1.2 (2) of EN 1994-1-1: 2004. By default, the distribution defined by clause 5.4.1.2 of EN 1994-1-1: 2004 is used.
- For the calculation of the cross-section properties (second moment of areas, elastic or plastic moment), the User may choose to consider the rebars of the slab which are in compression, according to clause 6.2.1.2

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	18	Rev.: B

of EN 1994-1-1: 2004. By default, this option is not activated. To be valid, this option assumes that the buckling of rebars is avoided by enough shear rebars.

According to clause 3.3.6 (2) of EN 1992-1-1:2004, it is possible to choose a value of the Young's modulus for the steel of rebars between 195 and 210 GPa. A default value of 210 GPa is considered.

(3) Definition of the beam model:

The User may adapt the fineness of the model used to analyse the beam (see **Annex C**). Following parameters are available:

- maximal distance between two adjacent nodes (default value 1 m),
- minimal number of nodes per usual span (default value 100),
- minimal number of nodes per cantilever span (default value 50).

(4) Load cases:

Following parameters are required to assess the modular ratio associated with each load case of composite beams:

- the creep multipliers for permanent load and for shrinkage (these values are not modifiable),
- the age of concrete of the slab and of the partial encasement of beams, for the self-weight (G1), the other permanent loads (G2) and shrinkage (SH). The values for the latter are not modifiable,

(5) Shear buckling:

The User may define the default value of the parameter η , that is used for the verification of beam to shear buckling (see § 8.3). This value should be between 1 and 1,2.

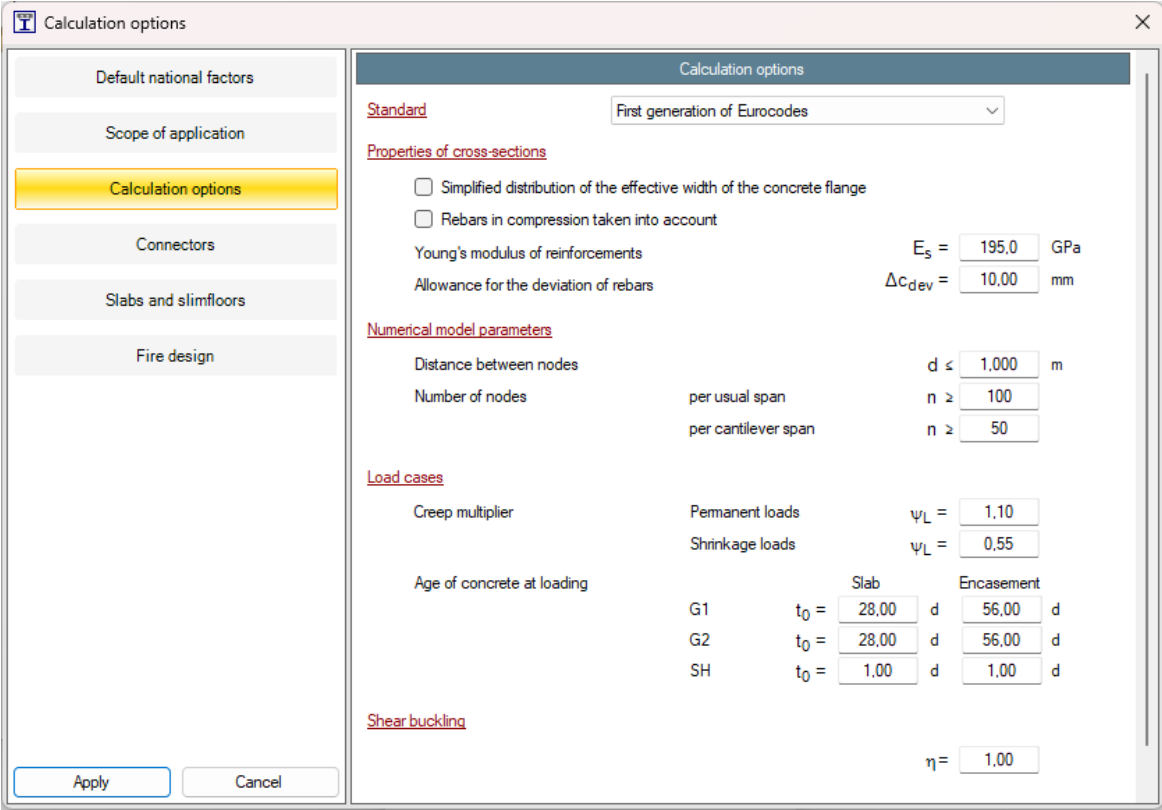
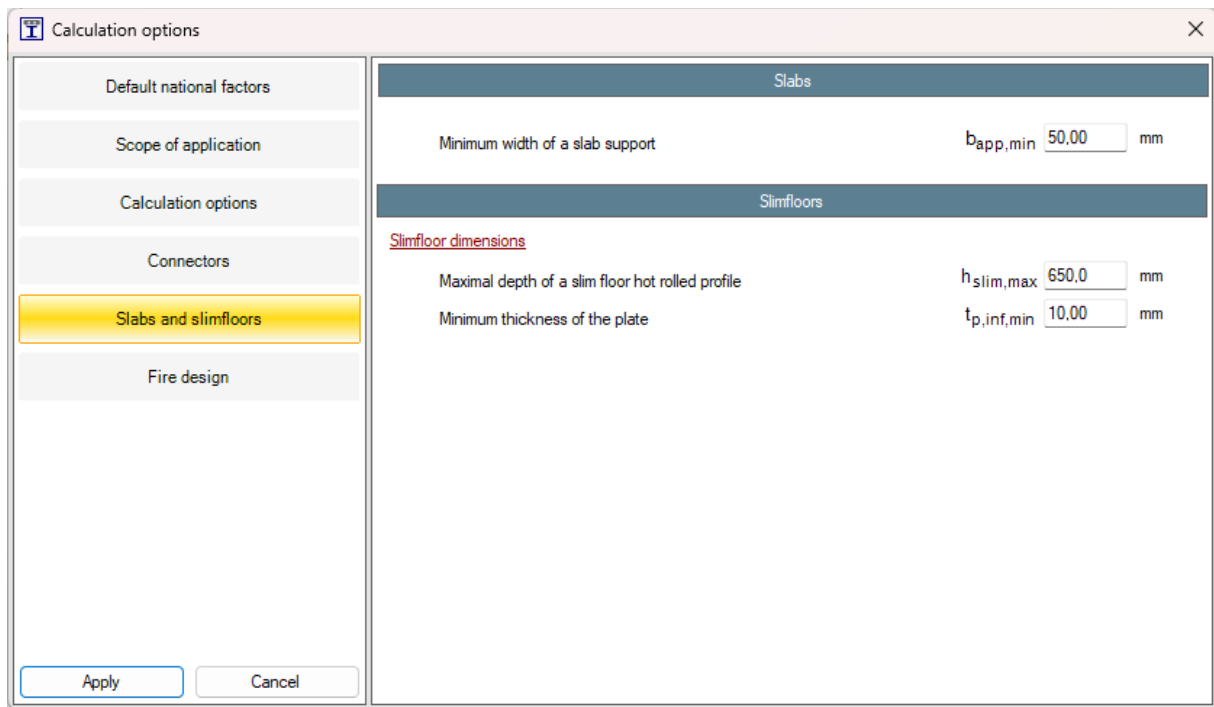


Figure 5: Definition of the calculation options



Calculation options

Default national factors

Scope of application

Calculation options

Connectors

Slabs and slimfloors

Fire design

Slabs

Minimum width of a slab support $b_{app,min}$ 50.00 mm

Slimfloors

Slimfloor dimensions

Maximal depth of a slim floor hot rolled profile $h_{slim,max}$ 650.0 mm

Minimum thickness of the plate $t_{p,inf,min}$ 10.00 mm

Apply Cancel

Figure 7: Options for slabs and slim-floors

3.3.7 Fire design

- (1) This window gives the parameters used in **ABC-PMX** for the fire design of beams. Except otherwise mentioned, these parameters may not be modified. Following parameters are given:
- Stefan Boltzman constant,
 - emissivity of concrete,
 - emissivity of fire,
 - convection factor (below the slab where the fire occurs),
 - convection factor (above the slab, without fire),
 - correction factor for the shadow effect (modifiable),
 - form factor (modifiable),
 - reference temperature (before the start of the fire),
 - maximal temperature.

Default national factors

Scope of application

Calculation options

Connectors

Slabs and slimfloors

Fire design

Apply

Cancel

Fire design

Constants

Boltzman constant

$\sigma = 5,67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$

Emissivity of concrete

$\varepsilon_c = 1$

Emissivity of fire

$\varepsilon_f = 1$

Convection factor

$\alpha_c = 25 \text{ W/m}^2 \text{ K}$

Convection factor above the slab

$\alpha_{cc} = 4 \text{ W/m}^2 \text{ K}$

Correction factor for the shadow effect

$k_{sh} = 1$

Form factor

$\phi = 1$

Parameters

Reference temperature

$\theta_0 = 20 \text{ }^\circ\text{C}$

Maximal temperature

1200 $^\circ\text{C}$

Figure 8: Fire design options

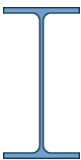
4 DEFINITION OF A BEAM

4.1 Type of the beam

4.1.1 Types of cross-sections

(1) Following types of cross-sections, as given in (2) to (5), are available in ABC-PMX.

(2) **Steel profiles below the slab, without connection to the slab**
 The first set of sections concerns steel profiles placed below the concrete slab without a connection to the slab. The steel element may be a hot-rolled profile (SHR type in Table 4) or may be obtained by welding of steel plates, for either bi-symmetrical (SWB type) or mono-symmetrical sections (SWM type). The design of these beams is carried out according to EN 1993-1-1 [4].
 For bi-symmetrical sections, either hot-rolled or welded, a partial encasement by concrete between the flanges will be available (see Table 5). For these sections, the composite action of the concrete between the flanges is taken into account and the design is carried out according to EN 1994-1-1 [7] but neglecting the contribution of the concrete slab.
 This type of cross-sections will be denoted hereafter in this document as **usual steel sections**, with or without partial concrete encasement.



SHR

hot rolled section



SWB

bi-symmetrical welded section



SWM

mono-symmetrical welded section

Table 4: Types of steel section without connection to the slab

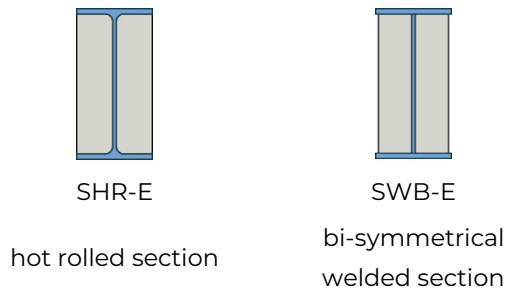


Table 5: Types of steel section with partial concrete encasement but without connection to the slab



Even for these steel profiles, a slab is to be defined. The beams in the software are always assumed to be part of a floor system.

According to the country, the option for welded section may not be available.

(3) **Steel profiles below the slab, with a connection to the slab**

The second set of sections concerns steel profiles placed below the concrete slab with a connection between them, which allows to take benefit from the composite action. The steel element may be a hot-rolled profile (CHR type in **Table 6**) or may be obtained by welding of steel plates, for either bi-symmetrical (CWB type) or mono-symmetrical sections (CWM type). The design of these beams is carried out according to EN 1994-1-1 [7].

A partial encasement by concrete between the flanges will also be available (see **Table 7**) for bi-symmetrical sections.

This type of cross-sections will be denoted hereafter in this document as **usual composite sections**, with or without partial concrete encasement.

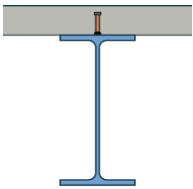
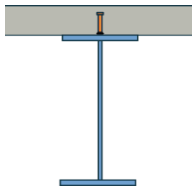
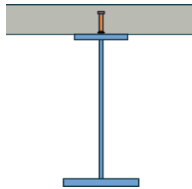
		
CHR	CWB	CWM
hot rolled section	bi-symmetrical welded section	mono-symmetrical welded section

Table 6: Types of steel section with a connection to the slab

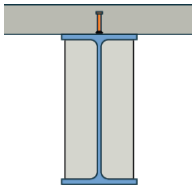
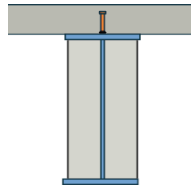
	
CHR-E	CWB-E
hot rolled section	bi-symmetrical welded section

Table 7: Types of steel section with partial concrete encasement and with a connection to the slab

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	23	Rev.: B

(4) **Slim floor steel sections (without connection to the slab)**

The third set of sections deals with steel profiles encased in a concrete slab, denominated as slim floors sections. No connection is provided between the steel profile and the concrete, so that no composite action is developed. The steel elements are obtained from hot-rolled profiles only. To have the possibility to build this type of floors, the lower flange or plate should be larger than the upper flange. For the SFB type (see **Table 8**), a steel plate is welded below the lower flange of a hot rolled profile. For the IFB sections, the hot-rolled profile is first cut longitudinally, to create a tee section. A steel plate is then welded to the tee section. For the IFB-A sections, the steel plate constitutes the lower flange of the section whereas for the IFB-B sections, it is the upper flange. The last type of section (SAB) is obtained by cutting in the longitudinal direction the upper flange of a profile to reduce its width.

This type of profiles is designed according to specific methods detailed in the present specifications, based on the principles of Eurocode 3.

This type of cross-sections will be denoted hereafter in this document as **(non-composite or) steel slim floor sections**.

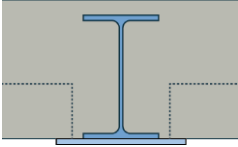
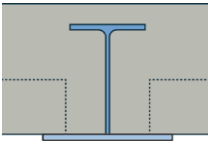
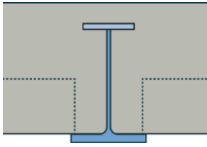
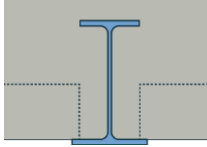
Non composite slim floors			
			
SFB	IFB – A	IFB – B	SAB

Table 8: Non-composite sections for slim-floors

(5) **Slim floor composite sections (with connection to the slab)**

The fourth set of sections deals with steel profiles encased in a slim concrete floor with a connection, which provides a composite action. These sections are denominated as composite slim floors sections. The same four types of steel cross-section are considered as for non-composite slim floor sections, together with three types of connectors (**Table 9**):

- composite dowels, by a steel reinforcement passing through the web (Group A);
- studs welded on the upper flange of the steel profile (Group B);
- studs welded on both sides of the profile's web (Group C).

This type of cross-sections will be denoted hereafter in this document as **composite slim floor sections**. Composite slim floor sections are possible only for intermediate beams. In slim floors, edge beams are calculated as steel sections.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	24	Rev.: B

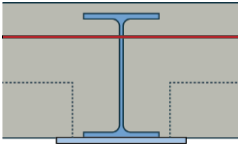
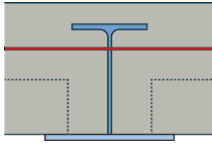
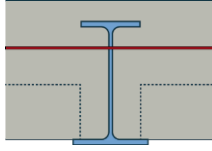
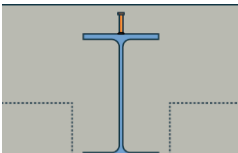
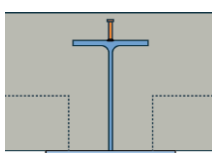
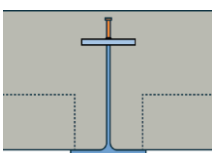
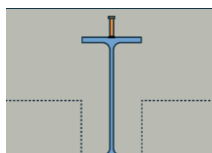
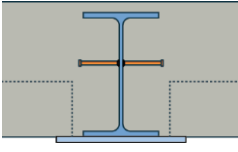
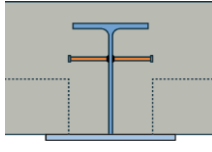
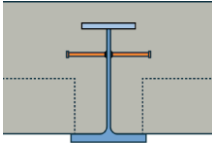
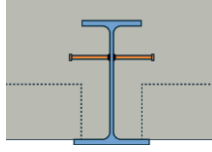
Composite slim floors			
Group A – Composite dowels			
		No composite dowel possible for CA-IFB-B	
CA-SFB	CA-IFB-A	CA-IFB-B	CA-SAB
Group B – Studs welded on the upper flange			
			
CB-SFB	CB-IFB-A	CB-IFB-B	CB-SAB
Group C – Studs welded on the web			
			
CC-SFB	CC-IFB-A	CC-IFB-B	CC-SAB

Table 9: Composite sections for slim-floors



Slim floor beams are not available in this version of ABC-PMX. They will be released in a later version.

4.1.2 Definition of the type in ABC-PMX

- (1) The definition of a new beam starts by the selection of its type among the following choices (**Figure 9**):
- steel beam,
 - steel beam with partial encasement,
 - composite beam,
 - composite beam with a partial encasement of the steel profile,
 - slim floor steel beams,
 - slim floor composite beams.

Once the type of the beam has been chosen, it is not possible to modify it.

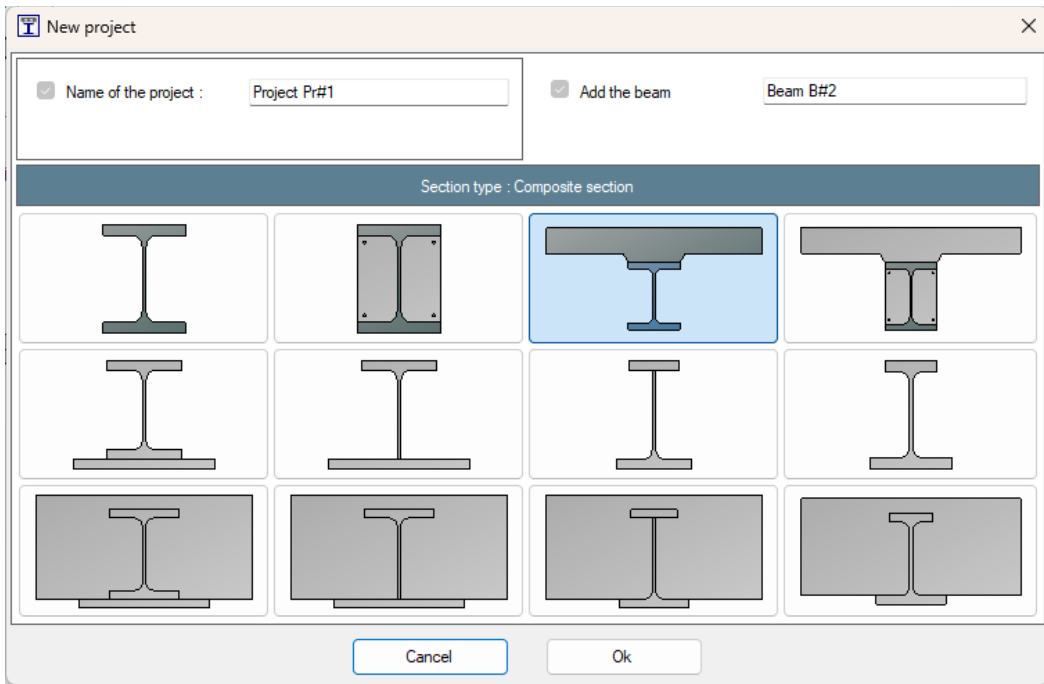


Figure 9: Definition of the type of a beam

4.2 Identification of the beam

- (1) The identification of a beam is defined in the associated window (**Figure 10**), which is opened by clicking on the following button of the main toolbar (**§ 3.1.3**):

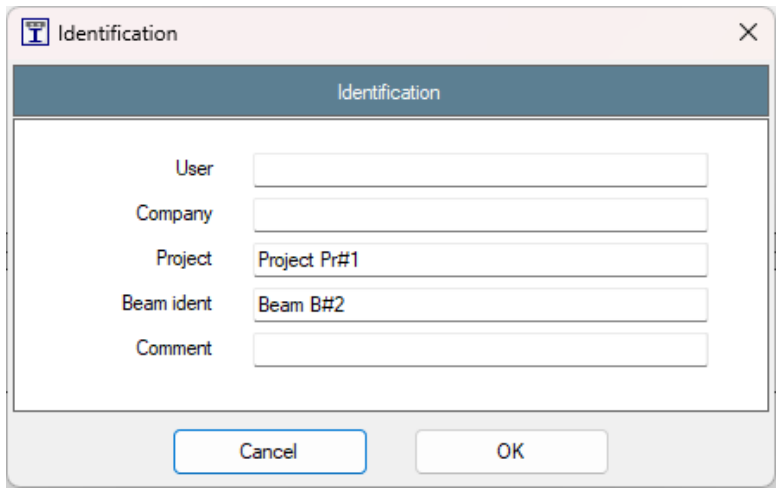
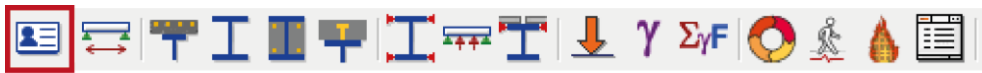


Figure 10: Beam identification

4.3 Geometrical description of the beam

- (1) The main parameters of a beam are defined in the associated window (**Figure 11**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

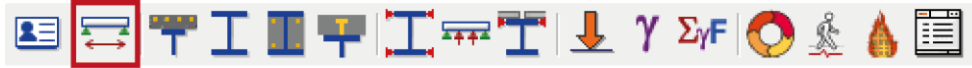
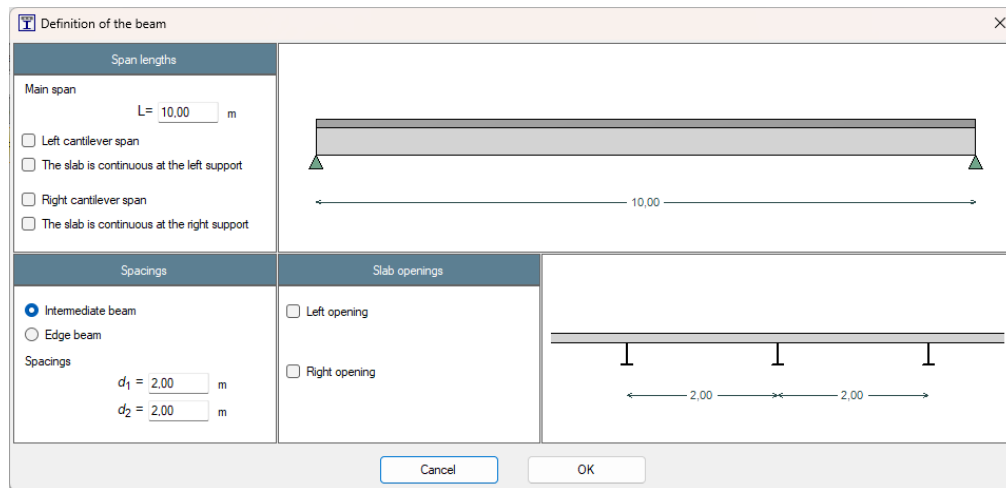



Figure 11: General parameters for the definition of a beam

- (2) A beam has to be defined either as an intermediate beam or as an edge beam (**Figure 12** for usual sections and **Figure 13** for slim floor sections).

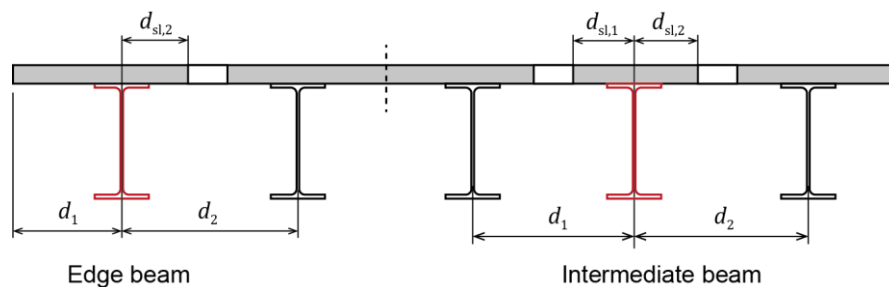


Figure 12: Definition of edge beams and intermediate beams for usual sections

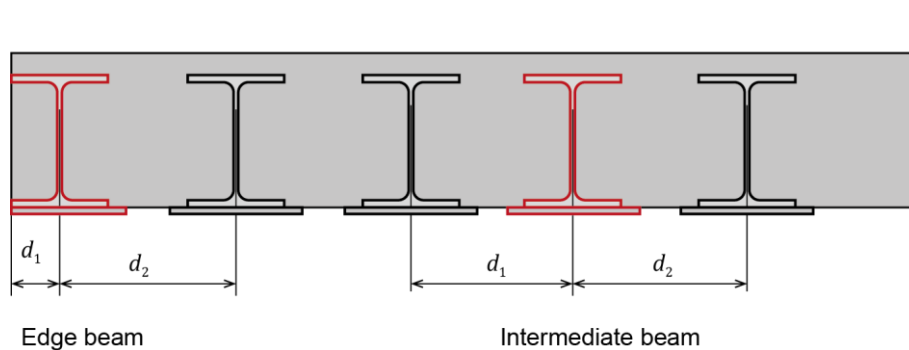


Figure 13: Definition of edge beams and intermediate beams for slim floor sections

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	27	Rev.: B

(3) The geometry of the main span of the beam is defined as follows (see **Figure 12** to **Figure 14**):

- For intermediate beams:
 - L is the length of the main span
 - d_1 is the spacing of the beam to the left beam
 - d_2 is the spacing of the beam to the right beam
- For edge beams:
 - L is the length of the main span
 - d_1 is the spacing of the beam to the slab edge
 - d_2 is the spacing of the beam to the adjacent beam

L should be at least 5 m: $L \geq L_{\min} = 5 \text{ m}$. It is possible to modify $L_{\min}$ in the options of the software in Expert mode.

The minimum spacing of intermediate beams is given by: $d_{\min} = \max\{1 \text{ m}; 2 \sum b_f; 2 \sum b_p\}$

For an edge beam, if d_1 is set to zero by the User, the following values will be considered in the calculations:

- for usual beams: $d_1 = b_{fs} / 2$
- for slim floor beams: $d_1 = b_s / 2$, where b_s is the section width.

(4) For slim floors, the distance d_1 for edge beams is always assumed to be at the edge of the profile.

(5) For cantilever spans, the following additional parameters are required:

L_g is the length of the left cantilever span

L_d is the length of the right cantilever span

The length of a cantilever span should fulfil the following conditions:

$$500 \text{ mm} \leq L_g \text{ or } L_d \leq 0,3 L \quad (1)$$

Cantilever spans are not available for slim floor steel or composite beams.

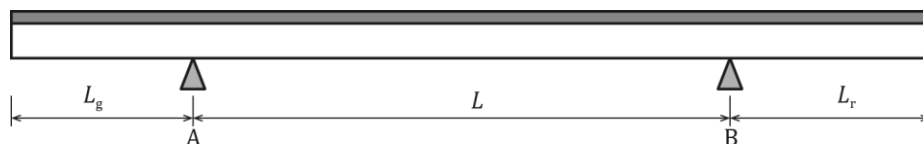


Figure 14 : Spans of the beam

(6) In the central span only, it is possible to define the presence of slab openings. A slab opening will be defined simply by the distance d_{sl} of the opening edge to the beam axis. The presence of a slab opening will be possible only between two adjacent beams. So that, the following parameters will be defined by the User for slab openings (see **Figure 12**):

- For intermediate beams:
 - $d_{sl,1}$ distance of a slab opening edge to the left
 - $d_{sl,2}$ distance of a slab opening edge to the right
- For edge beams:
 - $d_{sl,2}$ distance of a slab opening edge to the side of the adjacent beam



Slab openings are not possible on the cantilever side of the slab, in the case of edge beams.

(7) There are no slab openings for beams with slim floor sections.

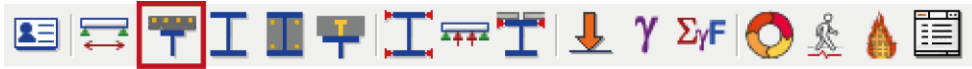
(8) At any support of the main span without adjacent cantilever span, the User should define, for composite beams with usual sections, if the slab is continuous at the support. See § 11.4 for the effect of the continuity of the slab at a simple support.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	28	Rev.: B

4.4 Concrete slab above steel profiles

4.4.1 General

- (1) For usual steel or composite beams, with or without partial encasement, the slab above the steel profile should be defined, through the associated window (**Figure 15**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):



- (2) For steel beams (non-composite beams), the parameters of the slab are used only for the automatic calculation of the slab weight.

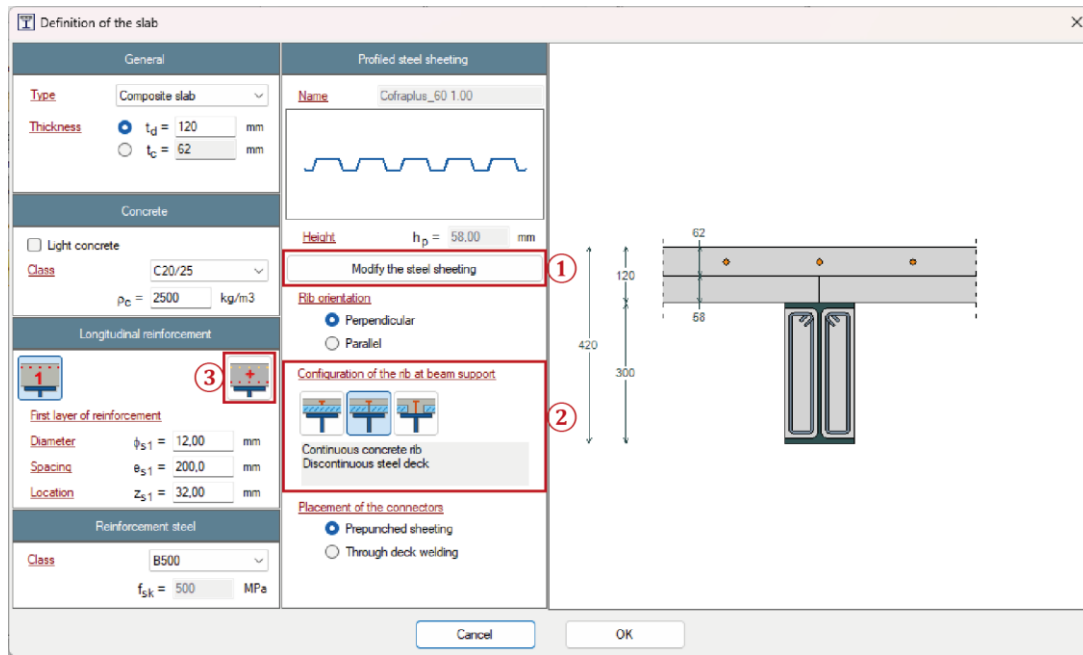


Figure 15: Definition of a slab

4.4.2 Definition of the slab

- (1) For usual steel or composite sections, steel profiles are below the slab (cross-section from **Table 4** to **Table 7**) and three configurations of slabs are possible:

- solid slabs;
- solid slabs with precast elements;
- composite slabs.

- (2) The parameters for a solid slab are as follows (**Figure 16**):

- t_d total thickness of the slab;
- t_h thickness of the haunch (by default = 0).

The following requirement should be fulfilled:

$$t_{c,min} \leq t_d \quad (2)$$

where: $t_{c,min}$ is the minimum thickness of the active part of slab, with $t_{c,min} = 50$ mm.

$t_{c,min}$ is a parameter of the software (cf. § 3.3.3).

- (3) Haunches are available only for solid slabs.

The width of the concrete haunch (b_h) will be assumed to be equal to the width of the upper flange ($b_h = b_{fs}$). For the resistance of the slab, the haunch will be neglected. For the calculation of the mass and for

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	29	Rev.: B

graphical purposes, the edges of the haunches will be assumed to have a slope defined by its angle to the vertical axis $\theta_h = 30^\circ$. The thickness of the haunch should fulfil the following criterion:

$$t_h \leq 0,5 t_d \quad (3)$$

The value of θ_h is an option parameter of the software (cf. § 3.3.3).

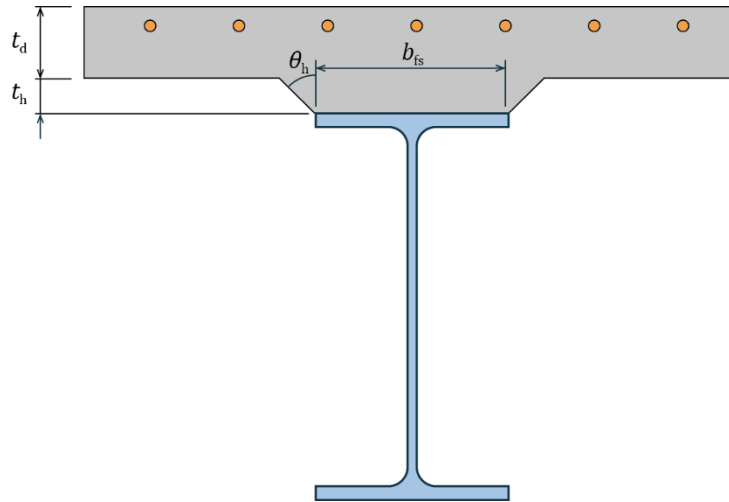


Figure 16 : Parameters for concrete solid slabs

- (4) For solid slabs with precast elements, it will not be possible to define a concrete haunch. The parameters for this configuration are as follows:

- t_d total thickness of the slab;
- t_{pc} thickness of the precast elements;
- t_j thickness of the joint between precast elements that is filled with concrete.

The following requirements should be fulfilled:

$$\begin{aligned} t_{c,min} &\leq t_d \\ 10 \text{ mm} &\leq t_{pc} \leq t_d - 40 \text{ mm} \\ 0 &\leq t_j \leq t_{pc} \end{aligned} \quad (4)$$

- (5) A composite slab is constituted from a profiled steel sheeting, above which the concrete is poured. The parameters for this configuration of slabs are as follows:

- t_d total thickness of the slab;
- orientation of the profiled steel sheeting, either longitudinal or transversal.

The profiled steel sheeting may be modified by clicking on the button “modify the steel sheeting” (see ① in **Figure 15**). The associated window and the additional parameters for the steel profiled sheeting are described in chapter 4.5.

For composite slabs, the following condition should be checked:

$$t_c = t_d - h_p \geq t_{c,min} \quad (5)$$

where: t_c is the thickness of the slab above the profiled steel sheeting;

h_p is the height of the sheeting (ignoring the potential upper stiffener – see § 4.5.1);

$t_{c,min}$ see (2).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	30	Rev.: B

4.4.3 Configurations of composite slabs at beam support

- (1) For composite slabs above steel profiles (steel or composite usual sections), the User will define the configuration at beam supports as given in this chapter.
- (2) The same configuration will be considered for all the beam, including the cases with multiple spans.
- (3) Composite beams with ribs perpendicular to the beam axis can have one of the three following configurations (see **Figure 17**), to be defined by the User (see ② in **Figure 15**):

- Configuration PE1:

The concrete rib and the steel sheeting are both continuous above the beam

With this configuration:

- the shear resistance of the connector is calculated with the reduction factor k_t according to **§ 6.7.2 (3)**;
- the steel profiled sheeting is considered for the resistance to longitudinal shear, according to **§ 9.5.7**.

- Configuration PE2 (default configuration):

The concrete rib is continuous, but the steel sheeting is not.

With this configuration:

- the shear resistance of the connector is calculated with the reduction factor k_t according to **§ 6.7.2 (3)**;
- the steel profiled sheeting is not considered for the resistance to longitudinal shear, see **§ 9.5.6**;

- Configuration PE3:

The steel sheeting and the concrete rib are both interrupted at each end of the upper flange, so that a continuous parallel concrete flange is formed above the steel flange.

With this configuration:

- the shear resistance of the connector is calculated with the reduction factor k_t according to **§ 6.7.2 (2)**, considering a representative width of the concrete flange equal to:

$$b_0 = b_{fs} - 2 b_{app,min}$$

where $b_{app,min}$ is the minimum width of the slab support, which is a parameter of the software (see **§ 3.3.6**);

- the steel profiled sheeting is not considered for the resistance to longitudinal shear.

To be compatible with this solution, the flange thickness should not be lower than:

$$b_{fs} \geq 50 \text{ mm} + 2 b_{app,min} \quad (6)$$

where 50 mm is the minimal width of the concrete trough, according to 6.6.5.8 (2) of EN 1994-1-1.

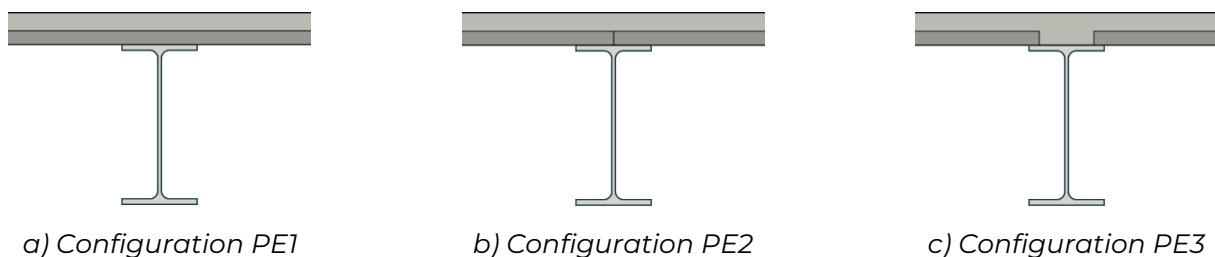


Figure 17 : Configurations of perpendicular steel sheeting across a beam

- (4) For composite beams with ribs parallel to the beam axis, the sheeting is always assumed to be continuous over the flange, so that the concrete rib above the flange has the same dimensions as other ribs (see **Figure 18**). In practice, if the sheeting is not continuous, the concrete rib above the flange has a width larger than the other ribs, which is more favourable for the shear resistance of the stud.

The shear resistance of the connector is calculated with the reduction factor k_t according to **§ 6.7.2 (2)**.

- (5) **Annex A** gives the specific configuration of the composite slabs when Cofraplus 220 decking is used.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	31	Rev.: B

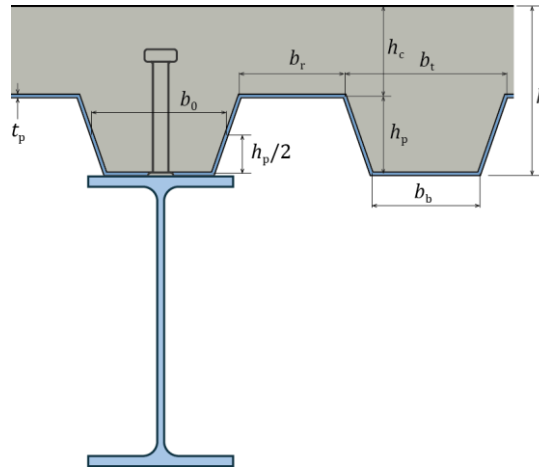


Figure 18 : Configuration of parallel steel sheeting above a beam

- (6) For usual composite cross-sections, with ribs perpendicular to the beam axis, the User should select one of the following options for the connection:

- pre-punched steel sheeting (default choice),
- through deck welding of the connectors.

This choice is requested only for perpendicular sheeting with continuous ribs on the beam support (configuration PE1 and PE2 according to **(3)**).

4.4.4 Concrete

- (1) The concrete of the slab should be defined by the User through the following parameters:

- the concrete class,
- the option specifying light weight concrete,
- the density of concrete.

- (2) See § 5.2 for the properties of concrete.

4.4.5 Definition of the longitudinal reinforcement in the slab

- (1) For beams with multiple spans (i.e. with cantilevers – see § 4.3), longitudinal reinforcement of the slab should be defined by the User. By default, longitudinal reinforcements are considered for the stiffness and the resistance of the cross-section only when they are under tension. They may also be considered when in compression, see § 6.5.5 (1).

- (2) The longitudinal reinforcement consists of at least one layer of steel rebars. When the thickness of the slab is sufficient, a second layer may be defined (see button ③ in **Figure 15**). By default, only one layer is proposed.

- (3) A layer of steel rebars is defined by the following parameters (**Figure 19**):

e_{si} spacing of rebars axes;

d_{si} diameter of the rebars (to be selected among the following list: 6, 8, 10, 12, 14, 16, 20, 25 and 32 mm);

z_{si} location of the layer axis, measured from the upper side of the slab.

The location z_i of the rebars in the axes of the cross-section (see § 1.4) is given by:

$$z_i = t_h + t_d - z_{si}$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	32	Rev.: B

These parameters should fulfil the following conditions:

$$2 d_{si} \leq e_{si} \leq e_{si,max} = 500 \text{ mm} \quad (7)$$

$$d_{si} < z_{si} < t_c - d_{si}$$

See § 6.2 for t_c ; See § 4.4.6 for the conditions on concrete cover.

- (4) For each layer, the number of rebars located in the effective width of the concrete flange, b_{eff} (see § 6.1), is calculated by:

$$n_{si} = b_{eff} / e_{si} \quad (n_{si} \text{ can be a real value in the calculations})$$

The area of steel rebars accounted for the bending moment resistance of the composite cross-section is obtained by:

$$A_{si} = n_{si} \pi d_{si}^2 / 4 \quad (8)$$

- (5) The total area of the longitudinal reinforcement is given by:

$$A_s = \sum A_{si} \quad (9)$$

The ratio of reinforcement of the slab is assessed as follows:

$$\rho_s = \frac{A_s}{b_{eff} t_c} \quad (10)$$

See § 6.2 for t_c .

- (6) The class of reinforcing steel is to be chosen among the following list: B450, B500.

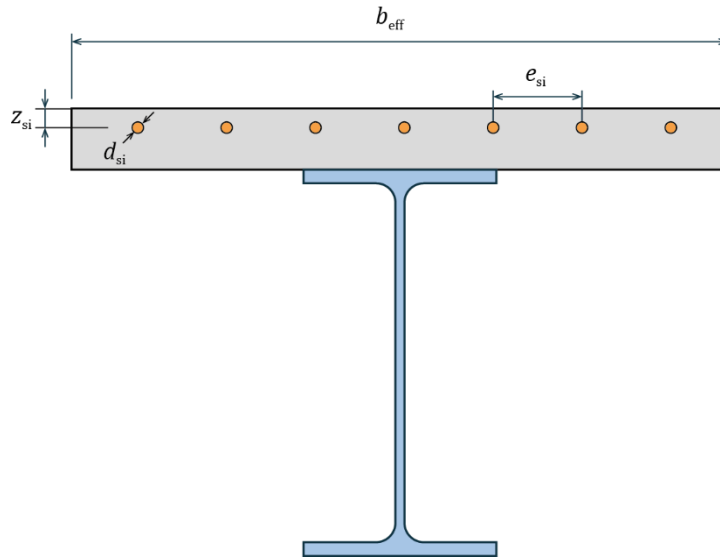


Figure 19 : Definition of a layer of steel reinforcement

4.4.6 Requirements for the concrete cover

- (1) A control of the minimum concrete cover according to EN 1992, 4.4.1 is performed by **ABC-PMX**. Only a warning message is displayed if the conditions are not fulfilled. The control does not deal with environment conditions but only with bond requirement.
- (2) Any layer should fulfil the following conditions:

$$\frac{d_{si}}{2} + c_{min} + \Delta c_{dev} \leq z_{si} \leq t_c - \left(\frac{d_{si}}{2} + c_{min} + \Delta c_{dev} \right) \quad (11)$$

where: Δc_{dev} is the allowance for the deviation of rebars; $\Delta c_{dev} = 10 \text{ mm}$ by default, it can be modified in the software settings (see § 3.3.4);

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	33	Rev.: B

c_{min} is the minimum concrete cover, which is obtained by:

$$c_{min} = \max\{10\text{ mm}; c_{min,b}\}$$

$c_{min,b}$ is the minimum concrete cover for bond requirement, given by: $c_{min,b} = d_{si}$



Requirements for protection against corrosion are not covered.

4.4.7 Definition of the transverse reinforcement

- (1) For composite beams, the transverse reinforcement is automatically defined by the software, in order to resist the longitudinal shear – see § 9.5.
Transverse reinforcements are not required for the design of other configurations.
- (2) The transverse reinforcement required to resist the transversal bending in the slab is not treated by the software.

4.5 Profiled steel sheeting

4.5.1 General

- (1) For a composite slab, the profiled steel sheeting may be modified by the User by clicking on the button “modify the steel sheeting” (see ① in Figure 15) in the window for the definition of the slab (see § 4.4.2).
- (2) A profiled steel sheeting may be defined directly from the database or from its parameters (Figure 20).

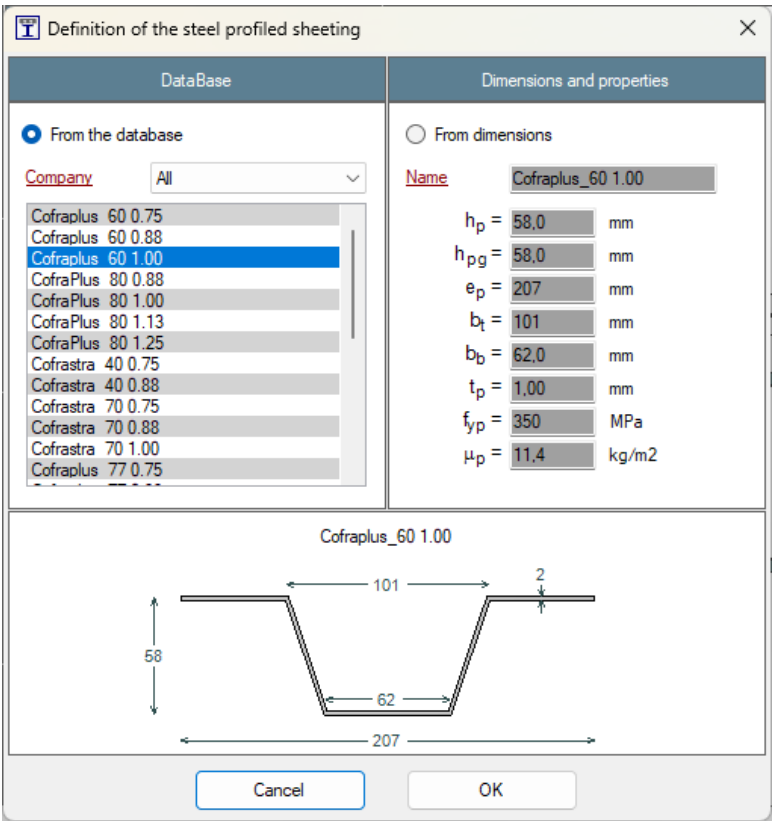


Figure 20 : Definition of the steel profiled sheeting of a composite slab

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	34	Rev.: B

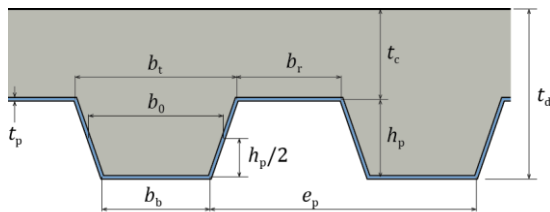
4.5.2 Geometry

(1) The geometry of a profiled steel sheeting is defined by the following parameters (**Figure 21**):

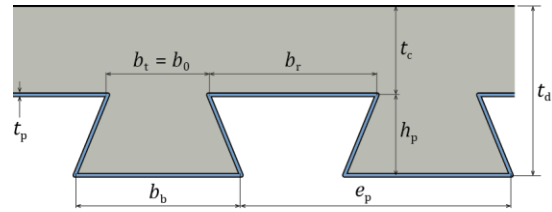
t_p	thickness of the sheet
e_p	spacing between the axes of two adjacent ribs
h_p	height of the sheet, ignoring any upper stiffener
h_{pg}	height of the sheet, including any upper stiffener
b_b	width of a rib at its bottom side
b_t	width of a rib at its upper side



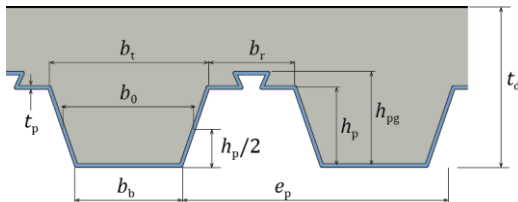
If $h_p = h_{pg}$, no upper stiffener is present.



a) Open trough



b) Re-entrant trough



c) Open trough and upper stiffener

Figure 21 : Dimensions of profiled steel sheeting

(2) From the initial parameters in **(1)**, additional ones are calculated as follows:

b_r	width of the sheet between two adjacent ribs: $b_r = e_p - b_t$
b_m	average width of a rib: $b_m = (b_t + b_b) / 2$
b_0	representative width of a rib, obtained as follows:
▪ for open troughs:	$b_0 = b_m$ (width at mid height)
▪ for re-entrant troughs:	$b_0 = b_t$ (width at the top side)

4.5.3 Other parameters

(1) A profiled steel sheeting is also described by the following parameters:

- μ_p surface mass of the sheet (in kg/m²);
- f_p yield strength of the steel of the sheet (in MPa);
- I_{pu} second moment of area per unit of width.



I_{pu} is used when the flexural stiffness of the deck will be used for the verification of the resistance of the steel profile to lateral torsional buckling in construction phase (see § 8.5.5 (2)) and for the check of the steel deck (see § 8.5.7).

4.6 Steel cross-sections

4.6.1 Definition of the steel profile

(1) For usual steel or composite beams, with or without partial encasement, the steel profile should be defined, through the associated window (**Figure 22**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

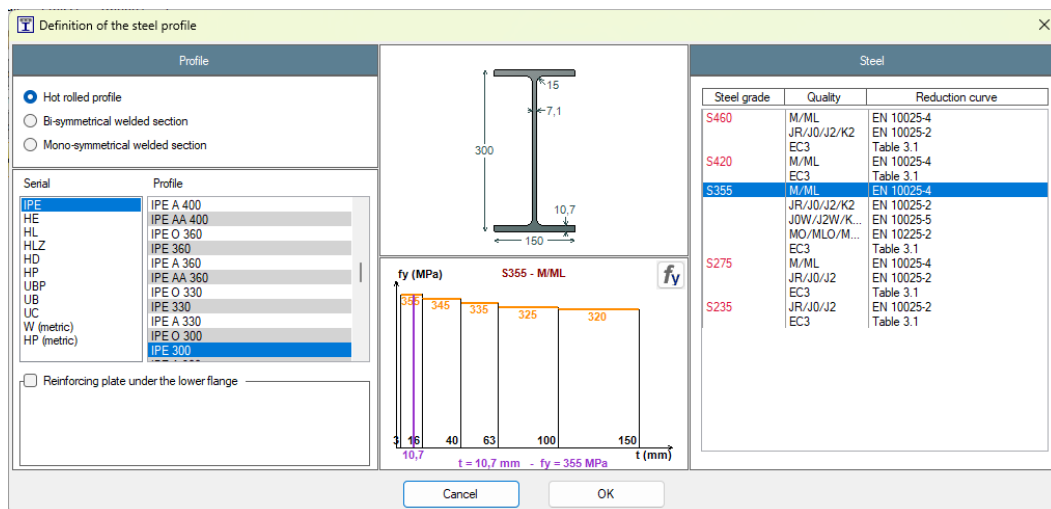
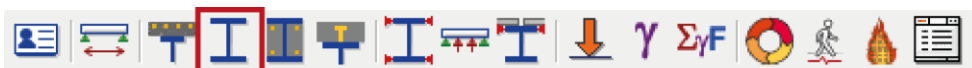


Figure 22: Definition of a slab

4.6.2 General description of a steel cross-section

(1) Any steel profile is described by the following geometrical parameters:

- h_s total depth of the section
- h_b depth of the parent profile (when a part or all the section is obtained from a hot rolled profile)
- b_{fs} width of the upper flange
- t_{fs} thickness of the upper flange
- b_{fi} width of the lower flange
- t_{fi} thickness of the lower flange
- r_{cs} radius of the upper fillet
- r_{ci} radius of the lower fillet

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	36	Rev.: B

- h_w depth of the web (accounted between flanges)
 t_w thickness of the web
 d_w clear depth of the web (accounted between fillets)
 b_p width of the plate (hot-rolled profile with plate, slim floors SFB and IFB)
 t_p thickness of the plate (hot-rolled profile with plate, slim floors SFB and IFB)
 a thickness of the weld throat, for welded sections. The same thickness is used for both flanges.

(2) The following relationships are verified by these parameters:

$$\begin{aligned}
 h_s &= t_{fs} + h_w + t_{fi} + t_p \\
 h_w &= d_w + r_{cs} + r_{ci} + 2\sqrt{2}a
 \end{aligned}
 \tag{12}$$

The parameters are given in the next chapter for each type of section.

4.6.3 Hot rolled profile

(1) For a hot rolled profile, the parameters of the section are shown in **Figure 23 a**. In this case, the following formulas are applied:

$$\begin{aligned}
 b_{fs} &= b_{fi} \\
 t_{fs} &= t_{fi} \\
 r_{cs} &= r_{ci} \\
 h_s &= h_b \\
 a &= 0
 \end{aligned}$$

(2) In the software, hot rolled profiles are defined from the ArcelorMittal database.

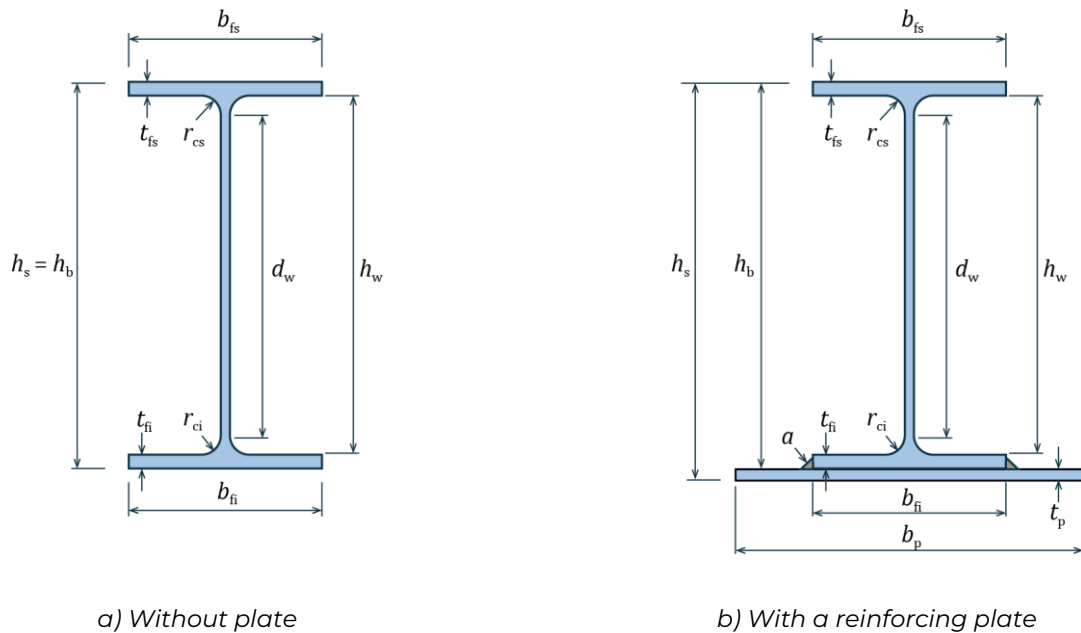


Figure 23 : Dimensions of a hot rolled profile

(3) For hot rolled profiles, it is possible to define a reinforcing plate located below the lower flange (**Figure 23 b**). This plate is defined by:

its dimensions b_p and t_p ;

its steel, which may be chosen independently from the steel of the profile.

The reinforcing plate is assumed to cover the full length of the beam.

	Project: ABCPMX-II Software - DAL-22-003					
	Product reference: DRV/PMX/MT/001					
	Date:	19/05/2025	Authors:	POM	Page:	37
					Rev.:	B

4.6.4 Bi-symmetrical welded section

- (1) For a bi-symmetrical welded section, the parameters of the section are shown in **Figure 24**. They are defined directly by the User.
- (2) They fulfil the following formulas:

$$\begin{aligned}
 b_{fs} &= b_{fi} \\
 t_{fs} &= t_{fi} \\
 r_{cs} &= r_{ci} = 0 \\
 b_p &= t_p = 0 \\
 h_s &= h_w + t_{fs} + t_{fi}
 \end{aligned}$$

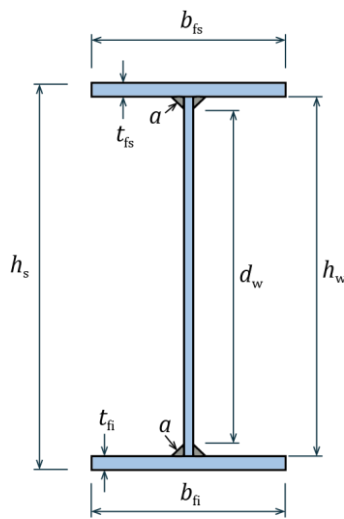


Figure 24 : Dimensions of a bi-symmetrical welded section

- (2) The following scope of application is considered for welded sections:

$$\begin{aligned}
 100 &\leq h_w < 1500 \text{ mm} \\
 100 &\leq b_f < 500 \text{ mm} \\
 6 &\leq t_f \leq 100 \text{ mm and } 3 \leq t_w < 50 \text{ mm} \\
 a &\geq 3 \text{ mm}
 \end{aligned}$$

4.6.5 Mono-symmetrical welded section

- (1) For a mono-symmetrical welded section, the parameters of the section are shown in **Figure 25**. They are defined directly by the User.
- (2) They fulfil the following formulas:

$$\begin{aligned}
 b_{fs} &\neq b_{fi} \\
 t_{fs} &\neq t_{fi} \\
 r_{cs} &= r_{ci} = 0 \\
 b_p &= t_p = 0 \\
 h_s &= h_w + t_{fs} + t_{fi}
 \end{aligned}$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	38	Rev.: B

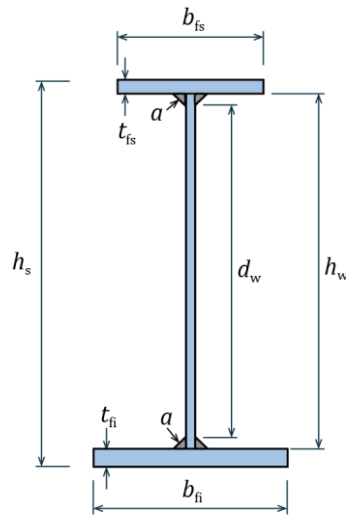


Figure 25 : Dimensions of a mono-symmetrical welded section

- (2) The areas of the flanges are given by:

$$\begin{aligned} A_{fs} &= b_{fs} t_{fs} \\ A_{fi} &= b_{fi} t_{fi} \end{aligned} \quad (13)$$

For mono-symmetric sections in Normal mode, the following criterion should be fulfilled (see § 3.3.3):

$$1 \leq A_{fi} / A_{fs} \leq 3 \quad (14)$$

- (3) § 4.6.4 (3) applies.

4.6.6 Steel

- (1) The steel grade of the profile should be selected by the User.
- (2) For hot-rolled profiles with reinforcing plates, the steel grade of the plate should also be chosen. Both steel grades are independent.
- (3) See § 5.1 for the steel properties.

4.7 Connection for usual composite beams

4.7.1 General

- (1) For usual composite beams, with or without partial encasement, the connection between the steel profile and the slab should be defined, through the associated window (**Figure 26**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

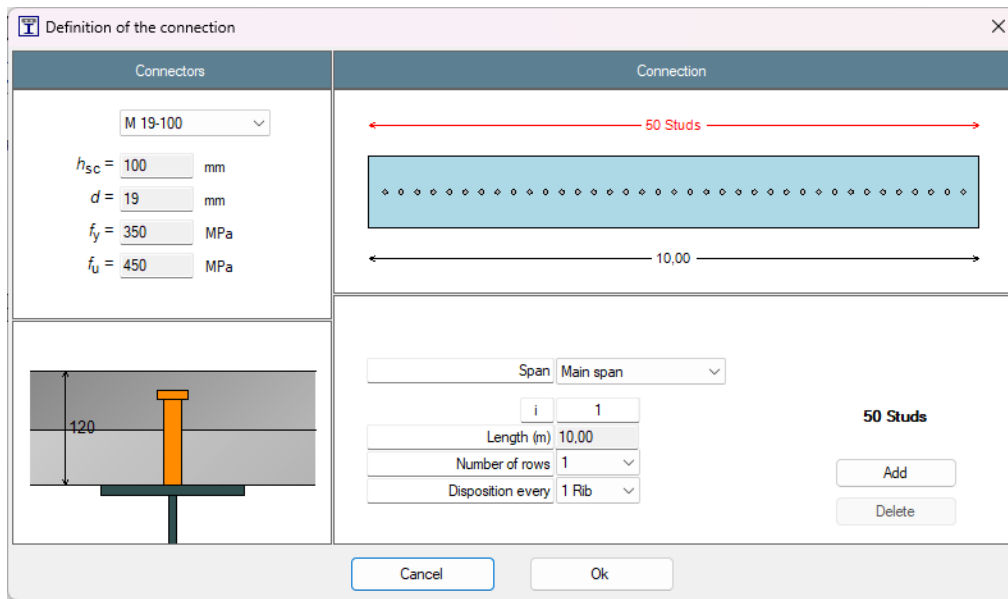
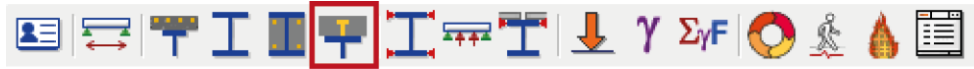


Figure 26: Definition of the connection

- (2) The definition of the connection consists of the definition of the welded studs and the connection arrangement, as follows.

4.7.2 Welded studs

- (1) The parameters describing a welded stud are as follows (**Figure 27**):

h_{sc} height of the stud
 d diameter of the shank
 f_{usc} ultimate strength of the steel

The dimensions of a stud should fulfil the following condition (see § 3.3.5):

$$h_{sc}/d \geq 3 \quad (\text{first generation of Eurocodes});$$

$$h_{sc}/d \geq 3,9 \quad (\text{second generation of Eurocodes}).$$

- (2) All the studs used in a beam are identical.
(3) The studs are defined directly from the stud's database.
(4) Studs should fulfil the following conditions for the minimum concrete coverage:

$$h_{sc} \leq t_d - c_{\min} \quad (15)$$

$c_{\min}$ is the minimal concrete coverage defined as follows:

- $c_{\min} = 20$ mm according to EN 1994-1-1: 2005, 6.6.5.2 (2);
- $c_{\min} = 15$ mm according to EN 1994-1-1: 2024, 8.6.10.2 (2).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	40	Rev.: B

If this condition is not fulfilled, a warning message is displayed. In any case, the condition $h_{sc} \leq t_d$ should be met.

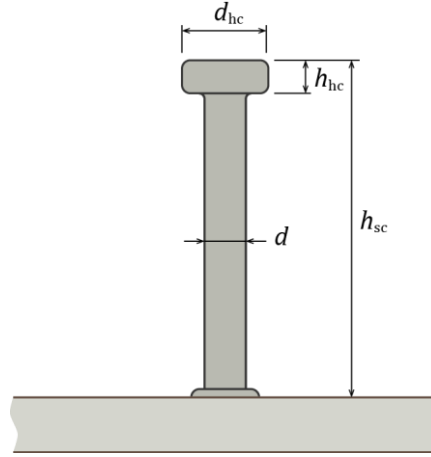


Figure 27 : Dimensions of a welded stud

4.7.3 Connection arrangement

- (1) Except in the case of an automatic process as described in § 9.4, the User should define the arrangement of the connection. The connection arrangement is continuous along each connection zone. 1 to 3 connection zones may be defined by the User.



The automatic design of the connection is not available in this version of ABC-PMX.

- (2) A connection arrangement on a connection zone (of index i) is defined by:
 - s_{xi} the longitudinal spacing between lines of connectors;
 - n_{ri} the number of connectors in each line.
- (3) For composite slabs with perpendicular ribs, with configurations PE1 or PE2 according to § 4.4.3, the spacing s_{xi} is equal to a multiple of the spacing e_p between the axes of two adjacent ribs (see § 4.5.1). In all other cases, s_{xi} should be defined directly by the User.
- (4) The maximum number of connectors in each row, $n_{ri,max}$, where $n_{ri} \leq n_{ri,max}$, is defined as follows:
 - for all types of slabs, the following relation should be fulfilled:

$$(n_{ri} - 1) s_{yi,min} + d + 2 e_{D,min} \leq b_{fs}$$
 - for all composite slabs with perpendicular ribs and a PE3 arrangement at beam support, the criterion is as follows:

$$(n_{ri} - 1) s_{yi,min} + d + 2 e_{D,min} \leq b_{fs} - 2 b_{app,min}$$
 - $n_{ri,max} = 2$ for composite slabs with perpendicular ribs using PE1 ou PE2 configurations;

where: $b_{app,min}$: see § 3.3.6 (2);

$s_{yi,min}$ is the minimum transverse spacing between two studs, see § 4.7.4 (3);

$e_{D,min}$ is the minimum distance of a connector to the edge of the flange, see § 4.7.4 (3).

4.7.4 Detailing of connectors

- (1) According to EN 1994-1-1, § 6.6.5.7, the following rules in (2) to (4) are used in the software for the spacing of headed studs (see § 3.3.5).
- (2) The longitudinal spacing s_{xi} between two lines of connectors should fulfil the following criteria:

$$s_{xi,min} = 5 d \leq s_{xi} \leq s_{xi,max} = \min\{800 \text{ mm} ; 6 t_d\} \quad (16)$$

where d is the diameter of the headed stud (see § 4.7.2)

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	41	Rev.: B

t_d is the total thickness of the slab (see § 4.4.2).

- (3) The transverse spacing s_{yi} between two adjacent connectors should fulfil the following criterion:

$$s_{yi,min} \leq s_{yi} \quad (17)$$

where: $s_{yi,min} = 2,5 d$ for solid slabs;

$s_{yi,min} = 4 d$ for composite slabs.

- (4) The distance of a connector to the edge of the flange should be less than $e_{D,min} = 20$ mm (acc. to EN 1994-1-1, 6.6.5.6 (2)).
- (5) According to EN 1994-1-1, 6.6.5.7 (5), the following criterion will be checked for usual composite beams, when the number of connectors per line, n_r , is equal or greater than 2:

$$d \leq 2,5 t_{fs} \quad (18)$$



when $n_r = 1$, it is assumed that the studs are in the axis of the web.

4.8 Compatibility of the connectors and the profiled steel sheeting

4.8.1 First generation of Eurocodes

- (1) The following criteria apply to usual composite sections.
- (2) For composite slab with transverse ribs and with the configurations PE1 or PE2 (see § 4.4.3), the following criteria should be fulfilled (see EN 1994-1-1, 6.6.4.2):
- the height of the steel profiled sheeting should not be greater than 85 mm: $h_p \leq 85$ mm;
 - the representative width of the rib, b_0 , is at least equal to h_p : $b_0 \geq h_p$;
 - for through deck welded studs (see § 4.5.3), the diameter of the stud should not be greater than 20 mm: $d \leq 20$ mm;
 - for pre-punched sheets, the diameter of the stud should not be greater than 22 mm: $d \leq 22$ mm.
- (3) For all composite slabs, the height of the stud above the steel profiled sheeting should not be lower than twice the diameter of the stud (see EN 1994-1-1, 6.6.5.8 (1)): $(h_{sc} - h_p) \geq 2 d$.
- (4) For all composite slabs, the minimum width of the ribs should not be lower than 50 mm (see EN 1994-1-1, 6.6.5.8 (2)): $\min\{b_t; b_b\} \geq 50$ mm

4.8.2 Second generation of Eurocodes

- (1) § 4.8.1 applies except the following criteria.
- (2) For composite slab with transverse ribs and with the configurations PE1 or PE2 (see § 4.4.3), the following conditions should be fulfilled (see prEN 1994-1-1, 8.6.9.2):
- the height of the steel profiled sheeting should not be greater than 105 mm: $h_p \leq 105$ mm;
 - minimum thickness of sheeting: $t_p \geq 0,7$ mm.

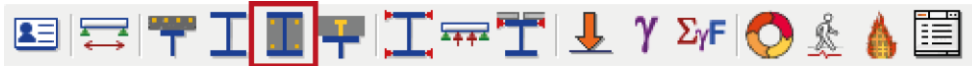


The second generation is not yet available in this version of ABC-PMX.

4.9 Partial concrete encasement

4.9.1 General

- (1) For usual steel or composite beams with partial encasement, the encasement should be defined, through the associated window (**Figure 28**), which is opened by clicking on the following button of the main toolbar (**§ 3.1.3**):



- (2) Partial concrete encasement of a steel profile is available for bisymmetrical steel profiles, either welded or hot rolled, i.e. for sections of type SHR-E, SWB-E, CHR-E and CWB-E (see **§ 4.1.1**).

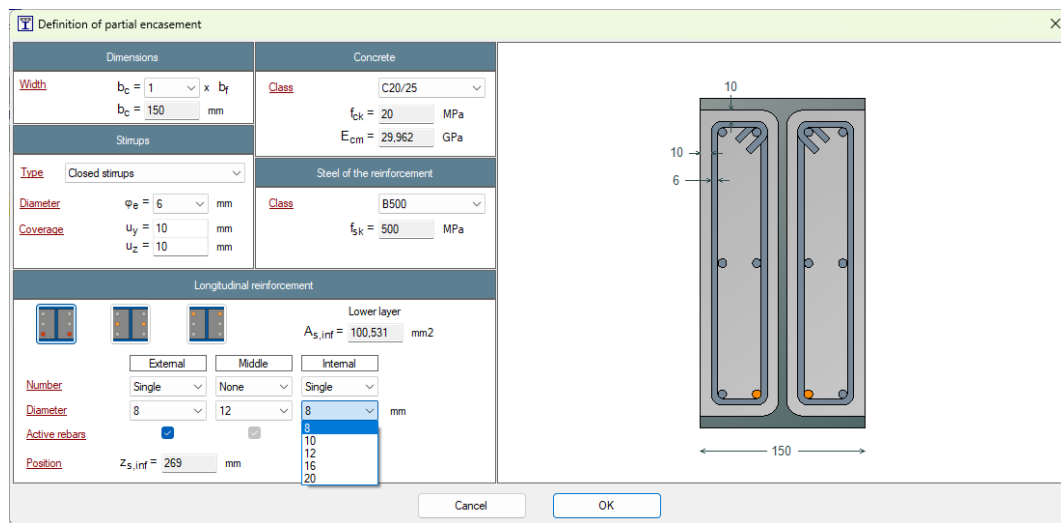


Figure 28: Definition of the partial encasement of a beam

4.9.2 Parameters

- (1) The partial concrete encasement of a steel profile should be defined by the parameters as described in **(2)** to **(12)**.
- (2) The total width of the concrete encasement is denoted b_c and should fulfil the following criterion:

$$0,8 b_{fs} \leq b_c \leq b_{fs} = b_{fi} \quad (19)$$

- (3) The concrete of the encasement is defined by its concrete class (acc. to **Table 14**) and by the mass of concrete per unit of volume, ρ_{ce} (in kg/m^3 , default value of 2500 kg/m^3). See **§ 5.2** for the properties of concrete.
- (4) The characteristic value of the compression strength of the concrete for the encasement is denoted f_{cek} (see **§ 5.2**). Only normal concretes are available for partial encasements.



Concrete for the encasement and concrete for the slab are defined separately and may be different.

- (5) The reinforcement of the encasement is defined by the stirrup, the longitudinal rebars and the concrete cover.
- (6) The diameter of the stirrups d_{est} is chosen among the following list: 6; 8, 10, 12, 14 and 16 mm. The location of the stirrup is defined by the concrete cover to the external side, c_{est} , and the concrete cover to the internal side of the flanges, c_{fst} . These concrete coverings are defined by the User.



Stirrups are not used for the resistance of the cross-section. They only influence the location of longitudinal rebars.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	43	Rev.: B

- (7) The longitudinal reinforcement of a chamber consists of one to three layers. The upper and lower layers are located at the extreme locations that fulfil the geometric interaction with stirrups and the concrete covering (**Figure 29**).
- (8) Each layer consists of one to three rebars (number n_{esi}) per chamber, of identical diameter d_{esi} . The diameter of rebars is to be selected among the following list: 6; 8, 10, 12, 14, 16, 20, 25 and 30 mm. Each layer may have a specific diameter.
- (9) The location of the lower layer (index $i = 1$) of reinforcement is automatically calculated as follows:

$$z_{es1} = -h_s + t_{fi} + c_{fst} + d_{est} + d_{es1} / 2 \quad (20)$$

- (9) The location of the upper layer (index $i = 3$) of reinforcement, if any, is automatically calculated as follows:

$$z_{es3} = -t_{fs} - c_{fst} - d_{est} - d_{es3} / 2 \quad (21)$$

- (10) The location of the middle layer (index $i = 2$) of reinforcement, if any, is defined by the User. It should fulfil:

$$|z_{es2} - z_{esi}| \geq (d_{es2} + d_{esi}) / 2 + \max\{d_{es2}, d_{esi}\} \quad (22)$$

- (11) The number of rebars n_{esi} in each layer i of reinforcement, should fulfil the following criterion:

$$c_{est} + 2 d_{est} + n_{esi} d_{esi} \leq (b_c - t_w) / 2 \quad (23)$$

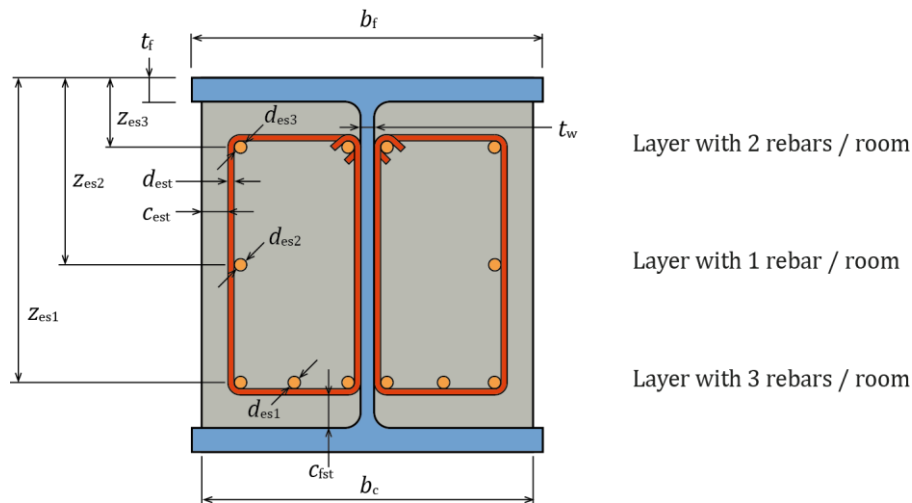


Figure 29 : Definition of the partial concrete encasement of steel profiles

- (12) The class of reinforcing steel is to be chosen among the following list: B450, B500.
The same steel properties apply to all the reinforcements of the partial concrete encasement.
See § 5.3 for the properties of the steel for reinforcements.
- (13) For the Ultimate limit state in normal situation, an option proposes to the User to take into account the concrete and reinforcement of the partial concrete encasement (see § 4.17.2). By default, the option is deactivated.



It is assumed that the partial encasements are meant mainly for fire protection.

- (14) For all rebars and especially for those in the corners of the stirrups, which are mandatory for constructive purposes, it is possible to specify if they are to be considered in the calculations (option “active rebar”).
- (15) The area of concrete in the partial encasement is given by:

$$A_{ce} = b_f h_a - A_a - (b_f - b_c) h_w \quad (24)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	44	Rev.: B

4.10 Lateral restraint of steel beams

4.10.1 General

- (1) Lateral restraints may be defined for usual steel beams, with or without partial concrete encasement. Lateral restraints may also be defined for usual composite beams, with or without partial concrete encasement, to deal with the construction stage only.
- (2) The lateral restraints are defined through the associated window (**Figure 30**Figure 28), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

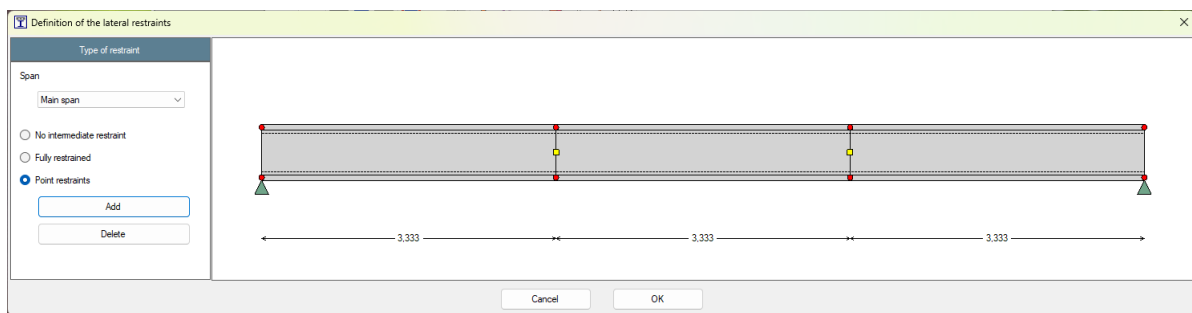
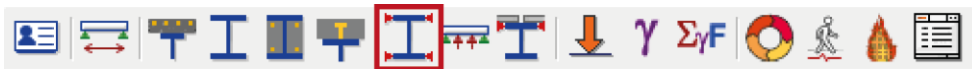


Figure 30 : Definition of the lateral restraints of a steel profile

- (3) Lateral restraints may consist of a continuous restraint of one or both flanges. It may also be constituted by a set of cross-sections with point lateral restraints (see § 4.10.2).
- (4) Lateral restraints are not used for the check of usual composite beams at ultimate limit states in normal conditions. The restraints defined by the User are used only for the checks at ultimate limit states for the construction stage of those beams.

For the construction stage of usual composite beams, if the slab is composite with perpendicular ribs, it is also possible to activate the profiled steel sheeting as a contribution to the lateral restraint of the beam (see § 4.11).

4.10.2 Point lateral restraints

- (1) Point lateral restraints are defined as follows, for each span:
 - n_{slr} number of sections with lateral restraint;
 - $x_{slr,i}$ location of the cross-section with lateral restraints, from the left end of the span;
 - $typ_{slr,i}$ type of restraint: both flanges, upper flange or lower flange (by default, both flanges)
- (2) By default, no intermediate lateral restraint is present.
- (3) Cross-sections at supports are always assumed to be laterally restrained, for both flanges.
- (4) No lateral restraints are assumed by default at the cross-sections with point props.

4.11 Lateral restraints of composite beams during the construction phase

4.11.1 General

- (1) For a beam with usual composite cross-section, with or without partial concrete encasement, the User can choose to provide a restraint against lateral torsional buckling by the stabilising action of the profiled steel sheeting. This option is available only when the slab is composite and when the ribs are perpendicular to

the beam axis. It is used only for the construction stage for unpropped or point propped configurations (see § 4.10).

- (2) The parameters defining the lateral restraints of the beam in construction phase by the profiled steel sheeting are defined through the associated window (**Figure 31**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

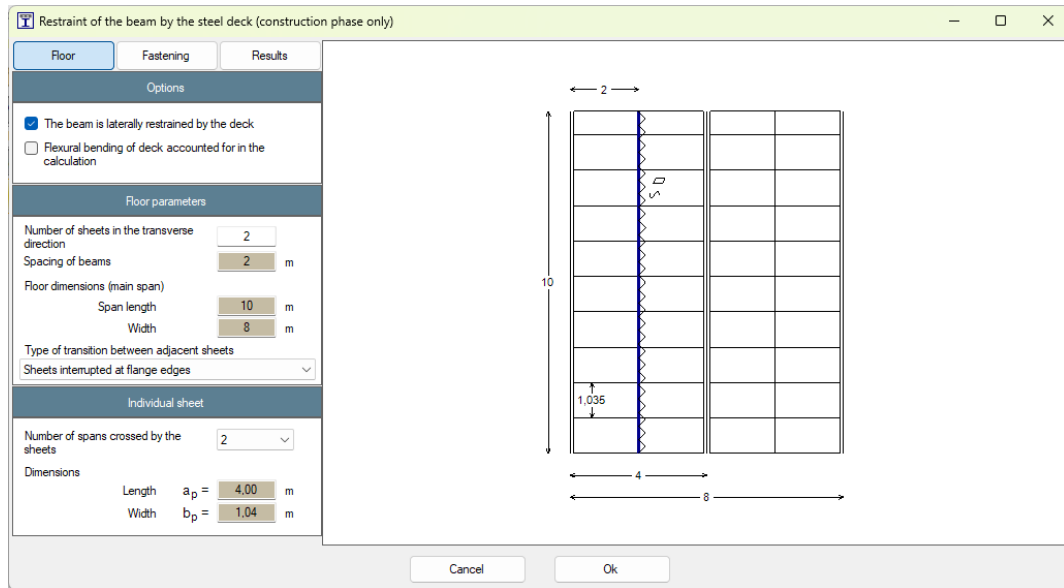


Figure 31 : Definition of the restraints of a profile in construction phase by the steel sheeting

4.11.2 Parameters of the restraint by the steel profiled sheeting

- (1) When the option is activated, the following parameters should be defined by the User (**Figure 32**):
- n_t total number of sheets of the deck in the direction perpendicular to the beam;
 - m number of spans crossed by an individual sheet, chosen in the list: 1, 2, 3, 4, 5;
 - t_{pr} thickness of the coating applied to the sheets (see § 8.5.7 (9)).
- (2) The width of a single sheet, b_p , is defined in the database for the steel profiled sheeting.
- (3) When $n_t > 1$, the User should define the type of transition between sheets at beam support, among one of the following options (**Figure 33**):
- overlapping of sheets;
 - sheets with a gap;
 - adjacent sheets without a gap.

This choice is to be done only for the configuration PE1 according to **Figure 17**. For the two other configurations, the automatic following association is done by **ABC-PMX**:

- configuration PE2 => adjacent sheets without a gap;
- configuration PE3 => sheets with a gap.

Moreover, for a configuration PE1, the value $n_t = 1$ is not consistent.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	46	Rev.: B

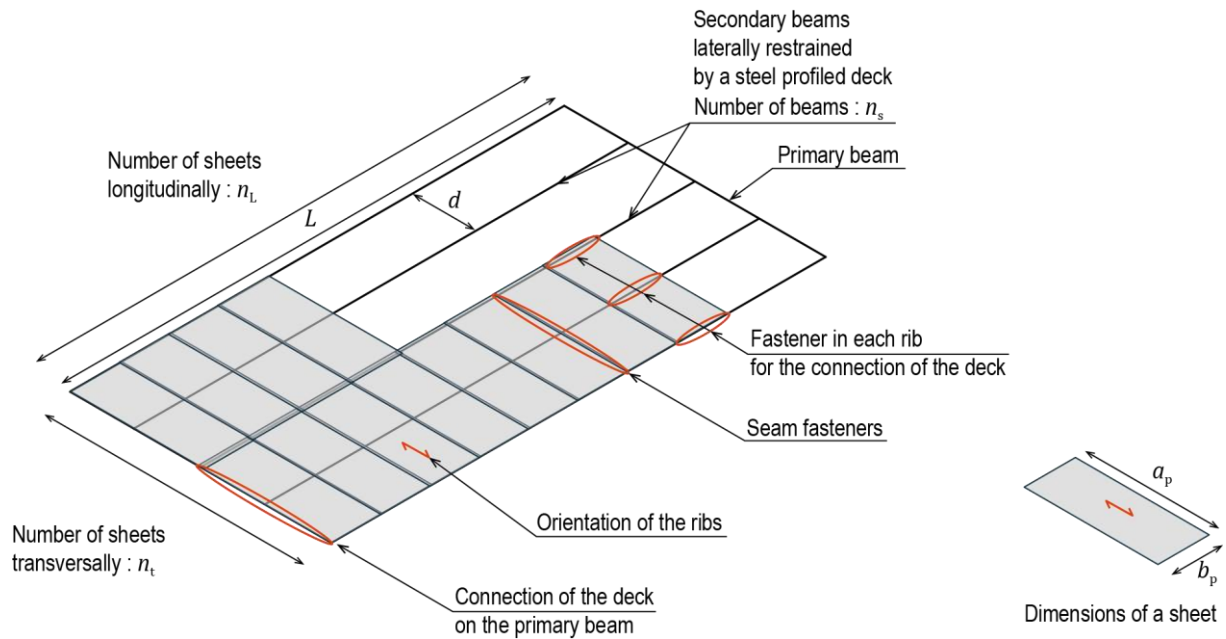


Figure 32 : Definition of the parameter of the steel decking

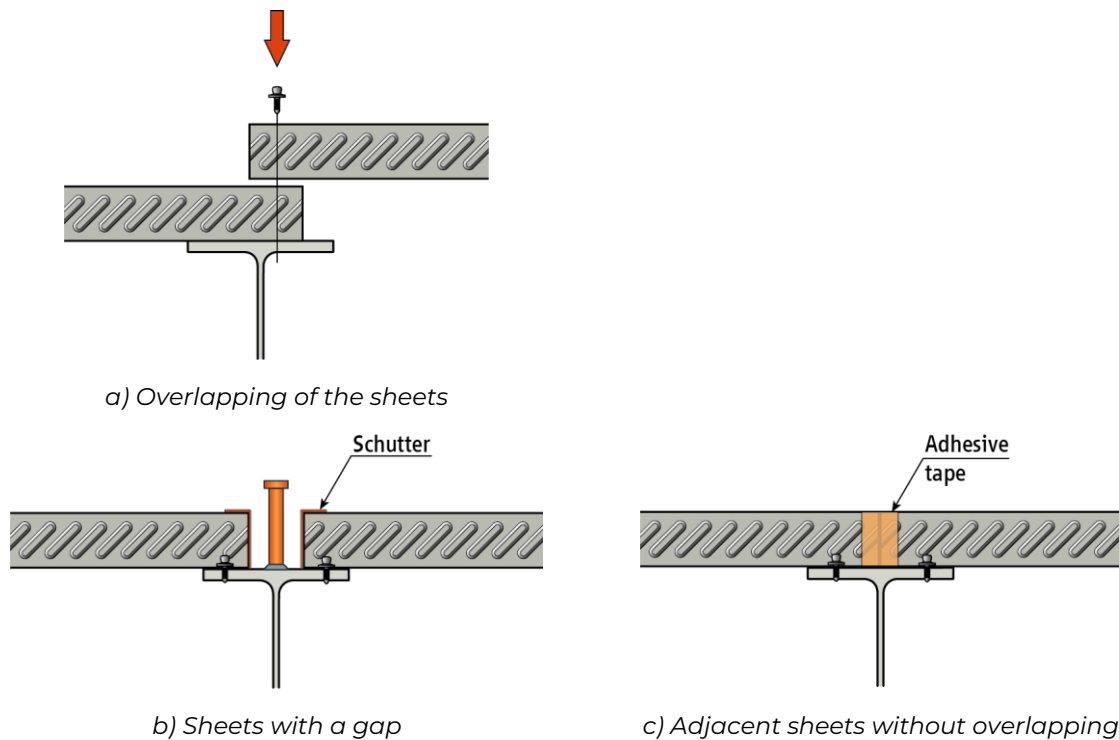


Figure 33: Transition between sheets at a beam support

- (4) The method applied to consider the effect of the profiled steel sheeting on the lateral torsional buckling of the beam (see **Annex E**) assumes that the steel deck connects together several regularly spaced secondary beams. The constant spacing of these beams is denoted d .

The distance d is obtained as follows:

- for edge beams (see § 4.1): $d = d_2$;
- for intermediate beams: $d = (d_1 + d_2) / 2$

If for intermediate beams $|d_1 - d_2| > 0,25 \max\{d_1 ; d_2\}$, the method is theoretically not valid. A warning message is displayed to notify the User that the floor is outside the scope of application of the method.

- (5) Parameters for the fastening of the sheeting to the beams are as follows. The User should define if the sheets are fastened on the beams by one fastener in each rib or one fastener every 2 ribs.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	47	Rev.: B

e_L is the longitudinal spacing between these fasteners (**Figure 34**):

$e_L = e_p$ if there is one fastener in each rib;

$e_L = 2 e_p$ if there is one fastener every two ribs.

The type of fasteners and their diameter are chosen according to **Table 10**.

Fastener	Type	Diameter d_{fb} (mm)	s_p (mm/kN)
Self-drilling screws	Normal washer	from 5,5 to 6,3	0,15
	Neoprene washer	from 5,5 to 6,3	0,35
Fired pins	washer+pin	from 3,7 to 4,8	0,10

Table 10: Definition of the beam-to-sheeting fasteners

- (5) Parameters for the fastening of the sheeting between adjacent sheets are as follows. The User should define e_c , the spacing of seam fasteners used for the connection between two adjacent sheets (**Figure 34**).

e_c should fulfil the following condition: $e_c \leq e_{c,max} = 500$ mm.

The type of fasteners and their diameter should be chosen according to **Table 11**.

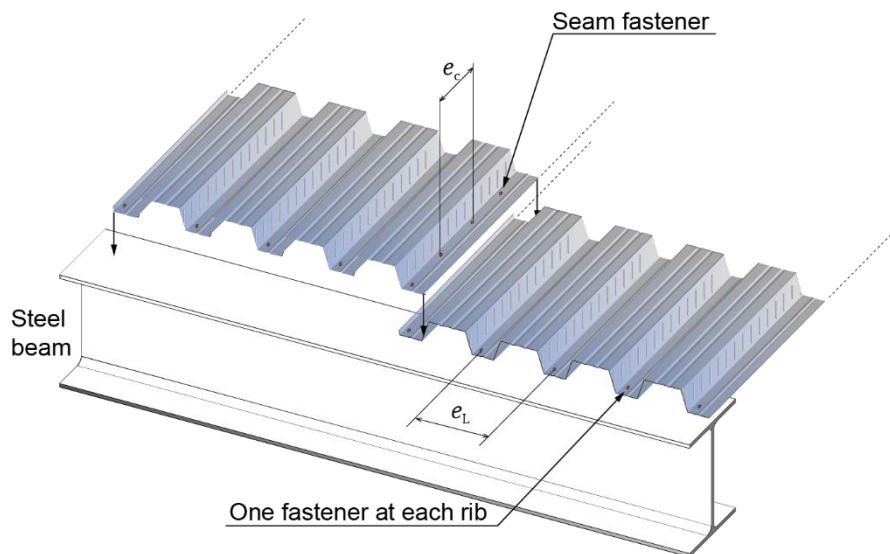


Figure 34: Sheet/girder fasteners and seam fasteners

Fastener	Diameter d_{fs} (mm)	s_s (mm/kN)
Self-drilling screws	4,1 à 4,8	0,25
Rivet	4,8	0,30

Table 11: Definition of the seam fasteners

- (6) The following additional parameters are calculated from the values defined by the User:

n_L is the number of sheets in the longitudinal direction, given by: $n_L = L / b_p$;

n_c is the number of seam fasteners along the deck span, excluding those which are fixed to the beams. It may be obtained by: $n_c = (d / e_c) - 1$;

n_f is the number of fasteners on the beam per individual sheet, which is given by:

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	48	Rev.: B

$$n_f = b_p / e_p \quad \text{if there is one fastener in each rib;}$$

$$n_f = b_p / (2 e_p) \quad \text{if there is one fastener every two ribs.}$$

- (7) As a calculation option, the User can choose to consider only the shear stiffness of the deck or the shear stiffness and the rotation restraint of the beam.

In the latter case, the second moment of area per unit of length, I_{pu} , is used (see § E.4 (2)). This parameter is included in the database for steel profiled sheeting (see § 4.4.3).

4.12 Propping

4.12.1 General

- (1) For all composite beams, the propping during the construction phase is defined through the associated window (**Figure 35**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

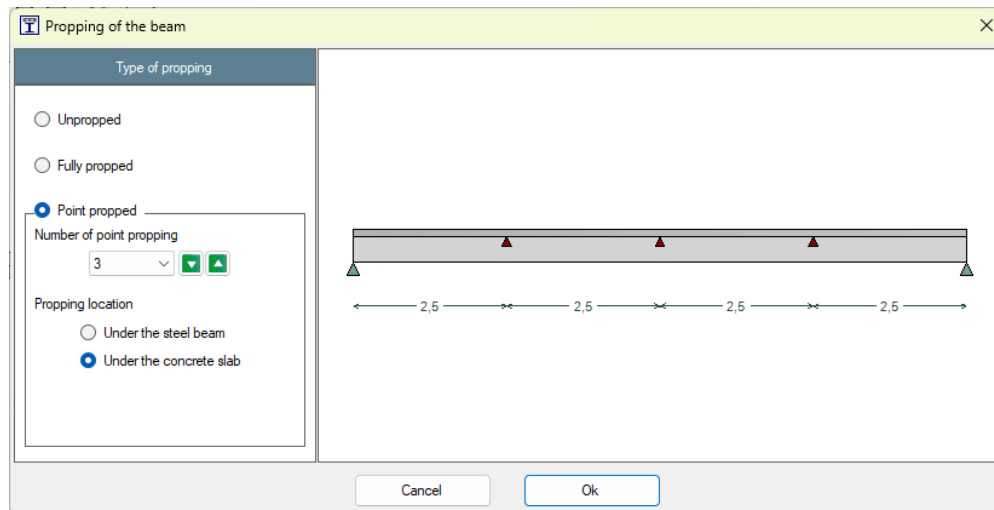


Figure 35: Definition of the propping

4.12.2 Effect of the propping on the beam design

- (1) For all beams having a composite cross-section (according to (3) for usual sections and (5) for slim floors of § 4.1.1), the User should define the type of propping considered at the construction stage (during the pouring of concrete), among the following list:
- un-propped beam (by default);
 - fully propped beam;
 - point propping of the beam, to be defined by the User.
- (2) For un-propped beams:
- the G_1 loads are resisted by the steel profile alone;
 - the resistance of the beam is checked at ultimate limit state during the construction stage, considering the steel cross-section, according to § 8;
 - the resistance of the beam is checked at ultimate limit state for the final stage, considering the composite action, according to § 8.
- (3) For fully propped beams:
- the G loads are resisted by the beam with composite effect between the concrete slab and the steel profile, considering a modular ratio for long term effects;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	49	Rev.: B

- the resistance of the beam is checked at ultimate limit state for the final stage only, considering the composite action, according to § 8.
- (4) For propped beams with temporary point supports:
- the G_1 loads are decomposed in two phases (see § 7.7.1): during a first period, for the pouring of concrete, the G_1 loads are supported by the steel profile alone ($G_{1,pp}$), considering a continuous beam with n_{pp} intermediate supports. A second period, after the removal of the propping, activates the composite action, with a modular ratio for long terms effects. The loads applied to this load case are the reactions at intermediate temporary support obtained from the first phase ($G_{1,c}$);
 - the resistance of the beam is checked at ultimate limit state during the construction stage, considering the steel cross-sections, according to § 8;
 - the resistance of the beam is checked at ultimate limit state for the final stage, considering the composite action, according to § 8.
- (5) For fully propped (see (3)) or partially propped (see (4)) beams, the time $t_{0,1c}$ when the temporary supports are removed, is assumed to be equal to 28 d.
- (6) If $t_{0,1c} < 28$ days, an additional stage at the time t_u should be considered for the verification at ultimate limit state, considering the composite actions with reduced properties of concrete. It is not covered by **ABC-PMX**.
- (7) For cross-section with partial concrete encasement, § 4.13.2 (4) applies.

4.12.3 Definition of the point propping

- (1) For the main span, the User can define the number of point-props, n_{pp} , that are uniformly distributed along the span length (Figure 36).
- (2) For cantilever spans, the User can only choose to prop the end of the span.

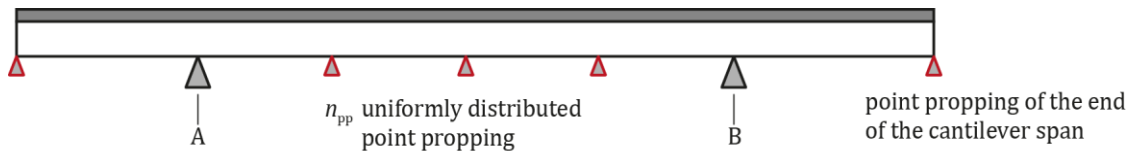


Figure 36 : Definition of the point props

- (3) For the main span, the location of the i^{th} temporary support to the left intermediate support of the beam (A) is given by:
- $$x_{pi} = i L / (n_{pp} + 1)$$
- (4) An option is provided to choose the location of the temporary supports, either below the lower flange or below the lower side of the slab. The latter option is a better solution for the stability and is set by default. No lateral torsional buckling occurs in this case.

4.13 Load cases

4.13.1 General

- (1) For all types of beams, the load cases are defined through the associated window (**Figure 37**), which is opened by clicking on the following button of the main toolbar (**§ 3.1.3**):

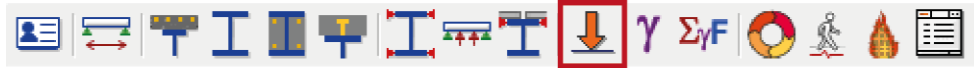
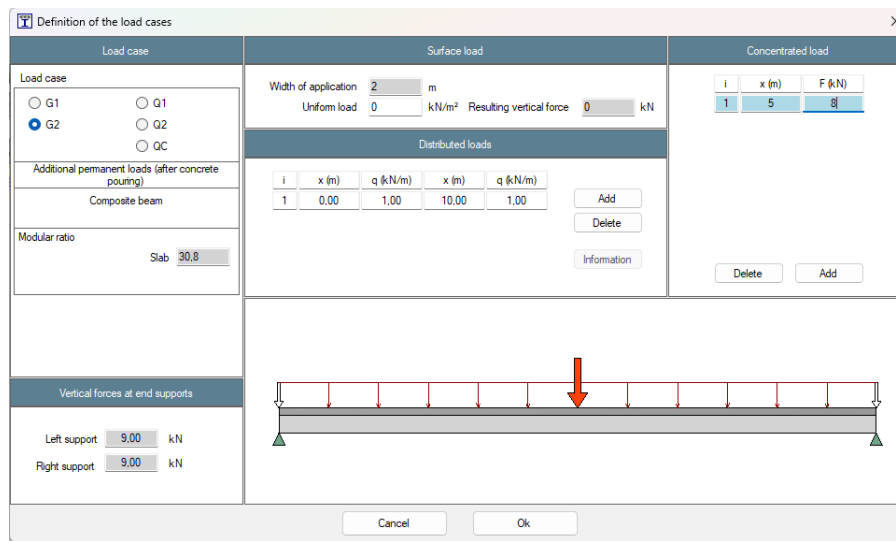



Figure 37: Definition of User load cases

- (2) For each load case, the state of the beam (composite or not) as well as the value of the associated modular ratios are displayed, when relevant.
- (3) For each load case, the vertical reaction forces at end support are given.

4.13.2 Type of loading

- (1) For each beam, it is possible to define 4 or 5 load cases, as follows:
- one load case Q_c for the construction phase (composite beams only);
 - one load case G for the permanent loads (steel beams);
 - one load case G_1 for the permanent loads which already act on the beam when concrete is poured (composite beams);
 - one load case G_2 for the additional permanent loads which act on the beam only after the concrete has been poured (composite beams);
 - two load cases Q_1 and Q_2 for the live loads.

G is defined only for usual steel sections or for fully propped composite beams. G_1 and G_2 are defined only for usual composite sections unpropped or with point proppings.

Concrete is always assumed to be poured in one phase only.



Part of the load cases is automatically defined by ABC-PMX. See § 7.6.

- (2) The load case Q_c is considered for beams with composite sections (according to (3) for usual sections and (5) for slim floors of § 4.1.1) and without continuous propping (see § 4.12).
- (3) Load case G_1 acts according to the propping – see § 4.12.2 (2) to (4).
Load case G_2 acts in the composite phase (long term actions).
- (4) For sections with partial concrete encasement, either steel or composite, it is assumed that the concrete of the encasement is poured at least 28 days before the installation on site and that it may contribute to

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	51	Rev.: B

the stiffness and the resistance of the cross-section (if the relevant option is activated – see § 4.9 (13) and § 4.17).

- (5) For composite sections, both load cases Q_1 and Q_2 act with composite action.
- (6) For load case G_2 , which only acts in composite phase, **ABC-PMX** uses the time $t_{0,G2}$ at first loading of permanent load case, in days, by default equal to 28 d.
- (7) For shrinkage load case (see § 7.7.1), the time $t_{0,SH}$ at first loading is assumed to be equal to $t_{0,SH} = 1$ d.
- (8) Time t of the concrete at the moment considered for the design checks is defined for a default value associated to a life time of 50 years, i.e. $t = 18\,250$ d.
- (9) See **Table 28** for the parameters associated to load cases.
- (10) See § 7.7.4 for the live loads arrangements. See § 7.6 for the list of load cases considered in the calculations.

4.13.3 Definition of the loadings

- (1) Any load case is defined by the following sets of forces, to be defined by the User (**Figure 38**):
 - a surface load Q_L ;
 - a set of point loads, defined by a location $x_{L,Fi}$ and a force F_{Li} . The location is measured from the left end of the span;
 - a set of distributed loads, defined by the locations $x_{L,1qi}$ and $x_{L,2qi}$ between which the load is applied and by the loads $q_{L,1i}$ and $q_{L,2i}$ at each end of the loaded segment.
- (2) By default, no load is applied, except for the dead load (load case G or G_1).
- (3) For the load case G or G_1 , the self-weight of the steel profile, the concrete slab, the partial concrete encasement and the steel profiled sheeting is automatically assessed (see § 7.7.2). The self-weight of reinforcements is not explicitly assessed and is assumed to be included in the weight of the concrete.

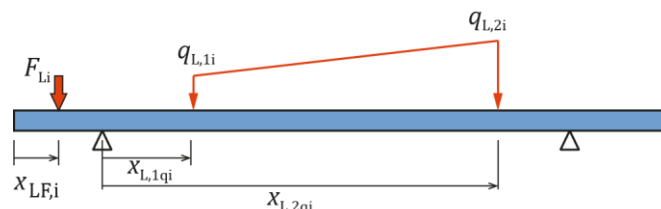
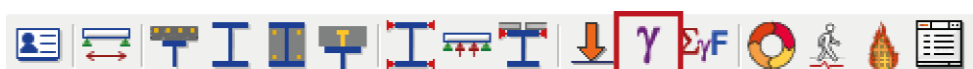


Figure 38 : Definition of loadings

4.14 Nationally Determined Parameters

4.14.1 General

- (1) For each composite beam analysed with the software, the User has the possibility to define the nationally determined parameters considered in the calculations. The default values are the values defined in the options of the software – see § 3.3.2.
- (2) The national parameters are defined through the associated window (Figure 39), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):



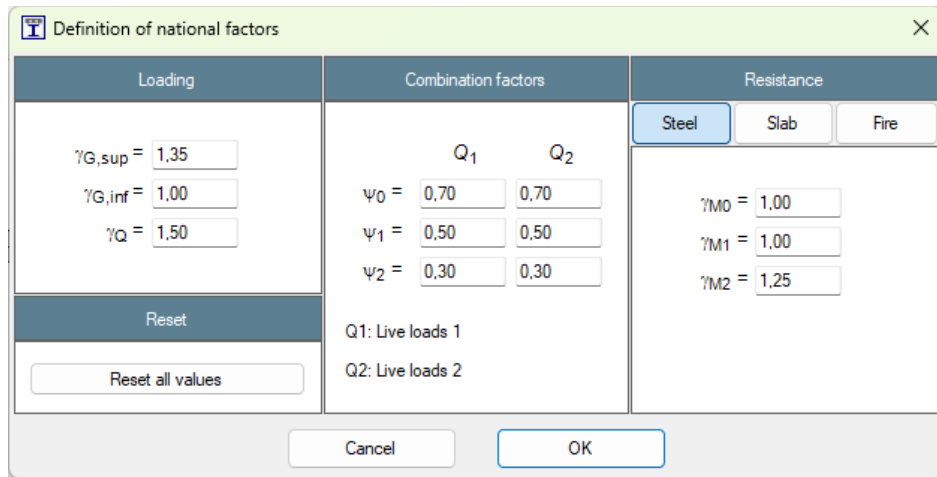


Figure 39: Definition of national parameters

4.14.2 Partial factors

- (1) The following partial factors on actions for ULS combinations are used:
 - $\gamma_{G,sup}$, $\gamma_{G,inf}$ for permanent actions;
 - γ_Q for variable actions;
- (2) The partial factors for design resistances in normal situations are:
 - γ_{M0} for the section resistance of the structural steel;
 - γ_{M1} for the element resistance of the structural steel;
 - γ_{M2} for the element resistance of the welds;
 - γ_C for the compression resistance of the concrete;
 - γ_V for the resistance of shear connectors;
 - γ_s for the resistance of reinforcement steel bars;
 - γ_p for the resistance of the profiled steel sheeting.
- (3) For the resistance of shear connectors, it is possible to define the use of one common partial factor γ_V in both formulas **(82)** and **(83)** or one separate value for each formula, denoted respectively γ_{Vs} and γ_{Vc} .
- (4) In fire situations, the following partial factors on resistances are considered:
 - $\gamma_{M,fi}$ for the resistance of structural steel (steel beams);
 - $\gamma_{M,fi,a}$ for the resistance of structural steel (composite beams);
 - $\gamma_{M,fi,c}$ for the compression resistance of the concrete;
 - $\gamma_{M,fi,V}$ for the resistance of shear connectors;
 - $\gamma_{M,fi,s}$ for the resistance of reinforcement steel bars.

4.14.3 Combination factors

- (1) For each live load case Q_1 and Q_2 , the three combination factors ψ_0 , ψ_1 and ψ_2 should be defined.

4.15 Combination of loads

4.15.1 General

- (1) For all types of beams, the combinations of loads are defined through the associated window (**Figure 40**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

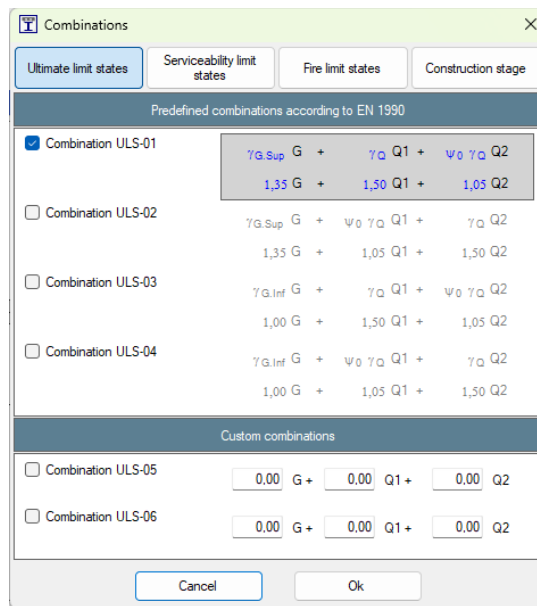
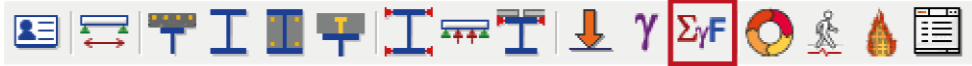


Figure 40: Definition of the combinations of loads

4.15.2 Ultimate Limit States (ULS) – Persistent or transient situations

- (1) It is possible to define up to 6 ULS load combinations: 4 predefined combinations proposed by the program and two custom combinations defined by the User.
- (2) By default, the four predefined following ULS load combinations are automatically generated and proposed by **ABC-PMX** (Expressions 6.10 of EN 1990 [1]):

- When the permanent loads have unfavourable effects:

$$Q_1 \text{ is the basic variable action: } \gamma_{G,sup} G + \gamma_Q Q_1 + \psi_{0,2} \gamma_Q Q_2$$

$$Q_2 \text{ is the basic variable action: } \gamma_{G,sup} G + \gamma_Q Q_2 + \psi_{0,1} \gamma_Q Q_1$$

Without variable actions, the two later resumes as follows: $\gamma_{G,sup} G$

- When the permanent loads have favourable effects:

$$Q_1 \text{ is the basic variable action: } \gamma_{G,inf} G + \gamma_Q Q_1 + \psi_{0,2} \gamma_Q Q_2$$

$$Q_2 \text{ is the basic variable action: } \gamma_{G,inf} G + \gamma_Q Q_2 + \psi_{0,1} \gamma_Q Q_1$$

Without variable actions, the two later resumes as follows: $\gamma_{G,inf} G$

For composite beams, unpropped or point propped during construction, G is replaced by $(G_1 + G_2)$ in the previous combinations.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	54	Rev.: B

- (3) For a composite beam at the construction stage, without continuous propping (see § 4.12), only one predefined load combination is considered:

$$\gamma_{G,\text{sup}} G_1 + \gamma_Q Q_c$$

It is also possible to add a custom combination defined by the User.

4.15.3 Serviceability Limit States (SLS)

- (1) It is possible to define up to 6 load combinations for the calculations at the Serviceability Limit States: 4 predefined combinations proposed by the program and two custom combinations defined by the User.
- (2) By default, the four predefined following SLS load combinations are proposed by the program (Expressions 6.14b of EN 1990 [1]), for characteristic combinations:

Q_1 is the basic variable action: $G + Q_1$

$$G + Q_1 + \psi_{0,2} Q_2$$

Q_2 is the basic variable action: $G + Q_2$

$$G + Q_2 + \psi_{0,1} Q_1$$

For composite beams, unpropped or point propped during construction, G is replaced by $(G_1 + G_2)$ in the previous combinations.

- (3) For a composite beam without continuous propping, only one predefined SLS load combination is considered at the construction stage:

$$G_1 + Q_c$$

It is also possible to add a custom combination defined by the User.

- (4) When a control of cracking is performed for beams with cantilevers, quasi-permanent SLS combinations are automatically considered for the calculation of the tension stresses in the reinforcements (see § 11.4.4). These combinations are defined by:

$$G + \psi_{2,1} Q_1 + \psi_{2,2} Q_2$$

4.15.4 Ultimate limit state – Fire situation

- (1) It is possible to define up to 5 combinations of loads for the limit states associated to the accidental fire situation. 3 combinations are predefined according to Equation (6.11a) of EN 1990:

$$G + \psi_{1,1} Q_1 + \psi_{2,2} Q_2$$

$$G + \psi_{2,1} Q_1 + \psi_{2,2} Q_2$$

$$G + \psi_{2,1} Q_1 + \psi_{1,2} Q_2$$

For composite beams, unpropped or point propped during construction, G is replaced by $(G_1 + G_2)$ in these combinations.

- (2) The last 2 possible combinations of loads are directly defined by the User.
- (3) By default, all 5 combinations of loads for the fire situation are deactivated. Fire calculations are carried out by the software only if at least one of the combinations of loads for the fire situation is activated.

4.16 Vibratory comfort assessment

4.16.1 General

- (1) For all types of beams, the parameters for the vibratory assessment of the beam are defined through the associated window (**Figure 41**), which is opened by clicking on the following button of the main toolbar (**§ 3.1.3**):

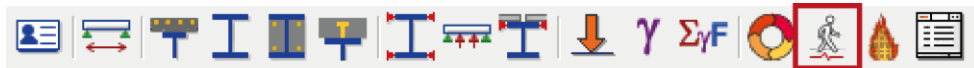
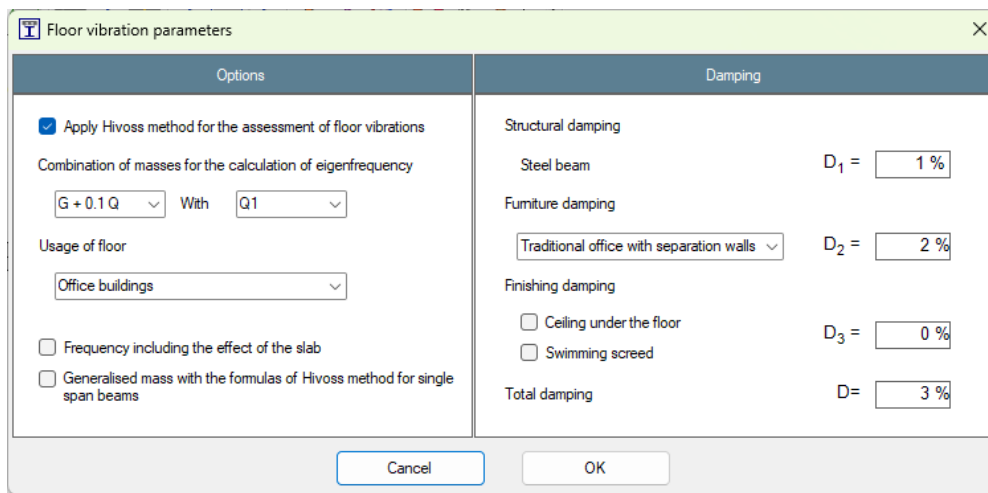



Figure 41: Definition of the parameter for the vibratory assessment

4.16.2 Parameters

- (1) The User should define the following parameters for the application of the Hivoss method (see **§ 11.3**):
- Application of the HiVoSS method (Yes by default);
 - Selection of the combination of mass to be considered, among the following list:
 - G
 - $G + 0,1 Q$
 - $G + 0,2 Q$
 - $G + 0,3 Q$
 - $G + 0,4 Q$
 - $G + 0,5 Q$
 where G means $G_1 + G_2$ when relevant and Q is to be chosen by the User as either Q_1 , Q_2 or $Q_1 + Q_2$;
 - Definition of the use of floors for the damping D_2 (by default office buildings) among the following list:
 - Critical areas,
 - Hospitals, surgeries,
 - Schools, training centres,
 - Residential buildings,
 - Office buildings,
 - Meeting rooms,
 - Senior citizen residential buildings,
 - Hotels,
 - Industrial workshops,
 - Sport facilities.
 - Presence of a ceiling under the floor (No by default). (If yes, 1% is added to D_3);
 - Presence of a swimming screed (No by default). (If yes, 1% is added to D_3).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	56	Rev.: B

- Definition of the type of furniture for the floor, among the following list:
 - Traditional office with separation walls,
 - Paperless office,
 - Open plan office (by default),
 - Library,
 - Houses,
 - Schools,
 - Gymnasium.

The latter choice influences the definition of damping D_2 . It is possible for the User to define its own value of D_2 damping.

The total value of the damping is displayed.

- (2) For single span beams, the User can specify if the modal mass is obtained by the simplified formula **(234)** or by the general method with formula **(235)**. See § 11.3.2.

4.17 Calculation options

4.17.1 General

- (1) For all types of beams, the options for the calculation of the beam are defined through the associated window (**Figure 42**), which is opened by clicking on the following button of the main toolbar (§ 3.1.3):

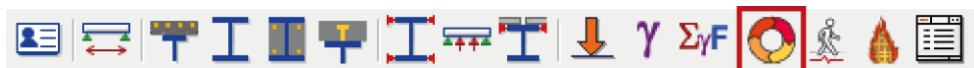
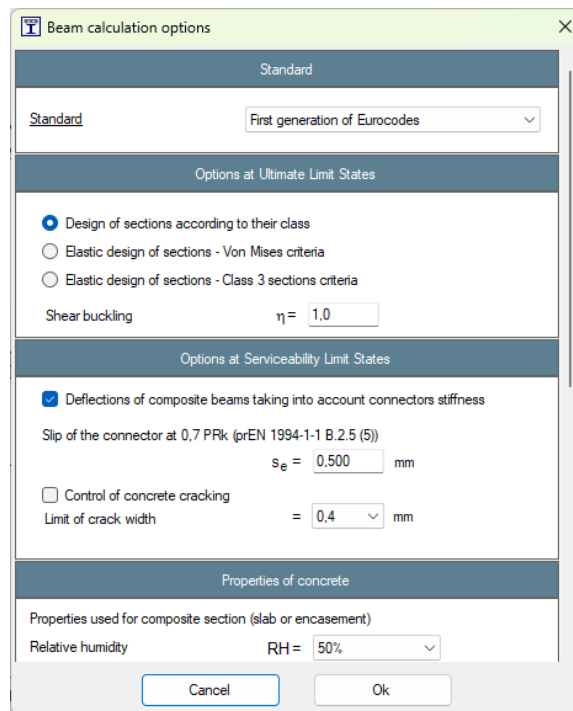



Figure 42: Calculation options

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	57	Rev.: B

4.17.2 Options

Following options may be defined by the User:

Ultimate Limit States:

The cross-sections may be designed according to one of the following options:

- they are designed according to their Class (by default);
- an elastic design using the von Mises criteria is used, whatever the class of the sections is;
- an elastic design based on Class 3 criteria is used, whatever the class of the sections is.

The value of the parameter η that is used for the check of the resistance to shear buckling (see § 8.3) may be defined. The default value is as defined in the software options (see § 3.3.4).

Serviceability Limit States:

The User can specify if the deflection of the beam should consider the flexibility of the connectors. In this case, the value of the parameter s_e should be defined. s_e is the slip of the connector at load $0,7 P_{Rk}$ (see prEN 1994-1-1:2024, B.2.5 (6)). See § 11.1 for the influence of s_e on the deflection of the beam.

The User can also specify if the control of cracking should be performed (see § 11.4). In this case, the value of the maximal width of cracks should be defined (to be chosen among the predefined values of Table 9.2 in EN 1994-1-1: 0,2 mm, 0,3 mm and 0,4 mm.).

Properties of concrete:

For the calculation of the section properties of composite beams, the User may define the following options for the concrete:

- the relative humidity RH , either 50% or 80% (see § 6.3);
- the deformation of concrete due to shrinkage ε_{sh} (see § 7.7.3). The default value is 325×10^{-6} , according to the value defined by Annex C of En 1994-1-1 for normal concrete in dry environments,
- the value of the Young's modulus for the reinforcements. By default, the same value as for the structural steel is considered (i.e. $E_s = E_a = 210\,000$ MPa), but a value between $190\,000$ MPa and E_a may be chosen (see § 5.3).

The values of ages t_0 and t (see § 6.3) are displayed.

Properties of cross-sections:

For the calculation of the section properties, the User may define:

- if a simplified distribution of the effective width of the slab is considered, either according to EN 1994-1-1, clause 5.4.1.2 (4) for the analysis or according to clause 6.1.2 (2) for the section resistances,
- if rebars are considered in compression. By default, they are ignored. The User should be aware that when the rebars in compression, requirements of EN 1992-1-1 9.2.1.2 (3) should be fulfilled in order to prevent the buckling of rebars. This option should not be selected with composite slabs.
- is the concrete and rebars of partial encasements should be considered in the calculations at room temperatures.

Parameters:

The User can define the value of the G-force, either 9,81 or 10 m/s².

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	58	Rev.: B

4.18 Fire design options

4.18.1 General

- (1) For all types of beams, the options for the calculation of the beam are defined through the associated window (**Figure 43**), which is opened by clicking on the following button of the main toolbar (**§ 3.1.3**):

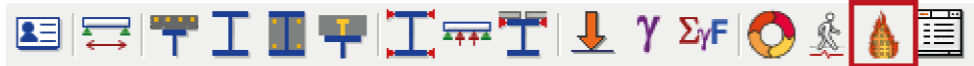
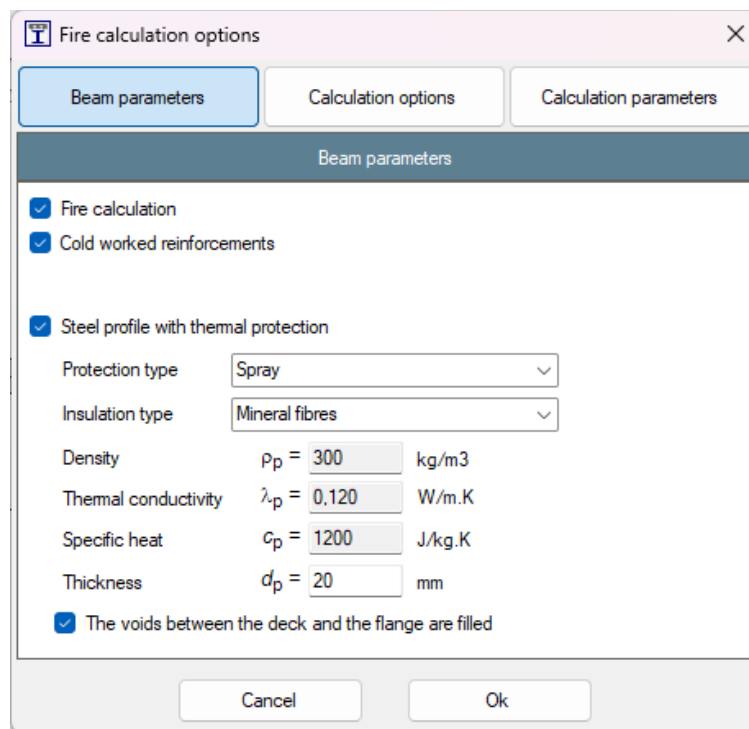



Figure 43: Fire design options

4.18.2 Beam parameters

- (1) The User selects the option to activate the fire design of the beam. When the option is activated, the User should define specific parameters.
- (2) For usual steel or composite sections, the steel profile can be either unprotected or protected. It is not possible to define a fire protection for partially encased profiles.
- (3) The protection of a steel profile may be obtained by:
 - spray;
 - intumescent paint;
 - boards (or panels).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	59	Rev.: B

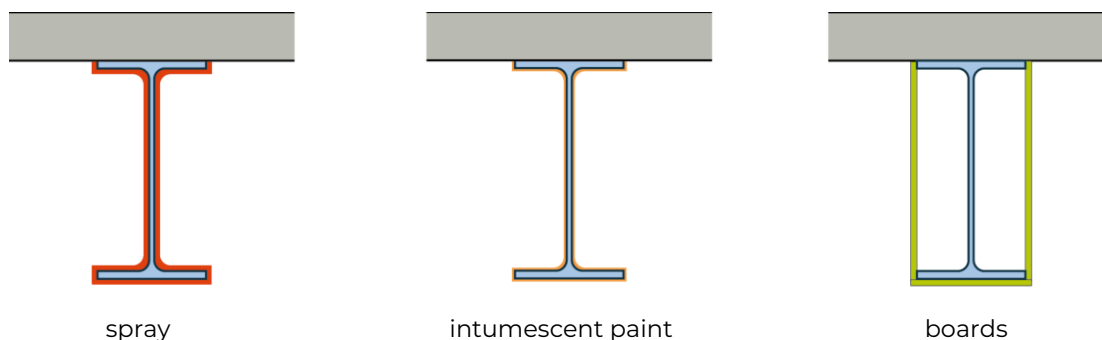


Figure 44: Type of fire protection

- (4) Only the generic properties for the insulation material are considered in the software. The thermal protection is assumed to be applied on all exposed sides of the profile.
- (5) According to the type of fire protection, the User should choose one of the predefined generic material given in **Table 12**. All properties of generic materials are assumed to be independent of the temperature. The thickness of the fire protection is denoted d_p .

Type of insulation	Density ρ_p [kg/m ³]	Thermal conductivity λ_p [W/mK]	Specific heat c_p [J/kgK]
Low density spray			
mineral fibres	300	0,12	1200
vermiculite	350	0,12	1200
High density spray			
perlite with cement	550	0,12	1100
perlite with plaster	650	0,12	1100
Intumescent paint	0	(a)	0
Boards:			
vermiculite or perlite with cement	800	0,2	1200
silicate fibre	600	0,15	1200
fibro-cement	800	0,15	1200
plaster	800	0,20	1700

(a) the value of λ_p is to be defined by the User in the range of [0,005 to 0,012]

Table 12: Thermal properties of insulation material

- (6) For composite slabs with perpendicular ribs, it is assumed that the voids between the deck and the steel profile are filled when the profile is protected.
- In Expert mode, it is possible to specify that these voids are not filled. In this case, a specific method for the calculation of the temperature in the profile is applied, see **Annex H**.
- (7) For unprotected sections, the User may define the type of finishing as carbon steel (CS) or as hot-dip galvanised steel (HDG).
- (8) The User should define if the reinforcements are made of cold worked (by default) or hot rolled steel – see **§ 10.7.3**.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	60	Rev.:	B

4.18.3 Calculation options

- (1) The User can choose to consider the reinforcements in compression.
- (2) For composite beams, the temperature of the slab may be obtained either with the tabulated method of EN 1994-1-2 (by default) or by a numerical analysis. See **Annex G**.
- (3) When a numerical analysis is performed for the temperature of the slab, the User may define:
 - the maximal size of the elements;
 - if the density of concrete is constant (by default) or if it depends on the temperature;
 - if the thermal conductivity of concrete is obtained according to the French national annex of EN 1994-1-2.
- (4) As an option, the User indicates if the concrete strength is reduced or not for temperature below 250°C.
- (5) The temperature of the slab is obtained in each element that represent the slab, defined either through the tabulated method or by the numerical method. The temperature of rebars that are considered for the resistance to negative bending moment may be obtained as follows:
 - as the average temperature of the elements that contains the rebar;
 - as the maximal temperature of the elements that contains the rebar;
 - as the temperature of the element where the axis is located.
- (6) For partially encased steel profiles, the User can choose to take into account the root fillets when applying the calculation method in Annex F of EN 1994-1-2.

4.18.4 Fire design parameters

- (1) Following factors used in the fire design are displayed:
 - Stefan Boltzman constant,
 - emissivity of concrete,
 - emissivity of fire (modifiable),
 - convection factor (below the slab where the fire occurs),
 - convection factor (above the slab, without fire),
 - correction factor for the shadow effect (modifiable),
 - form factor (modifiable),
 - reference temperature (before the start of the fire),
 - maximal temperature.
- (2) The user may define the time increment used in the iterative processes.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	61	Rev.: B

5 MATERIAL PROPERTIES

5.1 Structural steel

5.1.1 Structural steel for hot rolled profiles

- (1) Structural steel is defined by its grade. For hot rolled profiles, the available steel grades are defined according to the ArcelorMittal database (see § 3.2.5 (1)). The latter includes at least the list of usual steel grades as given in **Table 13**.
- (2) Following steel grades are available.

S235 S275 S355 S420 S460

Table 13: Usual steel grades

- (3) **ABCPMX** accepts higher steel grades in the database. For composite sections using profiles with such high strength steels, plastic design of the beam is not possible, and the elastic design is imposed, i.e. the sections will be treated as Class 3 cross-sections.
- (4) The functions that give the nominal yield strength f_y from the steel grade and the thickness is included in the database. These functions follow the rules of EN 10025 or of Table 3.1 of EN 1993-1-1 (according to the settings of the software, the latter may not be available).
- (5) For hot-rolled profiles, one common nominal yield strength f_y is considered uniformly for the whole profile. It is assessed from the greatest value of the flange thicknesses and the web thickness.

$$f_{yfs} = f_{yfi} = f_{yw} = f_y \quad (25)$$

where f_{yfs} , f_{yfi} and f_{yw} are the yield strength associated to the upper flange, to the lower flange and to the web, respectively.

- (7) The design yield strength of the structural steel is obtained by:

$$f_{yd} = f_y / \gamma_{M0} \quad (26)$$

5.1.2 Structural steel for welded sections

- (1) For welded sections, steel grade is selected among the list of steels in the steel database.
- (2) The nominal yield strength f_y is assessed separately from the steel grade, using the thickness of the upper flange, the lower flange and the web. Associated parameters are: f_{yfs} , f_{yfi} and f_{yw} .
- (3) The reduction functions of steel properties according to the thickness follow the rules of EN 10025 or, when the thickness does not exceed 80 mm, of Table 3.1 of EN 1993-1-1 (according to the settings of the software, the latter may not be available).

5.1.3 Structural steel for plates

- (1) For steel plates used in slim floor sections or as reinforcing plates for hot rolled profiles, the steel grade is defined from the list available in the steel database.
- (2) The nominal yield strength f_{ysp} is assessed from the steel grade, using the thickness of the plate, according to EN 10025 or to Table 3 of EN 1993-1-1.
- (3) The design yield strength of the steel plate is obtained by:

$$f_{yspd} = f_{ysp} / \gamma_{M0} \quad (27)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	62	Rev.: B

5.1.4 Other structural steel properties

- (1) According to EN 1993-1-1 [4], the following common properties are considered for all the types of structural steel:

E_a Young's modulus: $E_a = 210\,000\text{ MPa}$;
 ν_a Poisson factor: $\nu_a = 0,3$;
 G_a shear modulus: $G_a = 80\,770\text{ MPa}$;
 ρ_a density: $\rho_a = 7\,850\text{ kg/m}^3$.

- (2) The steel parameters ε_{fs} , ε_{fl} , ε_w and ε_p , for respectively the upper flange, the lower flange, the web and the steel plate, are obtained as follows, where f_y is in MPa:

$$\varepsilon = \sqrt{235 / f_y} \quad (28)$$

5.2 Concrete

- (1) Concrete is defined by:

- its type - see (2);
- its density - see (3);
- its class - see (4).

- (2) Two types of concrete are possible:

- Normal weight concrete (NWC)
- Light weight concrete (LWC)



Light weight concrete is not available for partial encasements.

- (3) The concrete density is denoted ρ_c . It should fulfil the following conditions:

- normal concrete: $2\,000\text{ kg/m}^3 \leq \rho_c$
- light concrete: $800 \leq \rho_c \leq 2\,200\text{ kg/m}^3$

- (4) The concrete is defined through its class, to be chosen among the list given in **Table 14** for NWC and in **Table 15** for LWC. When the prEN 1994-1-1 standard will be used (next generation), concrete classes of **Table 16** will also be proposed.

For slim floors composite sections, light weight concrete will not be available. Additionally, the use of C20/25 concrete class will not be possible with composite dowels connection.



Slim floor sections and the second generation of Eurocodes are not available in the version V1.1 of ABCPMX.

Concrete class	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
f_{ck} (MPa)	20	25	30	35	40	45	50

Table 14: Classes of normal weight concrete

Concrete class	LC20/22	LC25/28	LC30/33	LC35/38	LC40/44	LC45/50	LC50/55
f_{ck} (MPa)	20	25	30	35	40	45	50

Table 15: Classes of light weight concrete

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	63	Rev.: B

	NWC	LWC
Concrete class	C70/85	LC60/66
f_{ck} (MPa)	70	60

Table 16: Additional classes of concrete (next generation)

- (5) The mechanical properties for concrete, given in Table 3.1 of EN 1992-1-1 [3] for the normal weight concrete or in Table 11.3.1 of EN 1992-1-1 for the light weight concrete, are defined by:

f_{ck} is the characteristic value of the compression strength of the concrete “cylinder compressive test at 28 days”. Its value is obtained from the concrete class, see **Table 14**, **Table 15** and **Table 16**.

f_{cd} is the design compressive strength of the concrete, calculated by:

$$f_{cd} = f_{ck} / \gamma_c \quad (29)$$

E_{cm} is the mean secant modulus of elasticity of the concrete for short term loading. For normal concrete, its value is obtained by the following formula:

$$E_{cm} = 22 [f_{cm} / 10]^{0.3} \text{ with } f_{cm} \text{ in MPa} \quad (30)$$

For light weight concrete LWC, the mean secant modulus of elasticity is calculated by using the following formulae:

$$E_{lcm} = E_{cm} \eta_E \quad (31)$$

where: $\eta_E = (\rho_c / 2200)^2$ and E_{cm} is obtained from Formula (30).

f_{cm} is the mean value of concrete cylinder compressive strength. For normal concrete, its value is obtained by the following formula:

$$f_{cm} = f_{ck} + 8 \text{ MPa} \quad (32)$$

ρ_c is the concrete density (in kg/m³)

f_{ctm} is the mean tensile strength of the concrete, given by:

$$f_{ctm} = 0,3 [f_{ck}]^{2/3} \quad (33)$$

For light weight concrete LWC, the mean tensile strength is calculated by using the following formula:

$$f_{lctm} = f_{ctm} \eta_1 \quad (34)$$

where: $\eta_1 = 0,4 + 0,6 (\rho_c / 2200)$ and f_{ctm} is obtained from Formula (33).

- (6) The development of creep in concrete depends on the relative humidity RH , which is defined by the User (see § 4.17.2), from the two predefined values of 50% (interior of a building with controlled temperature) or 80% (exterior atmosphere). The default value is 50%.
- (7) The deformation for shrinkage ε_{sh} is defined in the calculation options (see § 4.17.2),.
- (8) For sections with partial concrete encasement, a specific class for the concrete between the flanges is selected. The characteristic value of the compressive strength of this concrete is denoted f_{cek} and all previous formulas for concrete are applied (see § 4.9).

5.3 Steel for reinforcements

- (1) The yield strength of rebars is denoted f_{sk} for the slab reinforcement and f_{sek} for the partial concrete encasement reinforcement.
- (2) The yield strength is 450 MPa for class B 450 and 500 MPa for class B 500.
- (3) Custom values may be defined by the User in Expert mode.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	64	Rev.:	B

- (4) The Young's modulus of steel rebars is E_s . By default, according to EN 1994-1-1, 3.2 (2), $E_s = E_a$. See § 4.17.2.
- (5) The design yield strength of reinforcement is obtained by:

$$\begin{aligned} f_{sd} &= f_{sk} / \gamma_s \\ f_{sed} &= f_{sek} / \gamma_s \end{aligned} \quad (35)$$

6 BASIC CALCULATIONS

6.1 Effective width of the concrete flange

6.1.1 General

- (1) The effective width of the concrete flange is calculated according to EN 1994-1-1, 5.4.1.2.
- (2) The geometric widths b_1 and b_2 of the concrete flange are given by:

- for an edge beam:

$$\begin{aligned} b_1 &= d_1 \\ b_2 &= d_2 / 2 \end{aligned}$$

- for an intermediate beam:

$$\begin{aligned} b_1 &= d_1 / 2 \\ b_2 &= d_2 / 2 \end{aligned}$$

See § 4.3 for the definitions of edge beams, intermediate beams, d_1 and d_2 .

- (3) If slab openings are defined, previous equations are replaced by:

- for an edge beam:

$$\begin{aligned} b_1 &= d_1 \\ b_2 &= \min\{d_{sl,2}; d_2 / 2\} \end{aligned}$$

- for an intermediate beam:

$$\begin{aligned} b_1 &= \min\{d_{sl,1}; d_1 / 2\} \\ b_2 &= \min\{d_{sl,2}; d_2 / 2\} \end{aligned}$$

See § 4.3 (6) for the definitions of edge beams, intermediate beams, $d_{sl,1}$ and $d_{sl,2}$.

- (4) For the calculation of the effective width of concrete flanges as given in § 6.1, the distance b_0 between the centres of the outstand shear connectors is taken equal to 0, according to clause 5.4.1.2 (9) of EN 1994-1-1.

6.1.2 Effective width of the concrete flange in the main span

- (1) The effective width at mid-span of the main span is given by the following formula:

$$\begin{aligned} b_{eff,m} &= b_{e1} + b_{e2} \\ b_{e1} &= \min\{L_e/8; b_1\} \\ b_{e2} &= \min\{L_e/8; b_2\} \end{aligned} \quad (36)$$

where: b_1 and b_2 are the geometric widths, according to § 6.1.1;

L_e is the distance between the points of zero bending moment.

- (2) In Formula (36), the distance L_e is obtained as follows:

- for a single span beam without cantilever: $L_e = L$;
- for a beam with one cantilever span: $L_e = 0,85 L$;
- for a beam with two cantilever spans: $L_e = 0,7 L$.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	65	Rev.:	B

- (3) At support A or B, if there is no adjacent span, the effective width of the concrete flange is given by:

$$b_{\text{eff},s} = \beta_1 b_{e1} + \beta_2 b_{e2} \quad (37)$$

where: b_{e1} and b_{e2} are defined by Formula (36);

β_1 and β_2 are defined by:

$$\beta_i = \min\{1; 0,55 + 0,025 L_e / b_{ei}\} \quad (38)$$

L_e see (2);

index s: A or B.

- (4) At support A or B, if there is an adjacent cantilever span, the effective width of the concrete flange is given by Formula (36), where L_e is calculated by:

$$L_e = 2 L_{cs} \quad (39)$$

where L_{cs} is the length of the adjacent cantilever span, i.e. either L_g or L_d (see § 4.3 (5)).

- (5) For any cross-section in the main span located at a distance d_x of the nearest support s (A or B) lower than $L / 4$, the effective width of the concrete flange is obtained by (Figure 45):

$$b_{\text{eff}} = b_{\text{eff},s} + 4 \frac{d_x}{L} (b_{\text{eff},m} - b_{\text{eff},s}) \quad (40)$$

- (5) For any cross-section in the main span located at a distance d_x of the nearest support s (A or B) greater or equal to $L / 4$, the effective width of the concrete flange is obtained by:

$$b_{\text{eff}} = b_{\text{eff},m} \quad (41)$$

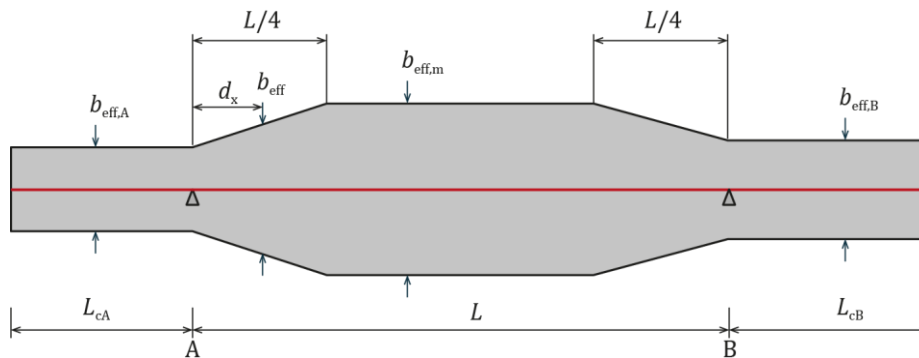


Figure 45 : Effective width of the concrete flange

6.1.3 Effective width of the concrete flange in the cantilever spans

- (1) The effective width of the concrete flange is constant along any cantilever span A or B and is given by clause (4) in § 6.1.2.

6.1.4 Simplified models for the effective width

- (1) An option is proposed in order to use a simplified distribution of the effective width along the beam, as an alternative to the model shown in Figure 45 – see § 4.17.2.
- (2) If this option is activated, the effective width of the concrete flange considered for the global analysis is assumed constant along the main span (Figure 46) and equal to $b_{\text{eff},m}$ (according to EN 1994-1-1, 5.4.1.2 (4)).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	66	Rev.: B

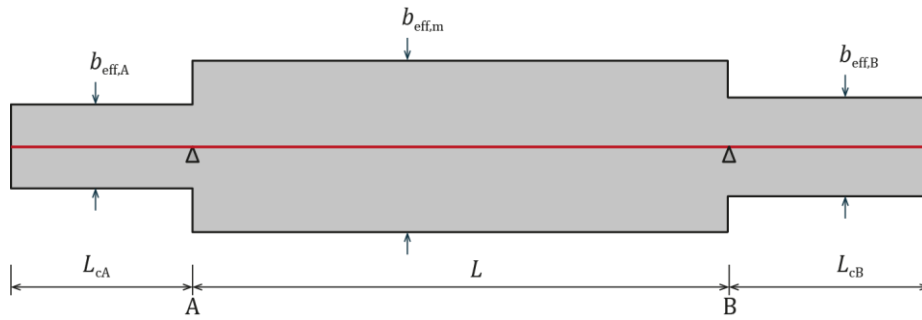


Figure 46 : Simplified effective width of the concrete flange for the global analysis

- (3) If the option is activated, the effective width of the concrete flange considered for the verification of cross-sections is obtained as follows (**Figure 47** and **Figure 48**) (according to EN 1994-1-1, 6.1.2 (2)).

For any cross-section in the main span located at a distance d_x of the nearest support s (A or B) lower than $0,15 L$, the effective width of the concrete flange is equal to $b_{eff,s}$, if there is a cantilever span at the support. For all other cross-sections of the main span, the effective width is equal to $b_{eff,m}$. For cantilever spans, **§ 6.1.3** applies.

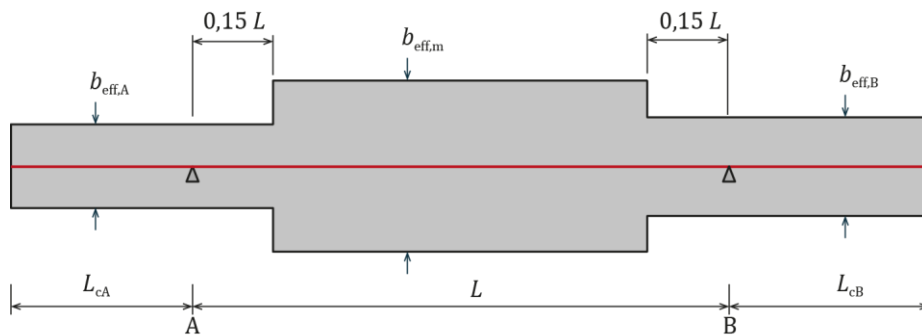


Figure 47: Simplified effective width of the concrete flange for the verification of cross-sections – example with two cantilever spans

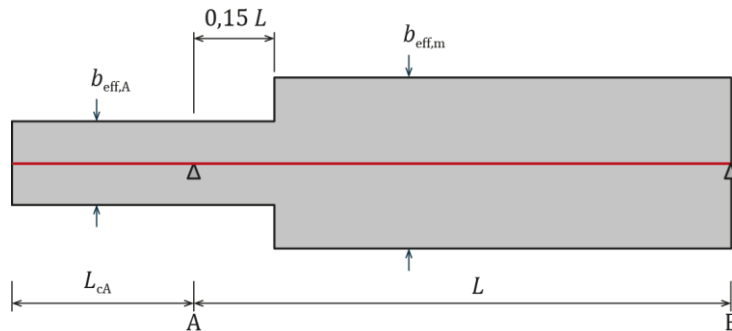


Figure 48: Simplified effective width of the concrete flange for the verification of cross-sections – example with one cantilever span

6.2 Active part of the concrete slab for composite sections

6.2.1 General

- (1) The slab participates to the resistance of the composite beams under bending moments. The participating area of the slab is defined by the effective width as given in § 6.1 and the thickness of the active part, as described here after.

6.2.2 Usual sections

- (1) The thickness t_c of the active part of the slab is obtained as follows:

- solid slabs: $t_c = t_d$
- solid slabs with precast elements: $t_c = t_d - t_{pc} + t_j$
- composite slabs: $t_c = t_d - h_p$ for ribs parallel to the beam axis;
 $t_c = t_d - h_{pg}$ for ribs perpendicular to the beam axis.

where: h_p and h_{pg} : see § 4.5.1.

t_j : see § 4.4.2 (4).

6.3 Modular ratio

6.3.1 Eurocode generation 1

- (1) A modular ratio is associated to each load case which is applied on the beam in a composite stage. The modular ratio is obtained with the following formula (see § 5.4.2.2 of EN 1994-1-1):

$$n_L = n_0 (1 + \psi_L \varphi_t) \quad (42)$$

where: n_0 is the modular ratio for short term loading, defined by: $n_0 = E_a / E_{cm}$

E_a is the Young's modulus of steel (see § 5.1.1)

E_{cm} is the secant modulus of concrete (see § 5.2)

ψ_L is the creep multiplier depending on the type of loading, which is given by **Table 17**.

Type of loading	ψ_L
Permanent loads	1,1
Shrinkage	0,55
Live loads	0

Table 17: Values of the creep multiplier

φ_t is the creep coefficient depending on the age t of concrete at the time considered for the design and on the age t_0 at the time of loading (see § 4.13.2).

$$\varphi_t = \varphi(t, t_0) \quad (43)$$



A modular ratio is also calculated for the concrete of the partial concrete encasement, if any, using the same formulas. For composite beams with partial concrete encasement, two separate values of modular ratios are thus used, assuming the same age t and separate concrete properties and t_0 values.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	68	Rev.: B

- (2) For normal concrete, φ_t is obtained according to Annex B of EN 1992-1-1 [3], as follows:

$$\varphi(t, t_0) = \varphi_0 \beta_c(t, t_0) \quad (44)$$

where: φ_0 is the notional creep coefficient, calculated by:

$$\varphi_0 = \varphi_{RH} \beta(f_{cm}) \beta(t_0) \quad (45)$$

φ_{RH} is a factor which accounts for the effect of relative humidity on the notional creep coefficient, given by:

$$\begin{aligned} \varphi_{RH} &= 1 + \frac{1 - RH / 100}{0,1 \sqrt[3]{h_0}} && \text{for } f_{cm} \leq 35 \text{ MPa} \\ \varphi_{RH} &= \alpha_2 \left[1 + \alpha_1 \frac{1 - RH / 100}{0,1 \sqrt[3]{h_0}} \right] && \text{for } f_{cm} > 35 \text{ MPa} \end{aligned} \quad (46)$$

RH is the relative humidity of the ambient environment, in %, see § 5.2 (4);

h_0 is the notional size of the member (see § 6.3.2);

f_{cm} : see § 5.2;

$\beta(f_{cm})$ is a factor which takes into account the concrete strength (see (3));

$\beta(t_0)$ is a factor which takes into account the age of concrete at loading (see (4));

$\beta_c(t, t_0)$ is a coefficient which accounts for the development of creep with time (see (5));

RH is the relative humidity of the ambient atmosphere, see § 5.2 (4);

t_0 is the age of concrete at loading in days, see § 4.13.2;

t is the age of concrete at the time considered for the design, see § 4.13.2.

- (3) $\beta(f_{cm})$ is obtained by:

$$\beta(f_{cm}) = \frac{16,8}{\sqrt{f_{cm}}} \quad (47)$$

- (4) $\beta(t_0)$ is given by:

$$\beta(t_0) = \frac{1}{(0,1 + t_0^{0,20})} \quad (48)$$

- (5) $\beta_c(t, t_0)$ is obtained by:

$$\beta_c(t, t_0) = \left[\frac{t - t_0}{\beta_H + t - t_0} \right]^{0,3} \quad (49)$$

where β_H is a coefficient depending on the relative humidity and on the notional size of the member, given by:

$$\beta_H = 1,5 \left[1 + \left(\frac{3 RH}{250} \right)^{18} \right] h_0 + 250 \leq 1500 \quad \text{for } f_{cm} \leq 35 \text{ MPa} \quad (50)$$

$$\beta_H = 1,5 \left[1 + \left(\frac{3 RH}{250} \right)^{18} \right] h_0 + 250 \alpha_3 \leq 1500 \alpha_3 \quad \text{for } f_{cm} > 35 \text{ MPa}$$

α_i factors are given by:

$$\alpha_1 = \left(\frac{35}{f_{cm}} \right)^{0,7} \quad \alpha_2 = \left(\frac{35}{f_{cm}} \right)^{0,2} \quad \alpha_3 = \left(\frac{35}{f_{cm}} \right)^{0,5} \quad (51)$$

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	69	Rev.:	B

6.3.2 Notional size of a member

- (1) The general formula to assess the notional size of a member is as follows:

$$h_0 = \frac{2 A_c}{u} \quad (52)$$

where A_c is the area of the concrete member;

u is the perimeter of the member in contact with the atmosphere.

- (2) For a solid slab without haunch, with or without a prefabricated slab, h_0 is calculated by:

$$h_0 = \frac{1}{\frac{1}{t_d} - \frac{b_{fs}}{2 b_{eff} t_d}} \quad (53)$$

- (3) For a solid slab with a haunch, h_0 is calculated by:

$$h_0 = \frac{b_{eff} t_d + (b_f + t_h \tan \theta_h) t_h}{b_{eff} - \frac{1}{2} b_f + t_h \left(\frac{1 - \sin \theta_h}{\cos \theta_h} \right)} \quad (54)$$

- (4) For a composite slab with longitudinal ribs, h_0 is calculated by:

$$h_0 = 2 t_c \left(1 + \frac{h_p b_m}{e_p t_c} \right) \quad (55)$$

See § 4.5.1 for the geometric parameters of the profiled steel sheeting.



For composite slabs, it is assumed that the side of concrete in contact with the steel profiled sheeting is not in contact with the atmosphere.

- (5) For a composite slab with perpendicular ribs, h_0 is calculated by:

$$h_0 = 2 t_c \quad (56)$$

- (6) For the concrete of a partial encasement, h_0 is calculated by:

$$h_0 = A_{ce} / h_w \quad (57)$$

where A_{ce} is the area of concrete in the two chambers of the partial encasement, as given by Formula (24).

6.4 Classification of cross-sections

6.4.1 General

- (1) The software does not cover Class 4 cross-sections.
- (2) Steel cross-sections are classified according to EN 1993-1-1 Table 5.2.
- (3) Composite cross-sections, including sections with concrete encasement, are classified according to EN 1994-1-1, 5.5.1.
- (4) The classification is carried out considering pure bending only.
- (5) The final Class of a cross-section is the maximum class of the flanges and of the web.

6.4.2 Usual steel sections

- (1) The class of a flange under compression is obtained from its geometrical slenderness c / t according to Table 18, where:

c is the outstand width, given by: $c = (b_f - t_w) / 2 - r_c$;

t is the flange thickness t_f ;

ε is the steel parameter, either equal to ε_{fs} or to ε_{fi} see § 5.1.4 (2).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	70	Rev.: B

Class	Criteria
1	$c / t \leq 9 \varepsilon$
2	$c / t \leq 10 \varepsilon$
3	$c / t \leq 14 \varepsilon$
4	otherwise

Table 18: Classification of an outstand panel under pure compression

(2) The upper flange is considered as in compression if one of the following conditions is fulfilled:

- under positive moments:
 - plastic distribution of stresses: $t_{fs} \leq -z_{pl}$
 - elastic distribution of stresses: $t_{fs} \leq -z_{el}$
- under negative moments:
 - plastic distribution of stresses: $0 \geq -z_{pl}$
 - elastic distribution of stresses: $0 \geq -z_{el}$

The lower flange is always considered in tension under positive bending moment and in compression under negative bending moment.

(3) The class of the web under full or partial compression, is obtained from its geometrical slenderness c / t and the bending factor α or ψ , according to **Table 19**, where:

c / t is given by: $c / t = d_w / t_w$

α is the fraction of the web that is under compression considering a plastic distribution of the stresses. It is calculated as given in **(4)**;

ψ is the ratio of stresses at edges of the web, as given in **(5)**, considering an elastic distribution;

ε is the steel factor, equal to ε_w – see **§ 5.1.4 (2)**.

Class	Criteria
1	$\alpha > 0,5$ $\frac{c}{t} \leq \frac{396\varepsilon}{13\alpha - 1}$
	$\alpha \leq 0,5$ $\frac{c}{t} \leq \frac{36\varepsilon}{\alpha}$
2	$\alpha > 0,5$ $\frac{c}{t} \leq \frac{456\varepsilon}{13\alpha - 1}$
	$\alpha \leq 0,5$ $\frac{c}{t} \leq \frac{41,5\varepsilon}{\alpha}$
3	$\psi > -1$ $\frac{c}{t} \leq \frac{42\varepsilon}{0,67 + 0,33\psi}$
	$\psi \leq -1$ $c / t \leq 62 \varepsilon (1 - \psi) \sqrt{-\psi}$
4	otherwise

Table 19: Classification of an internal panel under compression and bending – First generation of EN

(4) Under positive bending moment, α is obtained as follows, from the location of the plastic neutral axis z_{pl} of the cross-section, as calculated in **§ 6.5.1** or **6.5.3**:

- if $z_{pl} \geq -(t_{fs} + r_{cs})$: the web is fully in tension and its class is 1;
- if $z_{pl} \leq -(t_{fs} + r_{cs} + d_w)$: the web is fully under compression and $\alpha = 1$;
- otherwise:

$$\alpha = \frac{-z_{pl} - t_{fs} - r_{cs}}{d_w} \quad (58)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	71	Rev.: B



See § 1.4 for the definition of the axes.

Under negative bending moment, α is obtained as follows:

- if $z_{pl} \geq -(t_{fs} + r_{cs})$: the web is fully in compression and $\alpha = 1$;
- if $z_{pl} \leq -(t_{fs} + r_{cs} + d_w)$: the web is fully under tension and its class is 1;
- otherwise :

$$\alpha = \frac{h_s - t_{fi} - t_p - r_{ci} + z_{pl}}{d_w} \quad (59)$$

- (5) Under positive bending moment and elastic distribution of stresses, the stress ratio ψ is given by:

$$\psi = \frac{z_{el} + h_s - t_{fi} - r_{ci}}{z_{el} + t_{fs} + r_{cs}} \quad (60)$$

Under negative bending moment, it is obtained by:

$$\psi = \frac{z_{el} + t_{fs} + r_{cs}}{z_{el} + h_s - t_{fi} - r_{ci}} \quad (61)$$

- (6) For the classification of the web, the sign of bending moment is obtained as follows:

- for the global analysis, positive moments are assumed in the uncracked segments of the beam and negative moments in the cracked segments – see § 7.4;
- for the section analysis (i.e. verification at ULS or at SLS), the sign of the moment, obtained from the global analysis and the relevant combination of actions, is considered.

6.4.3 Usual steel sections with partial concrete encasement

- (1) The class of a flange under compression is obtained from its geometrical slenderness c / t according to **Table 20**, where:

c / t is calculated according to § 6.4.2 (1);

ε is the steel parameter, either equal to ε_{fs} or to ε_{fi} see § 5.1.4 (2).

A flange is assumed in compression when the conditions in § 6.4.2 (2) are met.

Class	Criteria
1	$c / t \leq 9 \varepsilon$
2	$c / t \leq 14 \varepsilon$
3	$c / t \leq 20 \varepsilon$
4	otherwise

Table 20: Classification of an outstand panel under pure compression



According to Table 5.2 of EN 1994-1-1.

- (2) The class of the web is obtained according to § 6.4.2 (3). Any Class 3 web with partial concrete encasement will be considered as Class 2.

6.4.4 Usual composite sections

- (1) The class of a flange under compression is obtained according to § 6.4.2 (1) and (2).



For the upper flange, the positive effect of the connection according to EN 1994-1-1, 5.5.2 (1) is not considered in ABCPMX.

- (2) The class of the web is obtained from § 6.4.2 (3), where the plastic and elastic neutral axes are obtained according to § 6.5.5.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	72	Rev.: B

6.4.5 Usual composite sections with partial concrete encasement

- (1) The class of a flange under compression is obtained according to **§ 6.4.3 (1)**.
- (2) The class of the web is obtained from **§ 6.4.3 (2)**, where the plastic and elastic neutral axes are obtained according to **§ 6.5.6**.

6.5 Section bending properties

6.5.1 General

- (1) The properties of cross-sections are assessed through a general method based on the discretisation of the section.
- (2) The following chapters give the principles for the calculation of section properties. They also provide simplified formulas for limited scopes of application (for validation purposes).
- (3) For all types of cross-sections including concrete, the plastic bending moment is assessed first by assuming a full shear connection.
- (4) The procedure to assess the plastic bending moment with partial shear connection is given in the relevant chapters.
- (5) For cross-sections with partial concrete encasement, it is always assumed, for any cross-section, that the full connection of the encasement is obtained.



The full connection is assumed according to clause 6.3.2 (1) of EN 1994-1-1.

The design of the connection between the profile and the partial encasements is not covered by the software.

6.5.2 Steel sections

- (1) The location of the elastic neutral axis of any steel cross-section is obtained with the following formula:

$$z_{el} = \frac{\sum A_i z_i}{A_a} \quad (62)$$

where: A_i is the area of any elementary part which constitutes the cross-section;

z_i is the location of the centroid of any elementary part;

A is the area of the cross-section which is given by:

$$A_a = \sum A_i \quad (63)$$

The formulas for A_i and z_i are given in **Table 21**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	73	Rev.: B

Elementary part	A_i	z_i	I_i
1 – Upper flange	$b_{fs} t_{fs}$	$-t_{fs} / 2$	$b_{fs} t_{fs}^3 / 12$
2 – Upper root	$r_{cs} (t_w + (2 - \pi/2) r_{cs})$	$-t_{fs} - z_2$	$t_w r_{cs}^3 / 12 + k_{cl} r_{cs}^4$
3 – Web between roots	$d_w t_w$	$-(t_{fs} + r_{cs} + d_w / 2)$	$t_w d_w^3 / 12$
4 – Lower root	$r_{ci} (t_w + (2 - \pi/2) r_{ci})$	$-(h_s - t_p - t_{fi} - z_4)$	$t_w r_{ci}^3 / 12 + k_{cl} r_{ci}^4$
5 – Lower flange	$b_{fi} t_{fi}$	$-(h_s - t_p - t_{fi} / 2)$	$b_{fi} t_{fi}^3 / 12$
6 – Lower plate (*)	$b_p t_p$	$-h_s + t_p / 2$	$b_p t_p^3 / 12$
where : $k_{cl} = 0,0151$ $k_{cG} = (10 - 3\pi) / (12 - 3\pi)$ $z_{cs} = k_{cG} r_{cs}$			
$z_2 = \frac{t_w + z_{cs} (4 - \pi)}{4 - \pi + 2 t_w / r_{cs}}$ $z_4 = \frac{t_w + z_{ci} (4 - \pi)}{4 - \pi + 2 t_w / r_{ci}}$			
(*) elementary part no 6 is considered for hot rolled profiles with reinforcing plates			

Table 21: Properties of elementary parts of a cross-section

- (2) The second moment of area of the cross-section about its strong axis $y - y$ is calculated by the following formula:

$$I_{y,a} = \sum I_i + \sum A_i (z_i - z_{el})^2 \quad (64)$$

- (3) The plastic properties of the cross-section are assessed by considering the following assumptions:

- the area of the steel profile is stressed to its characteristic design yield strength in tension and in compression; for hot-rolled profiles, a uniform value is assumed whereas specific values are considered for each flange and the web, for welded sections;
- the area of the lower plate (if any) is stressed to its characteristic design yield strength in tension and in compression;
- when MV interaction is considered, the yield strength of the web (including the fillets) is reduced accordingly.

The characteristic resistance of the bending moment $M_{pl,a,Rk}$ is obtained considering the nominal yield strength of steel. The design plastic bending resistance $M_{pl,a,Rd}$ is obtained by:

$$M_{pl,a,Rd} = M_{pl,a,Rk} / \gamma_{M0} \quad (65)$$

- (4) For usual steel sections, the elastic characteristic resistance to bending moment is obtained as follows:

$$M_{el,a,Rk} = \frac{I}{\max\{z_s, z_i\}} f_y \quad (66)$$

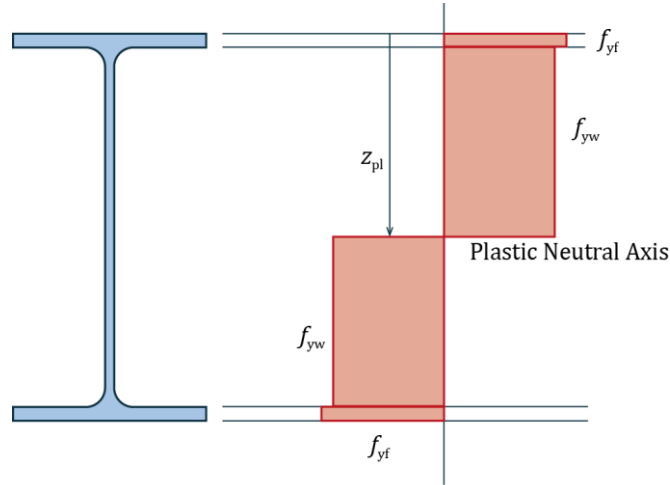
where: $z_s = |z_{el}|$

$$z_i = |h_s - z_{el}|$$

z_{el} is the location of the elastic neutral axis.

The design elastic bending resistance $M_{el,Rd}$ is obtained by:

$$M_{el,a,Rd} = M_{el,a,Rk} / \gamma_{M0} \quad (67)$$



In pure bending, for hot-rolled profiles: $f_{yw} = f_{yf} = f_y$

For MV interaction, f_{yw} is calculated according § 8.4

Figure 49: General principles for the plastic properties of a usual steel cross-section

6.5.3 Simplified methods for symmetrical steel sections

- (1) For symmetrical usual steel sections, either hot-rolled or welded (sections SHR and SWB – see **Table 4**), the location of elastic and plastic neutral axes are given by:

$$z_{el} = z_{pl} = -h_s / 2 \quad (68)$$

- (2) The area of the cross-section is given by:

$$A_a = 2 b_f t_f + h_w t_w + (4 - \pi) r_c^2 \quad (69)$$

- (3) The second moment of area is given by:

$$I_{y,a} = \frac{1}{12} [b_f h_s^3 - (b_f - t_w)(h_s - 2 t_f)^3] + 2 k_{cl} r_c^4 + (4 - \pi) \frac{r_c^2}{2} \left(\frac{h_s}{2} - t_f - k_{cG} r_c \right)^2 \quad (70)$$

- (4) The plastic bending modulus is given by:

$$W_{pl,a} = b_f t_f (h_s - t_f) + h_w^2 t_w + (4 - \pi) r_c^2 (h_s/2 - t_f - k_{cG} r_c) \quad (71)$$

where: k_{cG} see **Table 21**.

$$M_{pl,a,Rd} = W_{pl,a} f_y / \gamma_{M0}$$

6.5.4 Usual steel sections with partial concrete encasement

- (1) If the partial concrete encasement is considered in the calculations (see § 4.9 (13) and § 4.17.2), the bending properties of usual steel sections with partial concrete encasement are assessed by an iterative procedure (Annex B), with the following assumptions (**Figure 50**):

- the part of concrete in tension area is ignored;
- the longitudinal steel reinforcement (active rebars only, see § 4.9) in the tension area is considered;
- the longitudinal steel reinforcement (active rebars only) in the compression area is considered if the associated option is activated (see § 4.17.2);
- both positive and negative moments are considered separately.

- (2) Elastic section properties are assessed by considering the modular ratio (see § 6.2) associated to the relevant load case and to the concrete of the encasement (see § 4.9).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	75	Rev.: B

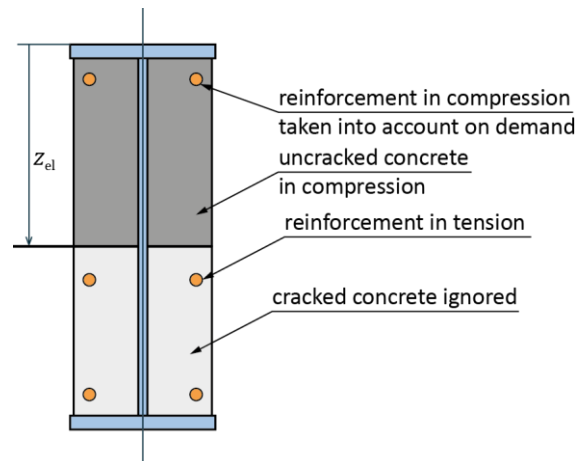


Figure 50: Elastic bending properties of steel section with partial concrete encasement under positive bending moment

6.5.5 Usual composite sections

- (1) For the calculation of the elastic and plastic properties of usual composite cross-sections, the concrete in tension and the reinforcement in compression (by default) are neglected. The concrete of haunches is also neglected.



The User has the possibility to activate the reinforcement in compression – see § 4.17.2.

If the User activates the reinforcement in compression, the following warning message will be displayed: when reinforcements in compression are accounted for in the calculation of the bending properties, the requirements of EN 1992-1-1, 9.2.1.2 (3) should be fulfilled.

If the slab contains only one layer of reinforcement, reinforcement in compression is disregarded, whatever the option of the User is.

- (2) When a partial shear connection is used, the plastic properties are obtained considering an equivalent active part of the slab, defined as follows:

$$t_{c,\eta} = \frac{\eta N_{cx}}{0,85 b_{eff} f_{cd}} \quad \text{first generation of Eurocodes} \quad (72)$$

where: η and N_{cx} : see § 9.2.2.

6.5.6 Usual composite sections with partial concrete encasement

- (1) For the calculation of the elastic and plastic properties of usual composite sections with partial concrete encasement, § 6.5.5 applies.

6.5.7 Reduction of the plastic resistance for composite section with deep plastic neutral axis

- (1) For usual composite cross-sections, with or without partial concrete encasement, and with a steel profile of grade S420 or S460, the plastic bending resistance for positive bending moment should be reduced as follows:

$$M'_{pl,Rd} = \beta M_{pl,Rd} \quad (73)$$

where: $M_{pl,Rd}$ is the plastic bending resistance of the cross-section;

β is the reduction factor, as given in (2).



According to EN 1994-1-1, 6.2.1.2 (2).

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	76	Rev.:	B

(2) The reduction factor β is given by:

$$\beta = 1 \quad \text{if } z_{plc} / h_t \leq 0,15 \quad (74)$$

$$\beta = 1 - \frac{0,15}{0,25} \left(\frac{z_{plc}}{h_t} - 0,15 \right) \quad \text{if } 0,15 < z_{plc} / h_t \leq 0,40$$

where: h_t is the total depth of the composite section, given by: $h_t = h_s + t_h + t_d$;

z_{plc} is the distance between the plastic neutral axis and the top fiber of the cross-section, obtained

by: $z_{plc} = t_h + t_d - z_{pl}$.

(3) If $z_{plc} > 0,4 h_t$, the plastic design of sections is not possible. In this case, Class 3 design criteria will be used.

6.6 Other properties of sections

6.6.1 Shear area of usual steel and composite sections

(1) The shear area of a hot rolled profile is given by (acc. to EN 1993-1-1, 6.2.6 (3)):

$$A_v = \max\{A - 2 b_f t_f + (t_w + 2r_c) t_f ; \eta h_w t_w\} \quad (75)$$

where: A is given by formula (69);

$$b_f = b_{fs} = b_{fi}$$

$$t_f = t_{fs} = t_{fi}$$

$$r_c = r_{cs} = r_{ci}$$

η : see § 4.17.2 and § 8.3.

(2) The shear area of welded section is given by (idem):

$$A_v = \eta h_w t_w \quad (76)$$

(3) The same shear areas are used for steel and for composite sections.

6.6.2 Torsion and warping of steel profiles

(1) The torsion constant of hot rolled steel profile is obtained by the following formula (cf. Annex B.2.2 of [22]):

$$I_t = \frac{1}{3} [2 (b_f - 0,63 t_f) t_f^3 + h_w t_w^3] + 2 \alpha D^4 \quad (77)$$

where:

$$\alpha = -0,042 + 0,2204 \frac{t_w}{t_f} + 0,1355 \frac{r}{t_f} - 0,0865 \frac{r t_w}{t_f^2} - 0,0725 \left(\frac{t_w}{t_f} \right)^2$$

$$D = \frac{(t_f + r)^2 + r t_w + t_w^2 / 4}{2 r + t_f}$$

(2) The torsion constant I_t of a welded section is obtained by the following formula (cf. Annex A.2 of [22]):

$$I_t = I_{tfs} + I_{tfi} + I_{tw} + I_{tns} + I_{tni} \quad (78)$$

where:

$$I_{tfs} = \frac{b_{fs} t_{fs}^3}{3} \left(1 - 0,63 \frac{t_{fs}}{b_{fs}} \left(1 - \frac{t_{fs}^4}{12 b_{fs}^4} \right) \right)$$

$$I_{tfi} = \frac{b_{fi} t_{fi}^3}{3} \left(1 - 0,63 \frac{t_{fi}}{b_{fi}} \left(1 - \frac{t_{fi}^4}{12 b_{fi}^4} \right) \right)$$

$$I_{tw} = \frac{h_w t_w^3}{3}$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	77	Rev.: B

$$I_{tns} = 0,15 \frac{t_w}{t_{fs}} \left[\frac{t_{fs}^2 + t_w^2/4}{t_{fs}} \right]^4$$

$$I_{tni} = 0,15 \frac{t_w}{t_{fi}} \left[\frac{t_{fi}^2 + t_w^2/4}{t_{fi}} \right]^4$$

- (3) The warping constant of hot rolled steel profiles is given by:

$$I_w = \frac{b_f^3 t_f}{24} (h_s - t_f)^2 \quad (79)$$

- (4) The warping constant of welded steel sections is given by:

$$I_w = \frac{I_{fzi} I_{fzs}}{I_{fzi} + I_{fzs}} \left(h_s - \frac{t_{fs}}{2} - \frac{t_{fi}}{2} \right)^2 \quad (80)$$

where:

$$I_{fzs} = \frac{t_{fs} b_{fs}^3}{12}$$

$$I_{fzi} = \frac{t_{fi} b_{fi}^3}{12}$$

6.7 Shear resistance of welded studs

6.7.1 General

- (1) The shear resistance of welded studs is calculated according to the type of sections and the configuration of slab (see **Table 22**).

Type of sections and configuration of slab	Formulas of P_{Rd}
Usual composite section	
Solid slabs	see § 6.7.2 (1)
Composite slab with longitudinal ribs	see § 6.7.2 (2) with reduction factor k_ℓ
Composite slab with transverse ribs and a Cofraplus 220 sheeting	formulas for solid slabs in § 6.7.2 (1)
Composite slabs with transverse ribs and common sheetings	
continuous concrete rib ⁽¹⁾	see § 6.7.2 (3) with reduction factor k_t
discontinuous concrete rib ⁽²⁾	see § 6.7.2 (2)

Table 22: Calculation of the shear resistance of studs

6.7.2 First generation of Eurocodes

- (1) The shear resistance P_{Rd} of a welded stud in a solid slab is obtained with the following formula, according to § 6.6.3.1 of EN 1994-1-1 [7]:

$$P_{Rd} = \min\{P_{Rd,s}; P_{Rd,c}\} \quad (81)$$

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	78	Rev.:	B

where: $P_{Rd,s}$ is the shear resistance associated with the failure of the shank of the stud, given by:

$$P_{Rd,s} = 0,8 \frac{f_{u,sc} \pi d^2}{4 \gamma_{Vs}} \quad (82)$$

$P_{Rd,c}$ is the shear resistance associated with the failure of the concrete, given by:

$$P_{Rd,c} = 0,29 \frac{\alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_{Vc}} \quad (83)$$

d diameter of the shank (see § 4.7.2);

$f_{u,sc}$ ultimate tensile strength of the steel of the stud (see § 4.7.2);

f_{ck} characteristic value of the cylinder compressive strength of concrete at 28 days (see § 5.2);

E_{cm} secant modulus of elasticity of concrete (see § 5.2);

α parameter defined as follows:

$$\alpha = 1 \quad \text{for } h_{sc} / d > 4$$

$$\alpha = 0,2 \left(\frac{h_{sc}}{d} + 1 \right) \quad \text{for } 3 \leq h_{sc} / d \leq 4$$



Formula (81) is applied to welded studs in solid slabs with or without haunch and with or without prefabricated part.

- (2) The shear resistance P_{Rd} of a welded stud in a composite slab with ribs parallel to the beam axis (**Figure 18**) is obtained by the following formula, according to § 6.6.4.1 of EN 1994-1-1:

$$P_{Rd} = k_{\ell} \min\{P_{Rd,s}; P_{Rd,c}\} \quad (84)$$

where: $P_{Rd,s}$ and $P_{Rd,c}$ are as defined by Formulas (82) and (83) respectively;

k_{ℓ} is a reduction factor to take account of the effect of the longitudinal rib, given by:

$$k_{\ell} = 0,6 \frac{b_0}{h_p} \left(\frac{h_{sc}}{h_p} - 1 \right) \quad \text{but } k_{\ell} \leq 1,0 \quad (85)$$

h_p is the height of the steel profiled decking (§ 4.5.1);

h_{sc} is the height of the stud;

b_0 is the representative width of a rib (see § 4.5.1 (4)).

(2) is also applied for composite slabs with perpendicular ribs, when the configuration at beam support is PE3 – see § 4.4.3. In this case, $b_0 = b_{fs} - 2 b_{app,min}$.

- (3) The shear resistance P_{Rd} of a welded stud in a composite slab with ribs perpendicular to the beam axis, with PE1 or PE2 configurations – see § 4.4.3, is obtained by the following formula, according to § 6.6.5.1 of EN 1994-1-1:

$$P_{Rd} = k_t \min\{P_{Rd,s}; P_{Rd,c}\} \quad (86)$$

where: $P_{Rd,s}$ and $P_{Rd,c}$ are as defined by Formulas (82) and (83) respectively;

k_t is a reduction factor to take account of the effect of the longitudinal rib, given by:

$$k_t = \frac{0,7}{\sqrt{n_r}} \frac{b_0}{h_p} \left(\frac{h_{sc}}{h_p} - 1 \right) \quad \text{but } k_t \leq k_{t,max} \quad (87)$$

h_p is the height of the steel profiled decking (§ 4.5.1);

h_{sc} is the height of the stud;

n_r is the number of studs in the through, which should not be greater than 2;

$k_{t,max}$ is the maximal value of the reduction factor k_t , which is given by **Table 23**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	79	Rev.: B

Number of studs per rib	t (mm) Thickness of the steel sheeting	Through deck welding $d \leq 20$ mm	Prepunched steel sheeting $d = 19$ or 22 mm
$n_r = 1$	$\leq 1,0$	0,85	0,75
	$> 1,0$	1,0	0,75
$n_r = 2$	$\leq 1,0$	0,70	0,60
	$> 1,0$	0,80	0,60

Table 23: Upper limits of the reduction factor k_t

(4) The scope of application for Formula (86) is as follows:

- through deck welding: the diameter of the shank is not greater than 20 mm: $d \leq 20$ mm
- prepunched sheeting: $19 \text{ mm} \leq d \leq 22 \text{ mm}$
- the height of the sheeting is lower than 85 mm: $h_p \leq 85 \text{ mm}$
- the width b_0 of the rib should be at least equal to h_p : $b_0 \geq h_p$
- the number of studs in the rib is not greater than 2: $n_r \leq 2$

Other configurations are out of the scope of the **ABC-PMX**.

6.8 Calculation of elastic stresses – Usual Sections

6.8.1 Principles

- (1) For each cross-section, elastic stresses are assessed, if required, at predefined stress points.
- (2) Under the action of a bending moment M_{Ed} , the normal stresses at a stress point i of the structural steel profile are obtained by:

$$\sigma_i = M_{Ed} (z_i - z_{el}) / I_y \quad (88)$$

where: I_y is the second moment of area of the cross-section;

z_{el} is the location of the neutral axis;

z_i is the location of the point i .

For composite sections, properties of the transformed cross-section with the relevant modular ratios, are used.

- (3) For composite cross-section under the action of a bending moment M_{Ed} , the normal stresses at a stress point i of the steel reinforcement are obtained by:

$$\sigma_i = M_{Ed} \frac{(z_i - z_{el})}{n_s I_{y,h}} \quad (89)$$

where n_s is the modular ratio for the reinforcement, given by $n_s = E_a / E_s$;

$I_{y,h}$ is the second moment of area of the transformed cross-section.

- (4) For composite cross-section under the action of a bending moment M_{Ed} , the normal stresses at a stress point i in concrete is obtained by:

$$\sigma_i = M_{Ed} \frac{(z_i - z_{el})}{n_c I_{y,h}} \quad (90)$$

where n_c is the modular ratio for concrete (either partial concrete encasement or slab), obtained considering the load case;

$I_{y,h}$ is the second moment of area of the transformed cross-section.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	80	Rev.: B

- (5) Under the action of a shear force V_{Ed} , the shear stresses at the stress points are obtained by:

$$\tau_i = \frac{V_{Ed} S_i}{t_w I_y} \quad \text{at stress points in the web of the structural steel profile} \quad (91)$$

$$\tau_i = 0 \quad \text{at other stress points}$$

where: S_i is the first moment of area associated to the stress point;

t_w is the thickness of the web;

I_y is the second moment of area of the cross-section.

For composite sections, S_i and I_y are obtained considering the transformed cross-section with the relevant modular ratios.

- (6) For composite sections, elastic stresses are calculated separately for each load case, considering the relevant modular ratio and the associated properties of the transformed cross-section.
- (7) For composite sections, stresses in concrete and in reinforcement are calculated for each load case considering the relevant properties under positive bending moments, both in tension and in compression. After combinations, either at serviceability limit state or ultimate limit state, the following process are applied:
- if the stresses in the concrete slab are in compression, the bending moment hypothesis is kept and the calculations are unchanged;
 - if the stresses in the concrete slab are in tension, the bending moment hypothesis is not correct, and the calculations are carried out again considering negative bending moments.

6.8.2 Usual steel sections

- (1) For usual steel cross-sections without partial concrete encasement, the following 5 stress points are defined in the vertical axis of the section (**Figure 51**):
- at the external (point ①) and internal (②) sides of the upper flange;
 - at the centroid of the section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.

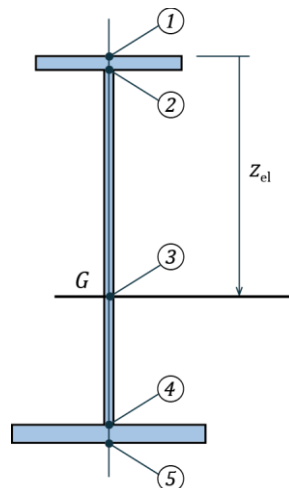


Figure 51 : Calculation of elastic stresses for usual steel sections

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	81	Rev.: B

- (2) The location of the five stress points are given in **Table 24**.

i	z_i
①	$z_1 = 0$
②	$z_2 = -t_{fs}$
③	$z_3 = z_{ela}$
④	$z_4 = -h_s + t_{fi}$
⑤	$z_5 = -h_s$

Table 24: Location of the stress points of usual steel sections

- (3) Under the action of an axial force N_{Ed} , the normal stresses at these 5 points are obtained by:

$$\sigma_i = N_{Ed} / A_a \quad (92)$$

where: A_a is the area of the cross-section.

- (4) Under the action of a bending moment M_{Ed} , the normal stresses at the 5 stress points are obtained with Formula (88), where I_y is the second moment of area of the section, $I_{y,a}$ and the location z_i are as given in **Table 24**.
- (5) Under the action of a shear force V_{Ed} , the shear stresses are obtained by Formula (91) for the stress point in the web ② to ④, where S_i is the first moment of area as given in **Table 25**.

i	S_i
②	$S_2 = b_{fs} t_{fs} (z_{el} - t_{fs} / 2)$
③	$S_3 = b_{fs} t_{fs} (z_{el} - t_{fs} / 2) + (z_{el} - t_{fs})^2 t_w / 2$
④	$S_4 = b_{fi} t_{fi} (z_{el} - (h_s - t_{fi} / 2))$

Table 25: First moments of areas for the calculation of shear stresses

6.8.3 Usual steel sections with partial concrete encasement

- (1) For usual steel cross-sections with partial concrete encasement, the stress points comprise the five same points as in § 6.8.2 (1), plus the following points in the concrete and the reinforcements in the encasement (**Figure 52**):
- at the upper (point ⑥) and lower (⑦) sides of the concrete encasement;
 - at the first, second and third layers of reinforcements, if any (points ⑧, ⑨ and ⑩ respectively).



Only active reinforcement (see § 4.9.2) are considered for the check of stresses (see § 8.2.2).

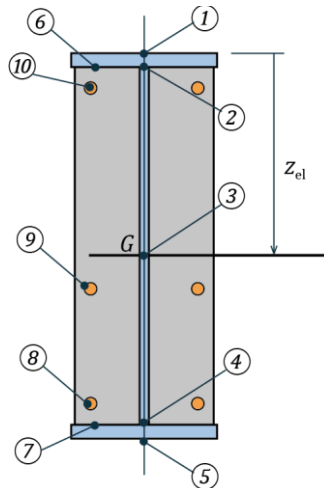


Figure 52 : Calculation of elastic stresses for usual steel sections with partial concrete encasement

- (2) The location of the five stress points in the steel profile are given in **Table 24**. The location of the other stress points are given in **Table 26**.

i	z_i
⑥	$z_6 = -t_{fs}$
⑦	$z_7 = -h_s + t_{fi}$
⑧	$z_8 = z_{es1}$
⑨	$z_9 = z_{es2}$
⑩	$z_{10} = z_{es3}$

Table 26: Location of the points for stress calculations

- (3) Under the action of a bending moment M_{Ed} , the normal stresses are obtained:
- by Formula **(88)**, for the stress points ① to ⑤ (structural steel);
 - by Formula **(89)**, for the stress points ⑧ to ⑩ (steel reinforcement);
 - by Formula **(90)**, for the stress points ⑥ and ⑦ (concrete), with the modular ratio n_{ec} for the concrete of the encasement and the associated load case (see § 6.3);
- where, in these formulas, $I_{y,h}$ is the second moment of area of the transformed section, and the location z_i is given in (2).
- (4) Under the action of a shear force V_{Ed} , the same formula as in § 6.8.2 (5) applies. No shear stress is considered neither in concrete nor in steel reinforcement.

6.8.4 Usual composite sections

- (1) For usual composite cross-sections without partial concrete encasement, the 8 stress points are defined as follows (**Figure 53**):
- at the external (point ①) and internal (②) sides of the upper flange;
 - at the centroid of the steel section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.
 - at the upper side of the concrete face (point ⑪);
 - at the first and second layers of longitudinal reinforcement of the slab, if any (points ⑫ and ⑬).

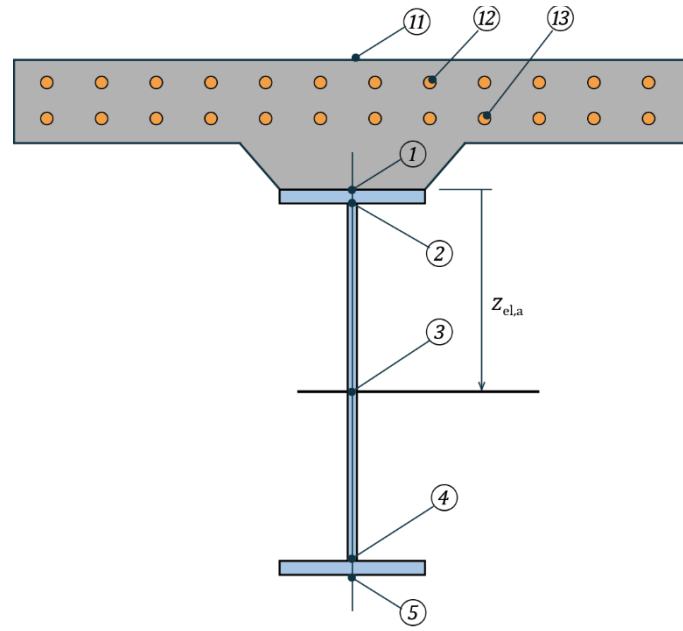


Figure 53 : Calculation of elastic stresses for usual composite sections

- (2) The location of the five stress points in the steel profile are given in **Table 24**. The location of the other stress points are given in **Table 27**.

i	z_i
⑪	$z_{11} = t_d + t_h$
⑫	$z_{12} = z_{s1}$
⑬	$z_{13} = z_{s2}$

Table 27: Location of the stress points in the slab of usual composite sections

- (3) Under the action of a bending moment M_{Ed} , the normal stresses are obtained:
- by Formula (88), for the stress points ① to ⑤ (structural steel);
 - by Formula (89), for the stress points ⑫ and ⑬ (steel reinforcement);
 - by Formula (90), for the stress point ⑪ (concrete), with the modular ratio n_c for the concrete of the slab and the associated load case (see § 6.3);

where, in these formulas, $I_{y,h}$ is the second moment of area of the transformed section, and the location z_i is given in (2).

- (4) Under the action of a shear force V_{Ed} , the same formula as in § 6.8.2 (5) applies. No shear stress is considered neither in concrete nor in steel reinforcement.
- (5) Under the axial force N_{sh} due to the shrinkage of concrete, the normal stresses in the stress points are obtained by:

$$\sigma_i = N_{sh} / A_h \quad \text{for points ① to ⑤ (structural steel);} \quad (93)$$

$$\sigma_i = \frac{N_{sh}}{n_s A_h} \quad \text{for the stress points ⑫ and ⑬ (steel reinforcement);} \quad (94)$$

$$\sigma_i = N_{sh} \left(\frac{1}{n_c A_h} - \frac{1}{A_c} \right) \quad \text{for the stress point ⑪ (concrete)} \quad (95)$$

where: n_c is the modular ratio for the concrete of the slab and the associated load case (see § 6.3);

A_h is the area of the transformed section;

A_c is the area of the slab;

N_{sh} is the axial force in the slab due to shrinkage, defined by § 7.7.3 (4).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	84	Rev.: B

6.8.5 Usual composite section with partial concrete encasement

(1) For usual composite cross-sections with partial concrete encasement, the 13 stress points are defined as follows (**Figure 54**):

- at the external (point ①) and internal (②) sides of the upper flange,
- at the centroid of the steel section (③),
- at the external (point ⑤) and internal (④) sides of the lower flange,
- at the upper (point ⑥) and lower (⑦) sides of the concrete encasement,
- at the first, second and third layers of reinforcements in the partial encasement, if any (points ⑧, ⑨ and ⑩ respectively),
- at the upper side of the concrete face (point ⑪),
- at the first and second layers of longitudinal reinforcement of the slab, if any (points ⑫ and ⑬).



Only active reinforcement (see § 4.9.2) are considered for the check of stresses (see § 8.2.2).

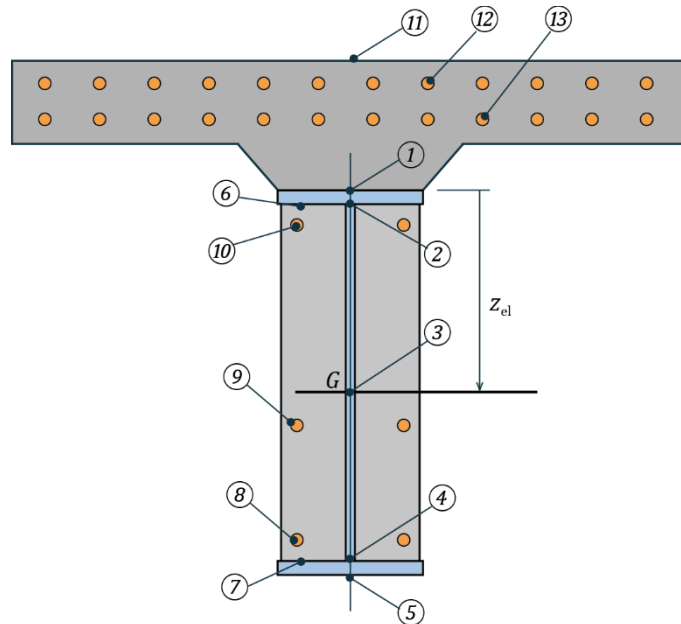


Figure 54: Calculation of elastic stresses for usual composite sections with partial concrete encasement

- (2) The location of the five stress points in the steel profile (① to ⑤) are given in **Table 24**. The location of the 5 stress points of the encasement (⑥ to ⑩) are given in **Table 26**. The location of the other stress points for the slab (⑪ to ⑬) are given in **Table 27**.
- (3) Under the action of a bending moment M_{Ed} , the normal stresses are obtained:
- by Formula (88), for the stress points ① to ⑤ (structural steel);
 - by Formula (89), for the stress points ⑧ to ⑩, ⑫ and ⑬ (steel reinforcement);
 - by Formula (90), for the stress point ⑪ (concrete), with the modular ratio n_c for the concrete of the slab and the associated load case (see § 6.3);
 - by Formula (90), for the stress points ⑥ and ⑦ (concrete), with the modular ratio n_{ec} for the concrete of the encasement and the associated load case (see § 6.3).
- where, in these formulas, $I_{y,h}$ is the second moment of area of the transformed section, and the location z_i is given in (2).
- (4) Under the action of a shear force V_{Ed} , the same formula as in § 6.8.2 (5) applies. No shear stress is considered neither in concrete nor in steel reinforcement.
- (5) Under the axial force N_{sh} due to the shrinkage of the slab concrete (defined by § 7.7.3 (4)), the normal stresses in the stress points are obtained by:

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	85	Rev.: B

Formula **(93)** for points ① to ⑤ (structural steel);

Formula **(94)** for the stress points ⑫ and ⑬ (steel reinforcement);

Formula **(95)** for the stress point ⑪ (concrete of the slab);

$$\sigma_i = \frac{N_{sh}}{n_{ce} A_h} \quad \text{for the stress points ⑥ and ⑦ (concrete of the encasement)} \quad (96)$$

where: n_{ce} is the modular ratio for the concrete of the encasement (see § 6.3);

A_h is the area of the transformed section;

A_c is the area of the slab;

N_{sh} is the axial force in the slab due to shrinkage,.

- (6) Under the axial force N_{she} due to the shrinkage of the encasement concrete (defined by § 7.7.3 (5)), the normal stresses in the stress points are obtained by:

$\sigma_i = N_{she} / A_h$ for points ① to ⑤ (structural steel);

$\sigma_i = \frac{N_{she}}{n_s A_h}$ for the stress points ⑫ and ⑬ (steel reinforcement);

$\sigma_i = \frac{N_{she}}{n_c A_h}$ for the stress point ⑪ (concrete of the slab);

$$\sigma_i = N_{she} \left(\frac{1}{n_{ce} A_h} - \frac{1}{A_{ce}} \right) \quad \text{for the stress points ⑥ and ⑦ (concrete of the encasement)} \quad (97)$$

6.8.6 Equivalent stresses in the steel profile

- (1) For the points in the structural steel section, the equivalent von Mises stresses, combining the effect of normal and shear stresses, are obtained as follows:

$$\sigma_{eq,i} = \sqrt{\sigma_i^2 + 3 \tau_i^2} \quad (98)$$

6.8.7 Effect of concrete cracking

- (1) In composite sections where the concrete is assumed to be cracked (i.e. $M_{Ed} < 0$), stresses in the slab reinforcements increase due to the effects of tension stiffening of concrete between cracks. The additional stresses due to this effect are given by (according to EN 1994-1-1, 7.4.3 (3)):

$$\Delta\sigma_s = \frac{0,4 f_{ctm}}{\alpha_{st} \rho_s} \quad (99)$$

where: f_{ctm} is the mean tensile stress of concrete, see § 5.2 (3);

ρ_s is the ratio of reinforcement, see § 4.4.5 (5);

α_{st} is obtained by:

$$\alpha_{st} = \frac{A I}{A_a I_a} \quad (100)$$

A and I are the area and the second moment of area, respectively, of the transformed composite section, calculated considering negative bending moment, according to § 6.5.5 or § 6.5.6;

A_a and I_a are the area and the second moment of area, respectively, of the steel section.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	86	Rev.: B

7 GLOBAL ANALYSIS

7.1 General

- (1) Global analysis is carried out with a Finite Element module (see **Annex C**), using elastic beam elements with two degrees of freedom at each node (vertical displacement and rotation).
- (2) All load cases are assessed considering the state of concrete (if any) and the associated modular ratio.
- (3) Deflections are calculated under elastic assumptions. For composite beams, they take into account the loading sequence.

7.2 Design sections

- (1) Shear forces, bending moments and deflections are calculated at design sections along the beam. Design checks are also performed at each of these design sections.
- (2) The design sections correspond to the discretisation of the beam used in the Finite Elements analysis (**Annex C**).

7.3 Critical sections

- (1) Critical sections are defined as follows, according to clause 6.1.1 (4) of EN 1994-1-1:

- sections where the bending moment is maximum;
- at supports;
- at point load locations;
- at point propping.



*Sections where the bending moment is maximum are not known when design sections are established first.
The vertical shear force is always assessed considering both LHS and RHS sides of the cross-sections.*

- (2) ULS verifications are carried out at each design section.

7.4 Cracked and uncracked segments of the slab

- (1) § 7.4 applies to usual composite sections.
- (1) For a beam with a single span, the full length of the beam is assumed to be uncracked.
- (2) For a beam with a main span and one or two cantilever spans, the segment of the beam where the concrete is cracked is obtained by a simplified method as follows. The full length of the cantilever spans is assumed to be with cracked concrete. For the main span, it is assumed that 15% of the span on each side of each internal support (i.e. with a cantilever span) are assumed to be cracked (**Figure 55**).

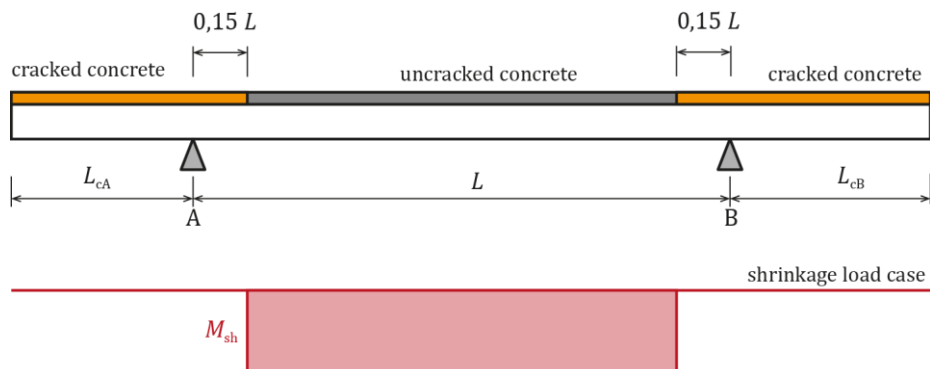


Figure 55: Simplified determination of the cracked and uncracked concrete segments of the slab

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	87	Rev.: B

7.5 Cracked parts of the encasement concrete

- (1) This chapter applies to usual section with partial concrete encasement.
- (2) The properties of the elements are assessed according to § 6.5.4 or 6.5.6, considering a negative bending moment in the section where the slab would be cracked and positive bending moment where the slab would be uncracked.

7.6 Load cases

- (1) The load cases considered in the global analysis are described in **Table 28**.

Load case	Phase	ψ_L	t_0	n_L
Q_c (except for fully propped beams)	Steel	/	/	/
G for steel beams	Steel	/	/	/
G for fully propped composite beams	Composite	1,1	$t_{0,G1}$	long term
G_1 for unpropped composite beams	Steel	/	/	/
G_{1pp} for point propped comp. beams	Steel	/	/	/
G_{1c} for point propped comp. beams	Composite	1,1	$t_{0,G1c}$	long term
G_2 (unpropped or point propped)	Composite	1,1	$t_{0,G2}$	long term
Q_1	Steel or Composite	0	/	short term
Q_2	Steel or Composite	0	/	short term
SH shrinkage (composite beams only)	Composite	0,55	$t_{0,SH}$	long term
EM Eigen mode for vibration	Steel or Composite	0	/	short term

Table 28: Load cases considered in the global analysis

Shrinkage load case is assessed only for beams with usual composite sections, if an elastic design is carried out or if required for the calculation of deflections.

Load cases G_{1pp} and G_{1c} are detailed in § 7.7.1.

- (2) The global analysis is carried out for each load case by Finite Element Analysis.
- (3) For each load case (except the eigen mode), the calculation sheet gives:
 - the internal bending moment and shear force along the beam,
 - the reaction at supports,
 - the maximal deflection of each span.

7.7 Specific load cases

7.7.1 Permanent load case G_1

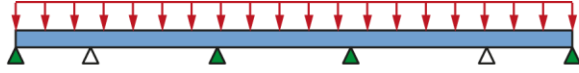
- (1) Load case G_1 is assessed for composite beams, either unpropped or with point proppings. It includes the self-weight loads for concrete, profiled sheeting and steel, as described in § 7.7.2.
- (2) For beams with point propping (see § 4.12), the load case G_1 is decomposed as follows (**Figure 56**):

$$G_1 = G_{1,pp} + G_{1,c}$$

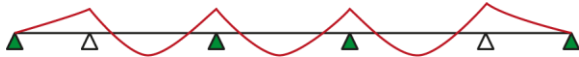
where: $G_{1,pp}$ corresponds to all loads applied on the beam and resisted by steel only (no composite action), considering the temporary intermediate supports;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	88	Rev.: B

$G_{1,c}$ is the load case corresponding to the state of the beam after the point propping has been removed. The reaction forces at temporary supports obtained from $G_{1,pp}$ are introduced downwards on the beam, which then acts compositely.



Application of the loads G_1 on the steel beam with point proppings



Global analysis for load case $G_{1,pp}$
Steel properties of the cross-sections
=> Reactions at proppings



Application of the reaction forces on the composite beam without proppings



Global analysis for load case $G_{1,c}$
Transformed composite properties of the cross-sections

Figure 56: Specific load cases for beams with point propping

7.7.2 Self-weight

- (1) The “self-weight” loads, associated to the load case G or G_1 , is automatically be assessed by **ABC-PMX**, as a uniformly distributed load along the beam.
- (2) The distributed load for the self-weight of the beam is obtained by:

$$q_{sw} = g (\rho_a A_a + \rho_c A_c + \mu_p d_c + \rho_{ec} A_{ec}) \quad (101)$$

where: g is the acceleration of gravity (see § 4.17.2),

ρ_a is the density of steel (see § 5.1.1),

ρ_c is the density of the slab concrete (see § 5.2),

ρ_{ec} is the density of the encasement concrete, if any (see § 5.2),

A_a is the area of the steel profile,

A_c is the area of the concrete slab – see (3),

A_{ec} is the area of the concrete in partial encasement – see (4),

μ_p is the surface mass of the steel profile sheeting that is supported by the beam (see § 4.5.3),

$\mu_p = 0$ for solid slabs,

d_c is the width of the slab, supported by the beam: $d_c = d_1 + d_2$

d_1 and d_2 are the influence widths of the slab, obtained by (see § 4.1):

$d_1 = b_1$ for an edge beam,

$d_1 = b_1 / 2$ for an intermediate beam,

$d_2 = b_2 / 2$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	89	Rev.: B

(3) A_c is calculated as follows:

- for solid slabs (with or without prefabricated parts):

$$A_c = d_c t_d + t_h (b_{fs} + t_h \tan(\theta_h) / 2)$$

- for composite slabs:

$$A_c = d_c t_c \left(1 + \frac{b_m h_p}{e_p t_c} \right)$$

$$\text{where } t_c = t_d - h_p$$

(4) A_{ec} is equal to 0 for usual cross-sections without partial concrete encasements. For usual cross-sections with partial concrete encasement, A_{ec} is given by § 4.9 (14).

7.7.3 Shrinkage load case

- The shrinkage load case is automatically assessed for the calculation of the deflection and for the calculation of elastic stresses, when elastic design is used.
- The shrinkage load case is used only for composite beams.
- The shrinkage load case consists of a normal force N_{sh} and a bending moment M_{sh} (Figure 55) applied to the uncracked segment of the main span (see § 7.4).
- For usual composite section without partial concrete encasement, the applied shrinkage bending moment, M_{sh} , due to the shrinkage in the slab, is obtained as follows:

$$M_{sh} = N_{sh} z_{sh} \quad (102)$$

where: N_{sh} is the normal tension force in the slab due to shrinkage, given by:

$$N_{sh} = \frac{E_a}{n_{sh}} b_{eff} t_c \varepsilon_{sh} \quad (103)$$

b_{eff} is the effective width of the slab at mid-span – see § 6.1;

t_c is the thickness of active part of the slab – see § 6.2;

E_a is the Young's modulus of structural steel – see § 5.1.1;

n_{sh} is the modular ratio for the shrinkage load case – see § 6.2;

ε_{sh} is the shrinkage deformation – see § 5.2 (5);

z_{sh} is the distance between the centroid of the active part of the slab and the centroid of the composite cross-section, considering n_{sh} as the modular ratio.

- For sections with partial concrete encasement, the effect of shrinkage in the concrete of the encasement is not considered.

7.7.4 Live load arrangements

- When the beam comprises one or two cantilever spans, the live loads Q_1 and Q_2 are split in three separate load cases associated to three loading arrangements as given in Table 29.

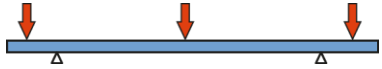
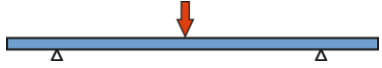
Arrangement	Q_1	Q_2	Description	Figure
#1	$Q_{1\#1}$	$Q_{2\#1}$	All spans are loaded	
#2	$Q_{1\#2}$	$Q_{2\#2}$	Only the main span is loaded	
#3	$Q_{1\#3}$	$Q_{2\#3}$	Only the cantilever spans are loaded	

Table 29: Live load arrangements

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	90	Rev.:	B

- (2) The same principle of live load arrangement is considered for load case Q_c with point propping, for the verification of the resistance to lateral torsional buckling.

7.8 Calculation of internal forces, moments and deflections

- (1) For each load case, **ABC-PMX** assesses:
- the internal shear force and bending moment at each design section;
 - the deflection of each span.
- This calculation is carried out through Finite Element analysis (**Annex C**).

8 VERIFICATIONS OF BEAMS AT ULTIMATE LIMIT STATES

8.1 General

- (1) Chapter **8** applies to beams with steel or composite sections, either with or without partial concrete encasement. It also applies to beams with usual composite sections in construction stage.
- (2) For each design section, following checks are performed by the software:
- resistance of design sections to bending moment – see **§ 8.2**;
 - resistance of design sections to shear force – see **§ 8.3**;
 - resistance to MV interaction – see **§ 8.4**;
 - resistance to lateral torsional buckling, when relevant – see **§ 8.5**.
- (3) Sections are classified before the verifications –see **§ 6.4**. Class 4 cross-sections are out of the scope of **ABC-PMX**.
- (4) For continuous beams with usual composite sections, requirements for the reinforcement of the slab at supports apply if a plastic design is used, see **§ 8.6**.
- (5) For welded profiles, a design of the weld is given in **§ 8.7**.
- (6) Resistance to transverse forces is not checked by **ABC-PMX**. For concentrated loads, including reactions at supports, either in the construction phase or in the final composite one, the User should review the detailing of the beam and check the local stability of the web if required.

8.2 Verification of the resistance to bending moment

8.2.1 Usual steel sections

- (1) For each design section, the following criterion is checked:

$$\Gamma_M = \frac{M_{Ed}}{M_{Rd}} \leq 1 \quad (104)$$

where: M_{Ed} is the design bending moment in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

M_{Rd} is the design resistance of the cross-section to bending moment. According to the type and the class of cross-section, M_{Rd} is obtained according to the references mentioned in **(2)** to **(4)**.

- (2) For Class 1 or 2 cross-sections, M_{Rd} is equal to the plastic bending resistance $M_{pl,a,Rd}$, which is obtained according to **§ 6.5.1 (4)** or **6.5.3 (4)**.
- (3) For Class 3 sections, M_{Rd} is equal to the elastic bending resistance $M_{el,a,Rd}$, which is obtained according to **§ 6.5.1 (5)** or **6.5.3 (5)**.
- (4) If elastic design is imposed, clause **(3)** applies whatever the class of the cross-section is.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	91	Rev.: B

8.2.2 Usual steel sections with partial concrete encasement

- (1) For each design section of class 1 or 2, criterion **(104)** is used, where M_{Rd} is equal to the plastic resistance of the cross-section, obtained according to § 6.5.
- (2) For each cross-section of class 3, following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for the structural steel, points ① and ⑤;} \quad (105)$$

$$\Gamma_{Mce} = \frac{|\sigma_i|}{f_{ced}} \quad \text{for the encasement concrete, points ⑥ and ⑦;} \quad (106)$$

$$\Gamma_{Mse} = \frac{|\sigma_i|}{f_{sed}} \quad \text{for the reinforcement active rebars, points ⑧ to ⑩};$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mce}; \Gamma_{Mse}\} \leq 1$$

where the normal stresses σ_i are calculated according to § 6.8.3;

f_{yd} is the design yield strength of the structural steel, see **Table 30** and **(3)**;

f_{ced} is the design compressive strength of the encasement concrete, obtained from § 5.2 considering the characteristic strength f_{cek} ;

f_{sed} is the design yield strength of the reinforcing steel in concrete encasement, see § 5.3.



See § 4.9.2 for the definition of active rebars of the partial encasement.

- (3) For the criterion Γ_{Ma} , f_{yd} is obtained as given in **Table 30**.

i	f_{yd}
①	f_{yfs} / γ_{M0}
②	$\min\{f_{yfs}; f_{yw}\} / \gamma_{M0}$
③	f_{yw} / γ_{M0}
④	$\min\{f_{yfi}; f_{yw}\} / \gamma_{M0}$
⑤	f_{yfi} / γ_{M0}

Table 30: Design yield strengths in steel sections

- (4) If elastic design is imposed, clause **(2)** applies whatever the class of the cross-section is.

8.2.3 Usual composite sections

- (1) For each design section of class 1 or 2, criterion **(104)** applies, where M_{Rd} is equal to the plastic resistance of the composite cross-section, obtained according to § 6.5.5, taking into account the actual degree of shear connection η of the cross-section. η is obtained according to § 9.2.2 (1), where ΣP_{Rdi} is the sum of the resistance of all studs located between the cross-section and the support (the lowest value for the two supports being taken into account). For a support with an adjacent span, ΣP_{Rdi} is replaced by $\Sigma P_{Rdi} - A_s f_{sd}$, see § 9.2.2 (2).
- (2) For each cross-section of class 3, the following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for the structural steel, points ① and ⑤;} \quad (106)$$

$$\Gamma_{Mc} = \frac{|\sigma_i|}{f_{cd}} \quad \text{for the concrete of the slab, point ⑪};$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	92	Rev.: B

$$\Gamma_{Ms} = \frac{|\sigma_i|}{f_{sd}} \quad \text{for the reinforcement rebars, points (12) and (13); (*)}$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mc}; \Gamma_{Ms}\} \leq 1$$

where the normal stresses σ_i are calculated according to § 6.8.3;

f_{yd} is the design yield strength of the structural steel, see § 8.2.2 (3);

f_{cd} is the design compressive strength of the concrete, obtained from § 5.2 considering the characteristic strength f_{ck} ;

f_{sd} is the design yield strength of the slab reinforcement, see § 5.3.

- (3) The criterion Γ_{Ms} is applied for reinforcement in tension. It is also applied to reinforcement in compression if the associated option has been activated.
- (4) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

8.2.4 Usual composite section with partial concrete encasement

- (1) For each design section of class 1 or 2, criterion (104) where M_{Rd} is equal to the plastic resistance of the cross-section, obtained according to § 6.5.6, taking into account the degree of connection as described in § 8.2.3 (1).
- (2) For each cross-section of class 3, the following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for the structural steel, points (1) and (5);} \quad (107)$$

$$\Gamma_{Mce} = \frac{|\sigma_i|}{f_{ced}} \quad \text{for the encasement concrete, points (6) and (7);}$$

$$\Gamma_{Mse} = \frac{|\sigma_i|}{f_{sed}} \quad \text{for the reinforcement active rebars, points (8) to (10);}$$

$$\Gamma_{Mc} = \frac{|\sigma_i|}{f_{cd}} \quad \text{for the concrete of the slab, point (11);}$$

$$\Gamma_{Ms} = \frac{|\sigma_i|}{f_{sd}} \quad \text{for the reinforcement rebars, points (12) and (13);}$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mc}; \Gamma_{Ms}; \Gamma_{Mce}; \Gamma_{Mse}\} \leq 1$$

where the normal stresses σ_i are calculated according to § 6.8.3;

f_{yd} is the design yield strength of the structural steel, see § 8.2.2 (3);

f_{cd} is the design compressive strength of the concrete, obtained from § 5.2 considering the characteristic strength f_{ck} ;

f_{sd} is the design yield strength of the slab reinforcement, see § 5.3;

f_{ced} is the design compressive strength of the encasement concrete, obtained from § 5.2 considering the characteristic strength f_{cek} ;

f_{sed} is the design yield strength of the reinforcing steel in concrete encasement, see § 5.3.



See § 4.9.2 for the definition of active rebars of the partial encasement.

- (3) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	93	Rev.:	B

8.3 Verification of the resistance to shear force

8.3.1 General criterion

- (1) Except when an elastic design is imposed, the following criterion is checked for each design section:

$$\Gamma_V = \frac{V_{Ed}}{V_{Rd}} \leq 1 \quad (108)$$

where: V_{Ed} is the design shear force in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

V_{Rd} is the design resistance of the cross-section to shear force, as given in § 8.3.2 or 8.3.3.

- (2) When an elastic design based on the von Mises criteria is imposed (see § 4.17.2), the following criterion is applied, for all sections:

$$\Gamma_V = \frac{\tau_{Ed}}{f_{yw} / (\sqrt{3} \gamma_{M0})} \quad (109)$$

where: τ_{Ed} is the maximal shear stress at points ②, ③ and ④, obtained according to § 6.8.

8.3.2 Usual cross-sections without partial concrete encasement

- (1) For steel or composite sections without partial concrete encasement, V_{Rd} is calculated as follows.

- (2) Webs are considered as compact when the following criterion is fulfilled (acc. to EN 1993-1-5, 5.1 (2)):

$$\frac{h_w}{t_w} \leq 72 \frac{\varepsilon_w}{\eta} \quad (110)$$

where: η is the parameter as defined by EN 1993-1-5: 2005, clause 5.1 (2). See § 4.17.2 for the value of η used in **ABC-PMX**.

ε_w is the steel factor see § 5.1.4 (2).

- (3) For compact webs, V_{Rd} is equal to the plastic resistance and is obtained by:

$$V_{Rd} = V_{pl,Rd} = \frac{A_v f_{yw}}{\sqrt{3} \gamma_{M0}} \quad (111)$$

where: A_v is the shear area – see § 6.6.

- (4) For slender webs, V_{Rd} is equal to the resistance to shear buckling and is obtained by:

$$V_{Rd} = V_{b,Rd} = \chi_w \frac{h_w t_w f_{yw}}{\sqrt{3} \gamma_{M1}} \quad (112)$$

where: χ_w is the reduction factor for shear buckling, which is given in **Table 31**.

$\bar{\lambda}_w$ is the reduced slenderness for shear buckling, which is calculated by:

$$\bar{\lambda}_w = \frac{h_w}{t_w} \frac{1}{37,4 \varepsilon_w \sqrt{k_\tau}} \quad (113)$$

$$k_\tau = 5,34$$



It is assumed that the beam has no intermediate transverse stiffener and that it has non rigid posts at supports.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	94	Rev.: B

$\bar{\lambda}_w$	Main span with 2 adjacent cantilever spans Cantilever spans	Main span without adjacent cantilevers spans or with one cantilever span
$\bar{\lambda}_w \leq 0,83 / \eta$	η	η
$0,83 / \eta < \bar{\lambda}_w \leq 1,08$	$0,83 / \bar{\lambda}_w$	$0,83 / \bar{\lambda}_w$
$\bar{\lambda}_w > 1,08$	$1,37 / (0,7 + \bar{\lambda}_w)$	$0,83 / \bar{\lambda}_w$

Table 31: Reduction factor for shear buckling

8.3.3 Usual cross-sections with partial concrete encasement

- (1) For steel or composite sections with partial concrete encasement, V_{Rd} is calculated as follows.
- (2) Webs are considered as compact when the following criterion is fulfilled (acc. to EN 1994-1-1, 6.3.1 (2)):

$$\frac{d_w}{t_w} \leq 124 \varepsilon_w \quad (114)$$

- (3) For compact webs, V_{Rd} is equal to the plastic resistance and is obtained by Formula (111) in 8.3.2 (3).



The contribution of the concrete of the partial concrete encasement is neglected. The shear force is assumed to be resisted by the steel section alone.

- (4) Slender webs are outside the scope of application of the software.

8.4 Verification of the resistance to MV interaction – First generation of EN

8.4.1 General

- (1) Tree MV criteria are possible according to the situation:
 - the plastic MV interaction criterion (see § 8.4.2);
 - the EN 1993-1-5 MV interaction criterion (see § 8.4.3);
 - the elastic MV interaction criterion (see § 8.4.4).
- (2) The plastic criterion is applied to all steel sections with compact webs, whatever their class is, except if an elastic calculation with von Mises criteria is used.
- (3) The EN 1993-1-5 criterion is used for:
 - steel or composite sections with slender webs;
 - composite sections with Class 3 cross-sections.
- (4) The elastic criterion is used when an elastic design with von Mises principle is used.

8.4.2 General criterion for usual sections with compact webs

- (1) The definition of compact webs is given in § 8.3.2 (2) for cross-sections without partial concrete encasement and in § 8.3.3 (2) for cross-sections with partial concrete encasement.
- (2) For each design section with a compact web, the following criterion is checked for the MV interaction:

$$\Gamma_{MV} = \frac{M_{Ed}}{M_{V,Rd}} \leq 1 \quad (115)$$

where: M_{Ed} is the design bending moment in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

$M_{V,Rd}$ is the plastic design resistance of the cross-section to bending moment, taking account of the presence of the shear force.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	95	Rev.: B



$M_{V,Rd}$ is the plastic shear resistance of the cross-section whatever its class is.

- (3) $M_{V,Rd}$ is obtained to § 6.5, by considering a reduced yield strength for the web, given by:

$$\begin{aligned} f_{y,w} &= (1 - \rho) f_y & \text{when } V_{Ed} > 0,5 V_{pl,Rd} \\ f_{y,w} &= f_y & \text{when } V_{Ed} \leq 0,5 V_{pl,Rd} \end{aligned} \quad (116)$$

where : $V_{pl,Rd}$ is the plastic shear resistance of the web, given by § 8.3.2 (3);

$$\rho = \left(2 \frac{V_{Ed}}{V_{pl,Rd}} - 1 \right)^2$$

For usual composite section, $M_{V,Rd}$ is assessed taking into account the actual degree of shear connection, as given in § 8.2.3 (1) or 8.2.4 (1).

8.4.3 General criterion for usual section with slender webs

- (1) For slender webs of sections without partial concrete encasement, the following criterion is used, according to EN 1993-1-5, § 7.1:

$$\begin{aligned} \Gamma_{MV} &= \frac{M_{Ed}}{M_{pl,Rd}} + \left(1 - \frac{M_{f,Rd}}{M_{pl,Rd}} \right) \left(2 \frac{V_{Ed}}{V_{b,Rd}} - 1 \right)^2 \leq 1 & \begin{aligned} &\text{if } M_{Ed} > M_{f,Rd} \\ &\text{and if } V_{Ed} > 0,5 V_{b,Rd} \end{aligned} \\ \Gamma_{MV} &= \frac{M_{Ed}}{M_{pl,Rd}} \leq 1 & \begin{aligned} &\text{if } M_{Ed} \leq M_{f,Rd} \\ &\text{or if } V_{Ed} \leq 0,5 V_{b,Rd} \end{aligned} \end{aligned} \quad (117)$$

where: $M_{pl,Rd}$ is the plastic resistance of the composite gross cross-section, whatever its class is;

$M_{f,Rd}$ is the plastic resistance of the composite cross-section neglecting the contribution of the web;

$V_{b,Rd}$ is the resistance to shear buckling, according to § 8.3.2 (4).



As Class 4 cross-sections are outside the scope of the software, the gross area of the flanges is always fully effective.

In ABC-PMX, following limitation is applied: $\Gamma_{MV} \leq \Gamma_M$.

8.4.4 Elastic design

- (1) When an elastic design is imposed, the following criterion is applied to cross-sections:

$$\Gamma_{MV} = \frac{\sigma_{eq,Ed}}{f_{yw} / \gamma_{M0}} \leq 1 \quad (118)$$

where: $\sigma_{eq,Ed}$ is the maximal equivalent von Mises stress at all points of structural steel, obtained according to § 6.8.6 for usual sections.

- (2) When an elastic design is imposed to usual cross-sections with slender webs, § 8.4.3 applies.



There is no elastic design for slender webs, which would consider the shear buckling.

8.5 Resistance to lateral torsional buckling

8.5.1 General

- (1) Resistance to lateral torsional buckling is not checked if both flanges are continuously restrained (see § 4.10.1).

8.5.2 Beams with usual steel sections

- (1) The beam is divided in segments. Segments are delimited by:

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	96	Rev.: B

- the ends of the beam (for cantilever spans);
- the supports;
- the cross-sections with both flanges laterally restrained.

In the simple case of a simply supported beam with no lateral intermediate restraints, only one segment is considered for the full beam.

- (2) The resistance of each segment i to lateral torsional buckling is checked through the following criterion:

$$\Gamma_{LTi} = \frac{M_{Ed}}{M_{b,Rd}} \leq 1 \quad (119)$$

where: M_{Ed} is the maximum bending moment along the segment;

$M_{b,Rd}$ is the buckling resistance, as given in **(3)**.

- (3) The buckling resistance $M_{b,Rd}$ is obtained by the following equation, according to EN 1993-1-1, 6.3.2.1:

$$M_{b,Rd} = \chi_{LT} M_{Rk} / \gamma_{M1} \quad (120)$$

where: M_{Rk} is the characteristic bending resistance (i.e. obtained by neglecting the partial factors) – see **§ 6.5.2**;

γ_{M1} : see **§ 4.14.2**;

χ_{LT} is the reduction factor for lateral torsional buckling.

- (4) χ_{LT} is obtained by:

$$\chi_{LT} = \frac{1}{\Phi_{LT} + \sqrt{\Phi_{LT}^2 - \bar{\lambda}_{LT}^2}} \leq 1 \quad (121)$$

where: $\Phi_{LT} = 0,5 \left[1 + \alpha_{LT} (\bar{\lambda}_{LT} - 0,2) + \bar{\lambda}_{LT}^2 \right]$

α_{LT} is the imperfection factor;

$\bar{\lambda}_{LT}$ is the relative slenderness of the beam, given by:

$$\bar{\lambda}_{LT} = \sqrt{\frac{M_{Rk}}{M_{cr}}} \quad (122)$$

M_{cr} is the critical moment for lateral torsional buckling.

- (5) α_{LT} is given by **Table 32** as a function of the buckling curve, the latter being given in **Table 33** from the geometry of the cross-section.

Buckling curve	a	b	c	d
Imperfection factor α_{LT}	0,21	0,34	0,49	0,76

Table 32: Imperfection factor

Cross-sections	Limits	Buckling curve
Hot rolled profiles	$h_s / b_f \leq 2$	a
	$h_s / b_f > 2$	b
Welded sections	$h_s / b_f \leq 2$	c
	$h_s / b_f > 2$	d

Table 33: Reduction curves



For welded sections, b_f is taken as the width of upper flange if $M_{Ed} > 0$ and as the width of the lower flange otherwise.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	97	Rev.: B

- (6) The critical moment M_{cr} is obtained by:

$$M_{cr} = \alpha_{cr} M_{Ed} \quad (123)$$

where: M_{Ed} : see (2);

α_{cr} is the critical amplification factor, obtained by use of the module for critical analysis (see **Annex D**), considering the full beam.

8.5.3 Beams with usual steel sections and partial concrete encasement

- (1) For beams with usual steel sections and partial concrete encasement, § 8.5.2 applies, where M_{Rk} is obtained according to § 6.5.4 and M_{cr} according to (2).
- (2) The critical moment M_{cr} is calculated according to § 8.5.2 (6), where α_{cr} is calculated using the the module for critical analysis (**Annex D**), with the following section properties (see [17]):

- second moment of area about the weak axis:

$$I_z = I_{z,a} + \frac{h_w b_c^3}{12 n_0}$$

- torsion inertia:

$$I_t = I_{t,a} + 0,1 I_{t,c} \frac{G_c}{G_a}$$

where: $G_c = 0,3 E_a / n_x$

$$I_{t,c} = \frac{1}{3} \left(1 - 0,63 \frac{b_c}{h_w} \right) h_w b_c^3$$

n_x is the modular ratio to be considered. By default, in the software, $n_x = n_0$ (according to [17]). It will be possible in the settings of the software to skip to the long-term modular ratio (according to clause 6.4.2 (9) of EN 1994-1-1).

- warping factor:

$$I_w = I_{w,a}$$

where: n_0 is the modular ratio for short term actions, see § 6.3.

$I_{z,a}$, $I_{t,a}$, and $I_{w,a}$ are the section properties of the steel section.

8.5.4 Beams with usual composite sections – composite stage – First generation of EN

- (1) For single span beams, without negative bending moment, it is assumed that the slab is a part of a floor which prevents its lateral instability, so that the beam is not prone to lateral torsional buckling.
- (2) For beams with cantilever spans, negative bending moments are formed near the supports, which leads to lower flanges being compressed. The beams should thus be checked against LT buckling, as described below, according to the method given by EN 1993-1-1, 6.3.2.4.
- (3) The beam is separated in segments delimited by:
- the ends of the beam (for cantilever spans);
 - the supports.
- (4) The resistance of each segment i to lateral torsional buckling is checked through the following criterion:

$$\Gamma_{LTi} = \frac{M_{Ed}}{M_{b,Rd}} \leq 1 \quad (124)$$

where: M_{Ed} is the maximum negative bending moment along the segment;

$M_{b,Rd}$ is the buckling resistance, as given in (5).

- (5) The buckling resistance $M_{b,Rd}$ is obtained by the following equation, according to EN 1993-1-1, 6.3.2.4 (2):

$$M_{b,Rd} = k_{f,B} \chi_{LT} M_{el,Rk} / \gamma_{M1} \text{ and } M_{b,Rd} \leq M_{el,Rd} \quad (125)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	98	Rev.: B

where: $M_{el,Rd}$ is the design elastic bending resistance obtained according to § 6.5 where the partial factor for steel γ_{M0} is replaced by γ_{M1} ;

γ_{M1} : see § 4.14.2;

χ_{LT} is the reduction factor for lateral torsional buckling, calculated according to § 8.5.2 (4), considering the reduced slenderness $\bar{\lambda}_f$ given in (6) and the imperfection factors as given in (7);

$k_{f,B}$ is a modification factor, where: $k_{f,B} = 1$

(6) The reduced slenderness $\bar{\lambda}_f$ is obtained by:

$$\bar{\lambda}_f = \frac{k_C L_c}{i_{f,z} \lambda_1} \quad (126)$$

where: $k_C L_c$ is the buckling length of the segment, see (9);

$$\lambda_1 = \pi \sqrt{E / f_y} = 93,9 \varepsilon$$

$i_{f,z}$ is the radius of gyration of the cross-section constituted by the lower flange, the lower root fillet if any and 1/3 of the compressed part of the web, see (8).

(7) α_{LT} is given by Table 32 as a function of the buckling curve, chosen as follows:

- curve *d* for welded sections;
- curve *c* for hot rolled sections.

(8) $i_{f,z}$ is given by:

$$i_{f,z} = \sqrt{\frac{I_{fz}}{A_{fz}}} \quad (127)$$

where:

$$I_{fz} = \frac{b_{fi}^3 t_{fi}}{12} + \frac{t_w^3 h_{wc}}{12} + (4 - \pi) \frac{r_{ci}^2}{2} \left(\frac{t_w}{2} + k_{cG} r_{ci} \right)^2$$

$$A_{fz} = b_{fi} t_{fi} + h_{wc} t_w + (4 - \pi) \frac{r_{ci}^2}{2}$$

h_{wc} is the 1/3 of the compressed part of the web, which is given by:

$$h_{wc} = (h_s - t_{fi} - z_{na}) / 3$$

z_{na} is the location of the neutral axis, obtained as follows: $z_{na} = z_{el}$;

z_{el} is the location of the elastic neutral axil of the composite section under negative bending moment;

k_{cG} : see Table 21.

(9) The buckling length $k_C L_c$ is obtained as follows:

$$k_C L_c = \pi \sqrt{\frac{EI}{\alpha_{cr} N_{Ed}}} \quad (128)$$

where: EI is the flexural stiffness of the composite section under negative bending moments. For composite sections with partial concrete encasement, I will be assessed considering the modular ratio n_x as defined in § 8.5.3 (2).

N_{Ed} is the maximum axial force in the lower flange along the segment, obtained by:

$$N_{Ed} = \sigma_i b_{fi} t_{fi} \quad (129)$$

σ_i is the elastic stress at mid-thickness of the lower flange under the relevant combination of load, obtained according to § 6.8., where $z_s = -h_s + t_{fi} / 2$;

α_{cr} is the critical amplification factor, calculated with the LTBeamN module, considering the column constituted by the lower flange loaded by the axial forces obtained with Formula (147).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	99	Rev.: B

8.5.5 Beams with usual composite sections – construction stage

- (1) For beams with usual composite sections, if point lateral restraints have been defined for the construction stage, § 8.5.1 (cross-section without partial concrete encasement) or § 8.5.3 (cross-section without partial concrete encasement) applies.
- (2) If a restraint by the steel profiled sheeting is considered (see § 4.11), the same chapters apply, but where the critical moment M_{cr} is calculated taking into account the presence of the sheeting – see Annex E. In this case, the profiled steel sheeting and its connection are checked according to § 8.5.7.
- (3) For beams with point propping, specific load arrangements are considered for the load case Q_c – see § 7.7.4.

8.5.6 Beams with usual composite sections and partial concrete encasement

- (1) For a usual composite beam with partial concrete encasement in the final stage, it is assumed that no lateral torsional buckling may occur, even in negative bending segments of the beam. In this case, no check is carried out.

8.5.7 Verification of the profiled steel sheeting used as lateral restraint in the construction phase

- (1) This chapter describes the verifications carried out by **ABC-PMX** at the ultimate limit states for the steel profiled sheeting used as a lateral restraint against lateral torsional buckling in the construction phase (see § 8.5.5). It also gives the verifications performed for the fastening of the steel sheeting.
- (2) The maximum shear force in the shear panel for the stabilisation of a beam is given by the following formula (according to Annex B of the ECCS Recommendations [14]):

$$V_{m,Ed} = \frac{\pi}{\mu_{cr} - 1} S_{act} \frac{e_0}{L} \quad (130)$$



Attention should be paid to the error in the formula of the ECCS Recommendations. Formula (130) here above takes into account the adequate correction – cf. [21].

where L is the span length of the beams (**Figure 32**);

S_{act} is the shear stiffness of the deck – see § E.3;

e_0 is the equivalent horizontal imperfection of the flange restrained by the deck, which may be obtained by:

$$e_0 = \alpha_{LT} (\bar{\lambda}_{LT} - 0,2) \frac{W_{el}}{A} \quad (131)$$

W_{el} is the elastic bending modulus of the steel profile cross-section;

A is the area of the steel profile cross-section;

μ_{cr} is minimum amplifier to reach the elastic critical buckling moment, defined by:

$$M_{cr} = \mu_{cr} M_{Ed}$$

M_{Ed} is the design bending moment in the beam;

M_{cr} is the elastic critical moment for lateral torsional buckling (see § E.2 (1));

$\bar{\lambda}_{LT}$ is the reduced slenderness for lateral torsional buckling, defined by:

$$\bar{\lambda}_{LT} = \sqrt{\frac{M_{Rk}}{M_{cr}}}$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	100	Rev.: B

M_{Rk} is the characteristic resistance in bending of the cross-section, calculated as follows:

$M_{Rk} = W_{pl} f_y$ for Class 1 or 2 cross-sections;

$M_{Rk} = W_{el} f_y$ for Class 3 cross-sections.

W_{pl} is the plastic bending modulus of the profile cross-section;

W_{el} is the elastic bending modulus of the profile cross-section;

f_y is the yield strength of steel;

α_{LT} is the imperfection factor, determined according to **Table 32** from the buckling curve. The choice of the latter is done according to **Table 33** from the geometry of the cross-section.



e_0 is an equivalent imperfection, which considers the geometric imperfection of the beam and other destabilising phenomena, such as initial stresses. Formula (130) for the calculation of the shear force of the steel deck is conservatively based on this equivalent imperfection e_0 .

- (3) The resistance of the horizontal shear panel under the action of the stabilising forces is checked through the following criterion:

$$\Gamma_{pm} = \frac{V_{m,Ed}}{V_{m,Rd}} \leq 1 \quad (132)$$

where $V_{m,Rd}$ is the shear resistance of the panel, which is given by the following Formula:

$$V_{m,Rd} = n_c F_{s,Rd} + \frac{\beta_1}{\beta_3} F_{p,Rd} \quad (133)$$

$F_{p,Rd}$ is the design strength of an individual sheet fastener (cf. (4) ou (5));

$F_{s,Rd}$ is the design strength of an individual seam fastener (cf. (6));

n_c is the number of seam fasteners per side lap (excluding those which passes through both sheets and the supporting beam);

β_1 is a coefficient whose value is given by **Table 48** in **Annex E** from n_f ;

n_f is the number of sheet-to-beam fasteners per sheet width;

β_3 is a factor which is equal to 1 when seam fasteners are in the troughs. On the other hand, it is given by the following formula when seam fasteners are in the crests:

$$\beta_3 = \frac{n_f - 1}{n_f} \quad (134)$$

- (4) For the deck-to-beam fastening, the resistance of a self-drilling screw is given by the following formula (according to the bearing resistance defined by Table 8.2 in EN 1993-1-3):

$$F_{p,Rd} = \alpha d_{fb} t_p f_{up} / \gamma_{M2} \quad (135)$$

where d_{fb} is the nominal diameter of the screw (see **Table 10** in § 4.11.2);

t_p is the design thickness of the fastened sheet;

f_{up} is the ultimate strength of the sheet;

γ_{M2} is the partial factor for the resistance of fasteners; its value in France is $\gamma_{M2} = 1,25$;

α is a factor defined as follows:

- if $t_{fs} = t_p$: $\alpha = \min\{2,1; 3,2 \sqrt{t_p / d_{fb}}\}$
- if $t_{fs} \geq 2,5 t_p$ and $t_p < 1,0$ mm: $\alpha = \min\{2,1; 3,2 \sqrt{t_p / d_{fb}}\}$
- if $t_{fs} \geq 2,5 t_p$ and $t_p \geq 1,0$ mm : $\alpha = 2,1$
- if $t_p < t_{fs} < 2,5 t_p$: α is obtained by linear interpolation

t_{fs} is the thickness of the beam upper flange where the screw is fastened.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	101	Rev.: B

The range of validity of Formula **(130)** is defined by:

$$e_1 \geq 3 d_{fb}$$

$$e_L \geq 3 d_{fb}$$

$$3,0 \text{ mm} \leq d_{fb} \leq 8 \text{ mm}$$

where e_1 is the spacing of the centre of the hole to the closer edge of the deck.

- (5) When fired pins are used, the following formula is used to assess $F_{p,Rd}$ (according to Table 8.3 in EN 1993-1-3):

$$F_{p,Rd} = 3,2 d_{fb} t_p f_{up} / \gamma_{M2} \quad (136)$$

where d_{fb} is the nominal diameter of the pin (see **Table 10** in § 4.11.2);

The range of validity of Formula **(136)** is as follows:

$$e_1 \geq 4,5 d_{fb}$$

$$e_L \geq 4,5 d_{fb}$$

$$3,7 \text{ mm} \leq d_{fb} \leq 6 \text{ mm}$$

$$f_{up} \leq 550 \text{ MPa.}$$

- (6) The resistance of a self-drilling screw or of a rivet used for the seaming of two adjacent steel sheets with the same thickness, is obtained as follows, according to Table 8.2 in EN 1993-1-3 (this criterion is associated to the bearing resistance of the sheeting):

$$F_{s,Rd} = \alpha d_{fs} t_p f_{up} / \gamma_{M2} \quad (137)$$

where d_{fs} is the nominal diameter of the screw (see **Table 11** in § 4.11.2);

t_p is the design thickness of the fastened sheet;

f_{up} is the ultimate strength of the sheet;

γ_{M2} is the partial factor for the resistance of fasteners; its value in France is $\gamma_{M2} = 1,25$;

α is a factor defined by:

$$\alpha = \min \left\{ 2,1; 3,2 \sqrt{t_p / d_{fs}} \right\}$$

- (7) Formula **(130)** for the calculation of $V_{m,Rd}$ assumes that the ultimate limit state of the steel deck is obtained by bearing of fasteners. It thus requires preventing the other failure modes, which are considered as non permissible modes. According to Annex B of ECCS Recommendations **[14]**, the three following criteria are checked by the software, as given in **(8)** to **(10)**.

- (8) The sheet-to-beam fastener resistance is checked through the following criterion:

$$0,6 \frac{d}{e_L} F_{p,Rd} \geq V_{m,Rd} \quad (138)$$

where d is the span length of the steel sheets (equal to spacing between secondary beams – see **Figure 32**);

e_L is the spacing between sheet-to-beam fasteners (see **Figure 34**);

- (9) The resistance of the sheeting to end failure is checked through the following criterion:

$$F_{p,end,Rd} \geq V_{m,Rd} \quad (139)$$

where $F_{p,end,Rd}$ is the resistance of the steel profiled sheets at the deck end, given by;

$$F_{p,end,Rd} = k_n d \frac{t_p^{1,5} f_{yp}}{e_p^{0,5} \gamma_p} \quad (140)$$

e_p the rib spacing of the sheets (§ 4.5.1);

f_{yp} is the yield strength of the sheets;

t_p is the design thickness of the sheets, obtained by: $t_p = t_{nom} - t_{pr}$;

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	102	Rev.:	B

t_{nom} is the nominal thickness of the sheets;

t_{pr} is the thickness of the coating applied to the sheets (§ 4.11.2);

k_n is a factor obtained as follows:

$k_n = 0,9$ when a fastener is used in every rib at the end of the deck,

$k_n = 0,3$ when a fastener is used in every two ribs at the end of the deck;

γ_p is the partial factor for the resistance of the steel deck (see § 4.14.2).

- (10) The resistance to shear buckling of the sheeting is checked through the following criterion:

$$V_{b,Rd} \frac{d}{L} \geq V_{m,Rd} \quad (141)$$

where $V_{b,Rd}$ is the resistance of the steel deck to shear buckling;

d and L are as given in **Figure 32**.

- (11) When the criterion of formula (142) is satisfied, the shear buckling resistance of the steel deck only depends on the global shear buckling.

$$\frac{b_r}{t_p} \leq 86,7 \varepsilon \quad \text{with:} \quad \varepsilon = \sqrt{\frac{235}{f_{yp}}} \quad (142)$$

In this case, the shear buckling resistance is obtained by:

$$V_{b,Rd} = V_{bg,Rd} \quad (143)$$

When the criterion of formula (142) is not satisfied, the effect of local buckling is taken into account, with the following formula:

$$V_{b,Rd} = V_{bred,Rd} = \frac{V_{bg,Rd} V_{bl,Rd}}{V_{bg,Rd} + V_{bl,Rd}} \quad (144)$$

where $V_{bg,Rd}$ is the resistance of the steel deck to global shear buckling (see (12));

$V_{bl,Rd}$ is the resistance of the steel deck to local shear buckling (see (13));

b_r and t_p – see **Figure 57**.

- (12) The resistance to global shear resistance of the steel deck is obtained by:

$$V_{bg,Rd} = k_{ng} \frac{L}{d^2} D_x^{0,25} D_y^{0,75} / \gamma_p \quad (145)$$

where: k_{ng} is a factor taking into account the influence of the fastening, obtained as follows:

$k_{ng} = 28,8$ when a fastener is used in every rib at the end of the deck;

$k_{ng} = 14,4$ when a fastener is used in every two ribs at the end of the deck;

D_x and D_y are the stiffnesses of the deck obtained by:

$$D_x = \frac{E_a t_p^3 e_p}{12 (1 - \nu^2) u}$$

$$D_y = \frac{E_a I_p}{e_p}$$

u is the perimeter length of a complete single corrugation of pitch e_p , which may be calculated by (see **Figure 57** for the geometrical parameters) :

$$u = b_b + b_r + \sqrt{4 h_p^2 + (b_t - b_b)^2}$$

I_p is the second moment of area of a single corrugation about its neutral axis: $I_p = I_{pu} e_p$

I_{pu} see § 4.5.3;

E_a is the Young's modulus of steel.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	103	Rev.: B

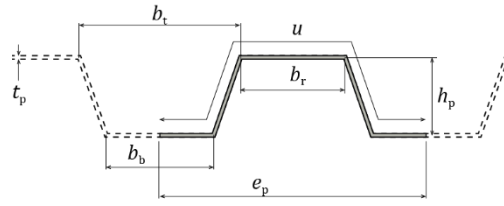


Figure 57: Perimeter length of a single corrugation

- (13) The resistance to local shear resistance of the steel deck is obtained by:

$$V_{bl,Rd} = 4,83 E_a \left(\frac{t_p}{b_r} \right)^2 L t_p / \gamma_p \quad (146)$$

where : b_r and t_p – see **Figure 57**;

L - see **Figure 32**;

E_a - see **§ 5.1.4**.

8.6 Reinforcement of usual composite cross-section under negative bending moment

- (1) For a composite beam with usual composite cross-sections, with or without partial concrete encasement, the longitudinal reinforcement of the slab should fulfil the criterion given in **(2)**, when negative bending moments occur in the beam and a plastic resistance is used for the verification of cross-sections (see EN 1994-1-1:2005, clause 5.5.1 (4)).
- (2) For the cases specified in **(1)**, the reinforcement ratio ρ_s should fulfil the following criterion:

$$\rho_s = \frac{A_s}{A_c} \geq \rho_{s,min} = \delta \frac{f_y}{235} \frac{f_{ctm}}{f_{sk}} \sqrt{k_c} \quad (147)$$

where: A_s is the area of the longitudinal reinforcement – see **§ 4.4.5 (5)**;

A_c is the effective area of the concrete flange, given by: $A_c = b_{eff} t_c$;

b_{eff} : see **§ 6.1**;

t_c is thickness of the active part of the concrete flange, see **§ 6.2**;

f_y is the nominal yield strength of the structural steel, see **§ 5.1.1**;

f_{sk} is the characteristic yield strength of the reinforcement, see **§ 5.3**;

f_{ctm} is the mean tensile strength of the concrete, see **§ 5.2**;

δ is equal to 1 for Class 2 cross-sections, and to 1,1 for Class 1 sections;

k_c is a factor which takes into account the stress distribution within the section immediately prior to cracking. It is given by:

$$k_c = \min \left\{ 1; 0,3 + \frac{2 z_0}{2 z_0 + t_c} \right\} \quad (148)$$

z_0 is the vertical distance between the centroids of the uncracked concrete flange and the uncracked composite section, calculated using the short term modular ratio, n_0 – see **§ 6.3**. It is obtained by:

$$z_0 = t_h + t_d - \frac{t_c}{2} - z_{elh} \quad (149)$$

z_{elh} is the location of the elastic neutral axis of the transformed composite cross-section, calculated according to **§ 6.5**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	104	Rev.: B

8.7 Design of welds for welded sections

- (1) For usual steel or composite sections with welded sections, the thickness of weld throats will be designed to fulfil the following criterion (see [15]), assuming a continuous fillet weld on both sides of the web:

$$a_w \geq \frac{g_{z,ELU,Ed}}{2 f_{vw,d}} = \frac{\sqrt{3}}{2} \frac{g_{z,ELU,Ed}}{f_u} \beta_w \gamma_{M2} \quad (150)$$

where: a_w is the thickness of throats for the welds between the flanges and the web;

$g_{z,ELU,Ed}$ is the cumulated shear flow in the welds for the relevant ULS combination of load;

f_u is the nominal ultimate tensile strength of the weaker part joined; For hot rolled profiles obtained from the ArcelorMittal database, f_u will be read from this base. For other cases, f_u will be obtained from **Table 35**;

β_w is a correlation factor given in **Table 34**;

γ_{M2} see § 4.14.2.

Steel grades	S235	S275	S355	S420, S460, higher steel grades
β_w	0,8	0,85	0,9	1,0

Table 34: Correlation factor

Steel grades	S235	S275	S355	S420	S460 and higher steel grades
f_u [MPa]	360	370	470	540	570

Table 35: Ultimate limit strength

- (2) The shear flow in welds for a section in a non-composite stage is given by:

- For the upper flange:

$$g_{zs,Ed} = \frac{V_{Ed} b_{fs} t_{fs} |z_{fs} - z_{ela}|}{I_{y,a}} \quad (151)$$

- For the lower flange:

$$g_{zi,Ed} = \frac{V_{Ed} b_{fi} t_{fi} |z_{fi} - z_{ela}|}{I_{y,a}} \quad (152)$$

where: z_{ela} is the location of the elastic neutral axis;

$I_{y,a}$ is the second moment of area of the steel cross-section;

z_{fs} and z_{fi} are the locations of the centroids of the upper and lower flanges respectively, see **Table 21**;

- (3) The shear flow in welds for a section in a composite stage is given by:

- For the upper flange, if the centroid is outside the slab ($z_{elh} < t_h$)

$$g_{zs,Ed} = \frac{V_{Ed} \left(b_{fs} t_{fs} |z_{fs} - z_{el,h}| + \frac{b_{eff} t_c}{n} |t_h + t_d - t_c / 2 - z_{elh}| \right)}{I_{y,h}} \quad (153)$$

- For the upper flange, if the centroid is in the slab ($z_{elh} \geq t_h$)

$$g_{zs,Ed} = \frac{V_{Ed} \left(b_{fs} t_{fs} |z_{fs} - z_{el,h}| + \frac{b_{eff} z_{elh}^2}{2 n} \right)}{I_{y,h}} \quad (154)$$

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	105	Rev.:	B

- For the lower flange:

$$g_{zi,Ed} = \frac{V_{Ed} b_{fi} t_{fi} |z_{fi} - z_{elh}|}{I_{y,h}} \quad (155)$$

where: z_{elh} is the location of the elastic neutral axis of the composite transformed cross-section;

$I_{y,h}$ is the second moment of area of this section.

- (4) Following additional criterion will be applied for the automatic design of welds:

$$\max\{3; \sqrt{t_{\max}} - 0,5 \text{ mm}\} \leq a_w \leq 0,7 t_{\min} \quad (156)$$

where: $t_{\max} = t_{fs}$

$t_{\min} = t_w$

9 DESIGN OF THE CONNECTION BETWEEN THE SLAB AND THE STEEL PROFILE

9.1 Introduction

- Chapter **9** applies to the connection between the steel profile and the slab, for beams with usual composite section, with or without partial concrete encasement.
- For the connection, following types of design are available in the software:
 - plastic design, according to **§ 9.2**, applied when a plastic design of cross-sections is performed, i.e. all cross-sections are of Class 1 or 2;
 - elastic design, according to **§ 9.3**, applied when an elastic design of cross-sections is carried out, either because at least one section is of Class 3 or because the elastic design option has been selected.
- When plastic design is applied, full or partial connection will be possible for the positive bending moment segments of the beam. In negative moment segments, only full connection is considered.
- The software will propose in a later version, for a limited scope of application, the automatic design of the connection – see **§ 9.4**.
- The software automatically designs the transverse reinforcement required to resist the longitudinal shear in the connection – see **§ 9.5**.

9.2 Plastic design of the connection

9.2.1 General

- This chapter applies only to beams with usual composite section, with or without partial concrete encasement, where all the cross-sections are Class 1 or 2 (see **§ 6.4**) and the elastic design option is not selected.

9.2.2 Degree of shear connection

- For usual composite cross-section without partial concrete encasement, the degree of shear connection of the main span in positive bending is obtained as follows:

$$\eta = \frac{\sum P_{Rdi}}{N_{cx}} \quad (157)$$

where: N_{cx} is the maximal longitudinal shear force in the connection, given by:

$$N_{cx} = \min\{N_{c,Rd}; N_{a,Rd}\} \quad (158)$$

$N_{c,Rd}$ is the resistance to compression of the slab, given by:

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	106	Rev.: B

$$N_{c,Rd} = 0,85 b_{eff} t_c f_{cd} \quad (\text{first generation of EN}) \quad (159)$$

$$N_{c,Rd} = b_{eff} t_c f_{cd} \quad (\text{second generation of EN})$$

$N_{a,Rd}$ is the axial resistance of the steel profile, given by:

- for hot rolled profiles:

$$N_{a,Rd} = A_a f_{yd} \quad (160)$$

- for welded profiles:

$$N_{a,Rd} = (b_{fs} t_{fs} f_{yfs} + h_w t_w f_{yw} + b_{fi} t_{fi} f_{yfi}) / \gamma_{M0} \quad (161)$$

ΣP_{Rdi} is the sum of the resistance of all the studs located between the point of 0 moment in the bending moment diagram and the cross-section with the maximal positive bending moment.

- (2) As a simplification for beams with adjacent spans, ΣP_{Rdi} in Formula (157) will be replaced by:

$$\Sigma P_{Rdi} - A_s f_{sd} \quad (162)$$

where: A_s is the area of the longitudinal reinforcement at the intermediate support;

ΣP_{Rdi} is the sum of the resistance of all studs located between this intermediate support and the cross-section with the maximal positive bending moment.



By this simplification, the full connection in negative bending zone is automatically obtained.

- (3) For a beam with a usual composite section with partial concrete encasement, the degree of shear connection is calculated according to (1), where Formula (158) is replaced by:

$$N_{cx} = \min\{N_{c,Rd}; N_{a,Rd} + N_{es,Rd}\} \quad (163)$$

where: $N_{es,Rd}$ is the resistance to tension of the steel reinforcement of the partial concrete encasement, given by:

$$N_{es,Rd} = \Sigma A_{esi} f_{sed} \quad (164)$$

9.2.3 Limitation of partial shear connection

- (1) Partial shear connection will be possible for the segment of the main span in positive bending moment. Additionally, partial connection will be possible only if the welded studs fulfil the following condition:

$$h_{sc} / d \geq 4 \quad (165)$$



This condition is required to obtain ductile connectors, as defined in EN 1994-1-1 § 6.6.1.1 (5).

- (2) Partial shear connection should fulfil the following criterion:

$$\eta \geq \eta_0 \geq \eta_{min} \quad (166)$$

where η is the degree of shear connection, as calculated in § 9.2.2;

$$\eta_{min} = 0,4 ;$$

η_0 is the minimum degree of shear connection, as given in (3) to (5).

- (3) For steel sections with equal flanges, η_0 is given by:

$$\text{if } L_e \leq 25 \text{ m} \quad \eta_0 = 1 - \frac{355}{f_y} (0,75 - 0,03 L_e) \quad (167)$$

$$\text{if } L_e > 25 \text{ m} \quad \eta_0 = 1$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	107	Rev.: B

where L_e is obtained according to § 6.1.2 (2);

f_y is the minimum value of f_{yfs} , f_{yfi} and f_{yw} .

- (4) For steel profiles having a lower flange with an area equal to three times the area of the upper flange, η_0 is given by:

$$\text{if } L_e \leq 20 \text{ m} \quad \eta_0 = 1 - \frac{355}{f_y} (0,30 - 0,015 L_e) \quad (168)$$

$$\text{if } L_e > 20 \text{ m} \quad \eta_0 = 1$$

- (5) For steel profiles having a bottom flange with an area exceeding the area of the upper flange but less than three times that area, η_0 is obtained by a linear interpolation from Formulas (167) and (168).

- (6) For steel profiles outside this scope, partial shear connection is not possible (i.e. $\eta_0 = 1$).



In normal mode, the steel profiles are always within this scope – see § 4.6.5.

- (7) If the option for the second generation of Eurocodes is selected, Formula (169) applies, instead of Formula (166):

$$\eta \geq \eta_0 \rho_m^2 k_{up} \geq \eta_{min} \quad (169)$$

where: $\eta_{min} = 0,4$;

η_0 as defined in (3) to (5);

k_{up} is a factor taking into account the method of construction, given by:

$k_{up} = 1$ if the beam is propped during construction (fully propped or with point propping);

$k_{up} = 1 - \rho_{up}$ if the beam is unpropped;

ρ_{up} is defined as follows:

$$\rho_{up} = \min\{0,15; M_{a,Ed} / M_{pl,Rd}\} \text{ if } M_{Ed} / M_{Rd}(\eta) \leq 0,95$$

$$\rho_{up} = 0 \text{ otherwise;}$$

M_{Ed} is the maximal positive bending moment along the beam;

$M_{a,Ed}$ is the part of M_{Ed} that is applied to the structural steel before the composite action is achieved, due to load case G_1 ;

$M_{pl,Rd}$ is the bending moment resistance for full shear connection, obtained according to § 6.5;

$M_{Rd}(\eta)$ is the design moment bending resistance, considering the applied shear connection at the point of maximal bending moment.

ρ_m is the utilisation ratio, defined by:

$$\rho_m = \frac{M_{Ed}}{0,95 M_{Rd}(\eta)} \quad \text{but } 0,8 \leq \rho_m \leq 1 \quad (170)$$



$M_{a,Ed} = 0$ in case of fully propped beams.

9.3 Elastic design of the connection

- (1) Elastic design of the shear connection between the slab and the steel profile is applied when:

- some cross-section along the beam are of Class 3;
- the minimum degree of shear connection is not met;
- or the elastic design option has been selected by the User.

- (2) For each cross-section, the shear connection should fulfil the following criterion:

$$\Gamma_{cx} = \frac{v_{L,Ed}}{v_{L,Rd}} \leq 1 \quad (171)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	108	Rev.: B

where: $v_{L,Ed}$ is the design longitudinal shear force per unit of length (see **(3)**);

$v_{L,Rd}$ is the design longitudinal shear resistance per unit of length (see **(5)**).

(3) $v_{L,Ed}$ is given by:

$$v_{L,Ed} = \sum v_{L,Ed,i} \quad (172)$$

where: $v_{L,Ed,i}$ is the design longitudinal shear force per unit of length for the load case i where the beam acts compositely (see **(4)**), taking into account the relevant combination factors.

(4) In positive bending segments, $v_{L,Ed,i}$ is given by:

$$v_{L,Ed,i} = \frac{V_{Ed,i} S_{h,i}}{I_{h,i}} \quad (173)$$

where: $V_{Ed,i}$ is the design shear force for the load case i ;

$I_{h,i}$ is the second moment of area of the composite transformed cross-section, considering the modular ratio for the concrete of the slab n_i , according to § 6.3;

$S_{h,i}$ is the first moment of area of the composite transformed cross-section constituted by the concrete slab, about the elastic neutral axis, given by:

$$S_{h,i} = \frac{b_{eff} t_c}{n_i} \left(t_h + t_p + \frac{t_c}{2} - z_{el,i} \right) \quad (174)$$

$z_{el,i}$ is the location of the neutral axis of transformed composite section.

(5) In positive bending moment, if the reinforcements under compression are taken into account (see § 6.5.5 **(1)**), Formula **(174)** in **(4)** is replaced by:

$$S_{h,i} = \frac{b_{eff} t_c}{n_i} \left(t_h + t_p + \frac{t_c}{2} - z_{el,i} \right) + \sum \frac{A_{si}}{n_s n_i} (n_i - n_s) (z_i - z_{el,i}) \quad (175)$$

where: $n_s = E_a / E_s$.

(6) In negative bending moments, **(3)** and **(4)** apply, where Formula **(174)** is replaced by:

$$S_{h,i} = \sum \frac{A_{si}}{n_s} (z_i - z_{el,i}) \quad (176)$$

(7) $v_{L,Rd}$ is given by:

$$v_{L,Rd} = \frac{n_r P_{Rd}}{s_x} \quad (177)$$

where: P_{Rd} is the design shear resistance of a connector, according to § 6.6.2;

n_r is the number of shear connectors per line;

s_x is the spacing between two lines of connectors (see § 4.7.3).

9.4 Automatic design of the connection



Automatic design of the connection is not available in this version of ABC-PMX.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	109	Rev.:	B

9.5 Design of transverse reinforcement for usual composite sections

9.5.1 General and assumptions

- (1) The software automatically assesses the section of transverse reinforcement that is required to resist the longitudinal shear in the connection.
- (2) For solid slabs, two layers are assumed, one below and the other one above the head of the studs. For composite slabs, only one layer is considered.
- (3) For solid slabs with haunches, a third bended layer is assumed to be placed in the haunch, when the following condition is met (**Figure 58 b**):

$$h_{sc} - 70 \text{ mm} \leq t_h \quad (178)$$



This criterion assumes:

- the depth of the stud head is 10 mm;
- the lower layer of reinforcement should be 30 mm below the head;
- a concrete coverage below the layer is 20 mm;
- the diameter of reinforcement is 10 mm.

In this case, the potential area of shear failure d-d is assumed.

- (4) The transverse reinforcement area is assumed to be uniformly distributed along each connection zone.

9.5.2 Minimum transverse reinforcement

- (1) The area of the transverse reinforcement should fulfil the following criterion, according to EN 1992-1-1, § 9.2.2 (5):

$$\rho_t \leq \rho_{t,\min} = 0,08 \frac{\sqrt{f_{ck}}}{f_{ys}} \quad (179)$$

where: f_{ck} see § 5.2;

f_{ys} see § 5.3;

ρ_t is the ratio of transverse reinforcement, obtained by:

$$\rho_t = \frac{A_b + A_t}{s_f t_c} \quad \text{for solid slabs}$$

$$\rho_t = \frac{A_t}{s_f t_c} \quad \text{for composite slabs}$$

9.5.3 Configurations of slabs and potential areas of shear failure

- (1) The potential areas of shear failure considered in the calculation depend on the configuration of the slab, according to **Figure 58**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	110	Rev.: B

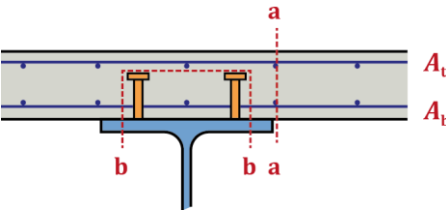
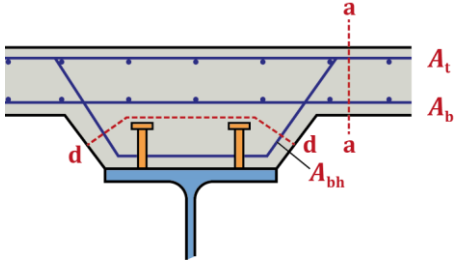
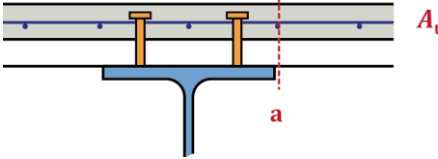
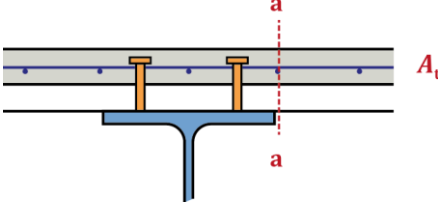
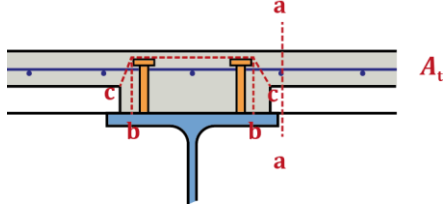
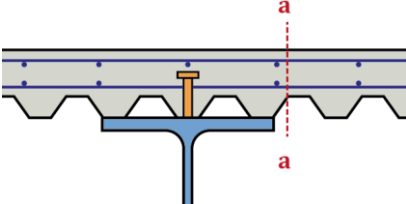
	
a) solid slabs	b) solid slabs with haunches
	
c) solid slabs with prefabricated parts	d) composite slabs, perpendicular continuous ribs
	
e) composite slabs, perpendicular non-continuous ribs	f) composite slabs, longitudinal ribs

Figure 58: Potential areas of shear failure

9.5.4 Longitudinal shear force

- (1) In each connection zone, the longitudinal shear force per unit length to be resisted by the transverse reinforcement is given by:

$$v_{cx,Ed} = \frac{n_r P_{Rd}}{s_x} \quad (180)$$

where: n_r is the number of connectors per file (perpendicularly to the beam axis);

s_x is the longitudinal spacing of studs files;

P_{Rd} is the shear resistance of a connector – see § 6.6.2.



This formula assumes a capacity design of the transverse reinforcement, to resist the maximal shear force in the connection. This formula is very close to the actual design force in the case of partial shear connection. It is also accurate in the case of full shear connection where the number of studs has been optimised.

- (2) The longitudinal shear force per unit of length in the potential area of shear failure is then obtained by:

$$v_{sf,Ed} = k_{sf} v_{cx,Ed} \quad (181)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	111	Rev.: B

where: k_{sf} is the distribution factor, given by:

$$\text{for a-a shear surfaces: } k_{sf} = \max \left\{ \frac{b_{ei} - b_0 / 2}{b_{eff,m}} \right\} \quad \text{in positive bending zone}$$

$$k_{sf} = \max \left\{ \frac{\beta_i b_{ei} - b_0 / 2}{b_{eff,s}} \right\} \quad \text{in negative bending zone}$$

$$\text{for b-b shear surfaces: } k_{sf} = 1$$

$$\text{for c-c shear surfaces: } k_{sf} = 1$$

$$\text{for d-d shear surfaces: } k_{sf} = 1$$

b_0 is the spacing between the two edge studs in a file, obtained by:

$$b_0 = (n_r - 1) b_{0,min}$$

$b_{0,min}$ is the minimum transverse spacing of studs:

$$b_{0,min} = 4 d \text{ for composite slabs;}$$

$$b_{0,min} = 2,5 d \text{ for solid slabs.}$$

$b_{eff,m}$ and $b_{eff,s}$ are the effective widths of the concrete slab, respectively at mid-span and at support, see § 6.1.2;

b_{ei} : see § 6.1.2.

(3) The shear stress in the potential area of shear failure is then obtained by:

$$\tau_{Ed} = \frac{v_{sf,Ed}}{h_f} = k_{sf} \frac{n_r P_{Rd}}{h_f S_x} \quad (182)$$

where: h_f is the developed length of the potential area of shear failure, given by:

$$\text{for a-a shear surface: } h_f = t_c$$

$$\text{for b-b shear surface: } h_f = 2 h_{sc} + b_d$$

$$\text{for c-c shear surface: } h_f = 2 d_{sc} + b_d \quad (\text{intermediate beam})$$

$$h_f = h_{sc} + d_{sc} + b_d \quad (\text{edge beam})$$

$$\text{for d-d shear surface: } h_f = b_0 + 2 \frac{b_f - b_0 + h_{cs} \tan \theta_h}{\sqrt{1 + (\tan \theta_h)^2}}$$

where: $b_d = b_0 + d$

b_0 see (2);

$$d_{sc} = \sqrt{d_1^2 + h_1^2}$$

$$h_1 = h_{sc} - h_p$$

d, h_{sc} see **Figure 27**;

$$d_1 = (b_t - b_d)/2$$

for profiled steel sheeting with longitudinal ribs

$$d_1 = (b_{fs} - b_d)/2 - b_{app,min}$$

for profiled steel sheeting with perpendicular ribs with PE3 configurations (see § 4.4.3).

b_t, h_p : see § 4.5.1;

$b_{app,min}$: see § 3.3.6.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	112	Rev.: B

9.5.5 Verification of the concrete crushing

- (1) The inclination angle of the compression concrete strut is sought from the criterion for the resistance to compression of the concrete strut, according to 6.2.4 (4) of EN 1992-1-1. This criterion is given by:

$$\tau_{Ed} \leq \nu f_{cd} \sin \theta_f \cos \theta_f \quad (183)$$

where: f_{cd} : see § 5.2;

ν is a reduction factor that takes into account the cracking of concrete in shear, defined by formula (6.6N) of EN 1992-1-1:

$$\nu = 0,6 (1 - f_{ck}/250)$$

For the second generation of Eurocodes, $\nu = 0,5$;

θ_f is the compression strut inclination angle.

- (2) For the determination of the compression strut inclination angle θ_f , the following formula will be used, in order to minimise the angle:

$$\theta_f = \frac{1}{2} \arcsin \left(\frac{2 \tau_{Ed}}{\nu f_{cd}} \right) \quad (184)$$

with: $\theta_{f,min} \leq \theta_f \leq \theta_{f,max}$

where the smallest possible value, $\theta_{f,min}$, is given by:

- if the connection zone is fully in positive bending moment: $\theta_{f,min} = 27^\circ$;
- if the connection zone is partly in negative bending moment: $\theta_{f,min} = 36^\circ$;

The maximum value of this angle is: $\theta_{f,max} = 45^\circ$.

- (3) Once the inclination angle θ_f of the concrete compression struts is defined, the criterion for the resistance of these strut is calculated by:

$$\Gamma_{sf} = \frac{\tau_{Ed}}{\nu f_{cd} \sin \theta_f \cos \theta_f} \leq 1 \quad (185)$$

9.5.6 Design of transverse reinforcement

- (1) From the value of the inclination angle as defined in § 9.5.5, reinforcement is designed to fulfil the following criterion:

$$\frac{A_{sf}}{s_f} f_{sd} \geq \frac{\tau_{Ed} h_f}{\cotan \theta_f} \quad (186)$$

where: A_{sf} / s_f is the area of reinforcement by length unit;

f_{sd} : see § 5.3.

- (2) For configurations of slabs with at least two layers of transverse reinforcement, the area of top layer per length unit is denoted A_t and the area of bottom layer per length unit is denoted A_b . For haunches, a third layer may be required and is denoted A_{bh} . See **Table 36** for the configurations of layers of reinforcement and potential areas of shear failure.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	113	Rev.: B

	Potential areas of shear failure	Solid slab without prefabricated part	Solid slab with haunch	Other configurations
Intermediate beam	a-a	$A_t + A_b$	$A_t + A_b$	A_t
	b-b	$2 A_b$	-	-
	d-d	-	$2 A_{bh}$	-
Edge beam	a-a	$A_t + A_b$	$A_t + A_b$	A_t
	b-b	A_b	-	-
	d-d	-	$2 A_{bh}$	-

Table 36: Design of transverse reinforcement

9.5.7 Design of transverse reinforcement with continuous profiled steel sheetings

- (1) For composite slabs with continuous steel profiled sheeting (configuration PE1 according to § 4.4.3), the contribution of the profiled deck to the resistance to longitudinal shear is taken into account as follows, according to 6.6.6.4 (4) of EN 1994-1-1. In this case, Formula (186) will be replaced by:

$$\frac{A_{sf}}{s_f} f_{sd} \geq \frac{\tau_{Ed} h_f}{\cotan \theta_f} - A_{pe} f_{pd} \quad \text{and} \quad \frac{A_{sf}}{s_f} f_{sd} \geq 0 \quad (187)$$

where: f_{pd} is the design value of the yield strength for the steel profiled deck, given by:

$$f_{pd} = f_p / \gamma_p$$

A_{pe} is the area of the profiled deck per length unit, given by:

$$A_{pe} = \frac{t_p}{e_p} \left(e_p + b_b - b_t + \sqrt{(b_b - b_t)^2 + 4 h_p^2} \right)$$

t_p , b_b , b_t , h_p and e_p : see § 4.5.1;

f_p : see § 4.5.3;

γ_p : see § 4.14.2.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	114	Rev.: B

10 FIRE DESIGN

10.1 General

- (1) When the associated option is chosen (see § 4.18.2), **ABC-PMX** assesses the standard fire resistance, assuming fire temperatures according to ISO 834 curve.
- (2) Fire calculations are performed only if fire combinations are defined by the User (see § 4.15.4).
- (3) See § 4.18 for the definition of the options and parameters for the fire design.

10.2 Principles for beams without partial concrete encasement

- (1) The fire calculations in the software follow the steps 1 to 5 (**Figure 59**):
 1. Design effect of actions in fire situation
 2. Cross-section classification
Steps 3 to 4 are iterative, starting from the initial time and considering a time increment:
 3. Section temperature at time t
 4. Design resistance criteria for bending moments, lateral torsional buckling and vertical shear, considering the calculated temperature at time t
 5. Steps 3 and 4 are repeated until the first design criterion reaches its limit, which gives the standard fire resistance.
- (2) The design resistance step (step no 4) will be carried for the following predefined times:
 - 30 min;
 - 60 min;
 - 90 min;
 - 120 min;
 - 180 min;
 - 240 min.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	115	Rev.: B

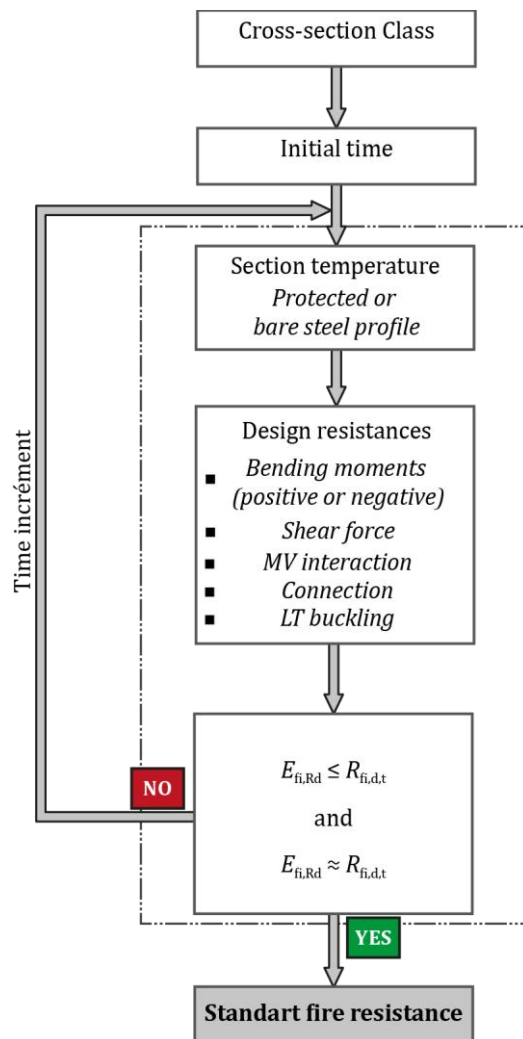


Figure 59: Calculation flowchart for the fire design in ABC-PMX – Beam without partial concrete encasement

10.3 Principles for beams with partial concrete encasement

- (1) For steel or composite beams with partial concrete encasement, the fire calculations in the software follow the steps 1 to 5 (**Figure 60**):
 1. Design effect of actions in fire situation
 2. Cross-section classification
 - Step 3 is incremental, starting from the initial time and considering a time increment:
 3. Design resistance criteria for bending moments, lateral torsional buckling and vertical shear, considering the resistance at time t
 4. Step 3 is repeated until the first design criterion reaches its limit, which gives the standard fire resistance.
- (2) The design resistance step (step no 3) will be carried for the following predefined times:
 - 30 min;
 - 60 min;
 - 90 min;
 - 120 min;
 - 180 min.
- (3) The design resistance at time t is obtained according to the simplified method of the Annex F of EN 1994-1-2, as described in **Annex F**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	116	Rev.: B

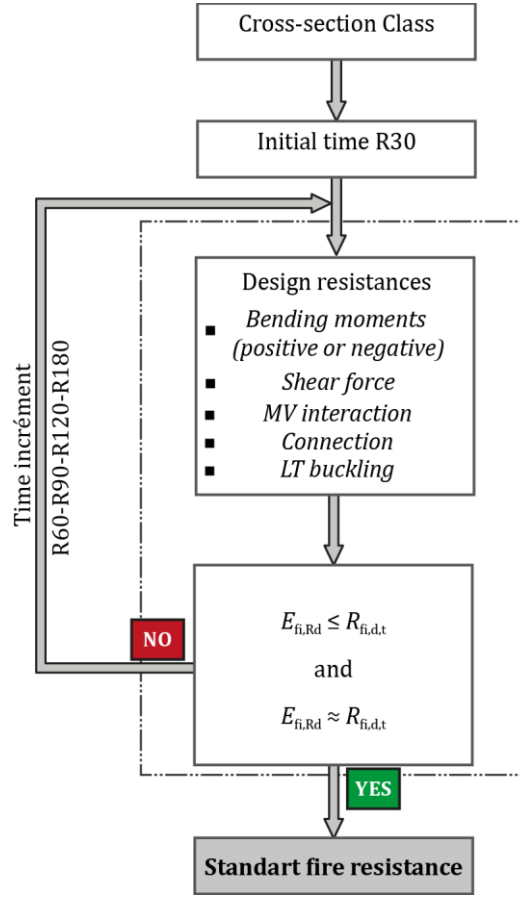


Figure 60: Calculation flowchart for the fire design in ABC-PMX – Beam with partial concrete encasement

10.4 Section properties - Classification

- (1) Sections are classified according to § 6.4, where the steel parameter ε is replaced by:

$$\varepsilon = 0,85 \sqrt{235/f_y} \quad (188)$$

where f_y is the yield strength of steel.

10.5 Heating of the section

10.5.1 Usual unprotected steel sections

- (1) A uniform temperature distribution is assumed over the cross-section.
- (2) The temperature (°C) is calculated at time t_i . The process starts at time t_0 where $\theta(t_0) = 20^\circ\text{C}$.
- (3) For unprotected beams, the variation of steel temperature is obtained according to EN 1993-1-2 § 4.2.5.1, as follows:

$$\Delta\theta_a(t_i) = \frac{k_{sh} (A_m/V)}{c_a \rho_a} \dot{h}_{net,d} \Delta t \quad (189)$$

where: A_m/V is the section factor of the unprotected beam (m^{-1}), given by § 10.6.1;

k_{sh} is the shadow factor, given in (4);

c_a is the steel specific heat [in J/(kg.K)], given as a function of the temperature in (5);

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	117	Rev.: B

ρ_a is the unit mass of steel [in kg/m³], which is assumed to be independent of the temperature (see § 5.1.4);

$\dot{h}_{net,d}$ is the increment of total received heat flux (W/m²), which is given in (6);

Δt is the time increment [in s] – see § 10.3

In Expert mode, the User may define Δt (see § 4.18.4). The default value is 5 s, and according to EN 1994-1-2, 4.3.4.2.2 (5), Δt should not be taken greater than 5 s when assessing formula (189).

- (4) The shadow factor k_{sh} is obtained according to EN 1993-1-2 § 4.2.5.1, as follows:

$$k_{sh} = 0,9 \frac{(A_m/V)_b}{(A_m/V)} \quad (190)$$

where $(A_m/V)_b$ is the box value of the section factor - see § 10.6.1 (6).

- (5) The steel specific heat c_a is calculated according to EN 1993-1-2 § 3.4.1.2, as follows:

- for $20^\circ\text{C} \leq \theta_a < 600^\circ\text{C}$

$$c_a = 425 + 0,773 \theta_a - \frac{1,69}{10^3} \theta_a^2 + \frac{2,22}{10^6} \theta_a^3 \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- for $600^\circ\text{C} \leq \theta_a < 735^\circ\text{C}$

$$c_a = 666 + \frac{13\,002}{738 - \theta_a} \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- for $735^\circ\text{C} \leq \theta_a < 900^\circ\text{C}$

$$c_a = 545 + \frac{17\,820}{\theta_a - 731} \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- for $900^\circ\text{C} \leq \theta_a < 1200^\circ\text{C}$

$$c_a = 650 \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- (6) The increment of total received heat flux $\dot{h}_{net,d}$ is composed of the radiative flux $\dot{h}_{net,r}$ and the convective flux $\dot{h}_{net,c}$ (see § 3.1 of EN 1991-1-2):

$$\dot{h}_{net,d} = \dot{h}_{net,r} + \dot{h}_{net,c} \quad (191)$$

where:

$$\dot{h}_{net,r} = \Phi \varepsilon_m \varepsilon_f \sigma [(\theta_r + 273)^4 - (\theta_m + 273)^4]$$

$$\dot{h}_{net,c} = \alpha_c (\theta_g - \theta_m)$$

σ is the Stefan-Boltzman constant ($\sigma = 5,67 \times 10^{-8} \text{ W/m}^2\text{K}^4$) – see § 4.18.4;

ε_f is the emissivity of fire; by default $\varepsilon_f = 1$ – see § 4.18.4;

ε_m is the emissivity of the steel, see (9);

θ_g is the gas temperature ($^\circ\text{C}$) – see (7);

θ_m is the surface temperature of the member ($^\circ\text{C}$), which is assumed equal to θ_a ;

θ_r is effective radiation temperature of the fire environment ($^\circ\text{C}$), which is assumed equal to θ_g ;

Φ is the view factor (see (8));

α_c is the coefficient of heat transfer by convection ($\text{W/m}^2\text{K}$). The default value is $\alpha_c = 25 \text{ W/m}^2\text{K}$. – see § 4.18.4.

- (7) The temperature of hot gases at time t_i , θ_g , is given by the following formula (according to 3.2.1 (1) in EN 1991-1-2):

$$\theta_g = 20 + 345 \log_{10}(8 t_i + 1) \quad (192)$$

where t_i is the time expressed in minutes.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	118	Rev.:	B

- (8) The view factor Φ is by default equal to 1. The User can specify another value in the range of [0; 1] – see § 4.18.4.
- (9) The emissivity of steel is taken from **Table 37**.

Type of steel	$\varepsilon_m (\theta \leq 500^\circ\text{C})$	$\varepsilon_m (\theta > 500^\circ\text{C})$
Carbon steel	0,7	0,7
Hot-dip galvanised steel	0,35	0,7

Table 37: Emissivity of steel



See Table 5.1 in prEN 1994-1-2:2024

10.5.2 Usual protected steel sections

- (1) A uniform temperature distribution is assumed over the cross-section.
- (2) The computation of the steel temperature θ_a at time t_i is carried out according to § 10.5.1 (2), where the variation of steel temperature is obtained according to EN 1993-1-2 § 4.2.5.2, as follows:

$$\Delta\theta_a(t_i) = \frac{\lambda_p (A_p/V)}{d_p c_a \rho_a} \frac{1}{1 + \phi/3} (\theta_g - \theta_a) \Delta t - (e^{\phi/10} - 1) \Delta\theta_g \quad (193)$$

with: $c_a, \rho_a, \theta_g, \theta_a, \Delta t$: see § 10.5.1;

λ_p is the thermal conductivity of the insulating element (W/m K) – see **Table 12**;

A_p/V is the section factor of the protected section (m^{-1}) – see § 10.6.2;

$\Delta\theta_g$ is the increase of the ambient gas temperature ($^\circ\text{C}$) during the time interval Δt ;

ϕ is defined by:

$$\phi = \frac{c_p \rho_p}{c_a \rho_a} d_p \frac{A_p}{V} \quad (194)$$

d_p is the thickness of the fire protection material (m);

c_p is the specific heat of the insulating material (J/kg K) – see **Table 12**;

ρ_p is the unit mass of the insulating material (kg/m^3) – see **Table 12**.

Δt is the time increment [in s] – see § 10.3

In Expert mode, the User may define Δt (see § 4.18.4). The default value is 10 s, and according to EN 1994-1-2, 4.3.4.2.2 (8), Δt should not be taken greater than 30 s when assessing formula (193).



For beams with composite slabs where the ribs are perpendicular to the beam axis, the voids between the slab and the upper flange are assumed to be filled. The **Annex H** gives the procedure which is applied when these voids are not filled.

10.5.3 Usual unprotected composite sections

- (1) According to EN 1994-1-2, 4.3.4.2.2, usual unprotected composite sections are divided in the four following parts:
- the lower flange;
 - the web (including root fillet for hot rolled profiles);
 - the upper flange;
 - the slab.
- (2) The temperature in each part is calculated independently, with the following assumptions:
- the temperature in each steel part is uniform;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	119	Rev.: B

- no heat transfer takes place between these different parts nor between the upper flange and the concrete slab;
- the temperature of the slab varies within the thickness and is uniform along the width.

(3) The temperature in a steel part is calculated according to § 10.5.1 (2) and (3), where, in Formula (189):

- the shadow factor k_{sh} is calculated as follows (see § 4.18.4 for the parameters):

$$k_{sh} = 0,9 \frac{t_{fs} + t_{fi} + b_{fi}/2 + \sqrt{h_w^2 + (b_{fs} - b_{fi})^2/4}}{h_w + b_{fi} + b_{fs}/2 + t_{fs} + t_{fi} - t_w} \quad (195)$$

- the section factors are obtained from (4) to (6).

(4) The section factor of the lower flange is calculated by:

$$\frac{A_m}{V} = 2 \frac{b_{fi} + t_{fi}}{b_{fi} t_{fi}} \quad (196)$$

(5) The section factor of the upper flange is calculated by the following formula if its upper side is covered by the slab, according to the conditions in § 10.6.1 (5):

$$\frac{A_m}{V} = \frac{b_{fs} + 2 t_{fs}}{b_{fs} t_{fs}} \quad (197)$$

If the conditions for the coverage by the slab are not met, the section factor is obtained by:

$$\frac{A_m}{V} = 2 \frac{b_{fs} + t_{fs}}{b_{fs} t_{fs}} \quad (198)$$

(6) The section factor for the web is obtained by:

$$\frac{A_m}{V} = \frac{2 d_w + \pi (r_{cs} + r_{ci})}{h_w t_w + (4 - \pi)(r_{cs}^2 + r_{ci}^2)/2} \quad (199)$$

(7) The temperature in the slab will be obtained with simplified tabulated methods as given in § 10.5.4 and in § 10.5.5, or using a numerical calculation, as described in § 10.5.6. The choice between these methods is according to the calculation options.

10.5.4 Slab – Simplified method – First generation of EN

(1) The temperature in plain slabs, with or without prefabricated parts, assuming the following assumptions:

- the slab is divided in up to 14 layers of uniform thickness, with a thickness of 5 mm in the first 60 mm, an additional layer of 20 mm if needed, and a last one with the remaining thickness, if needed (**Figure 61**);
- for normal concretes, the temperature in each of these layers is as given in **Table 38**, according to Table D.5 of EN 1994-1-2;
- for light weight concrete, the temperature in each layer will be as given in **Table 38**, but reduced by a factor of 0,9;
- when an empty cell is present in **Table 38**, the temperature of the layer is equal to the temperature of hot gases.

(2) The temperature in composite slabs will be assessed by applying (1), using the effective depth of the slab $h_{c,eff}$, obtained as follows:

- if $h_p/t_c \leq 1,5$:

$$h_{c,eff} = t_c + 0,5 h_p \frac{b_b + b_t}{e_p} \quad (200)$$

- if $h_p/t_c > 1,5$:

$$h_{c,eff} = t_c \left(1 + 0,75 \frac{b_b + b_t}{e_p} \right) \quad (201)$$

where: b_b , b_t , h_p and e_p are the dimensions of the profiled steel sheeting, as defined in § 4.5.1;

t_c see § 6.2.2.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	120	Rev.: B

	Thickness	zTop	Temperature in concrete layers (°C) after a fire duration of					
	(mm)	(mm)	30'	60'	90'	120'	180'	240'
1	5	5	535	705				
2	5	10	470	642	738			
3	5	15	415	581	681	754		
4	5	20	350	525	627	697		
5	5	25	300	469	571	642	738	
6	5	30	250	421	519	591	689	740
7	5	35	210	374	473	542	635	700
8	5	40	180	327	428	493	590	670
9	5	45	160	289	387	454	549	645
10	5	50	140	250	345	415	508	550
11	5	55	125	200	294	369	469	520
12	5	60	110	175	271	342	430	495
13	20	80	80	140	220	270	330	395
14	≥ 20	≥100	60	100	160	210	260	305

Table 38: Temperature distribution in solid slabs with normal concretes

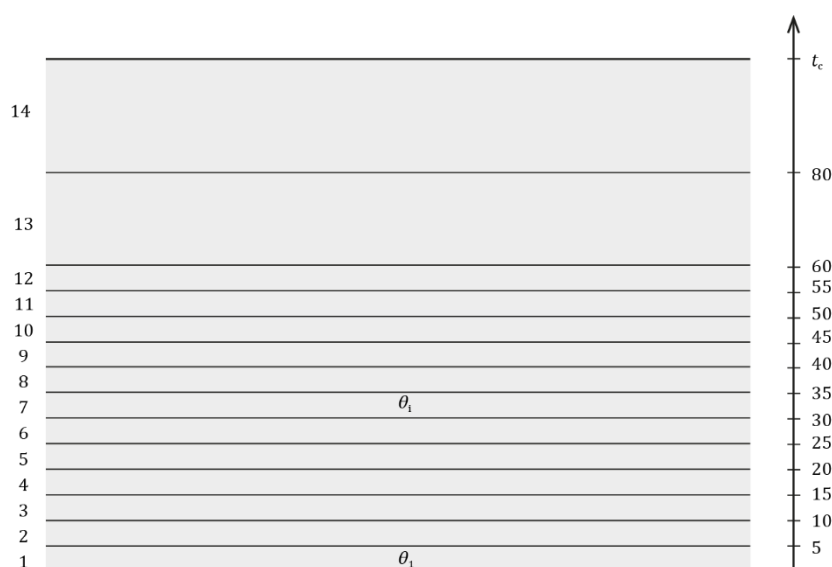


Figure 61: Temperature distribution in the slab

10.5.5 Slab – Simplified method – Second generation of EN

- (1) The same principles as in § 10.5.4 are used, but with **Table 39** (according to Table B.6 in prEN 1994-1-2:2023). This Table is not applicable to R240 calculations.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	121	Rev.: B

	Thickness	zTop	Temperature in concrete layers (°C) after a fire duration of				
	(mm)	(mm)	30'	60'	90'	120'	180'
	2,5	2,5	675	831	912	967	1042
	7,5	10	513	684	777	842	932
	10	20	363	531	629	698	797
	10	30	260	418	514	583	685
	10	40	187	331	423	491	591
	10	50	135	263	349	415	514
	10	60	101	209	290	352	448
	10	70	76	166	241	300	392
	10	80	59	133	200	256	344
	10	90	46	108	166	218	303
	10	100	37	89	138	186	267
	10	110	31	73	117	159	236
	10	120	27	61	100	137	209
	10	130	24	51	86	119	186
	10	140	23	44	74	105	166
	10	150	22	38	65	94	149

Table 39: Temperature distribution in solid slabs with normal concretes – Second generation of EN

- (2) For any slab with a thickness greater than 150 mm, the temperature in the slab will be assumed constant and uniform in the upper layer above 150 mm.

10.5.6 Slab – Numerical method

- (1) The temperature in solid slabs, with or without prefabricated parts, will be assessed by a Finite Difference Analysis (with assumptions as given in **Annex G**).

10.5.7 Usual protected composite sections

- (1) For protected composite section, the same principles as in § 10.5.3 (1) and (2) apply. For sections with board thermal protection, the steel profile is represented by one common part.
- (2) The temperature in a steel part is calculated according to § 10.5.2 (2), applying the section factors in (3) to (4) below.
- (3) For section with contour protection, the section factors of the parts are obtained according to § 10.5.3 (4) to (6).
- (4) For section with board thermal protection, the section factor of the steel profile is obtained according to § 10.6.2.
- (5) The temperature in the slab is obtained with simplified tabulated methods as given in § 10.5.4 and in § 10.5.5, or with a numerical calculation, as described in § 10.5.6. The choice between these methods is according to the calculation options.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	122	Rev.:	B

10.6 Section factors

10.6.1 Usual unprotected steel or composite sections

- (1) The section factor of an unprotected steel section is given by:

$$\frac{A_m}{V} = \frac{p}{A_a} \quad (202)$$

where: p is the perimeter of the section exposed to fire

A_a is the area of the steel profile (see § 6.5);

- (2) For a section exposed on all four sides, the perimeter p is given by:

$$p = 2 (h_s + b_{fs} + b_{fi} - t_w) + (\pi - 3) (r_{cs} + r_{ci}) \quad (203)$$

For hot rolled profiles, this equation may be simplified as follows:

$$p = 2 (h_s + 2 b_f - t_w + (\pi - 3) r)$$

- (3) For a section exposed on three sides only, the perimeter p is given by:

$$p = b_{fs} + 2 (h_s + b_{fi} - t_w) + (\pi - 3) (r_{cs} + r_{ci}) \quad (204)$$

- (4) An unprotected steel section will be considered as exposed on three sides if the upper side of the upper flange is covered by the slab as defined in (5).

- (5) The upper flange is covered by the slab if one of the following condition is met:

- the slab is a solid slab, with or without prefabricated parts;
- the slab is composite where the ratio of coverage of the upper side of the upper flange is not less than 0,85.

For composite slabs with transverse ribs with configurations PE 1 or PE2 (see **Figure 17**), the ratio of coverage of the upper flange is given by:

$$c_r = b_b / e_p \quad (205)$$

where b_b and e_p are defined in § 4.5.1.

For composite slabs with longitudinal ribs, the ratio of coverage of the upper flange is equal to:

$$c_r = b_b / b_{fs} \quad (206)$$

where b_{fs} is defined in § 4.6.2.

For composite slabs with transverse ribs with configuration PE3, the upper flange is assumed to be covered.

- (6) The box value of the section factor of an unprotected steel section is given by:

$$\left(\frac{A_m}{V} \right)_b = \frac{p_b}{A_a} \quad (207)$$

where: p_b is the box perimeter of the section exposed to fire

A_a is the area of the steel profile (see § 6.5);

- (7) For a section exposed on all four sides, the box perimeter p_b is given by:

$$p_b = 2 (h_s + \max\{b_{fs}; b_{fi}\}) \quad (208)$$

For hot-rolled profiles, this equation may be simplified as follows:

$$p_b = 2 (h_s + b_f)$$

- (8) For a section exposed on three sides only, the box perimeter p_b is given by:

$$p_b = 2 h_s + \max\{b_{fs}; b_{fi}\} \quad (209)$$

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	123	Rev.:	B

10.6.2 Usual protected steel or composite sections

- (1) The section factor of a protected steel section is given by:

$$\frac{A_p}{V} = \frac{p}{A_a} \quad (210)$$

where: p is the perimeter of the protected section exposed to fire

A_a is the area of the steel profile (see § 6.5);

- (2) For contour protection (i.e. spray or intumescent paint), the perimeter p is obtained according to § 10.6.1 (2) or (3).
- (2) For a section with board protection, the perimeter p is given by:

$$p = 2 h_s + \max\{b_{fs}; b_{fi}\} \quad (211)$$



In case of board protection, it is assumed that the upper side of the upper flange is always protected.

10.7 Reduction factors for mechanical properties

10.7.1 Construction steel

- (1) The steel properties vary when the temperature increases. These variations are taken into account through the following parameters:
- factor for the variation of the yield strength f_y : $k_{y,\theta}$
 - factor for the variation of the Young's modulus E_a : $k_{E,\theta}$
 - factor for the variation of the ultimate strength f_u : $k_{u,\theta}$
- (2) The values defined by Table 3.1 of EN 1993-1-2 are used by the software for factors $k_{y,\theta}$, $k_{u,\theta}$ and $k_{E,\theta}$ –see **Table 40**. For temperatures between the values of this table, linear interpolation is applied for the calculation of the temperature coefficients.

Steel temperature θ (°C)	Reduction factor $k_{y,\theta}$	Reduction factor $k_{E,\theta}$	Reduction factor $k_{u,\theta}$
20	1,00	1,000	1,25
100	1,00	1,000	1,25
200	1,00	0,900	1,25
300	1,00	0,800	1,25
400	1,00	0,700	1,00
500	0,78	0,600	0,78
600	0,47	0,310	0,47
700	0,23	0,130	0,23
800	0,11	0,090	0,11
900	0,06	0,0675	0,06
1000	0,04	0,0450	0,04
1100	0,02	0,0225	0,02
1200	0,00	0,000	0,00

Table 40: Temperature reduction factors applied to the steel properties

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	125	Rev.: B

10.7.2 Concrete

- (1) The concrete properties vary when the temperature increases. The resistance to compression is reduced by the factor $k_{c,\theta}$ as given in **Table 41** (according to Table 3.3 in EN 1994-1-2).

Temperature of concrete θ (°C)	Reduction factor $k_{c,\theta}$ for normal weight concrete	Reduction factor $k_{c,\theta}$ for light weight concrete
20	1,00	1,0
100	1,00	1,0
200	0,95	1,0
300	0,85	1,0
400	0,75	0,88
500	0,60	0,76
600	0,45	0,64
700	0,30	0,52
800	0,15	0,40
900	0,08	0,28
1000	0,04	0,16
1100	0,01	0,04
1200	0	0

Table 41: Temperature reduction factors applied to the concrete properties

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	126	Rev.: B

10.7.3 Reinforcing steel

- (1) The properties of steel reinforcement vary when the temperature increases. The yield strength is reduced by the factor $k_{s,\theta}$ as given in **Table 42**, according to 3.2.3 (3) of EN 1994-1-2 for cold worked reinforcing steels.

Steel temperature θ (°C)	Reduction factor $k_{s,\theta}$ for normal concrete	Reduction factor $k_{s,\theta}$ see (2)
	cold worked	hot rolled
20	1,00	1,00
100	1,00	1,00
200	1,00	1,00
300	1,00	1,00
400	0,94	1,00
500	0,67	0,78
600	0,40	0,47
700	0,12	0,23
800	0,11	0,11
900	0,08	0,06
1000	0,05	0,04
1100	0,03	0,02
1200	0	0

Table 42: Temperature reduction factors applied to the reinforcement properties



The values for hot-rolled rebars are identical to the values used for structural steel – see **Table 40**. They are used with the first generation of Eurocodes, according to EN 1994-1-2, 3.2.3 (2). With the second generation, these values are according to Table 5.4 in prEN 1994-1-2:2024.

10.8 Fire resistance criteria

10.8.1 Principles

- (1) The following checks are performed in fire situation:
- resistance of the cross-section to bending moment;
 - resistance of the cross-section to shear force;
 - when relevant, resistance of the cross-section to MV interaction;
 - resistance of the beam to lateral torsional buckling.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	127	Rev.:	B

10.8.2 Resistance to bending moment

- (1) The resistance of a cross-section to bending moment is checked through the following criterion:

$$\Gamma_{M,fi} = \frac{M_{fi,Ed}}{M_{fi,\theta,Rd}} \quad (212)$$

where: $M_{fi,Ed}$ is the bending moment under the relevant combination of load in fire situation;

$M_{fi,\theta,Rd}$ is the moment resistance of the cross-section at temperature θ .

10.8.3 Resistance to shear force

- (1) The resistance of a cross-section to shear force is checked through the following criterion:

$$\Gamma_{V,fi} = \frac{V_{fi,Ed}}{V_{fi,\theta,Rd}} \quad (213)$$

where: $V_{fi,Ed}$ is the shear force under the relevant fire combination of load;

$V_{fi,\theta,Rd}$ is the resistant shear force of the cross-section at temperature θ .

- (2) The resistance to shear force in fire situation considering a uniform temperature θ of the cross-section is given by (according to 4.2.3.3 (1) of EN 1993-1-2):

$$V_{fi,\theta,Rd} = k_{y,\theta} \frac{\gamma_{M0}}{\gamma_{Mfi}} V_{Rd} \quad (214)$$

where: V_{Rd} is the resistant shear force at room temperature;

$k_{y,\theta}$ is the reduction for the yield strength of steel at temperature θ as given in § 10.7.1;

γ_{Mfi} : see § 4.14.2.

10.8.4 Resistance to MV interaction

- (1) When $\Gamma_{V,fi} > 0,5$, the resistance to both bending moment and shear force, taking account of the influence of the shear force on the bending resistance, is checked through the following criterion, for Class 1, 2 or 3 sections:

$$\Gamma_{M,fi} = \frac{M_{fi,Ed}}{M_{V,fi,\theta,Rd}} \quad (215)$$

where: $M_{V,fi,Ed}$ is the resistant bending moment of the cross-section at temperature θ , taking into account the influence of the presence of V. It is obtained by reducing the yield strength of the web by $(1 - \rho_{fi})$, where ρ_{fi} is given by:

$$\rho_{fi} = (2 \Gamma_{V,fi} - 1)^2 \quad (216)$$



The yield resistance of the web is also reduced by $k_{y\theta}$.

10.8.5 Resistance to shear buckling

- (1) Resistance to shear buckling is checked when the following criterion is fulfilled (acc. to EN 1993-1-5, 5.1 (2)):

$$\frac{h_w}{t_w} > 72 \frac{\varepsilon_{fi,w}}{\eta} \quad (217)$$

where: η see § 3.3.4 (5).

$\varepsilon_{fi,w}$ is the steel factor for the web, see § 10.4.

h_w is the height of the web and t_w is its thickness.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	128	Rev.:	B

- (2) The resistance to shear buckling under the effect of a shear force V_{Ed} is checked through the following criterion:

$$\Gamma_{V,fi} = \frac{V_{fi,Ed}}{V_{b,fi,Rd}} \leq 1 \quad (218)$$

- (3) $V_{b,fi,Rd}$ is obtained by:

$$V_{b,fi,Rd} = \chi_w \frac{h_w t_w k_{y,\theta} f_{yw}}{\sqrt{3} \gamma_{M,fi}} \quad (219)$$

where: χ_w is the reduction factor for shear buckling, which is given in **Table 31**, using the reduced slenderness $\bar{\lambda}_{fi,w}$ – see **(4)**;

- (4) The reduced slenderness $\bar{\lambda}_{fi,w}$ is calculated by:

$$\bar{\lambda}_{fi,w} = \frac{h_w}{t_w} \frac{1}{37,4 \varepsilon_{fi,w} \sqrt{k_\tau}} \sqrt{\frac{k_{y,\theta}}{k_{E,\theta}}} \quad (220)$$

where: $k_{y,\theta}$ and $k_{E,\theta}$ are temperature reduction factors for structural steel, according to **Table 40**;

$$k_\tau = 5,34.$$



§ 10.8.5 is applied to all type of sections, by an engineering judgement of § E.4 from EN 1994-1-2.

10.8.6 Resistance to lateral torsional buckling

- (1) Resistance to lateral torsional buckling will be checked in fire situation only for usual steel section, either with or without partial concrete encasement.
- (2) For beams without partial concrete encasement, the resistance to lateral torsional buckling at temperature θ is checked by using the following criterion:

$$\Gamma_{LT,fi} = \frac{M_{Ed}}{M_{b,fi,Rd}} \leq 1 \quad (221)$$

where $M_{b,fi,Rd}$ is the bending resistance to lateral torsional buckling at temperature θ , obtained by:

$$M_{b,fi,Rd} = \chi_{LT,fi} k_{y,\theta} M_{Rk} / \gamma_{M,fi} \quad (222)$$

- (3) The characteristic bending resistance is obtained as follows:

- for classe 1 or 2 sections: $M_{Rk} = W_{pl} f_y$
- for classe 2 sections: $M_{Rk} = W_{el} f_y$

- (4) The reduction factor $\chi_{LT,fi}$ for lateral torsional buckling under fire situations is obtained as follows (see EN 1993-1-2, 4.2.3.3 (5)):

$$\chi_{LT,fi} = \frac{1}{\Phi_{LT,fi} + \sqrt{\Phi_{LT,fi}^2 - \bar{\lambda}_{LT,fi}^2}} \leq 1 \quad (223)$$

where: $\Phi_{LT,fi} = 0,5 \left[1 + \alpha_{LT,fi} \bar{\lambda}_{LT,fi} + \bar{\lambda}_{LT,fi}^2 \right]$;

$\alpha_{LT,fi}$ is the imperfection factor, obtained by: $\alpha_{LT,fi} = 0,65 \sqrt{235/f_y}$;

$\bar{\lambda}_{LT,fi}$ is the relative slenderness, calculated as follows:

$$\bar{\lambda}_{LT} = \sqrt{\frac{k_{y,\theta} M_{Rk}}{k_{E,\theta} M_{cr}}} \quad (224)$$

- (5) $k_{y,\theta}$ and $k_{E,\theta}$ are the temperature reduction factors for structural steel at temperature θ given by **Table 40**.
- (6) For beams with partial concrete encasement, no lateral torsional buckling checking is carried out (see § 4.3.4.3.4 (7) in EN 1994-1-2).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	129	Rev.: B

10.9 Verifications in fire situation

10.9.1 Usual steel sections

- (1) Beams with usual steel sections in fire situations are checked according to **§ 10.8.2** to **10.8.6**.
- (2) The resistant bending moment in fire situation considering a uniform temperature θ of the cross-section is given by (according to 4.2.3.3 (1) of EN 1993-1-2):

$$M_{fi,\theta,Rd} = k_{y,\theta} \frac{\gamma_{M0}}{\gamma_{Mfi}} M_{Rd} \quad (225)$$

where: M_{Rd} is the resistant bending moment at room temperature - $M_{Rd} = M_{pl,Rd}$ for Class 1 or 2 sections, and $M_{Rd} = M_{el,Rd}$ for Class 3 sections, where the class of the section is defined according to **§ 10.4**;
 $k_{y,\theta}$ is the reduction factor for the yield strength of steel at temperature θ as given in **§ 10.7.1**;
 γ_{Mfi} : see **§ 4.14.2**.

10.9.2 Usual composite sections

- (1) Beams with usual composite sections in fire situations are checked according to **§ 10.8.2** to **10.8.6**.
- (2) When **§ 10.8.2** (1) is applied, the resistant bending moment in fire situation of a usual composite section is obtained according to the principle of the plastic calculation in **§ 6.5.5**, where the following assumptions are considered:

- the area of the lower flange is stressed to the following yield strength:

$$k_{y,\theta} \frac{f_{y,fi}}{\gamma_{M,fi,a}}$$

where $k_{y,\theta}$ is obtained according to **§ 10.7.1** from the temperature in the lower flange;

- the area of the web, including the root fillet if any, is stressed to the following yield strength:

$$k_{y,\theta} (1 - \rho_{fi}) \frac{f_{y,w}}{\gamma_{M,fi,a}}$$

where $k_{y,\theta}$ is obtained according to **§ 10.7.1** from the temperature in the web;

- the area of the upper flange is stressed to the following yield strength:

$$k_{y,\theta} \frac{f_{y,fs}}{\gamma_{M,fi,a}}$$

where $k_{y,\theta}$ is obtained according to **§ 10.7.1** from the temperature in the upper flange;

- the area of each layer of the slab is stressed in compression to the following strength:

$$\alpha_{slab} k_{c,\theta} \frac{f_c}{\gamma_{M,fi,c}}$$

where $k_{c,\theta}$ is obtained according to **§ 10.7.2** from the temperature in the layer;

$$\alpha_{slab} = 1;$$

Starting from the top of the slab, only the layers which fulfil the following condition are considered in the calculation of the plastic moment:

$$\alpha_{slab} \sum_j A_j k_{c,\theta} f_{ck} / \gamma_{M,fi,c} \leq \eta_{fi} N_{cx,fi} \quad (226)$$

where: A_j is the area of the layer;

η_{fi} is the degree of shear connection, as given in **§ 10.10 (3)**;

$N_{cx,fi}$ is the longitudinal shear force in the connection for a full connection, as given in **§ 10.10 (2)**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	130	Rev.: B

- the area of a reinforcement layer is stressed in tension to the following strength:

$$k_{s,\theta} \frac{f_{sk}}{\gamma_{M,fi,s}}$$

where $k_{s,\theta}$ is obtained according to § 10.7 from the temperature in the layer, assuming that the latter is equal to the temperature of the slab layer where the reinforcement is located.

- (3) An option will be proposed for usual section composite without partial concrete encasement to assess the bending resistance according to Annex E of EN 1994-1-2 (Annex C of prEN 1994-1-2:2023). In this case, the plastic moment will be obtained applying (2) with the assumption that $k_{c,\theta} = 1$ as long as $\theta_c \leq 250^\circ\text{C}$.

10.9.3 Profiles with partial concrete encasement

- (1) The resistance of steel or composite beams with partial concrete encasement under fire situation is checked at the following time steps: 30 min; 60 min; 90 min; 120 min; 180 min.
- (2) The bending moment resistance under positive or negative moments is checked through the method given in Annex F.
- (3) The resistance to shear force is checked according to F.6.
- (4) The resistance of cross-section is limited by the depth of slab according to Table 43

Standart fire resistance	Minimum slab thickness $t_d \geq t_{d,min}$ [mm]
R30	60
R60	80
R90	100
R120	120
R180	150

Table 43: Minimum slab thickness

10.10 Connection

10.10.1 Shear resistance of connectors

- (1) The shear resistance of a welded stud in fire situation is obtained according to § 6.7.2, where (see § 4.3.4.2.5 in EN 1994-1-2):

- the partial factor γ_v is replaced by $\gamma_{v,fi}$;
- Formula (82) is replaced by:

$$P_{Rd,s} = 0,8^2 k_{u,\theta} \frac{f_{u,sc} \pi d^2}{4 \gamma_{M,fi,V}} \quad (227)$$

- Formula (83) is replaced by:

$$P_{Rd,c} = 0,29 k_{c,\theta} \frac{\alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_{M,fi,V}} \quad (228)$$

where, in Formulas (227) and (228):

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	131	Rev.: B

$k_{u,\theta}$ is the temperature reduction factor for steel, obtained from **Table 40**, considering a temperature of the connector θ_v equal to 80 % of the temperature of the upper flange;

$k_{c,\theta}$ is the temperature reduction factor for concrete, obtained from **Table 41**, considering a temperature of concrete θ_c equal to 40 % of the temperature of the upper flange.



The temperatures θ_v and θ_c are always taken greater or equal than the initial room temperature θ_{ref} .

10.10.2 Design of the connection

- (1) The longitudinal shear force in the connection along a critical segment of a usual composite beam without partial concrete encasement, for a full connection, is obtained by:

$$N_{cx,fi} = \min\{N_{c,fi,Rd}; N_{a,fi,Rd}\} \quad (229)$$

where: $N_{a,fi,Rd}$ is the axial resistance of the steel profile in fire situation, given by:

$$N_{a,fi,Rd} = \sum_j A_j k_{y,\theta} f_{yj} / \gamma_{M,fi} \quad (230)$$

the summation is carried out on the different parts j considered for the calculation of the temperature in the steel profile;

A_j is the area of part no. j ;

f_{yj} is the yield strength of part no. j ;

$k_{y,\theta}$ is temperature reduction factor, according to **Table 40**, considering the temperature in part no j ;

$N_{c,fi,Rd}$ is the axial resistance to compression of the slab in fire situation, given by:

$$N_{c,fi,Rd} = \alpha_{slab} \sum_j A_j k_{c,\theta} f_{ck} / \gamma_{c,fi} \quad (231)$$

the summation is carried out the different layers of slab j considered for the calculation of the temperature in the slab;

A_j is the area of part no j ;

f_{ck} is the resistance to compression of slab part no j ;

$k_{c,\theta}$ is the temperature reduction factor, according to **Table 41**, considering the temperature in the slab layer no j ;

$\alpha_{slab} = 1$.

- (2) The degree of shear connection in fire situation is obtained through the following equation:

$$\eta_{fi} = \frac{\sum P_{Rd,fi,i}}{N_{cx,fi}} \quad (232)$$

where: $N_{cx,fi}$ is the maximal longitudinal shear force in the connection, given by **(2)**:

$\sum P_{Rd,fi,i}$ is the sum of the resistance of all the studs $P_{Rd,fi}$ located between the point of 0 moment in the bending moment diagram and the cross-section with the maximal positive bending moment, calculated according to **(1)**.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	132	Rev.: B

11 VERIFICATIONS AT SERVICEABILITY LIMIT STATES

11.1 Deflections

- (1) The deflections of the beam are calculated through the Finite Element numerical analysis (see **Annex C**).
- (2) The calculation sheet reports, for each span:
 - the maximal deflection for each elementary load case;
 - the maximal deflection for each SLS combination of loads.

For multi-spans beams, the report gives the deflection in the main span and at the end of the cantilever spans.

- (3) For composite beams with partial degree of connection, the effect of a lack of interaction is considered, according to the method provided by the second generation of EN, by replacing the second moment of area of composite cross-sections in positive bending segments by the following one:

$$I_{yh,eff} = I_{y,ch} + I_{y,a} + \frac{A_{c,eff} A_a}{A_{c,eff} + A_a} z_{ch}^2 \quad (233)$$

where: $I_{y,ch}$ is the second moment of area of the transformed area of the slab, assuming uncracked concrete;

$I_{y,a}$ is the second moment of area of the steel profile (including the partial concrete encasement if any);

$A_{c,eff}$ is obtained by:

$$A_{c,eff} = A_c / n_{eff}$$

A_c is the area of the slab, given by: $A_c = b_{eff} t_c$

n_{eff} is the effective modular ratio, given by:

$$n_{eff} = n \left(1 + \frac{\pi^2 E_a A_c / n}{L_e^2 c_s} \right)$$

n is the modular ratio according to **§ 6.3**;

L_e is the distance between points of zero moment, as given in **§ 6.1.2 (2)**;

c_s is the stiffness of the shear connection, given by:

$$c_s = n_r k_{sc} / s_x$$

n_r is the number of studs per file, see **§ 4.7.3**;

s_x is the longitudinal spacing between files of studs, idem;

k_{sc} is the stiffness of an individual stud. According to prEN 1994-1-1:2024, B 2.5 (6), k_{sc} is obtained as follows:

$$k_{sc} = 0,7 P_{Rk} / s_e$$

where s_e is the slip of a connector at $0,7 P_{Rk}$. s_e is defined by the User (see **§ 4.17**), with a default value of 0,5 mm;

z_{ch} is the distance between the centroids of the concrete slab and of the steel section.



Formula (233) is applied specifically for each connection zone.
Reference: prEN 1994-1-1: 2024, 9.3.1 (5).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	133	Rev.: B

11.2 Eigen frequencies

- (1) Eigen frequencies of the beam is assessed by Finite Element analysis, using modal analysis.
- (2) Eigen frequencies of the beam are assessed for the following combinations of loads:
 - G
 - $G + 0,1 Q$
 - $G + 0,2 Q$
 - $G + 0,3 Q$
 - $G + 0,4 Q$
 - $G + 0,5 Q$
 where Q means Q_1 , Q_2 or $Q_1 + Q_2$.
- (3) Each force of any load case is transferred in the model as an equivalent mass by dividing the force by g .

11.3 Comfort to vibrations

11.3.1 General

- (1) The assessment of the comfort to vibrations is carried out according to the Hivoss method [13]. It implies the calculations of the dynamic properties of the beam (eigen-frequency and modal mass) and of the damping of the floor.

11.3.2 Dynamic properties of the floor

- (1) The eigen-frequency of the beam is assessed by the software according to § 11.2, considering the combination of masses as defined by the User, see § 4.16.2.
- (2) Considering Table 7-2 of [13], the modal mass of a single span beam may be obtained directly by:

$$m_{\text{mod}} = m_{\text{beam}} / 2 \quad (234)$$

where m_{beam} is the full mass of the beam considering the combination of masses.

The general formula to obtain the modal mass of the beam is as follows (according to § 7.4.5 in [13]):

$$m_{\text{mod}} = \sum_{\text{nodes } i} \left(\frac{\delta_i}{\delta_{\text{max}}} \right)^2 dm_i \quad (235)$$

where: δ_i is the deflection at node i in the eigenmode;

δ_{max} is the maximum value of the deflections of nodes in the eigenmode;

dm_i is the mass attributed to the node i .

- (3) The damping is obtained by summing up the appropriate values D_1 to D_3 , defined as follows (Table 7-4 of [13]):

- Structural damping D_1

According to the type of the beam:

- Steel beam: $D_1 = 1\%$
- Composite beam: $D_1 = 1\%$

- Damping for furniture D_2

According to the type of use defined by the user in the following list:

- Traditional office with partition walls: $D_2 = 2\%$
- Paperless office: $D_2 = 0\%$
- Open plan office: $D_2 = 1\%$
- Library: $D_2 = 1\%$
- Houses: $D_2 = 1\%$
- Schools: $D_2 = 2\%$
- Gymnastic: $D_2 = 0\%$
- User defined: D_2 between 0 and 5 %

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	134	Rev.: B

- Damping due to the finishing D_3

According to the type of finishing defined by the User in the following list:

- Ceiling under the floor: $D_3 = 1 \%$
- Free floating floor: $D_3 = 0 \%$
- Swimming screed: $D_3 = 1 \%$

Total damping: $D = D_1 + D_2 + D_3$

Limits: D is an integer value between $1\% \leq D \leq 9\%$

11.3.3 Determination of the OS-RMS₉₀ class

- (1) Considering the values of the eigen-frequency f , the modal mass m_{mod} and the damping D , the class of the $OS - RMS_{90}$ will be obtained from the numerical transcription of the Figure 7-4 to 7-12 of [13].

The $OS - RMS_{90}$ classes are defined according to the **Table 44**.

Class	OS-RMS ₉₀ limits
A	$OS-RMS_{90} \leq 0,1$
B	$0,1 < OS-RMS_{90} \leq 0,2$
C	$0,2 < OS-RMS_{90} \leq 0,8$
D	$0,8 < OS-RMS_{90} \leq 3,2$
E	$3,2 < OS-RMS_{90} \leq 12,8$
F	$12,8 < OS-RMS_{90} \leq 51,2$
G	$51,2 < OS-RMS_{90}$

Table 44: OS-RMS₉₀ Classes

11.3.4 Vibration performance assessment

- (1) According to $OS - RMS_{90}$ class and to the use of the floor defined by the User, a vibration assessment is given to the floor in the calculation report, between: recommended, critical, not recommended.

This assessment follows the rules defined in Table 7-5 of [13].

11.4 Cracking of concrete for beams with usual composite sections

11.4.1 General

- (1) **§ 11.4** applies to beams with usual composite sections, with or without partial concrete encasement.
- (2) At the support of beam with no adjacent span, if no limitation of concrete cracking is imposed (see **§ 4.17.2**) and if there is a continuity of the slab at the support (which is not accounted for in the design, see **§ 4.1 (5)**), **§ 11.4.2** applies.
- (3) At the support of beam with no adjacent span, if a limitation of concrete cracking is imposed (see **§ 4.17.2**) and if there is a continuity of the slab at the support (which is not accounted for in the design, see **§ 4.1 (5)**), **§ 11.4.3** applies.
- (4) At the support of beam with an adjacent span, if a limitation of concrete cracking is imposed (see **§ 4.17.2**), **§ 11.4.4** applies.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	135	Rev.: B

11.4.2 Minimum reinforcement for simply supported beams

- (1) The minimum ratio of longitudinal reinforcement at a simple support of the beam with slab continuity should fulfil the following criteria (according to EN 1994-1-1, 7.4.1 (4)):

$$\begin{aligned}\rho_s &\geq 0,4 \% \text{ for propped construction;} \\ \rho_s &\geq 0,2 \% \text{ for un-propped construction}\end{aligned}\quad (236)$$

where ρ_s is the ratio of longitudinal reinforcement – see § 4.4.5 (5)



Fully propped beams and point propped beams are considered as propped construction.

11.4.3 Minimum reinforcement for crack width limitation without direct loading

- (1) For a slab at the support of the beam with no adjacent span and with slab continuity, the crack width limitation is obtained by fulfilling the following condition (according to EN 1994-1-1, 7.4.2):

$$A_s \geq A_{s,min} = k_s k_c k f_{ct,eff} A_{ct} / \sigma_s \quad (237)$$

where: k is a factor which allows for the effect of non-uniform self-equilibrating stresses: $k = 0,8$;

k_s is a factor which allows for the effect of the reduction of the normal force of the concrete slab due to initial cracking and local slip of the shear connection: $k_s = 0,9$;

k_c is given by Formula (148) in § 8.6;

A_{ct} is the area of the tensile zone of concrete immediately prior to cracking, given by: $A_{ct} = b_{eff} t_c$;

t_c is the thickness of the active part of the concrete flange, see § 6.2;

b_{eff} : see § 6.1;

σ_s is the maximal stress permitted in the reinforcement immediately after cracking. It is obtained as the minimum value of f_{sk} the characteristic yield strength of the reinforcement (see § 5.3) and the value σ_{s1} obtained from clause (2).

- (2) σ_{s1} is obtained from the maximal thickness of concrete crack $w_{k,max}$ and from the diameter d_s of rebars, according to Table 45.

Stress in rebars σ_{s1} [MPa]	Diameter of rebars d_s [mm]		
	$w_{k,max} = 0,4$ mm	$w_{k,max} = 0,3$ mm	$w_{k,max} = 0,2$ mm
160	40	32	25
200	32	25	16
240	20	16	12
280	16	12	8
320	12	10	6
360	10	8	5
400	8	6	4
450	6	5	-

Table 45: Relation between stress in rebars and diameter of rebars

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	136	Rev.: B

11.4.4 Control of cracking due to direct loading

- (1) For beams with a continuity at support, i.e. where negative bending moments appear at support, the stresses in the reinforcements of the slab will be assessed according to § 6.8, including the effect of tension stiffening, and considering quasi-permanent SLS combinations of loads (see § 4.15.3 (4)).
- (2) From these calculations, the maximal value σ_s is obtained.
- (3) The maximal diameter of rebars is calculated from σ_s according to Table 45. The maximal spacing of rebars is calculated from σ_s according to Table 46.

Stress in rebars σ_{s1} [MPa]	Spacing of longitudinal rebars e_s [mm]		
	$w_{k,max} = 0,4$ mm	$w_{k,max} = 0,3$ mm	$w_{k,max} = 0,2$ mm
160	300	300	200
200	300	250	150
240	250	200	100
280	200	150	50
320	150	100	-
360	100	50	-

Table 46: Relation between stress in rebars and spacing of rebars

- (4) The calculation report displays these two values for rebars.

11.4.5 Minimum reinforcement for crack width limitation in partial concrete encasement

- (1) For beams with partial concrete encasement, the longitudinal reinforcement of the encasement (see § 4.9) should fulfil condition of Formula (187), according to EN 1994-1-1, 7.4.2 (6), where:
 - A_s is the area of the lower layer of reinforcement, given by: $A_s = A_{se,inf}$;
 - A_c is the area of the concrete of partial encasement, given by Formula (24);
 - $k_c = 0,6$;
 - $k = 0,8$;
 - f_{sk} is replaced by f_{sek} , the characteristic yield strength of the reinforcing steel used in partial encasement.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	137	Rev.: B

Annex A Specific configuration of composite slabs with Cofraplus 220

- (1) This Annex provides the details on the specific configuration of composite slabs using Cofraplus 220 profiled steel sheeting.
- (2) This Annex applies to usual steel or composite beams.
- (3) When a composite slab uses Cofraplus 220 profiled steel sheeting, the following specific details apply (**Figure 62**):
 - the ribs are perpendicular to the beam axis;
 - the steel sheeting is supported by wings that are either bolted or welded to the profile's web;
 - the steel sheeting upper side and the upper fibre of the profile are at the same level.
- (4) When composite slabs with Cofraplus 220 are used with usual composite beams, the presence of the ribs has no effect on the shear connection and the shear resistance of the connectors is calculated considering plain slabs.
- (5) The software does not cover the design of the support wings nor their link with the web.

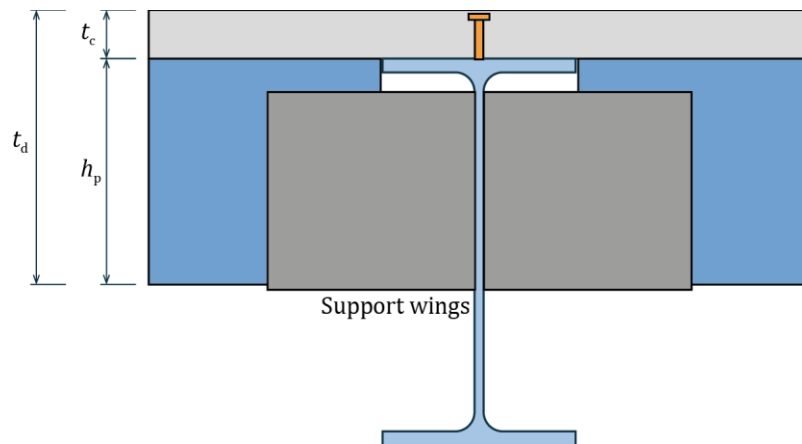


Figure 62: Specific configuration of a composite slabs with Cofraplus 220 sheeting

	Project: ABCPMX-II Software - DAL-22-003					
	Product reference: DRV/PMX/MT/001					
	Date:	19/05/2025	Authors:	POM	Page:	138
					Rev.:	B

Annex B Procedure for the calculation of section bending properties

B.1 General

- (1) A general procedure is applied for the calculation of the bending properties of all types of cross-sections. This procedure gives the values of:
 - the location of the elastic and plastic neutral axes;
 - the second moment of area of the transformed section for positive or negative bending moments;
 - the plastic resistance moment for positive or negative bending moments.
- (2) The procedure is based on a representation of the cross-section by section elements, with four types of elements, as described in this Annex and in **Figure 63**.

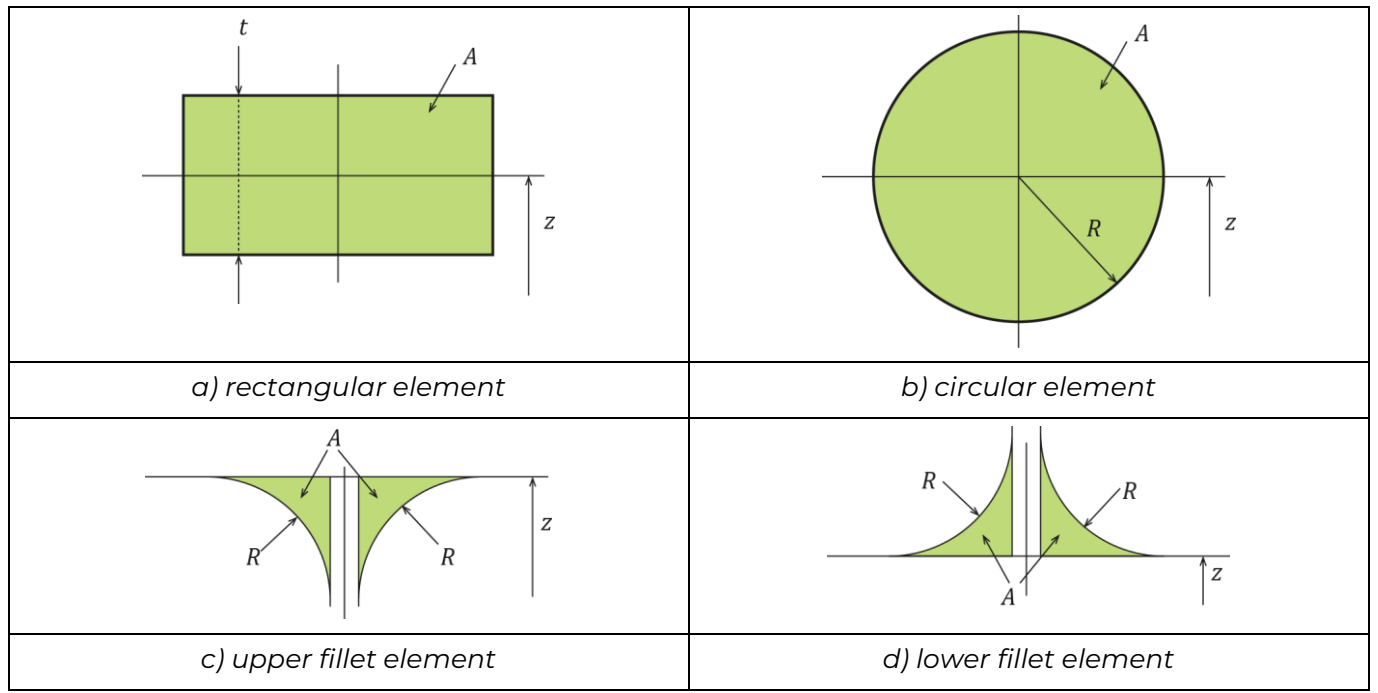


Figure 63: Type of elements

B.2 Section properties

Elastic properties

- (1) The location of the elastic neutral axis z_{el} is obtained by an iterative process, until the first moment of area of the section is equal to 0.
- (2) The first moment of area of the section about the yy axis at location z_{axe} is given by:

$$S_{axe} = \sum_i S_i(z_{axe}) \quad (238)$$

where: $S_i(z_{axe})$ is the first moment of area of the element no. i about this same axis.

- (3) Once the location of the elastic neutral axis has been obtained, the second moment of area of the transformed section about the y-y axis is given by:

$$I_{h,y} = \sum_i I_{y,h,i}(z_{el}) \quad (239)$$

where: $I_{y,h,i}(z_{axe})$ is the second moment of area of the element no. i about the neutral axis.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	139	Rev.: B

- (4) The area of the transformed section, $A_{h,y}$, is obtained by the following equation:

$$A_{h,y} = \sum_i \frac{A_i}{n_i} \quad (240)$$

- (4) The elastic moment of the section is obtained as follows:

$$M_{el} = I_y \times \min_i k_{\sigma,i} \quad (241)$$

where: $k_{\sigma,i}$ is the constraint ratio of the element no i .

Plastic properties

- (1) The location of the plastic neutral axis z_{pl} is obtained by an iterative process, until the axial resultant of the plastic stress diagram is equal to N_{Ed} , where N_{Ed} is the internal axial force in the cross-section. In the absence of MN interaction, $N_{Ed} = 0$.
- (2) The axial resultant of the plastic stress diagram, considering a plastic neutral axis at the location z_{axe} , is given by:

$$N_{pl,axe} = \sum_i N_{pl,i} (z_{axe}) \quad (242)$$

where: $N_{pl,i} (z_{axe})$ is the contribution of the element no i .

- (3) Once the location of the plastic neutral axis is known, the associated plastic moment is calculated with the following formula:

$$M_{pl} = \sum_i M_{pl,i} (z_{pl}) \quad (243)$$

where: $M_{pl,i} (z_{axe})$ is the contribution of the element no i .

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	140	Rev.: B

Annex C Procedure for the Finite Element calculations

C.1 Principles

- (1) All global analyses of the beam are carried out with Finite Elements numerical analysis, assuming an elastic behaviour.
- (2) The present Annex gives the general method for applying the Finite Element analyses in the software.

C.2 Numerical modelling

- (1) Nodes are imposed at each following section of the beam:
 - beam ends;
 - beam supports;
 - at point propping;
 - at point with point lateral restraints;
 - at support with beam continuity, point at 0,15 times the span length (which is the simplified threshold between positive and negative bending).
- (2) Nodes are added between the imposed nodes, to fulfil the following requirements:
 - the spacing between two adjacent nodes (i.e. the element length) should not be greater than $d_{el,max}$;
 - the minimum number of nodes per cantilever span should not be lower than $n_{can,min}$;
 - the minimum number of nodes per intermediate span should not be lower than $n_{int,min}$;

where $d_{el,max}$, $n_{can,min}$, $n_{int,min}$ are calculation options of the software that may be changed by the User (see § 3.3.4). Their default values are:

$$d_{el,max} = 1 \text{ m}$$

$$n_{can,min} = 5$$

$$n_{int,min} = 20$$

- (3) The number of nodes is denoted N_{nodes} .
All nodes are ranked by increasing order of their position x , measured from the left end of the beam. Node no 1 is on the left end and node no N_{nodes} is on the right end of the beam.
- (4) One beam element is placed between two adjacent nodes. The number of elements is denoted N_{elts} , where:

$$N_{elts} = N_{nodes} - 1$$

Between the nodes no i and no $i + 1$ is located the element no i .

- (5) Elements are assumed to be continuous all along the beam. There is no relaxation at nodes.
- (6) Elements are all aligned horizontally on the same line representing the beam axis. For composite beams, the variation of the vertical location of the centroid is not taken into account.
- (7) The main directions are defined as follows:
 - x is the horizontal direction of the beam axis, from the left to the right end;
 - z is the vertical upwards direction;
 - y is the horizontal direction that forms an orthonormal basis.
- (8) There are two degrees of freedom for each node: the vertical displacement u_z and the rotation r_y .
- (9) At each support of the beam, the degree of freedom u_z is fixed.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	141	Rev.: B

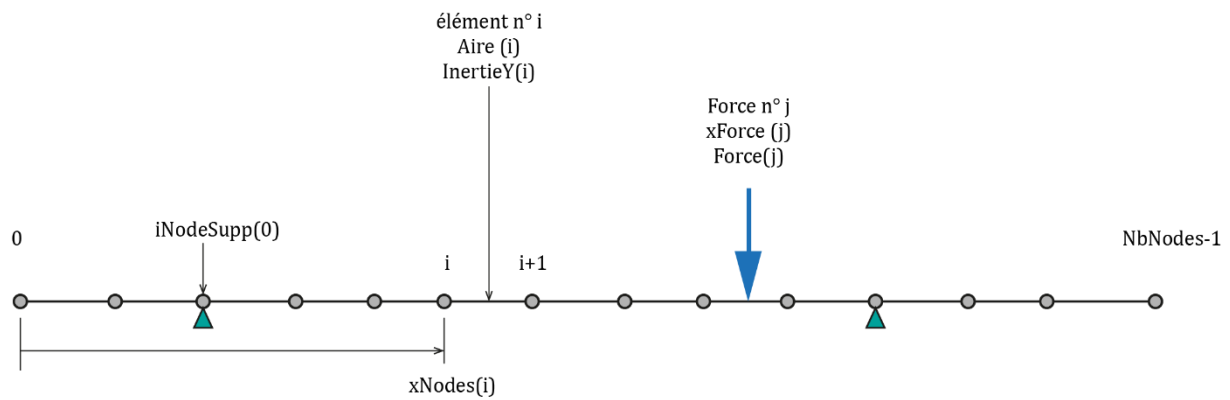


Figure 64: Nodes and elements along the beam

- (10) For each load case, the Finite Element numerical analysis will provide the following results, for each node:
- the vertical displacement u_z ;
 - the rotation u_y ;
 - the shear force, at LHS and RHS;
 - the bending moment, at LHS and RHS;
- (11) Additionally, the analysis will provide, for each support node, the vertical reaction R_Z .

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	142	Rev.: B

Annex D Critical moments for LT Buckling

D.1 Scope of application

- (1) This Annex describes the calculation of the critical moment for lateral torsional buckling of segments of beams which are located between two cross-sections, where both flanges are laterally restrained.

D.2 Critical moments

- (1) The critical moment to lateral torsional buckling of the beam segment will be assessed through the LTBeamN module.
- (2) The LTBeamN module is based on a separate DLL.
- (3) Each segment of beam will be represented in LTBeamN. At each end of the segment, the internal moment associated to the relevant combination of loads will be applied, as well as all loads applied on the segment, with the relevant combination factors.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	143	Rev.: B

Annex E Critical moment for LT Buckling taking account of the steel deck

E.1 Scope of application

- (1) This Annex applies to beams with usual composite cross-section (with or without partial concrete encasement), with a composite slab where the ribs are perpendicular to the beam.
- (2) The thickness of the profiled steel sheeting should be greater than 0,75 mm: $t_p \geq 0,75$ mm.

E.2 Calculation of the critical moment

- (1) The critical moment for LT Buckling, M_{cr} , is calculated by using the module of critical analysis (see **Annex D**), considering the shear and the flexural stiffnesses (respectively S_{act} –see § E.3 – and k_θ – see § E.4) provided by the steel profiled sheeting.

E.3 Shear rigidity of the steel profiled sheeting

General

- (1) The distributed shear stiffness of the steel profiled sheeting is obtained by:

$$S_{act} = \frac{L}{c} \quad (244)$$

where: L is the length of the span;

c is the total flexibility of the steel deck.

- (2) c is obtained by:

$$c = c_{1,1} + c_{1,2} + c_{2,1} + c_{2,2} \quad (245)$$

where: $c_{1,1}$ is the contribution of the steel profiled sheets distortion;

$c_{1,2}$ is the contribution of shear strain in the profiled deck;

$c_{2,1}$ is the contribution of the sheet-to-beam fasteners deformation;

$c_{2,2}$ is the contribution of the seam fasteners deformation.

Flexibilities due to the sheet deformation

- (1) The components $c_{1,j}$, associated to the sheet deformation, either due to the distortion or to the shear strain, are obtained with the following formulas:

$$c_{1,1} = \frac{L \alpha_5 K}{E_a d^2} \left(\frac{e_p}{t_p} \right)^{2,5} \quad (246)$$

$$c_{1,2} = \frac{2 L (1 + \nu_a) (1 + 2h_p/e_p)}{E_a t_p d} \quad (247)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	144	Rev.: B

where: L is the span length of the beam;

E_a and ν_a are respectively the Young's modulus and the Poisson's ratio for steel –see § 5.1.4;

h_p , t_p and e_p are respectively the height, the thickness and the rib spacing of the steel sheets (cf. § 4.5.1) ;

K is a sheeting constant, which depends on the geometry and the connection of the deck. K is equal to K_1 when a fastener is used in every rib and to K_2 when a fastener is used every two ribs.

K_1 and K_2 predefined for profiled steel sheeting from the database? To be discussed with ArcelorMittal.

Table 50 to **Table 53** give the values of K_1 and K_2 coefficients for any geometry of steel profiled sheeting with open rib, according to ECCS Recommendations, respectively for one fastener at every rib and at every two ribs.

Values K_1 and K_2 from ECCS Recommendations are valid only for open ribs, with $\theta_p \geq 0^\circ$. For profiled steel sheeting with re-entrant ribs, where $\theta_p < 0^\circ$, ECCS recommendations provides no values. It could be decided to limit the method to sheeting with open ribs only. Or potentially to sheetings with re-entrant ribs, where K_1 and K_2 are already known, by numerical simulations. See for example Table 4 in [16].

α_5 is a factor which accounts for the influence of the overlapping and the continuity of profiled sheets. It is obtained according to **Table 47**.

n_t Number of sheets in the transverse direction	Continuity of sheets at beam supports	
	With overlapping	Without overlapping
1 or 2	1,0 / m	1,0 / m
3	0,9 / m	1,0 / m
4	0,8 / m	1,0 / m
≥ 5	0,7 / m	1,0 / m
Coefficients m and n_t : see § 3.1		

Table 47: α_5 coefficient

Flexibilities due to the fasteners deformation

- (1) The components $c_{2,j}$, associated to the deformations of the beam-to-deck fasteners ($c_{2,1}$) or of the seam fasteners ($c_{2,2}$), are obtained with the following formulas:

$$c_{2,1} = 2 \frac{L s_p e_L}{d^2} \quad (248)$$

$$c_{2,2} = \frac{s_s s_p (n_L - 1)}{n_c s_p + \beta_1 s_s} \quad (249)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	145	Rev.: B

where: s_p is the slip of the beam-to-deck fasteners (see **Table 10**);

s_s is the slip of the seam fasteners (see **Table 11**);

e_L is the longitudinal spacing between beam-to-deck fasteners (see **Figure 34**);

n_L is the number of sheets in the direction of the beams;

n_c is the number of seam fasteners along the deck span, excluding those which are fixed to the secondary beam;

β_1 is a coefficient whose value is given by **Table 48**, as a function of the number of fasteners on the width of a profiled sheet.

n_f	β_1
2	1,0
3	1,0
4	1,04
5	1,13
6	1,22
7	1,33
8	1,45
9	1,56
10	1,68

Table 48: β_1 coefficient

Slip of fasteners

- (1) According to Table 5.1 of the ECCS recommendations **[14]**, the slip to be considered for the beam-to-sheeting fasteners is given in **Table 10** (see § 4.11.2), depending on the type of fastener. For seam fasteners, the values to be considered are given in **Table 11** (idem).

E.4 Stiffness of the torsional restraint

- (1) The stiffness of the torsional restraint, k_θ , opposing the torsional rotation $\Delta\theta_x$ of the beam, is calculated as follows (according to EN 1993-1-3:2005, clause 10.1.5.2):

$$k_\theta = \frac{1}{\frac{1}{k_{\theta,A}} + \frac{1}{k_{\theta,C}}} \quad (250)$$

where: $k_{\theta,A}$ is the rotational stiffness due to the beam-to-deck connection;

$k_{\theta,C}$ is the rotational stiffness corresponding to the flexural stiffness of the deck.

- (2) $k_{\theta,C}$ stiffness will be obtained with the following formula:

$$k_{\theta,C} = \frac{k E I_{pu}}{d} \quad (251)$$

where: I_{pu} is the effective second moment of area per unit width of the deck (see § 4.4.3);

d is the span of the deck, i.e. the spacing between beams;

k is a factor which takes into account the sheeting support conditions:

$k = 6$ for an intermediate beam;

$k = 3$ for an edge beam.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	146	Rev.:	B

(3) $k_{\theta,A}$ stiffness is obtained with the following formula:

$$k_{\theta,A} = k_{100} k_{ba} k_t k_{bR} k_g k_{bb} \quad (252)$$

where: k_{100} is the torsional stiffness for a flange width equal to 100 mm, which is given in Table 49.

Geometry of the deck	Fastener spacing	k_{100} (kNm/m)	b_{bmax} (mm)
$b_r \leq b_b$	$e_L = e_p^{(1)}$	5,2	40
	$e_L = 2 e_p^{(2)}$	3,1	40
$b_r > b_b$	$e_L = e_p^{(1)}$	3,1	120
	$e_L = 2 e_p^{(2)}$	2,0	120
(1) : One fastener in every rib			
(2) : One fastener in every two ribs			

Table 49: Values of k_{100} and b_{bmax}

k_{ba} is a factor which accounts for the actual flange width b_{fs} , as given by the following formulas:

$$k_{ba} = \left(\frac{b_{fs}}{100} \right)^2 \quad \text{when } 100 \text{ mm} \leq b_{fs} < 125 \text{ mm}$$

$$k_{ba} = 1,25 \left(\frac{b_{fs}}{100} \right) \quad \text{when } 125 \text{ mm} \leq b_{fs} \leq 200 \text{ mm}$$

Formula (251) has been calibrated for flange widths up to 200 mm. For flange widths above 200 mm, it is possible, from the safety point of view, to perform the calculations considering a width of 200 mm.

k_t is a coefficient taking into account the influence of the thickness of the deck, when this is different from 0,75 mm. k_t may be obtained by :

$$k_t = \left(\frac{t_{nom}}{0,75} \right)^{1,1} \quad \text{when } t_{nom} \geq 0,75 \text{ mm and } b_r \leq b_b$$

$$k_t = \left(\frac{t_{nom}}{0,75} \right)^{1,5} \quad \text{when } t_{nom} \geq 0,75 \text{ mm and } b_r > b_b$$

$$k_t = \left(\frac{t_{nom}}{0,75} \right)^{1,5} \quad \text{when } t_{nom} < 0,75 \text{ mm}$$

t_{nom} is the nominal thickness of the deck, in mm ;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	147	Rev.: B

k_{bR} is a coefficient associated to the spacing e_p between the ribs (in mm), obtained as follows:

$$k_{bR} = 1 \quad \text{when } e_p \leq 185 \text{ mm}$$

$$k_{bR} = \frac{185}{e_p} \quad \text{when } e_p > 185 \text{ mm}$$

k_g is a coefficient to consider the stabilising effect of gravity loads applied to the deck and transmitted to the beam. k_g is obtained as follows:

$$k_g = 1 + 0,08 (q_g - 1) \quad \text{when } t_{nom} = 0,75 \text{ mm and } b_r \leq b_b$$

$$k_g = 1 + 0,16 (q_g - 1) \quad \text{when } t_{nom} = 0,75 \text{ mm and } b_r > b_b$$

$$k_g = 1 + 0,095 (q_g - 1) \quad \text{when } t_{nom} \geq 1 \text{ mm}$$

q_g is the distributed load transmitted by the sheeting to the beam, expressed in kN/m, not exceeding 12 kN/m in the calculations. For composite slabs in the construction phase, this maximum value can be considered conservatively in the calculations with regards to the self-weight and construction loads;

For nominal thickness values between 0,75 and 1 mm, a linear interpolation can be used to calculate k_g ;

k_{bb} is a coefficient taking into account the width of the rib bottom, obtained by the following formula:

$$k_{bb} = \min\{1; \sqrt{b_{bmax} / b_b}\}$$

b_{bmax} is defined in **Table 49**.

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	148	Rev.:	B

K_1 coefficient for fasteners in every trough (acc. to [14])										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
0°	0,1	0,013	0,030	0,041	0,041	0,046	0,050	0,066	0,103	0,193
	0,2	0,042	0,096	0,131	0,142	0,142	0,153	0,199	0,311	0,602
	0,3	0,086	0,194	0,264	0,285	0,283	0,302	0,388	0,601	1,188
	0,4	0,144	0,323	0,438	0,473	0,468	0,494	0,629	0,972	1,935
	0,5	0,216	0,438	0,654	0,703	0,695	0,729	0,922	1,420	2,837
	0,6	0,302	0,674	0,911	0,980	0,965	1,008	1,266	1,938	3,892
	0,7	0,402	0,895	1,208	1,300	1,277	1,329	1,661	2,536	5,098
	0,8	0,516	1,146	1,546	1,662	1,631	1,692	2,107	3,208	6,453
5°	0,1	0,014	0,031	0,041	0,044	0,044	0,049	0,066	0,107	0,205
	0,2	0,050	0,099	0,128	0,134	0,132	0,146	0,198	0,336	0,652
	0,3	0,107	0,202	0,253	0,260	0,254	0,280	0,386	0,681	1,548
	0,4	0,188	0,338	0,413	0,417	0,404	0,448	0,629	1,158	2,639
	0,5	0,295	0,507	0,604	0,601	0,578	0,648	0,934	1,783	
	0,6	0,429	0,706	0,823	0,826	0,772	0,877	1,306	2,586	
	0,7	0,591	0,935	1,066	1,028	0,983	1,385	1,756	3,605	
	0,8	0,780	1,191	1,328	1,264	1,208	1,423	2,990	4,838	
10°	0,1	0,016	0,031	0,040	0,042	0,042	0,048	0,065	0,111	0,221
	0,2	0,056	0,101	0,123	0,125	0,123	0,139	0,200	0,366	0,873
	0,3	0,125	0,204	0,238	0,233	0,260	0,264	0,402	0,786	
	0,4	0,222	0,338	0,375	0,356	0,345	0,418	0,689	1,445	
	0,5	0,349	0,494	0,526	0,486	0,473	0,605	1,082	2,428	
	0,6	0,502	0,668	0,682	0,615	0,608	0,837	1,607		
	0,7	0,677	0,851	0,834	0,736	0,752	1,128	2,308		
	0,8	0,869	1,035	0,975	0,844	0,907	1,494	3,200		
15°	0,1	0,017	0,031	0,040	0,041	0,041	0,047	0,066	0,115	0,241
	0,2	0,062	0,102	0,118	0,115	0,113	0,134	0,209	0,403	
	0,3	0,139	0,202	0,218	0,204	0,200	0,254	0,440	0,945	
	0,4	0,244	0,321	0,325	0,293	0,294	0,414	0,796		
	0,5	0,370	0,448	0,426	0,371	0,396	0,636	1,329		
	0,6	0,508	0,568	0,508	0,434	0,513	0,941			
	0,7	0,646	0,668	0,561	0,483	0,664	1,349			
	0,8	0,768	0,735	0,578	0,527	0,861				
20°	0,1	0,018	0,032	0,039	0,039	0,039	0,046	0,066	0,111	0,276
	0,2	0,068	0,101	0,111	0,106	0,104	0,131	0,221	0,452	
	0,3	0,148	0,193	0,194	0,174	0,177	0,255	0,492		
	0,4	0,249	0,289	0,267	0,230	0,259	0,444	0,931		
	0,5	0,356	0,372	0,315	0,270	0,364	0,725			
	0,6	0,448	0,420	0,326	0,303	0,512				
	0,7	0,509	0,423	0,301	0,346					
	0,8	0,521	0,372	0,259	0,413					

Table 50: Values of K_1 for θ_p between 0° and 20°

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	149	Rev.:	B

K_1 coefficient for fasteners in every trough (acc. to [14])										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
25°	0,1	0,019	0,032	0,038	0,038	0,038	0,045	0,068	0,126	0,313
	0,2	0,072	0,099	0,103	0,095	0,095	0,129	0,236	0,513	
	0,3	0,151	0,178	0,166	0,144	0,160	0,268	0,557		
	0,4	0,238	0,244	0,204	0,176	0,247	0,494			
	0,5	0,306	0,272	0,203	0,204	0,376				
	0,6	0,333	0,248	0,172	0,241					
	0,7	0,300	0,174	0,142						
	0,8	0,204	0,081							
30°	0,1	0,020	0,032	0,037	0,036	0,036	0,044	0,070	0,133	
	0,2	0,075	0,095	0,094	0,084	0,087	0,132	0,256		
	0,3	0,148	0,157	0,135	0,116	0,152	0,291			
	0,4	0,208	0,186	0,139	0,139	0,253				
	0,5	0,226	0,161	0,112	0,176					
	0,6	0,180	0,089	0,093						
	0,7	0,077								
	0,8									
35 °	0,1	0,021	0,032	0,036	0,034	0,034	0,043	0,072	0,142	
	0,2	0,076	0,089	0,083	0,072	0,082	0,137	0,281		
	0,3	0,137	0,130	0,102	0,093	0,151				
	0,4	0,162	0,119	0,082	0,120					
	0,5	0,123	0,059							
	0,6	0,032								
	0,7									
	0,8									
40°	0,1	0,023	0,032	0,034	0,032	0,032	0,043	0,075		
	0,2	0,075	0,081	0,070	0,060	0,077	0,146			
	0,3	0,116	0,096	0,068	0,078					
	0,4	0,100	0,053	0,048						
	0,5	0,024								
	0,6									
	0,7									
	0,8									
45°	0,1	0,024	0,031	0,032	0,029	0,030	0,043	0,079		
	0,2	0,071	0,069	0,056	0,050	0,073				
	0,3	0,086	0,057	0,041						
	0,4	0,032								
	0,5									
	0,6									
	0,7									
	0,8									

Table 51: Values of K_1 for θ_p between 25° and 45°

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	150	Rev.:	B

<i>K₂ coefficient for fasteners in every two troughs (acc. to [14])</i>										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
0°	0,1	0,014	0,025	0,036	0,046	0,054	0,061	0,070	0,108	0,211
	0,2	0,031	0,065	0,099	0,129	0,151	0,169	0,206	0,318	0,649
	0,3	0,054	0,123	0,192	0,252	0,294	0,328	0,402	0,608	1,269
	0,4	0,084	0,202	0,316	0,414	0,482	0,535	0,653	0,968	2,056
	0,5	0,123	0,299	0,468	0,614	0,712	0,790	0,958	1,410	3,006
	0,6	0,169	0,415	0,649	0,846	0,982	1,090	1,318	1,928	4,113
	0,7	0,222	0,549	0,855	1,108	1,286	1,433	1,730	2,525	5,383
	0,8	0,284	0,699	1,086	1,398	1,398	1,818	2,196	3,198	6,811
5°	0,1	0,089	0,138	0,184	0,288	0,269	0,311	0,359	0,432	0,590
	0,2	0,300	0,433	0,564	0,690	0,810	0,934	1,091	1,358	2,046
	0,3	0,627	0,872	1,113	1,345	1,569	1,806	2,215	2,710	4,441
	0,4	1,076	1,453	1,826	2,187	2,535	2,910	3,446	4,498	8,057
	0,5	1,644	2,171	2,694	3,205	3,703	4,244	5,058	6,671	12,940
	0,6	2,280	2,961	3,639	4,313	4,999	5,797	6,971	9,571	
	0,7	2,961	3,803	4,620	5,443	6,347	7,479	9,206	13,010	
	0,8	3,802	4,838	5,788	6,612	7,701	9,257	11,760	17,200	
10°	0,1	0,091	0,140	0,186	0,229	0,270	0,312	0,362	0,440	0,627
	0,2	0,312	0,446	0,575	0,699	0,817	0,943	1,112	1,425	2,472
	0,3	0,665	0,907	1,144	1,370	1,589	1,835	2,204	2,979	
	0,4	1,156	1,529	1,891	2,239	2,578	2,984	3,655	5,251	
	0,5	1,793	2,313	2,819	3,305	3,782	4,397	5,519	7,872	
	0,6	2,533	3,206	3,858	4,509	5,192	6,096	7,875		
	0,7	3,334	4,148	4,949	5,780	6,737	8,112	10,820		
	0,8	4,236	5,170	6,051	7,066	8,404	10,470	12,590		
15°	0,1	0,093	0,142	0,198	0,231	0,271	0,313	0,364	0,448	0,682
	0,2	0,325	0,458	0,586	0,707	0,824	0,953	1,140	1,523	
	0,3	0,703	0,942	1,174	1,393	1,610	1,874	2,316	3,411	
	0,4	1,237	1,602	1,953	2,285	2,624	3,089	3,981		
	0,5	1,937	2,443	2,926	3,379	3,869	4,640	6,256		
	0,6	2,778	3,428	4,058	4,664	5,366	6,581			
	0,7	3,692	4,488	5,273	6,081	7,138	8,902			
	0,8	4,648	5,570	6,516	7,628	9,910				
20°	0,1	0,096	0,144	0,190	0,232	0,273	0,315	0,368	0,459	0,680
	0,2	0,339	0,472	0,597	0,716	0,832	0,966	1,177	1,659	
	0,3	0,743	0,976	1,204	1,416	1,633	1,927	2,461		
	0,4	1,317	1,673	2,009	2,325	2,679	3,246	3,840		
	0,5	2,075	2,559	3,011	3,436	3,993	4,969			
	0,6	3,006	3,625	4,194	4,752	5,588				
	0,7	4,042	4,789	5,494	6,272					
	0,8	5,122	6,013	6,883	7,861					

Table 52: Values of K_2 for θ_p between 0° and 20°

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	151	Rev.:	B

<i>K₂ coefficient for fasteners in every two troughs (acc. to [14])</i>										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
25°	0,1	0,098	0,147	0,192	0,234	0,274	0,317	0,373	0,475	0,665
	0,2	0,355	0,485	0,609	0,725	0,840	0,983	1,226	1,566	
	0,3	0,784	1,015	1,233	1,437	1,660	2,000	2,589		
	0,4	1,398	1,740	2,057	2,359	2,753	3,427			
	0,5	2,205	2,659	3,064	3,490	4,114				
	0,6	2,000	3,752	4,218	4,797					
	0,7	4,318	4,941	5,480						
	0,8	5,487	6,132							
30°	0,1	0,101	0,150	0,194	0,236	0,276	0,319	0,378	0,495	
	0,2	0,372	0,500	0,621	0,734	0,850	1,005	1,298		
	0,3	0,827	1,051	1,260	1,456	1,697	2,098			
	0,4	1,477	1,801	2,092	2,393	2,830				
	0,5	2,319	2,727	3,075	3,499					
	0,6	3,320	3,728	4,041						
	0,7	4,378								
	0,8									
35 °	0,1	0,105	0,153	0,197	0,238	0,278	0,322	0,385	0,525	
	0,2	0,390	0,516	0,634	0,744	0,862	1,035	1,329		
	0,3	0,872	1,088	1,284	1,548	1,741				
	0,4	1,553	1,849	2,105	2,412					
	0,5	2,400	2,713							
	0,6	3,278								
	0,7									
	0,8									
40°	0,1	0,109	0,156	0,200	0,241	0,280	0,325	0,394	0,569	
	0,2	0,411	0,538	0,647	0,753	0,878	1,077			
	0,3	0,919	1,122	1,301	1,496					
	0,4	1,614	1,859	2,085						
	0,5	2,376								
	0,6									
	0,7									
	0,8									
45°	0,1	0,114	0,160	0,203	0,243	0,282	0,329	0,409		
	0,2	0,434	0,553	0,661	0,764	0,899				
	0,3	0,965	1,148	1,306						
	0,4	1,634								
	0,5									
	0,6									
	0,7									
	0,8									

Table 53: Values of K_2 for θ_p between 25° and 45°

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	152	Rev.: B

Annex F Bending moment resistance in fire situation of partially encased profiles

F.1 Scope of application

- (1) This Annex describes the simplified method used in **ACB-PMX** to assess the fire design resistance at time t of cross-sections for steel or composite beams with partially encased profiles.
- (2) The simplified method of the Annex F of EN 1994-1-2:2005 are applied (Annex D in prEN 1994-1-2:2023).
- (3) The design resistance is calculated by iteration at the following time steps:
 - 30 min;
 - 60 min;
 - 90 min;
 - 120 min;
 - 180 min.
- (4) According to EN 1994-1-2: 2005, F.3 (1), the method is valid when the following conditions are fulfilled:

$$\begin{aligned}
 h &\geq h_{\text{lim}} \\
 b_c &\geq b_{c,\text{lim}} \\
 h b_c &\geq (h b_c)_{\text{lim}} \\
 t_f &\leq h/8
 \end{aligned}
 \tag{253}$$

where h_{lim} , $b_{c,\text{lim}}$ and $(h b_c)_{\text{lim}}$ are as given in **Table 54**.

Time step	$h_{\text{lim}}, b_{c,\text{lim}}$ (mm)	$(h b_c)_{\text{lim}}$ (mm ²)
R30	120	17 500
R60	150	24 000
R90	170	35 000
R120	200	50 000
R180	250	80 000

Table 54: Minimum cross-section dimensions

F.2 Bending resistance under positive moment

- (1) The positive bending resistant moment at a time step under fire situation is obtained using the plastic calculation according to the principles shown in **Figure 65** and as detailed below.

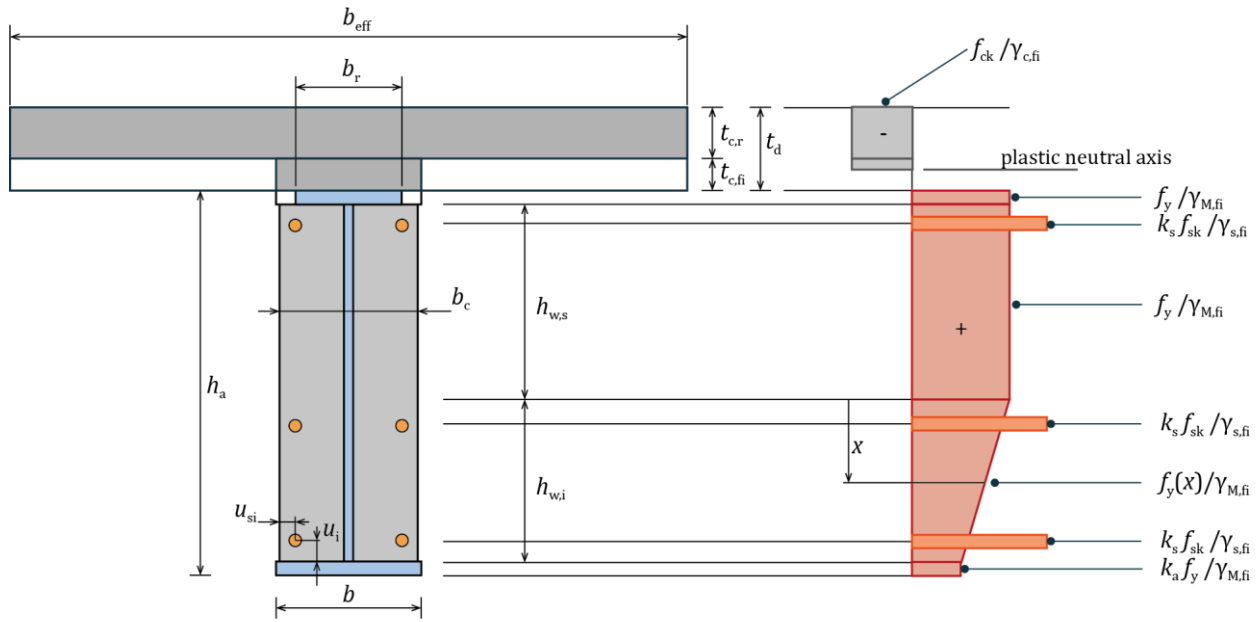


Figure 65 : Stress distribution for the calculation of the positive bending resistance

- (2) The lower flange is fully yielded to its reduced strength, given by:

$$k_a f_y / \gamma_{M,fi} \quad (254)$$

where: f_y is the yield strength of steel at room temperature, as defined in § 5;

$\gamma_{M,fi}$ is the partial factor, defined in § 4.14.2;

k_a is a reduction factor given in Table 55, where:

h_a is the height of the steel profile;

t_f is a thickness of the flanges;

b_c is the width of the concrete encasement.

Time step	Reduction factor k_a	$k_{a,min}$	$k_{a,max}$
R30	$\left[1,12 - \frac{84}{b_c} + \frac{h_a}{22 b_c} \right] a_0$	0,5	0,8
R60	$\left[0,21 - \frac{26}{b_c} + \frac{h_a}{24 b_c} \right] a_0$	0,12	0,4
R90	$\left[0,12 - \frac{17}{b_c} + \frac{h_a}{38 b_c} \right] a_0$	0,06	0,12
R120	$\left[0,1 - \frac{15}{b_c} + \frac{h_a}{40 b_c} \right] a_0$	0,05	0,1
R180	$\left[0,03 - \frac{3}{b_c} + \frac{h_a}{50 b_c} \right] a_0$	0,03	0,06
where:		$a_0 = 0,018 t_f + 0,7$	

Table 55: Reduction factor k_a

- (3) The web is divided in two parts. The upper part has a height $h_{w,s}$ and the lower part a height $h_{w,i}$. When $1 < h_a/b_c < 2$, $h_{w,i}$ is defined by Table 56. For other cases, it is defined by Formula (255), where a_1 , a_2 and $h_{w,i,min}$ are given in Table 57.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	154	Rev.: B

$$h_{w,i} = \frac{a_1}{b_c} + a_2 \frac{t_w}{b_c h_a} \quad \text{and} \quad h_{w,i} \leq h_{w,i,\min} \quad (255)$$

$h_{w,s}$ is given by: $h_{w,s} = h_w - h_{w,i}$

Time step	$h_{w,i}$	$h_{w,i,\min}$
R30	$h_{w,i} = 3600/b_c$	20
R60	$h_{w,i} = \frac{9\,500}{b_c} + 20\,000 \frac{t_w}{b_c h_a} \left(2 - \frac{h_a}{b_c}\right)$	30
R90	$h_{w,i} = \frac{14\,000}{b_c} + 75\,000 \frac{t_w}{b_c h_a} \left(\frac{86}{75} + 2 - \frac{h_a}{b_c}\right)$	40
R120	$h_{w,i} = \frac{23\,000}{b_c} + 110\,000 \frac{t_w}{b_c h_a} \left(\frac{7}{11} + 2 - \frac{h_a}{b_c}\right)$	45
R180	$h_{w,i} = \frac{35\,000}{b_c} + 250\,000 \frac{t_w}{b_c h_a} \left(\frac{15}{25} + 2 - \frac{h_a}{b_c}\right)$	55

Table 56: Height of the lower part of the web $h_{w,i}$ – case $1 < h_a/b_c < 2$

Range	Time step	a_1 (mm ²)	a_2 (mm ²)	$h_{w,i,min}$ (mm)
$h_a/b_c \leq 1$	R30	3 600	0	20
	R60	9 500	20 000	30
	R90	14 000	160 000	40
	R120	23 000	180 000	45
	R180	35 000	400 000	55
$h_a/b_c \geq 2$	R30	3 600	0	20
	R60	9 500	0	30
	R90	14 000	75 000	40
	R120	23 000	110 000	45
	R180	35 000	250 000	55

Table 57: Height of the lower part of the web $h_{w,i}$ (all parameters in mm)

- (4) The upper part of the web is yielded to its design strength, given by:

$$f_y / \gamma_{M,fi} \quad (256)$$

- (5) The lower part of the web is yielded to a trapezoidal strength, whose shape is given by:

$$f_y(x) / \gamma_{M,fi} \quad (257)$$

$$f_y(x) = f_y \left(1 - \frac{x}{h_{w,i}} (1 - k_a) \right)$$

where: k_a is as defined in **(2)**;

x is the distance of the fibre to frontier between the upper and lower parts of the web.

- (6) The effective width of the upper flange is yielded to its design strength, given by:

$$f_y / \gamma_{M,fi} \quad (258)$$

The effective width of the upper flange is given by:

$$b_r = b - 2b_{fi} \quad (259)$$

where b_{fi} is the width reduction, as given in **Table 58**.

Time step	b_{fi} (mm)
R30	$t_f/2 + (b - b_c)/2$
R60	$t_f/2 + 10 + (b - b_c)/2$
R90	$t_f/2 + 30 + (b - b_c)/2$
R120	$t_f/2 + 40 + (b - b_c)/2$
R180	$t_f/2 + 60 + (b - b_c)/2$

Table 58: Width reduction $b_{f,i}$ (in mm)

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	156	Rev.: B

- (7) The reinforcements in the concrete of the encasement are yielded to their reduced design strength, given by:

$$k_s f_{sk} / \gamma_{s,fi} \quad (260)$$

where: f_{sk} is the characteristic yield strength of the reinforcement, see § 5.3;

$\gamma_{s,fi}$ is the partial factor, defined in § 4.14.2;

k_s is a reduction factor, defined by:

$$k_s = \frac{(u a_3 + a_4) a_5}{\sqrt{A_m/V}} \quad \text{but } k_{s,min} \leq k_s \leq k_{s,max} \quad (261)$$

$$k_{s,min} = 0,1$$

$$k_{s,max} = 1$$

$$A_m = 2 h_a + b_c$$

$$V = h_a b_c$$

$$u = \frac{1}{\frac{1}{u_i} + \frac{1}{u_{si}} + \frac{1}{b_c - t_w - u_{si}}}$$

u_i is the distance (in mm) between the rebar axis and the inner face of the lower flange;

u_{si} is the distance (in mm) between the rebar axis and the outer face of the encasement;

a_3, a_4, a_5 are given in Table 59.

Time step	a_3	a_4	a_5
R30	0,062	0,16	0,126
R60	0,034	-0,04	0,101
R90	0,026	-0,154	0,090
R120	0,026	-0,284	0,082
R180	0,024	-0,562	0,076

Table 59: a_i factors for the calculation of the reduction factor k_s

- (7) A solid slab is divided in two parts:
- the upper part, whose dimensions are $b_{eff} \times t_{d,r}$;
 - the lower part, whose dimensions are $b \times t_{d,fi}$;

where : b_{eff} is the effective width of the slab, as given in § 6.1;

b is the width of the upper flange;

$t_{c,fi}$ is the thickness reduction of the solid slab, given in Table 60;

$t_{c,r}$ is the remaining thickness: $t_{c,r} = t_d - t_{c,fi}$.

- (8) Both parts of the slab are stressed in compression to their strength, given by:

$$f_{ck} / \gamma_{M,c,fi} \quad (262)$$

where: f_{ck} is the characteristic resistance to compression of concrete, see § 5.2;

$\gamma_{M,c,fi}$ is the partial factor for concrete in fire situations, defined in § 4.14.2.

Time step	$t_{c,fi}$ (mm)
R30	10
R60	20
R90	30
R120	40
R180	55

Table 60: Thickness reduction of a solid slab $t_{c,fi}$

- (9) For composite slabs with open profiles, disposed perpendicularly to the beam axis (configurations PE1 and PE2, see § 4.4.3), the thickness reduction $t_{c,fi}$ is applied from the upper face of the steel deck (Figure 66) (for a steel deck with or without an upper stiffener). The thickness of the remaining fully effective part is given by: $t_{c,r} = t_c - t_{c,fi} = t_d - h_p - t_{c,fi}$.

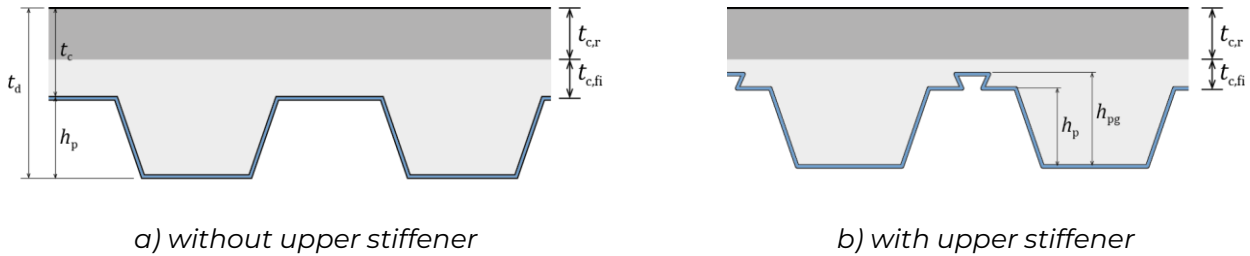


Figure 66 : Thickness reduction for composite slabs with perpendicular open profiles decks

- (10) For composite slabs with reintrant profiles (i.e. with $b_t < b_b$, see § 4.5.1 and Figure 21 b), disposed perpendicularly to the beam axis (configurations PE1 and PE2, see § 4.4.3), the thickness reduction $t_{c,fi}$ is applied from the lower face of the steel deck, but without being lower than the deck depth (Figure 67).

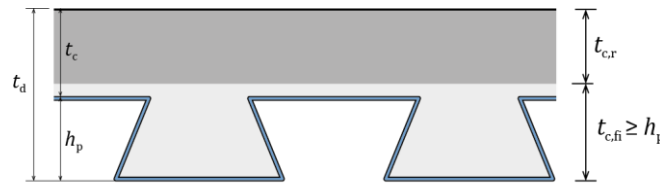


Figure 67 : Thickness reduction for composite slabs with perpendicular reintrant profile decks

- (11) For composite slabs with reintrant profiles, disposed parallel to the beam, the thickness reduction $t_{c,fi}$ is applied from the lower face of the steel deck. This also applies to composite slabs with perpendicular ribs, with configuration PE3 according to § 4.4.3.
- (12) For composite slabs with open profiles, disposed parallel to the beam, the thickness reduction $t_{c,fi}$ is applied considering the effective depth of the slab $h_{c,eff}$ defined in § 10.5.4 (2) (Figure 68).

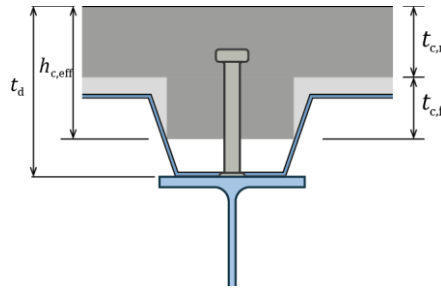


Figure 68 : Thickness reduction for composite slabs with parallel open profile decks

- (13) For solid slabs with prefabricated parts, the thickness reduction is applied from the lower face of the precast elements, but not lower that the joints filled with concrete (**Figure 69**).

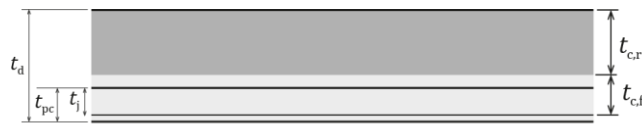


Figure 69 : Thickness reduction for solid slabs with precast elements

- (14) For solid slabs with haunches, the thickness reduction is applied from the lower face of the haunch (**Figure 70**).

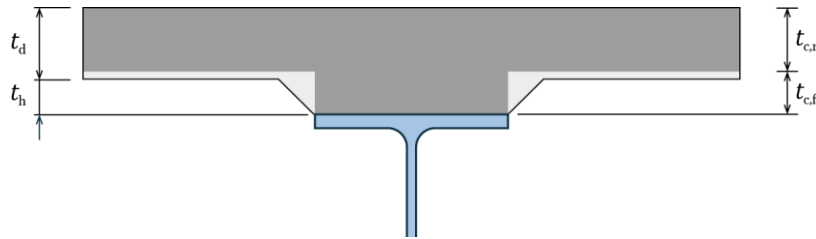


Figure 70 : Thickness reduction for solid slabs with precast elements

- (15) Clauses (8) to (14) apply only to composite beams. For non-composite beams with concrete encasement, the slab is ignored.

F.3 Longitudinal shear in the connection under positive moment

- (1) Under positive moment, the longitudinal shear force in the connection between the slab and the profile along a critical segment of a usual composite beam with partial concrete encasement, for a full connection, is obtained by:

$$N_{cx,fi} = \min\{N_{c,fi,Rd}; N_{a,fi,Rd} + N_{s,fi,Rd}\} \quad (263)$$

where: $N_{a,fi,Rd}$ is the axial resistance of the steel profile in fire situation, given by:

$$N_{a,fi,Rd} = [b_r t_{fs} + b_{fi} t_{fi} + t_w (h_{ws} + h_{wi} (1 + k_a))] f_{yj} / \gamma_{M,fi} \quad (264)$$

$N_{c,fi,Rd}$ is the axial resistance to compression of the slab in fire situation, given by:

$$N_{c,fi,Rd} = [b_{eff} t_{cr} + b_{fs} t_{c,fi}] f_{ck} / \gamma_{c,fi} \quad (265)$$

$N_{s,fi,Rd}$ is the axial resistance to tension of the reinforcement in the encasement in fire situation, given by:

$$N_{s,fi,Rd} = \sum A_{si} k_s f_{sk} / \gamma_{s,fi} \quad (266)$$

see **F.2** for the other parameters.



The critical length is accounted between the point of zero moment and the maximal positive bending moment.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	159	Rev.: B

F.4 Bending resistance under negative moment

- (1) The negative bending resistant moment at a time step under fire situation is obtained using plastic calculation according to the principles shown in **Figure 71** and as detailed below.

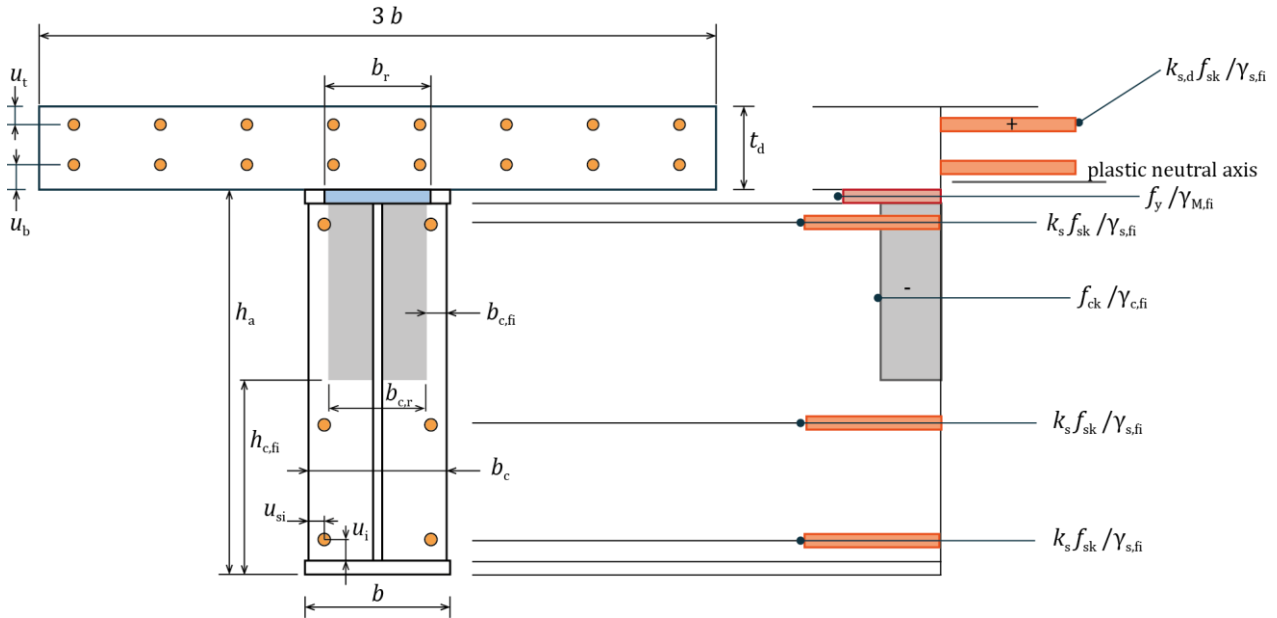


Figure 71 : Stress distribution for the calculation of the negative bending resistance

- (2) The contribution of the lower flange and the web is ignored.
- (3) The effective width of the upper flange is yielded to its design strength, given by Formula (258), where the effective width of the upper flange is given by Formula (259).
- (4) The contribution of the slab reinforcement in tension, located within an effective width limited to $3 \times b$, is taken into account by assuming that the rebars are yielded to their reduced design strength, given by:

$$k_{s,d} f_{sk} / \gamma_{s,fi} \quad (267)$$

where: f_{sk} is the characteristic yield strength of the reinforcement, see § 5.3;

$\gamma_{s,fi}$ is the partial factor, defined in § 4.14.2;

$k_{s,d}$ is a reduction factor, defined by Table 61 (where u is in mm);

u is the distance of the reinforcement layer to the closest outer face of concrete (u_t or u_b on **Figure 71**).

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	160	Rev.: B

Time step	$k_{s,d}$	$k_{s,d,min}$	$k_{s,d,max}$
R30	1	0	1
R60	$0,022 u + 0,34$		
R90	$0,0275 u - 0,1$		
R120	$0,022 u - 0,2$		
R180	$0,018 u - 0,26$		

Table 61: Reduction factor $k_{s,d}$

- (5) The effective part of the encasement concrete in compression is taken into account, assuming that it is stressed in compression to its strength, given by:

$$f_{cek} / \gamma_{c,fi} \quad (268)$$

where: f_{cek} is the characteristic resistance to compression of the encasement concrete, see § 5.2;

$\gamma_{c,fi}$ is the partial factor for concrete in fire situations, defined in § 4.14.2.

- (6) On each side of the concrete encasement, the ineffective width of concrete $b_{c,fi}$ is neglected. $b_{c,fi}$ is given by Table 62.

Time step	$b_{c,fi}$ (mm)	$b_{c,fi,min}$ (mm)
R30	25	25
R60	$60 - 0,15 b_c$	30
R90	$70 - 0,1 b_c$	35
R120	$75 - 0,1 b_c$	45
R180	$85 - 0,1 b_c$	55

Table 62: Ineffective width $b_{c,fi}$

- (7) The lower height of the concrete encasement $h_{c,fi}$ is ineffective and is neglected in the calculations. $h_{c,fi}$ is given by Table 63.

Time step	$h_{c,fi}$ (mm)	$h_{c,fi,min}$ (mm)
R30	25	25
R60	$165 - 0,4 b_c - 8 h_a/b_c$	30
R90	$220 - 0,5 b_c - 8 h_a/b_c$	45
R120	$290 - 0,6 b_c - 10 h_a/b_c$	55
R180	$360 - 0,7 b_c - 10 h_a/b_c$	65

Table 63: Ineffective height $h_{c,fi}$

- (8) The reinforcement of the encasement is taken into account either in tension or in compression, according to § F.2 (7).
- (9) The concrete of the slab is neglected.
- (10) For steel sections with partial concrete encasement, the contribution of the slab reinforcement is neglected.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	161	Rev.: B

F.5 Longitudinal shear in the connection under negative moment

- (1) Under positive moment, the longitudinal shear force in the connection between the slab and the profile along a critical segment of a usual composite beam with partial concrete encasement, for a full connection, is obtained by:

$$N_{cx,fi} = \min\{N_{sc,fi,Rd}; N_{ce,fi,Rd} + N_{se,fi,Rd} + N_{f,fi,Rd}\} \quad (269)$$

where: $N_{f,fi,Rd}$ is the axial resistance of the upper flange in fire situation, given by:

$$N_{f,fi,Rd} = b_r t_{fs} f_{yj} / \gamma_{M,fi} \quad (270)$$

$N_{ce,fi,Rd}$ is the axial resistance to compression of the encasement in fire situation, given by:

$$N_{ce,fi,Rd} = \left[(b_c - 2 b_{ci} - t_w)(h_w - h_{c,fi}) - (4 - \pi) \frac{r^2}{2} \right] f_{ck} / \gamma_{c,fi} \quad (271)$$

$N_{sc,fi,Rd}$ is the axial resistance to tension of the reinforcement in the slab in fire situation, given by:

$$N_{sc,fi,Rd} = \sum A_{si} k_{s,d} f_{sk} / \gamma_{s,fi} \quad (272)$$

$N_{se,fi,Rd}$ is the axial resistance to compression of the reinforcement in the encasement in fire situation, given by (only if the relevant option is activated):

$$N_{se,fi,Rd} = \sum A_{si} k_s f_{sk} / \gamma_{s,fi} \quad (273)$$

see **F.4** for the other parameters.



The critical length is accounted between the point of zero moment and the maximal negative bending moment.

F.6 Resistance to shear force

- (1) The shear resistance of the section is calculated from the following equation:

$$V_{pl,Rd,fi} = \frac{h_w t_w f_y}{\sqrt{3} \gamma_{M,fi}} \frac{h_{ws} + h_{wi} (1 + k_a) / 2}{h_w} \quad (274)$$

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	162	Rev.:	B

Annex G Numerical procedure for the assessment of the temperature in a solid slab in fire situation

G.1 Scope of application

- (1) This Annex describes the numerical model that may be used in the software to assess the heating of a solid slab under the effect of an ISO fire based underneath the slab

G.2 Assumptions

- (1) The slab of thickness t_c will be modelled by a set of n layers, each one of thickness t_{ci} (the thickness of layers may vary along the depth of the slab).

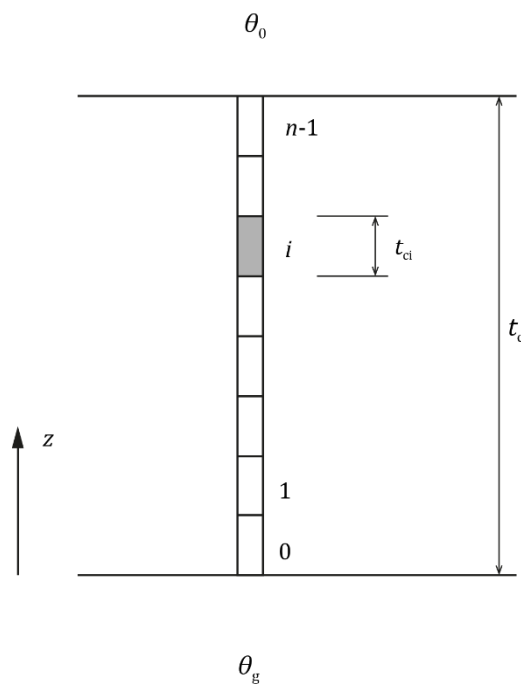


Figure 72 : Modelling of the slab

- (2) At time t , the temperature of gazes below the slab is denoted $\theta_g(t)$. It is obtained by Formula **(192)** in § 10.5.1 **(7)**. The temperature of gazes above the slab is $\theta_0 = 20^\circ\text{C}$.
- (3) The coefficients of heat transfer by convection are as follows:
- α_c on the exposed side of the slab (below);
 - α_{cc} on the non-exposed side of the slab (above);
 - see § 4.18.4 for α_c and α_{cc} .
- (4) The calculation will take into account the following parameters (see § 4.18.4):
- the emissivity of fire ε_f ;
 - the emissivity of concrete ε_c ;
 - the Stefan-Boltzmann constant σ ;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	163	Rev.: B

G.3 Properties of concrete

- (1) The density of concrete will be considered either as constant or as varying with its temperature (according to the option activated). If the density is assumed as constant, the value ρ_c defined by the User will be considered (see § 5.2 (3)). If it is assumed to vary with the temperature, the following equations will be used. With the first generation of EN, according to EN 1994-1-2: 2005, Formula (3.11):

$$\rho_c(\theta_c) = 2354 - 23,47 \frac{\theta_c}{100} \quad (275)$$

where ρ_c is in [kg/m³] and θ_c in [°C].

With the second generation of EN, according to prEN 1994-1-2 : 2024, Formulas (5.21) to (5.24):

$$\begin{aligned} \rho_c(\theta_c) &= \rho_{c0} & \text{for } 20^\circ\text{C} \leq \theta_c \leq 115^\circ\text{C} \\ \rho_c(\theta_c) &= \rho_{c0} \left(1 - 0,02 \frac{\theta_c - 115}{85} \right) & \text{for } 115^\circ\text{C} < \theta_c \leq 200^\circ\text{C} \\ \rho_c(\theta_c) &= \rho_{c0} \left(0,98 - 0,03 \frac{\theta_c - 200}{200} \right) & \text{for } 200^\circ\text{C} < \theta_c \leq 400^\circ\text{C} \\ \rho_c(\theta_c) &= \rho_{c0} \left(0,95 - 0,07 \frac{\theta_c - 400}{800} \right) & \text{for } 400^\circ\text{C} < \theta_c \leq 1200^\circ\text{C} \end{aligned} \quad (276)$$

where $\rho_{c0} = \rho_c$ as defined in 5.2 (3) for the normal situation.

- (2) The specific heat of concrete is a function of the temperature, which is obtained as follows (according to EN 1994-1-2 :2005 3.3.2 (4) (ou prEN 1994-1-2 :2024, 5.2.2.3 (1))):

$$\begin{aligned} c_c(\theta_c) &= 900 & \text{for } 20^\circ\text{C} \leq \theta_c \leq 100^\circ\text{C} \\ c_c(\theta_c) &= 900 + (\theta_c - 100) & \text{for } 100^\circ\text{C} < \theta_c \leq 200^\circ\text{C} \\ c_c(\theta_c) &= 1000 + (\theta_c - 100)/2 & \text{for } 200^\circ\text{C} < \theta_c \leq 400^\circ\text{C} \\ c_c(\theta_c) &= 1100 & \text{for } 400^\circ\text{C} < \theta_c \leq 1200^\circ\text{C} \end{aligned} \quad (277)$$

where ρ_c is in [J/kg.K] and θ_c in [°C].

- (3) In the case of normal weight concrete, the moisture content U will be taken into account as follows. With the first generation of EN, according to EN 1994-1-2 : 2005, 3.3.2 (8):

Maximal value of c_c^* at temperature 115°C, according to the value of U :

$$\begin{aligned} c_c^* &= 900 \text{ J/kg.K for } U = 0\% ; \\ c_c^* &= 2020 \text{ J/kg.K for } U = 3\% ; \\ c_c^* &= 5600 \text{ J/kg.K for } U = 10\% . \end{aligned}$$

For other values of U , the peak value c_c^* will be obtained by linear interpolation.

For temperatures between 100°C and 115 °C: linear interpolation of c_c between 900 J/kg.K and c_c^* ;

For temperatures between 115°C and 200°C: linear interpolation of c_c between c_c^* and $c_c(200^\circ\text{C})$.

With the second generation of EN, according to EN 1994-1-2 : 2024, 5.2.2.3 (3):

Constant value of c_c^* between 100°C and 115°C, according to the value of U :

$$\begin{aligned} c_c^* &= 900 \text{ J/kg.K for } U = 0\% ; \\ c_c^* &= 1470 \text{ J/kg.K for } U = 1,5\% ; \\ c_c^* &= 2020 \text{ J/kg.K for } U = 3\% ; \\ c_c^* &= 5600 \text{ J/kg.K for } U = 10\% . \end{aligned}$$

For other values of U , the peak value c_c^* will be obtained by linear interpolation.

For temperatures between 115°C and 200°C: linear interpolation of c_c between c_c^* and $c_c(200^\circ\text{C})$.

- (4) With the first generation of EN, the thermal conductivity of concrete will be obtained as follows.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	164	Rev.: B

For normal weight concretes, outside the French NA, according to the upper limit in EN 1994-1-2 :2005 3.3.2 (10):

$$\lambda_c(\theta_c) = 2 - 0,2451 \frac{\theta_c}{100} + 0,0107 \left(\frac{\theta_c}{100} \right)^2 \quad (278)$$

where λ_c is in [W/m.K] and θ_c in [°C].

For normal weight concretes, according to the French National Annex of EN 1994-1-2:

$$\begin{aligned} \lambda_c(\theta_c) &= 2 - 0,2451 \frac{\theta_c}{100} + 0,0107 \left(\frac{\theta_c}{100} \right)^2 && \text{for } 20^\circ\text{C} \leq \theta_c \leq 140^\circ\text{C} \\ \lambda_c(\theta_c) &= -0,02604 \theta_c + 5,324 && \text{for } 140^\circ\text{C} < \theta_c \leq 160^\circ\text{C} \\ \lambda_c(\theta_c) &= 1,36 - 0,136 \frac{\theta_c}{100} + 0,0057 \left(\frac{\theta_c}{100} \right)^2 && \text{for } 160^\circ\text{C} < \theta_c \leq 1200^\circ\text{C} \end{aligned} \quad (279)$$

For light weight concretes, according to EN 1994-1-2 :2005 3.3.2 (12), a constant value will be used for all temperatures: $\lambda_c(\theta_c) = 1,6 \text{ W/(m.K)}$.

(5) With the second generation of EN, the thermal conductivity of concrete will be obtained as follows.

For normal weight concretes, according to prEN 1994-1-2 :2024 5.2.2.2 (1), $\lambda_c(\theta_c)$ will be obtained with Formulas **(279)**.

For light weight concretes, according to prEN 1994-1-2 :2024 5.2.2.2 (2):

$$\begin{aligned} \lambda_c(\theta_c) &= 1 - \frac{\theta_c}{1600} && \text{for } 20^\circ\text{C} \leq \theta_c \leq 800^\circ\text{C} \\ \lambda_c(\theta_c) &= 0,5 && \text{for } 800^\circ\text{C} < \theta_c \leq 1200^\circ\text{C} \end{aligned} \quad (280)$$

G.4 Temperature in the slab under the effect of an ISO fire

(1) The heating of the slab is assessed incrementally. The temperature of the cell i at time t is denoted $\theta_i(t)$. The time increment between two steps is dt .

The temperature in the cell i at time $t+dt$ is obtained as follows:

$$\theta_i(t + dt) = \theta_i(t) + d\theta_i$$

where the variation of temperature in the cell, $d\theta_i$, is given by:

$$d\theta_i = \frac{q_{ti} + q_{bi}}{c_{ci} \rho_{ci} dz_i} \quad (281)$$

where: q_{ti} is the energy received by the top of cell (per unit length);

q_{bi} is the energy received by the bottom of the cell (per unit length);

c_{ci} is the specific heat of concrete, at temperature $\theta_i(t)$ – see G.3 (2) and (3);

ρ_{ci} is the density of concrete, at temperature $\theta_i(t)$ – see G.3 (1).

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	165	Rev.:	B

- (2) The energy q_{ti} received by the top of the cell during time increment dt is given by:

$$q_{ti} = \frac{\Delta\theta_{ti} dt}{\frac{\Delta\theta_{ti}}{q_{tbi}} + \frac{t_{ci}}{2\lambda_{ci}}} \quad (282)$$

where: q_{tbi} is the heat flux received through the upper boundary of the cell (per unit length);

λ_{ci} is the thermal conductivity of concrete, at temperature $\theta_i(t)$ – see G.3 (4) and (5);

$\Delta\theta_{ti}$ is the difference of temperature between the cell and the upper part, given by:

if the element is the last one ($i = n-1$): $\Delta\theta_{ti} = \theta_0 - \theta_i(t)$;

otherwise: $\Delta\theta_{ti} = \theta_{i+1}(t) - \theta_i(t)$

$\theta_{i+1}(t)$ is the temperature at time t in the cell $i+1$ above cell i .

- (3) If the cell i is at the top of the slab ($i = n-1$), its upper boundary is in contact with the gazes above the slab. The thermal flux q_{tbi} received through its upper boundary is given by:

$$q_{tbi} = q_{tbri} + q_{tbci} \quad (283)$$

where: q_{tbri} is the radiative heat flux per unit area, given by;

$$q_{tbri} = \varepsilon_c \sigma ((\theta_0 + 273)^4 - (\theta_i(t) + 273)^4) \quad (284)$$

q_{tbci} is the convective heat flux per unit area, given by:

$$q_{tbri} = \alpha_{cc} (\theta_0 - \theta_i(t)) \quad (285)$$

- (4) If the cell i is not at the top of the slab ($i < n-1$), its upper boundary is in contact with cell no $i+1$, and the thermal flux q_{tbi} received through its upper boundary is given by:

$$q_{tbi} = 2\lambda_{c(i+1)} \Delta\theta_{ti} / t_{c(i+1)} \quad (286)$$

- (5) The energy q_{bi} received by the bottom of the cell during time increment dt is given by:

$$q_{bi} = \frac{\Delta\theta_{bi} dt}{\frac{\Delta\theta_{bi}}{q_{bbi}} + \frac{t_{ci}}{2\lambda_{ci}}} \quad (287)$$

where: q_{bbi} is the heat flux received through the lower boundary of the cell (per unit length);

λ_{ci} is the thermal conductivity of concrete, at temperature $\theta_i(t)$ – see G.3 (4) and (5);

$\Delta\theta_{bi}$ is the difference of temperature between the cell and its lower part, given by:

if the cell is the first one ($i = 0$): $\Delta\theta_{bi} = \theta_g(t) - \theta_i(t)$;

otherwise: $\Delta\theta_{bi} = \theta_{i-1}(t) - \theta_i(t)$

$\theta_{i-1}(t)$ is the temperature at time t in the cell $i-1$ below cell i ;

$\theta_g(t)$ is the temperature of the hot gazes at time t , according to formula (192).

- (6) If the cell i is at the bottom of the slab ($i = 0$), its upper boundary is in contact with the hot gazes below the slab. The heat flux q_{bbi} received through its lower boundary is then given by:

$$q_{bbi} = q_{bbri} + q_{bbci} \quad (288)$$

where: q_{bbri} is the radiative heat flux per unit area, given by;

$$q_{bbri} = \varepsilon_f \varepsilon_c \sigma ((\theta_g(t) + 273)^4 - (\theta_i(t) + 273)^4) \quad (289)$$

q_{tbci} is the convective heat flux per unit area, given by:

$$q_{tbri} = \alpha_c (\theta_g(t) - \theta_i(t)) \quad (290)$$

- (7) If the cell i is not at the bottom of the slab ($i > 0$), its upper boundary is in contact with cell no $i-1$, and the heat flux q_{bbi} received through its lower boundary is given by:

$$q_{bbi} = 2\lambda_{c(i-1)} \Delta\theta_{bi} / t_{c(i-1)} \quad (291)$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	166	Rev.: B

Annex H Temperature in a steel profile with perpendicular composite slab, when the voids are unfilled

H.1 Scope of application

- (1) This Annex describes the analytical method that is applied to assess the temperature of a steel profile, with a composite slab whose ribs are perpendicular and when the voids between the deck and the upper flange are not filled.
- (2) The method described in this annex is valid in the following scope:
 - a composite slab is used with continuous perpendicular ribs;
 - the steel profile is protected either by spray or by intumescent paint (see § 4.18.2). Unprotected profiles and profiles protected by boards are outside the scope of application;
 - the voids between the steel deck and the upper flange of the profile are not filled;
 - the method is applicable to both steel beam and composite beams.

H.2 Temperature of gazes in the voids

- (1) The temperature in the voids between the deck and the upper flange is lower than the temperature of gazes outside the voids. The temperature in the voids, θ_{gv} , is obtained as follows:

$$\text{for } t_i \leq 40 \text{ mn:} \quad \theta_{gv} = 20 + 345 c_{red1} \log_{10}(8 t_i + 1) \quad (292)$$

$$\text{for } 40 < t_i \leq 120 \text{ mn:} \quad \theta_{gv} = \theta_{g40} + \frac{(\theta_{g120} - \theta_{g40})}{80} (t_i - 40)$$

where: t_i is the time expressed in minutes;

θ_{g40} is the temperature of gazes in the voids after 40 minutes, given by:

$$\theta_{g40} = 20 + 345 c_{red1} \log_{10}(8 \times 40 + 1) = 20 + 864,7 c_{red1} \quad (293)$$

θ_{g120} is the temperature of gazes in the voids after 120 minutes, given by:

$$\theta_{g120} = 20 + 345 c_{red2} \log_{10}(8 \times 120 + 1) = 20 + 1029 c_{red2} \quad (294)$$

c_{red1} is a coefficient defined by:

$$c_{red1} = 0,71 + 0,125 \ln \phi_{void} \text{ but } c_{red1} \geq 0 \quad (295)$$

c_{red2} is a coefficient defined by:

$$c_{red2} = 0,80 + 0,110 \ln \phi_{void} \text{ but } c_{red2} \leq 0 \quad (296)$$

ϕ_{void} is the void parameter defined as follows:

$$\phi_{void} = 4 \times \arctan\left(\frac{2 h_p}{b_{fs} + 2 d_p}\right) \times \arctan\left(\frac{e_p + b_b - 2 b_t}{b_{fs} + 2 d_p}\right) \quad (297)$$

h_p is the height of the steel deck, see **Figure 21**;

b_t , b_l and b_r are the dimensions of the steel deck as given in **Figure 21**;

b_{fs} is the width of the upper flange;

d_p is the thickness of the fire protection (see § 4.18.2 (5)).

- (2) When $c_{red2} = 1$, the temperature of gazes in the voids are obtained according to Formula (192).



The temperature of gazes outside the voids is given by Formula (192).

	Project: ABCPMX-II Software - DAL-22-003							
	Product reference: DRV/PMX/MT/001							
	Date:	19/05/2025	Authors:	POM	Page:	167	Rev.:	B

H.3 Temperature of the upper flange

- (1) The variation of temperature in the upper flange between two time steps is obtained by the following formula:

$$\Delta\theta_a(t_i) = \frac{(A_m/V)}{c_a \rho_a} \dot{h}_{\text{net,d}} \Delta t \geq \Delta\theta_{a,\text{fp}}(t_i) \quad (298)$$

where: A_m/V is the section factor of the upper flange (m^{-1}), given in **(2)**;

c_a : see § 10.5.1 (5);

ρ_a is the unit mass of steel [in kg/m^3], which is assumed to be independent of the temperature (see § 5.1.4);

Δt is the time increment [in s] – see § 10.3;

$\dot{h}_{\text{net,d}}$ is the increment of total received heat flux (W/m^2), which is given in **(3)**;

$\Delta\theta_{a,\text{fp}}(t_i)$ is the increase of temperature of the protected upper flange, obtained according to **(4)**.

- (2) The section factor of the upper flange is given by:

$$A_m/V = \frac{b_t + b_b - b_r}{(b_t + b_b) t_{fs}} \quad (299)$$

where: t_{fs} is the thickness of the upper flange;

b_t , b_b and b_r are the dimensions of the steel deck as given in **Figure 21**;

- (3) The increment of total received heat flux $\dot{h}_{\text{net,d}}$ is obtained with Formula **(191)**, where:

ε_f is the emissivity of fire; taken as $\varepsilon_f = 0,7$;

ε_m is the emissivity of the steel, see § 10.5.1 (9);

θ_g is the gas temperature in the void ($^{\circ}\text{C}$) – see § H.2;

Φ is the view factor, with $\Phi = 1$;

α_c is the coefficient of heat transfer by convection ($\text{W}/\text{m}^2\text{K}$) – see § 4.18.4.

- (4) The increase of temperature of the protected upper flange, $\Delta\theta_{a,\text{fp}}(t_i)$, is obtained according to § 10.5.2, considering the following assumptions:

- the section factor of the protected section is obtained by:

$$A_{\text{pm}}/V = \frac{b_{fs} + 2 d_p}{b_{fs} t_{fs}} \quad (300)$$

- the temperature of gazes is obtained according to Formula **(192)**.

H.4 Temperature of the lower flange

- (1) The temperature of the lower flange, θ_i , is assessed according to § 10.5.2, with the section factor of the web as given in § 10.5.3 (4).

- (2) When the temperature of the upper flange, θ_s , obtained according to § H.3, is greater than 400°C , the temperature of the lower flange should be modified as follows.

$$\theta_{i,\text{mod}} = [1 + (\theta_s - \theta_0) a_\theta] \theta_i \quad (301)$$

where: θ_0 is the temperature given in **Table 65**;

a_θ is the correction factor given in **Table 64**;

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MT/001						
	Date:	19/05/2025	Authors:	POM	Page:	168	Rev.: B

Height of the steel profile (mm)	Fire resistance	$a_{\theta} \times 10^{-4}$		
		$\theta_i \leq 450\text{ °C}$	$\theta_i = 550\text{ °C}$	$\theta_i \geq 650\text{ °C}$
200	R30	1,4	0,65	0
	R60	3,2	1,8	0,75
	R90	4,6	2,8	1,5
	R120	5,6	3,8	2,0
300	R30	0	0	0
	R60	1,0	0	0
	R90	2,0	1,1	0,65
	R120	2,6	1,6	0,88
400	R30	0	0	0
	R60	0	0	0
	R90	0,6	0	0
	R120	1,2	0,76	0,35
≥ 500	R30-R120	0	0	0

Table 64: Correction factor a_{θ}

	$\theta_i \leq 450\text{ °C}$	$\theta_i = 550\text{ °C}$	$\theta_i \geq 650\text{ °C}$
$\theta_0\text{ (°C)}$	400	450	500

Table 65: Relation between stress in rebars and diameter of rebars

H.5 Temperature of the web

- (1) The temperature of the web is obtained as the maximum value of:
- the temperature calculated according to § 10.5.2, with the section factor of the web as given in § 10.5.3 (6);
 - the average temperature of the upper and lower flanges, as given in § H.3 and H.4.